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Claudio Gorodski

Smooth Manifolds

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TO MY SON, DAVID

Preface

The concept of smooth manifold is ubiquitous in Mathematics. Indeed smooth manifolds appear as Riemannian manifolds in differential geometry, spacetimes in general relativity, phase spaces and energy levels in mechanics, domains of definition of ODEs in dynamical systems, Riemann surfaces in the theory of complex analytic functions, Lie groups in algebra and geometry..., to name a few instances. The notion took some time to evolve until it reached its present form in H. Whitney's celebrated *Annals of Mathematics* paper in 1936 [Whi36]. Whitney's paper in fact represents a culmination of diverse historical developments which took place separately, each in a different domain, all striving to make the passage from the local to the global.

From the modern point of view, the initial goal of introducing smooth manifolds is to generalize the methods and results of differential and integral calculus, in special, the inverse and implicit mapping theorems, the theorem on existence, uniqueness, and regularity of ODEs and Stokes' theorem. As usual in Mathematics, once introduced such objects start to attract interest on their own and new structure is uncovered. The subject of Differential Topology studies smooth manifolds per se. Many important results about the topology of smooth manifolds were obtained in the 1950s and 1960s in the high dimensional range. For instance, there exist topological manifolds admitting several non-diffeomorphic smooth structures (Milnor [Mil56], in the case of S^7), and there exist topological manifolds admitting no smooth structure at all (Kervaire [Ker60]). Moreover, the generalized Poincaré conjecture (GPC: *Is every compact n-manifold with the homotopy type of the n-sphere equivalent to the n-sphere?*) was proved in dimensions bigger than 4, with increasing generality of assumptions about the manifold (smooth, combinatorial, topological), by Smale, Stallings, Zeeman, and Newman in the 1960s. Major developments in dimension 4 took place in the 1980s: Freedman gave a complete classification of simply-connected compact topological 4-manifolds, including a proof of GPC; and Donaldson provided major contributions to the question of determining which topological 4-manifolds admit smooth structures. Despite these breakthroughs, the classification of smooth structures on smoothable topological 4-manifolds remains largely *terra incognita*. In lower dimensions, every topological manifold is smoothable. The very important case of dimension 3 has seen tremendous development after the geometrization program of Thurston (late 1970s), Hamilton's Ricci flow (1981), and Perelman's Ricci flow with surgery (2003), including a proof of GPC by Perelman, and continues to

attract a lot of attention. We mention in closing that the topology of compact surfaces is a classical subject already tackled in the nineteenth century [GX13, App. D].

The aim of these notes is much more modest. They are intended as a beginning graduate-level textbook and, in principle, presume a reasonable understanding of calculus (including the proofs of the inverse and implicit mapping theorems and the formula of change of variables in multiple integrals, roughly corresponding to most of the contents in the first three chapters of [Spi65]), the statement of the existence and uniqueness theorems of ordinary differential equations as in [Hur70, Chap. 2, §5], basic linear algebra and some point set topology; some experience with fundamental groups and covering spaces is also of help. More important than the details about specific prerequisites, that students may or may not have, is the level of maturity that one has acquired by working with key examples in those mathematical theories. The text is also suitable for a supervised reading course. On the other hand, on account of the most prominent concise character of these notes, we hope that their style will stimulate students to develop perseverance in case they are used for independent study.

In writing this text, indeed I have adhered to the following viewpoint. While belaboring every possible detail in a textbook at first seems to make the reading easier, on the other hand, the majority of such details are neither surprising nor illuminating, and the critical points of proofs risk to be lost in a mess of fussy details. Distinguishing the relative significance of the parts and seeing the larger narrative arc could be the most important things to cultivate.

The notes grow out of the one-semester graduate course "Differentiable manifolds and Lie groups" that I taught at the University of São Paulo in 2001, 2008, 2013, 2015, 2018, and 2019. By design, it is thus expected that most of the text can be covered in a typical one-semester graduate course (perhaps omitting some of ▶ Sects. 2.5, 3.6, 3.7, 4.6 and 4.7, if needed). The main material is divided into four chapters. ▶ Chapter 1 introduces the basic language of smooth manifolds, culminating with the Frobenius theorem. ▶ Chapter 2 introduces the basic language of tensors. The most important construction therein is the exterior derivative of differential forms. ▶ Chapter 3 is a first encounter with Lie groups and their Lie algebras, in which homogeneous manifolds are also briefly discussed. Finally, ▶ Chap. 4 is about integration on manifolds and explains Stokes' theorem, de Rham cohomology, and some rudiments of Differential Topology. Two short appendices round up some results about fundamental groups and covering spaces, and a rough sketch of Hodge theory. Routine exercises are scattered throughout the text, which aim to help the reader to digest the material. More elaborate problems can be found in the final section of each chapter. Rarely a problem is needed in the main text; when that happens, we have endeavored to provide effective hints and/or facilitate the solution by making the problem into a guided problem. Needless to say, working arduously in problems is a

necessary (but not sufficient) condition to advance one's comprehension of a mathematical theory.

I am most indebted to the (dozens of) graduate students who took my courses and impelled me to write this set of notes. I must of course acknowledge the generous financial support from the Brazilian state agencies CNPq (grant 302882/2017-0) and FAPESP (grant 16/23746-6). I also wish to thank the anonymous referees for their constructive criticism of earlier versions of the manuscript. I am happy to express my gratitude to the editors of Springer Nature for their guidance during the preparation of the final version of the manuscript and, particularly, Robinson dos Santos for his constant encouragement to move this project forward. Special thanks go to Pedro Zühlke, who worked with me as a TA in 2015, and whose detailed reading of a previous version of the manuscript has substantially helped to improve the text. Any remaining errors are of course my own fault.

São Paulo, Brazil C. G.
April 2020

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Smooth Manifolds



The idea behind the notion of smooth manifold (of dimension n) is, as a first step, to describe points of a set locally by giving n real numbers—the so-called *local coordinates*. The fact that we can use more than one system of local coordinates allows us to consider sets with more complicated topology. As a second step, we want to construct analytical objects on those sets that will be *independent of local coordinates*. A motivation coming from Physics is to write the equations of Mathematical Physics independent from the chosen coordinates. In this chapter, we present the fundamental constructions of the theory of smooth manifolds and prove Frobenius' theorem.

Throughout this book, we will say that a map between open sets of Euclidean spaces is *smooth* to mean that it is infinitely differentiable, that is, its component functions admit all partial derivatives of every order defined everywhere on the domain of the map.

1.1 Submanifolds of Euclidean Spaces

In order to motivate the definition of an abstract smooth manifold, we first define submanifolds of Euclidean spaces. There are two natural, is some sense dual, ways of thinking about submanifolds of Euclidean spaces. Recall from Vector Calculus and Differential Geometry the ideas of equations and parameterizations. For instance, the unit 2-sphere in $\mathbb{R}^3$ can be described by the equation $x^2 + y^2 + z^2 = 1$. This equation imposes a relation on the coordinates x, y, z of $\mathbb{R}^3$ so that, locally, we can think that two coordinates are "free" and the third one is dependent on the first two, and we view the sphere as a "two-dimensional" object. Thinking of a more general smooth equation $F(x, y, z) = 0$, it is important to impose some kind of non-degeneracy condition on F (namely, the gradient of F must be non-zero) so that the solution set will not degenerate into a curve, a point or something even worse than that. On the other hand, the sphere can also be viewed as a parameterized set (surface of revolution) by using spherical coordinates:

$$\begin{cases} x = \cos\theta \sin\varphi \\ y = \sin\theta \sin\varphi \\ z = \cos\varphi. \end{cases}$$

Here the coordinates $\theta \in (0, 2\pi)$, $\varphi \in (0, \pi)$ do not cover the whole sphere and we have an instance of topology imposing need of multiple coordinate systems to cover the

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set. It is also true that some kind of non-degeneracy needs to be imposed on a general smooth parameterization f defined on an open subset of $\mathbb{R}^2$ with values in $\mathbb{R}^3$ so that its image will faithfully represent the idea of a 2-dimensional object that we have (namely, the partial derivatives of f must be linearly independent at every point).

We next formalize and extend these ideas. A smooth map $f : U \to \mathbb{R}^{n+k}$, where $U \subset \mathbb{R}^n$ is open, is called an *immersion at* p, where $p \in U$, if the differential $df_p : \mathbb{R}^n \to \mathbb{R}^{n+k}$ is injective. f is called simply an *immersion* if it is an immersion everywhere. An injective immersion will be called a *parameterization*.

A smooth map $F : W \to \mathbb{R}^k$, where $W \subset \mathbb{R}^{n+k}$ is open, is called a *submersion at* p, where $p \in W$, if $dF_p : \mathbb{R}^{n+k} \to \mathbb{R}^k$ is surjective. F is called simply a *submersion* if it is a submersion everywhere. For $z_0 \in \mathbb{R}^k$, if F is a submersion along the level set $F^{-1}(z_0)$, then z_0 is called a *regular value* of F (in particular, a point $z_0 \in \mathbb{R}^k$ *not* in the image of F is always a regular value!).

Images of parameterizations and inverse images of regular values are thus candidates to be submanifolds of Euclidean spaces. Next we would like to explain why the second class has stronger properties than the first one. The argument involves the implicit mapping theorem and how it is proved to be a consequence of the inverse mapping theorem.

Assume then z_0 is a regular value of F as above and $F^{-1}(z_0)$ is non-empty; write M for this set and consider $p \in M$. Then dF_p is surjective and, up to relabeling the coordinates, we may assume that $(d_2F)_p$, which is the restriction of dF_p to $\{0\} \oplus \mathbb{R}^k \subset \mathbb{R}^n \oplus \mathbb{R}^k = \mathbb{R}^{n+k}$, is an isomorphism onto $\mathbb{R}^k$. Write $p = (x_0, y_0)$ where $x_0 \in \mathbb{R}^n$, $y_0 \in \mathbb{R}^k$. Define a smooth map

$$\Phi : W \to \mathbb{R}^{n+k}, \qquad \Phi(x, y) = (x, F(x, y) - z_0).$$

Then the matrix of $d\Phi_{(x_0, y_0)}$ has block form

$$\begin{pmatrix} \mathrm{id}_{\mathbb{R}^n} & 0 \\ (d_1F)_p & (d_2F)_p \end{pmatrix}$$

so it is easily seen to be an isomorphism. The inverse mapping theorem (cf. [Spi65, Theorem 2–11]) implies that there exist open neighborhoods A, B of x_0, y_0 in $\mathbb{R}^n$, $\mathbb{R}^k$, respectively, such that Φ is a diffeomorphism of $U := A \times B$ onto an open subset of $\mathbb{R}^{n+k}$, i.e., Φ is a smooth bijective map onto its image and the inverse map is also smooth. Now the fundamental fact is that

$$\Phi(M \cap U) = (\mathbb{R}^n \times \{0\}) \cap \Phi(U),$$

as it follows from the form of Φ; namely, Φ "rectifies" M (Fig. 1.1).

1.1.1 Definition (i) A subset $M \subset \mathbb{R}^{n+k}$ will be called an *embedded submanifold of dimension n of* $\mathbb{R}^{n+k}$ if, for every $p \in M$, there exists a diffeomorphism Φ from an open

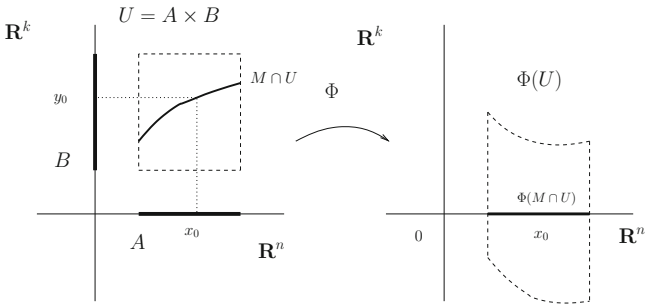


Fig. 1.1 Φ locally "rectifies" M

neighborhood U of p in $\mathbb{R}^{n+k}$ onto its image such that $\Phi(M \cap U) = (\mathbb{R}^n \times \{0\}) \cap \Phi(U)$. In this case we will say that (U, Φ) is a *local chart of* $\mathbb{R}^{n+k}$ *adapted to* M.

(ii) A pair (U, f) will be called a *parameterized submanifold of dimension n of* $\mathbb{R}^{n+k}$ if $U \subset \mathbb{R}^n$ is open and $f : U \to \mathbb{R}^{n+k}$ is an injective immersion.

□

Back to the above discussion, let

$$\varphi : M \cap U \to (\mathbb{R}^n \times \{0\}) \cap \Phi(U) \subset \mathbb{R}^n \times \{0\} = \mathbb{R}^n$$

be the restriction of Φ. Then $U_0 := \varphi(M \cap U)$ is an open subset of $\mathbb{R}^n$ and

$$\varphi^{-1} : U_0 \to \mathbb{R}^{n+k}$$

is the restriction of Φ^{-1} and indeed a parameterization of $M \cap U$. Note also that in this case φ^{-1} is a homeomorphism from U_0 onto $M \cap U$, where the latter is equipped with the topology induced from $\mathbb{R}^{n+k}$ (equivalently, the relative topology).

1.1.2 Exercise Check that $M \cap U$ is the graph of the smooth map $f : A \to B$, satisfying $f(x_0) = y_0$, given by $f = \text{proj}_{\mathbb{R}^k} \circ \varphi^{-1}$. ■

1.1.3 Example Let $(\mathbb{R}, f)$ be a parameterized submanifold of dimension 1 of $\mathbb{R}^2$, where $f : \mathbb{R} \to \mathbb{R}^2$ has image M described in Fig. 1.2. Then M is non-embedded. In fact no (sufficiently small) connected neighborhood of p in the induced topology from $\mathbb{R}^2$ can be homeomorphic to an interval of $\mathbb{R}$ (restrict such a homeomorphism to the complement of $\{p\}$ to get a contradiction). Note that f is not a homeomorphism onto its image. ■

1.1.4 Exercise Prove that the graph of a smooth map $f : U \to \mathbb{R}^k$, where $U \subset \mathbb{R}^n$ is open, is an embedded submanifold of dimension n of $\mathbb{R}^{n+k}$. (Hint: Use an argument inspired by Fig. 1.1.) ■

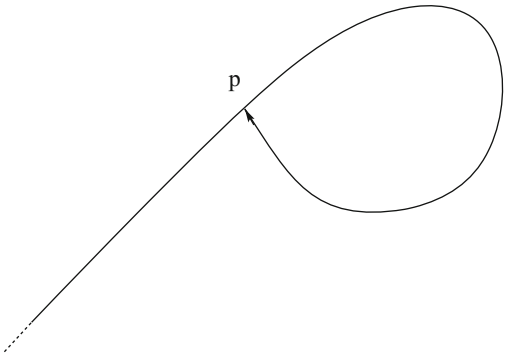


Fig. 1.2 A non-embedded submanifold of $\mathbb{R}^2$

1.1.5 Exercise Let $f, g : (0, 2\pi) \to \mathbb{R}^2$ be defined by

$$f(t) = (\sin t, \sin t \cos t), \qquad g(t) = (\sin t, -\sin t \cos t).$$

a. Check that f, g are injective immersions with the same image.
b. Sketch a drawing of their image.
c. Write a formula for $g^{-1} \circ f : (0, 2\pi) \to (0, 2\pi)$.
d. Deduce that the identity map $\mathrm{id} : \mathrm{im}\, f \to \mathrm{im}\, g$ is not continuous, where $\mathrm{im}\, f$ and $\mathrm{im}\, g$ are equipped with the topology induced from $\mathbb{R}$ via f and g, respectively.

■

The Algebra $C^\infty(M)$ of Real Smooth Functions on M

Let M be an embedded submanifold of $\mathbb{R}^{n+k}$.

1.1.6 Definition A function $f : M \to \mathbb{R}$ is said to be *smooth* at $p \in M$ if $f \circ \Phi^{-1} : \Phi(U) \cap \mathbb{R}^n \to \mathbb{R}$ is a smooth function for some adapted local chart (U, Φ) around p. □

1.1.7 Remark (i) The condition is independent of the choice of adapted local chart around p. Indeed if (V, Φ) is another one,

$$f \circ \Phi^{-1} = (f \circ \Psi^{-1}) \circ (\Psi \circ \Phi^{-1}),$$

where $\Psi \circ \Phi^{-1} : \Phi(U \cap V) \to \Psi(U \cap V)$ is a diffeomorphism and the claim follows from the chain rule for smooth maps between Euclidean spaces.

(ii) A smooth function on M is automatically continuous.

(iii) Let F be a smooth function defined on an open neighborhood of p in $\mathbb{R}^{n+k}$. Then the restriction of F to M is smooth at p.

1.1.8 Exercise Prove that a function $f : M \to \mathbb{R}$ is smooth if and only if for every $p \in M$ there exists an open neighborhood V of p in $\mathbb{R}^{n+k}$ and a smooth function $F : V \to \mathbb{R}$ such that $F|_{M\cap V} = f|_{M\cap V}$. ■

1.2 Definition of Abstract Smooth Manifold

Let M be a topological space. A *local chart* of M is a pair (U, φ), where U is an open subset of M and φ is a homeomorphism from U onto an open subset of $\mathbb{R}^n$. A local chart $\varphi : U \to \mathbb{R}^n$ introduces coordinates $(x_1, \ldots, x_n)$ on U, namely the component functions of φ, and that is why (U, φ) is also called a *system of local coordinates* or *coordinate system* on M.

A *(topological) atlas* for M is a family $\{(U_\alpha, \varphi_\alpha)\}$ of local charts of M, where the dimension n of the Euclidean space is fixed, whose domains cover M, namely $\bigcup U_\alpha = M$. If M admits an atlas, we say that M is *locally modeled on* $\mathbb{R}^n$ and M is a *topological manifold*. In this case, the number n is called the *dimension* of M and we write $\dim M = n$.

A *smooth atlas* is an atlas whose local charts satisfy the additional *compatibility condition*:

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta) \tag{1.2.1}$$

is smooth, for all α, β. A smooth atlas $\mathcal{A}$ defines a notion of smooth function on M as above, namely a function $f : M \to \mathbb{R}$ is *smooth* if $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth for all $(U, \varphi) \in \mathcal{A}$. We say that two atlases $\mathcal{A}, \mathcal{B}$ for M are *equivalent* if the local charts of one are compatible with those of the other, namely $\psi \circ \varphi^{-1}$ and $\varphi \circ \psi^{-1}$ are smooth for all $(U, \varphi) \in \mathcal{A}$, $(V, \psi) \in \mathcal{B}$. In this case, it is obvious that $\mathcal{A}$ and $\mathcal{B}$ define the same notion of smooth function on M.

A *smooth structure* on M is an equivalence class $[\mathcal{A}]$ of smooth atlases on M. Finally, a *smooth manifold* is a topological space M equipped with a smooth structure $[\mathcal{A}]$. In order to be able to do interesting analysis on M, we shall assume, as usual, that *the topology of M is Hausdorff and second-countable*. Occasionally, we will shorten "manifold of dimension n" to *n-manifold*.

1.2.2 Remark (a) It follows from general results in topology that (smooth) manifolds are metrizable. Indeed, manifolds are locally Euclidean and thus locally compact. A locally compact Hausdorff space is (completely) regular, and the Urysohn metrization theorem states that a second-countable regular space is metrizable [Mun00, Theorem 34.1].

(b) The condition of second-countability also rules out pathologies of the following kind. Consider $\mathbb{R}^2$ with the topology with basis of open sets $\{(a, b) \times \{c\} \mid a, b, c \in \mathbb{R}, a < b\}$. This topology is Hausdorff but not second-countable, and it is compatible with a structure of smooth manifold of dimension 1 (a continuum of real lines)! More important, second-countability of the topology of a smooth manifold ensures the existence of smooth partitions of unity (cf. ► Sect. 1.5).

1.2.3 Exercise Let M be a topological space. Prove that two smooth atlases $\mathcal{A}$ and $\mathcal{B}$ are equivalent if and only if their union $\mathcal{A} \cup \mathcal{B}$ is a smooth atlas. Deduce that every equivalence class of smooth atlases for M contains a unique representative which is *maximal* (i.e., not properly contained in any other smooth atlas in the same equivalence class). ■

Let M, N be smooth manifolds whose smooth structures are given by maximal atlases. A map $f : M \to N$ are called *smooth* if, for every $p \in M$, there exist local charts (U, φ), (V, ψ) of M, N around p, $f(p)$, respectively, such that $f(U) \subset V$ and $\psi \circ f \circ \varphi^{-1} : \varphi(U) \to \psi(V)$ is smooth. The map $\psi \circ f \circ \varphi^{-1}$ is sometimes referred to as the *local representation* or *local expression* of f with respect to the local charts (U, φ) and (V, ψ).

1.2.4 Remark (i) The definition is independent of the choice of local charts.
(ii) The definition is local in the sense that $f : M \to N$ is smooth if and only if its restriction to every open subset U of M is smooth (cf. Examples 1.2.7(vi) below).
(iii) A smooth map $M \to N$ is automatically continuous.

We have completed the definition of the category **DIFF**, whose objects are the smooth manifolds and whose morphisms are the smooth maps. An isomorphism in this category is usually called a *diffeomorphism*.

1.2.5 Exercise Let M be a smooth manifold with smooth atlas $\mathcal{A}$. Prove that any local chart $(U, \varphi) \in \mathcal{A}$ is a diffeomorphism onto its image. Conversely, prove any map $\tau : W \to \mathbb{R}^n$, where $n = \dim M$ and $W \subset M$ is open, which is a diffeomorphism onto its image belongs to a smooth atlas equivalent to $\mathcal{A}$; in particular, $(W, \tau) \in \mathcal{A}$ if $\mathcal{A}$ is maximal. ■

1.2.6 Remark In practice, explicitly written down atlases are finite (compare, for instance, Problem 1 and Example 1.2.9). However, in view of the last assertion in Exercise 1.2.5, it is often convenient to implicitly represent a smooth structure by a maximal atlas, and we shall usually do that.

1.2.7 Examples (i) $\mathbb{R}^n$ has a canonical atlas consisting only of one local chart, namely the identity map, which in fact is a *global* chart. This is the standard smooth structure on $\mathbb{R}^n$ with respect to which all definitions coincide with the usual ones. Unless explicit mention, we will always consider $\mathbb{R}^n$ with this smooth structure.
(ii) Any finite-dimensional real vector space V has a canonical structure of smooth manifold. In fact a linear isomorphism $V \cong \mathbb{R}^n$ defines a global chart and thus an atlas, and two such atlases are always equivalent since the transition map between their global charts is a linear isomorphism of $\mathbb{R}^n$ and hence smooth.
(iii) Submanifolds of Euclidean spaces (Definition 1.1.1(i)) are smooth manifolds. Namely, atlases are constructed by using restrictions of adapted charts. Note that the compatibility condition (1.2.1) is automatically satisfied.
(iv) Graphs of smooth maps defined on open subsets of $\mathbb{R}^n$ with values on $\mathbb{R}^k$ are smooth manifolds (cf. Exercise 1.1.4 and (iii)). More generally, a subset M of $\mathbb{R}^{n+k}$ with the

property that every one of its points admits an open neighborhood in M which is a graph as above is a smooth manifold.

(v) It follows from (iv) that the *n-sphere*

$$S^n = \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : x_1^2 + \cdots + x_{n+1}^2 = 1\}$$

is a smooth manifold.

(vi) If $\mathcal{A}$ is an atlas for M and $V \subset M$ is open, then $\mathcal{A}|_V := \{(V \cap U, \varphi|_{V\cap U}) : (U, \varphi) \in \mathcal{A}\}$ is an atlas for V. It follows that any open subset of a smooth manifold is a smooth manifold of the same dimension.

(vii) If M, N are smooth manifolds with atlases $\mathcal{A}$, $\mathcal{B}$, respectively, then $\mathcal{A} \times \mathcal{B}$ is an atlas for the Cartesian product $M \times N$ with the product topology, and hence $M \times N$ is canonically a smooth manifold of dimension $\dim M + \dim N$.

(viii) It follows from (v) and (vii) that the *n-torus*

$$T^n = S^1 \times \cdots \times S^1 \qquad (n \text{ factors})$$

is a smooth manifold.

(ix) The *general linear group* $GL(n, \mathbb{R})$ is the set of all non-singular $n \times n$ real matrices. Since the vector space $M(n, \mathbb{R})$ of $n \times n$ real matrices can be identified with a $\mathbb{R}^{n^2}$ and as such the determinant becomes a continuous function, $GL(n, \mathbb{R})$ can be viewed as the open subset of $\mathbb{R}^{n^2}$ where the determinant does not vanish and hence acquires the structure of a smooth manifold of dimension n^2. ■

The following two examples deserve a separate discussion.

1.2.8 Example The map $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^3$ is a homeomorphism, so it defines a local chart around any point of $\mathbb{R}$ and we can use it to define an atlas $\{f\}$ for $\mathbb{R}$; denote the resulting smooth manifold by $\tilde{\mathbb{R}}$. We claim that $\tilde{\mathbb{R}} \neq \mathbb{R}$ as smooth manifolds, because $C^\infty(\tilde{\mathbb{R}}) \neq C^\infty(\mathbb{R})$. In fact, $\mathrm{id} : \mathbb{R} \to \mathbb{R}$ is obviously smooth, but $\mathrm{id} : \tilde{\mathbb{R}} \to \mathbb{R}$ is not, because $\mathrm{id} \circ f^{-1} : \mathbb{R} \to \mathbb{R}$ maps x to $\sqrt[3]{x}$ so it is not differentiable at 0. On the other hand, $\tilde{\mathbb{R}}$ is diffeomorphic to $\mathbb{R}$. Indeed $f : \tilde{\mathbb{R}} \to \mathbb{R}$ defines a diffeomorphism since its local representation $\mathrm{id} \circ f \circ f^{-1}$ is the identity. ■

1.2.9 Example The *real projective space*, denoted $\mathbb{R}P^n$, as a set consists of all one-dimensional subspaces of $\mathbb{R}^{n+1}$. We introduce a structure of smooth manifold of dimension n on $\mathbb{R}P^n$. Each subspace is spanned by a non-zero vector $v \in \mathbb{R}^{n+1}$. Let U_i be the subset of $\mathbb{R}P^n$ specified by the condition that the i-th coordinate of v is not zero, where $i = 0, \dots, n$. Then $\{U_i\}_{i=0}^n$ covers $\mathbb{R}P^n$. Each line in U_i meets the hyperplane $x_i = 1$ in exactly one point, so there is a bijective map $\varphi_i : U_i \to \mathbb{R}^n \subset \mathbb{R}^{n+1}$, where we identify $\mathbb{R}^n$ with that hyperplane. For $i \neq j$, $\varphi_i(U_i \cap U_j) \subset \mathbb{R}^n \subset \mathbb{R}^{n+1}$ is precisely the open subset of the hyperplane $x_i = 1$ defined by $x_j \neq 0$, and

$$\varphi_j \circ \varphi_i^{-1} : \{x \in \mathbb{R}^{n+1} : x_i = 1,\ x_j \neq 0\} \to \{x \in \mathbb{R}^{n+1} : x_j = 1,\ x_i \neq 0\}$$

is the map

$$v \mapsto \frac{1}{x_j} v,$$

thus smooth. So far there is no topology in $\mathbb{R}P^n$, and we introduce one by declaring

$$\cup_{i=0}^{n} \{\varphi_i^{-1}(W) : W \subset \varphi_i(U_i) = \mathbb{R}^n \text{ is open}\}$$

to be a basis of open sets. It is clear that $\varnothing$ and M are open sets (since each U_i is open) and we have only to check that finite intersections of open sets are open. Let $W_i \subset \varphi_i(U_i)$ and $W_j \subset \varphi_j(U_j)$ be open. Then

$$\varphi_i^{-1}(W_i) \cap \varphi_j^{-1}(W_j) = \varphi_j^{-1}\left(\varphi_j \varphi_i^{-1}(W_i \cap \varphi_i(U_i \cap U_j)) \cap W_j\right).$$

Since $\varphi_j \varphi_i^{-1}$ is a homeomorphism, a necessary and sufficient condition for the left-hand side to describe an open set is that $\varphi_i(U_i \cap U_j)$ be open, and this does occur in this example. Now the topology is well-defined, second-countable, and the φ_i are homeomorphisms onto their images. It is also clear that the sets

$$\{\ell' \in \mathbb{R}P^n : \angle(\ell, \ell') < \epsilon\}$$

for $\ell \in \mathbb{R}P^n$ and $\epsilon > 0$ are open neighborhoods of ℓ. It follows that the topology is Hausdorff. Every $\ell \in \mathbb{R}P^n$ is represented by a $(n+1)$-tuple $(x_0 : \cdots : x_n)$, namely the coordinates of any vector $v \in \mathbb{R}^{n+1}$ spanning ℓ. These are called the *homogeneous coordinates* of ℓ. Note that the homogeneous coordinates of ℓ cannot all vanish and are defined up to multiplication by a non-zero scalar. ■

The argument in Example 1.2.9 is immediately generalized to prove the following proposition.

1.2.10 Proposition *Let M be a set and let n be a non-negative integer. A collection $\{(U_\alpha, \varphi_\alpha)\}$ of injective maps $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ satisfying:*

a. countably many domains U_α cover M;
b. $\varphi_\alpha(U_\alpha)$ is open for all α;
c. $\varphi_\alpha(U_\alpha \cap U_\beta)$ is open for all α, β; and
d. $\varphi_\beta \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is smooth for all α, β;

defines a second-countable topology and smooth structure on M (the Hausdorff condition is not automatic and must be checked in each case).

1.3 Tangent Space

As a motivation, we first discuss the case of an embedded submanifold M of $\mathbb{R}^{n+k}$. Fix $p \in M$ and take an adapted local chart (U, Φ) around p. Recall that we get a parameterization of M around p by setting $\varphi := \mathrm{proj}_{\mathbb{R}^n} \circ \Phi|_{M\cap U}$ and $U_0 = \varphi(M \cap U) \subset \mathbb{R}^n$ and taking

$$\varphi^{-1} : U_0 \to \mathbb{R}^{n+k}.$$

It is then natural to define the *tangent space* of M at p to be the image of the differential of the parameterization, namely

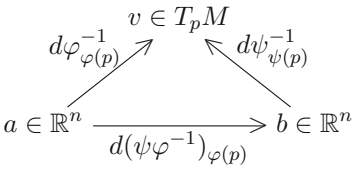
$$T_pM := d(\varphi^{-1})_{\varphi(p)}(\mathbb{R}^n).$$

If (V, Ψ) is another adapted local chart around p, $\psi := \mathrm{proj}_{\mathbb{R}^n} \circ \Psi|_{M\cap V}$, $V_0 = \psi(M \cap V) \subset \mathbb{R}^n$ and $\psi^{-1} : V_0 \to \mathbb{R}^{n+k}$ is the associated parameterization, then

$$\begin{aligned} d(\varphi^{-1})_{\varphi(p)}(\mathbb{R}^n) &= d(\psi^{-1})_{\psi(p)} d(\psi\varphi^{-1})_{\varphi(p)}(\mathbb{R}^n) \\ &= d(\psi^{-1})_{\varphi(p)}(\mathbb{R}^n) \end{aligned}$$

since $d(\psi\varphi^{-1})_{\varphi(p)} : \mathbb{R}^n \to \mathbb{R}^n$ is an isomorphism. It follows that T_pM is well-defined as a subspace (of dimension n) of $\mathbb{R}^{n+k}$.

Note that we have the following situation:



Namely, the tangent vector $v \in T_pM$ is represented by two different vectors $a, b \in \mathbb{R}^n$ which are related by the differential of the transition map. We can use this idea to generalize the construction of the tangent space to the case of an abstract smooth manifold.

Let M be a smooth manifold of dimension n and fix $p \in M$. Suppose that $\mathcal{A}$ is the maximal atlas defining the smooth structure of M. The *tangent space* of M at p is the set T_pM of all pairs (a, φ)—where $a \in \mathbb{R}^n$ and $(U, \varphi) \in \mathcal{A}$ is a local chart around p—quotiented by the equivalence relation

$$(a, \varphi) \sim (b, \psi) \quad \text{if and only if} \quad d(\psi \circ \varphi^{-1})_{\varphi(p)}(a) = b.$$

It follows from the chain rule in $\mathbb{R}^n$ that this is indeed an equivalence relation, and we denote the equivalence class of (a, φ) by $[a, \varphi]$. Each such equivalence class is called a *tangent vector* at p. For a fixed local chart (U, φ) around p, the map

$$a \in \mathbb{R}^n \mapsto [a, \varphi] \in T_pM$$

is a bijection, which we use to transfer the vector space structure of $\mathbb{R}^n$ to T_pM. This structure is well-defined, that is, independent of the choice of local chart around p, for if (V, ψ) is another one, then the linearity of $d(\psi \circ \varphi^{-1})_{\varphi(p)}$ implies that we get the same vector space operations on T_pM. Note that $\dim T_pM = \dim M$.

1.3.1 Exercise Let M be a smooth manifold and let $V \subset M$ be an open subset. Prove that there is a canonical isomorphism $T_pV \cong T_pM$ for all $p \in V$. ■

Let $(U, \varphi = (x_1, \ldots, x_n))$ be a local chart of M and denote by $\{e_1, \ldots, e_n\}$ the canonical basis of $\mathbb{R}^n$. The *coordinate vectors* at p with respect to this chart are defined to be

$$\frac{\partial}{\partial x_i}\Big|_p := [e_i, \varphi].$$

Note that

$$\left\{ \frac{\partial}{\partial x_1}\Big|_p, \ldots, \frac{\partial}{\partial x_n}\Big|_p \right\} \tag{1.3.2}$$

is a basis of T_pM.

In the case of $\mathbb{R}^n$, for each $p \in \mathbb{R}^n$ there is a canonical isomorphism $\mathbb{R}^n \to T_p\mathbb{R}^n$ given by

$$a \mapsto [a, \mathrm{id}], \tag{1.3.3}$$

where id is the identity map of $\mathbb{R}^n$. Usually we will make this identification without further comment. In particular, $T_p\mathbb{R}^n$ and $T_q\mathbb{R}^n$ are canonically isomorphic for every $p, q \in \mathbb{R}^n$. In the case of a general smooth manifold M, obviously there are no such canonical isomorphisms. Occasionally we shall denote by $(r_1, \ldots, r_n)$ the coordinates on $\mathbb{R}^n$ corresponding to id.

Tangent Vectors as Directional Derivatives

Let M be a smooth manifold and fix a point $p \in M$. For each tangent vector $v \in T_pM$ of the form $v = [a, \varphi]$, where $a \in \mathbb{R}^n$ and (U, φ) is a local chart of M, and for each $f \in C^\infty(U)$, we define the *directional derivative of f in the direction of v* to be the real number

$$\begin{aligned} v(f) &= \frac{d}{dt}\Big|_{t=0} (f \circ \varphi^{-1})(\varphi(p) + ta) \\ &= d(f \circ \varphi^{-1})_{\varphi(p)}(a). \end{aligned}$$

It is a simple consequence of the chain rule that this definition does not depend on the choice of representative of v.

In the case of $\mathbb{R}^n$, $\frac{\partial}{\partial r_i}\big|_p f$ is simply the partial derivative in the direction e_i, the ith vector in the canonical basis of $\mathbb{R}^n$. In general, if $\varphi = (x_1, \ldots, x_n)$, then $x_i \circ \varphi^{-1} = r_i$, so

$$v(x_i) = d(r_i)_{\varphi(p)}(a) = a_i,$$

where $a = \sum_{i=1}^n a_i e_i$. Since $v = [a, \varphi] = \sum_{i=1}^n a_i [e_i, \varphi]$, it follows that

$$v = \sum_{i=1}^n v(x_i) \frac{\partial}{\partial x_i}\Big|_p. \tag{1.3.4}$$

If v is a coordinate vector $\frac{\partial}{\partial x_i}|_p$ and $f \in C^\infty(U)$, we also write

$$\frac{\partial}{\partial x_i}\Big|_p f = \frac{\partial f}{\partial x_i}\Big|_p;$$

note that $\frac{\partial f}{\partial x_i} \in C^\infty(U)$. As a particular case of (1.3.4), take now v to be a coordinate vector of another local chart $(V, \psi = (y_1, \ldots, y_n))$ around p. Then

$$\frac{\partial}{\partial y_j}\Big|_p = \sum_{i=1}^n \frac{\partial x_i}{\partial y_j}\Big|_p \frac{\partial}{\partial x_i}\Big|_p.$$

Note that the preceding formula shows that, even if $x_1 = y_1$, we do not need to have $\frac{\partial}{\partial x_1} = \frac{\partial}{\partial y_1}$.

The Differential

Let $f : M \to N$ be a smooth map between smooth manifolds. Fix a point $p \in M$ and local charts (U, φ) of M around p, (V, ψ) of N around $q = f(p)$. The *differential* or *tangent map* of f at p is the linear map

$$df_p : T_pM \to T_qN$$

given by

$$[a, \varphi] \mapsto [d(\psi \circ f \circ \varphi^{-1})_{\varphi(p)}(a), \psi].$$

It is easy to check that this definition does not depend on the choices of local charts. Using the identification (1.3.3), one checks that $d\varphi_p : T_pM \to \mathbb{R}^m$ and $d\psi_q : T_qN \to \mathbb{R}^n$ are linear isomorphisms, where $m = \dim M$ and $n = \dim N$, and

$$df_p = (d\psi_q)^{-1} \circ d(\psi \circ f \circ \varphi^{-1})_{\varphi(p)} \circ d\varphi_p.$$

1.3.5 Proposition (Chain Rule) *Let M, N, P be smooth manifolds. If $f : M \to N$ and $g : N \to P$ are smooth maps, then $g \circ f : M \to P$ is a smooth map and*

$$d(g \circ f)_p = dg_{f(p)} \circ df_p$$

for all $p \in M$.

1.3.6 Exercise Prove Proposition 1.3.5. ■

If $f \in C^\infty(M, N)$, $g \in C^\infty(N)$, and $v \in T_pM$, then it is a simple matter of unraveling the definitions to check that

$$df_p(v)(g) = v(g \circ f).$$

Now (1.3.4) together with this equation gives that

$$\begin{aligned} df_p\left(\frac{\partial}{\partial x_j}\Big|_p\right) &= \sum_{i=1}^n df_p\left(\frac{\partial}{\partial x_j}\Big|_p\right)(y_i)\frac{\partial}{\partial y_i}\Big|_{f(p)} \\ &= \sum_{i=1}^n \frac{\partial(y_i \circ f)}{\partial x_j}\Big|_p \frac{\partial}{\partial y_i}\Big|_{f(p)}. \end{aligned}$$

The matrix

$$\left(\frac{\partial(y_i \circ f)}{\partial x_j}\Big|_p\right)$$

is called the *Jacobian matrix of f at p* relative to the given coordinate systems. Observe that the chain rule (Proposition 1.3.5) is equivalent to saying that the Jacobian matrix of $g \circ f$ at a point is the product of the Jacobian matrices of g and f at the appropriate points.

Consider now the case in which $N = \mathbb{R}$ and $f \in C^\infty(M)$. Then $df_p : T_pM \to T_{f(p)}\mathbb{R}$, and upon the identification between $T_{f(p)}\mathbb{R}$ and $\mathbb{R}$, we easily see that $df_p(v) = v(f)$. Applying this to $f = x_i$, where $(U, \varphi = (x_1, \ldots, x_n))$ is a local chart around p, and using again (1.3.4) shows that

$$\{dx_1|_p, \ldots, dx_n|_p\} \tag{1.3.7}$$

is the basis of T_pM^* dual to the basis (1.3.2), and hence

$$df_p = \sum_{i=1}^n df_p\left(\frac{\partial}{\partial x_i}\Big|_p\right) dx_i|_p = \sum_{i=1}^n \frac{\partial f}{\partial x_i}\Big|_p dx_i|_p.$$

Finally, we discuss smooth curves on M. A *smooth curve* in M is simply a smooth map $\gamma : (a, b) \to M$ where (a, b) is an interval of $\mathbb{R}$. One can also consider smooth curves γ in M defined on a closed interval $[a, b]$. This means that γ admits a smooth extension to an open interval $(a - \epsilon, b + \epsilon)$ for some $\epsilon > 0$.

If $\gamma : (a, b) \to M$ is a smooth curve, the *tangent vector* to γ at $t \in (a, b)$ is

$$\dot{\gamma}(t) = d\gamma_t \left(\frac{\partial}{\partial r}\Big|_t \right) \in T_{\gamma(t)} M,$$

where r is the canonical coordinate of $\mathbb{R}$. Note that an arbitrary vector $v \in T_p M$ can be considered to be the tangent vector at 0 to the curve $\gamma(t) = \varphi^{-1}(ta)$, where (U, φ) is a local chart around p with $\varphi(p) = 0$ and $d\varphi_p(v) = a$.

In the case in which $M = \mathbb{R}^n$, upon identifying $T_{\gamma(t)}\mathbb{R}^n$ and $\mathbb{R}^n$, it is easily seen that

$$\dot{\gamma}(t) = \lim_{h \to 0} \frac{\gamma(t + h) - \gamma(t)}{h}.$$

The Inverse Mapping Theorem

It is now straightforward to state and prove the inverse mapping theorem for smooth manifolds.

1.3.8 Theorem (Inverse Mapping Theorem) *Let $f : M \to N$ be a smooth map between two smooth manifolds M, N, and let $p \in M$ and $q = f(p)$. If $df_p : T_p M \to T_q N$ is an isomorphism, then there exists an open neighborhood W of p such that $f(W)$ is an open neighborhood of q and f restricts to a diffeomorphism from W onto $f(W)$.*

Proof The proof is really a transposition of the inverse mapping theorem for smooth mappings between Euclidean spaces to manifolds using local charts. Note that M and N have the same dimension, say, n. Take local charts (U, φ) of M around p and (V, ψ) of N around q such that $f(U) \subset V$. Set $\alpha = \psi \circ f \circ \varphi^{-1}$. Then $d\alpha_{\varphi(p)} : \mathbb{R}^n \to \mathbb{R}^n$ is an isomorphism. By the inverse mapping theorem for smooth mappings of $\mathbb{R}^n$ [Spi65, Theorem 2–11], there exists an open subset $\tilde{W} \subset \varphi(U)$ with $\varphi(p) \in \tilde{W}$ such that $\alpha(\tilde{W})$ is an open neighborhood of $\psi(q)$ and α restricts to a diffeomorphism from $\tilde{W}$ onto $\alpha(\tilde{W})$. It follows that $f = \psi^{-1} \circ \alpha \circ \varphi$ is a diffeomorphism from the open neighborhood $W = \varphi^{-1}(\tilde{W})$ of p onto the open neighborhood $\psi^{-1}(\alpha(\tilde{W}))$ of q. □

A smooth map $f : M \to N$ satisfying the conclusion of Theorem 1.3.8 at a point $p \in M$ is called a *local diffeomorphism* at p. It follows from the above and the chain rule that f is a local diffeomorphism at p if and only if $df_p : T_p M \to T_q N$ is an isomorphism. In this case, there exist local charts (U, φ) of M around p and (V, ψ) of N around $f(p)$ such that the local representation $\psi \circ f \circ \varphi^{-1}$ of f is the identity, owing to Exercise 1.2.5.

1.3.9 Exercise Let $f : M \to N$ be a smooth bijective map that is a local diffeomorphism everywhere. Show that f is a diffeomorphism. ■

1.4 Submanifolds of Smooth Manifolds

Similar to the situation of submanifolds of Euclidean spaces, some manifolds are contained in other manifolds in a natural way (compare Definition 1.1.1). Let N be a smooth manifold of dimension $n + k$. A subset M of N is called an *embedded submanifold* of N of dimension n if, for every $p \in M$, there exists a local chart (V, ψ) of N such that $p \in V$ and $\psi(V \cap M) = \psi(V) \cap \mathbb{R}^n$, where we identify $\mathbb{R}^n$ with $\mathbb{R}^n \times \{0\} \subset \mathbb{R}^n \times \mathbb{R}^k = \mathbb{R}^{n+k}$. We say that (V, ψ) is a local chart of N *adapted* to M. An embedded submanifold M of N is a smooth manifold in its own right, with respect to the relative topology, in a canonical way. In fact an atlas of M is furnished by the restrictions to M of those local charts of N that are adapted to M. Namely, if $\{(V_\alpha, \psi_\alpha)\}$ is an atlas of N consisting of adapted charts, then $\{(V_\alpha \cap M, \psi_\alpha|_{V_\alpha \cap M})\}$ becomes an atlas of M. Note that the compatibility condition for the local charts of M follows automatically from the compatibility condition for N.

1.4.1 Exercise Let N be a smooth manifold and let M be an embedded submanifold of N. Prove that T_pM is canonically isomorphic to a subspace of T_pN for every $p \in M$. ■

Immersions and Embeddings

Another class of submanifolds can be introduced as follows. Let $f : M \to N$ be a smooth map between smooth manifolds. The map f is called an *immersion* at $p \in M$ if $df_p : T_pM \to T_{f(p)}N$ is injective. If f is an immersion everywhere, it is simply called an *immersion*. Now call the pair (M, f) an *immersed submanifold* or simply a *submanifold* of N if $f : M \to N$ is an injective immersion.

Let M be an embedded submanifold of N and consider the inclusion $\iota : M \to N$. The existence of adapted local charts implies that ι can be locally represented around any point of M by the standard inclusion $x \mapsto (x, 0)$, $\mathbb{R}^n \to \mathbb{R}^{n+k}$. Since this map is an immersion, also ι is an immersion. It follows that (M, ι) is an immersed submanifold of N. This shows that every embedded submanifold of a smooth manifold is an immersed submanifold, but the converse is not true.

1.4.2 Example Let N be the 2-torus $T^2 = S^1 \times S^1$ viewed as an embedded submanifold of $\mathbb{R}^2 \times \mathbb{R}^2 = \mathbb{R}^4$ and consider the smooth map

$$F : \mathbb{R} \to \mathbb{R}^4, \qquad F(t) = (\cos at, \sin at, \cos bt, \sin bt),$$

where $a, b \in \mathbb{R}$ and $a \neq 0$. Note that the image of F lies in T^2. Denote by (r_1, r_2, r_3, r_4) the coordinates on $\mathbb{R}^4$. For each choice of $i \in \{1, 2\}$ and $j \in \{3, 4\}$, (r_i, r_j) gives a system of coordinates defined on an open subset of T^2, and in this way we obtain an atlas for T^2. We deduce that the induced map $f : \mathbb{R} \to T^2$ is smooth, because $r_i \circ f$ is simply the ith

component of F. Since N is an embedded submanifold of $\mathbb{R}^4$, we can consider $T_{f(t)}N$ to be a subspace of $\mathbb{R}^4$, and the tangent vector $f'(t) \in T_{f(t)}N$ is the usual derivative $F'(t)$. Since $F'(t)$ never vanishes, f is an immersion. Note that if b/a is an irrational number, then f is an injective map, so $(\mathbb{R}, f)$ is an immersed submanifold which we claim is *not* an embedded submanifold of T^2. In fact, the assumption on b/a implies that M is a dense subset of T^2, but an embedded submanifold of another manifold is always locally closed; here a subset A of a topological space X is called *locally closed* if every point of A admits a neighborhood V in X such that $A \cap V$ is closed in V. (If b/a is rational, one can use Proposition 1.4.6 to show that $f(\mathbb{R})$ is an embedded circle in T^2.) ■

We would like to further investigate the gap between immersed submanifolds and embedded submanifolds.

1.4.3 Lemma (Local Form of an Immersion) *Let M and N be smooth manifolds of dimensions n and $n+k$, respectively, and suppose that $f : M \to N$ is an immersion at $p \in M$. Then there exist local charts of M and N such that the local expression of f at p is the standard inclusion of $\mathbb{R}^n$ into $\mathbb{R}^{n+k}$.*

Proof Let (U, φ) and (V, ψ) be local charts of M and N around p and $q = f(p)$, respectively, such that $f(U) \subset V$, and set $\alpha = \psi \circ f \circ \varphi^{-1}$. Then $d\alpha_{\varphi(p)} : \mathbb{R}^n \to \mathbb{R}^{n+k}$ is injective, so, up to rearranging indices, we can assume that $d(\pi_1 \circ \alpha)_{\varphi(p)} = \pi_1 \circ d\alpha_{\varphi(p)} : \mathbb{R}^n \to \mathbb{R}^n$ is an isomorphism, where $\pi_1 : \mathbb{R}^{n+k} = \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ is the projection onto the first factor. By the inverse mapping theorem, by shrinking U, we can assume that $\pi_1 \circ \alpha$ is a diffeomorphism from $U_0 = \varphi(U)$ onto its image V_0; let $\beta : V_0 \to U_0$ be its smooth inverse. Now we can describe $\alpha(U_0)$ as being the graph of the smooth map $\gamma = \pi_2 \circ \alpha \circ \beta : V_0 \subset \mathbb{R}^n \to \mathbb{R}^k$, where $\pi_2 : \mathbb{R}^{n+k} = \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^k$ is the projection onto the second factor. By Exercise 1.1.4, $\alpha(U_0)$ is a submanifold of $\mathbb{R}^{n+k}$, and the map $\tau : V_0 \times \mathbb{R}^k \to V_0 \times \mathbb{R}^k$ given by $\tau(x, y) = (x, y - \gamma(x))$ is a diffeomorphism such that $\tau(\alpha(U_0)) = V_0 \times \{0\}$. Finally, we put $\tilde{\varphi} = \pi_1 \circ \alpha \circ \varphi$ and $\tilde{\psi} = \tau \circ \psi$, shrinking V if necessary. Then $(U, \tilde{\varphi})$ and $(V, \tilde{\psi})$ are local charts, and for $x \in \tilde{\varphi}(U) = V_0$ we have that

$$\begin{aligned}\tilde{\psi} \circ f \circ \tilde{\varphi}^{-1}(x) &= \tau \circ \psi \circ f \circ \varphi^{-1} \circ \beta(x) = \tau \circ \alpha \circ \beta(x) \\ &= \tau(x, \gamma(x)) = (x, 0).\end{aligned}$$

□

1.4.4 Proposition *If $f : M \to N$ is an immersion at $p \in M$, then there exists an open neighborhood U of p in M such that $f|_U$ is injective and $f(U)$ is an embedded submanifold of N of the same dimension as M.*

Proof The local injectivity of f at p is an immediate consequence of the fact that some local expression of f at p is the standard inclusion of $\mathbb{R}^n$ into $\mathbb{R}^{n+k}$, hence, injective. Moreover, in the course of proof of Lemma 1.4.3, we have produced a local chart $(V, \tilde{\psi})$ of N adapted to $f(U)$. □

A smooth map $f : M \to N$ is called an *embedding* if it is an immersion and a homeomorphism from M onto $f(M)$ with the induced topology.

1.4.5 Proposition *Let N be a smooth manifold. A subset $P \subset N$ is an embedded submanifold of N if and only if it is the image of an embedding.*

Proof Let $f : M \to N$ be an embedding with $P = f(M)$. To prove that P is an embedded submanifold of N, it suffices to check that it can be covered by open sets in the relative topology each of which is an embedded submanifold of N. Owing to Proposition 1.4.4, any point of P lies in a set of the form $f(U)$, where U is an open subset of M and $f(U)$ is an embedded submanifold of N. Since f is an open map into P with the relative topology, $f(U)$ is open in the relative topology and we are done. Conversely, if P is an embedded submanifold of N, it has the relative topology and thus the inclusion $\iota : P \to N$ is a homeomorphism onto its image. Moreover, we have seen above that ι is an immersion, whence it is an embedding. □

Recall that a continuous map between locally compact, Hausdorff topological spaces is called *proper* if the inverse image of a compact subset of the target space is a compact subset of the domain. It is known that proper maps are closed. Also, it is clear that if the domain is compact, then every continuous map is automatically proper. An embedded submanifold M of a smooth manifold N is called *properly embedded* if the inclusion map is proper.

1.4.6 Proposition *If $f : M \to N$ is an injective immersion which is also a proper map, then the image $f(M)$ is a properly embedded submanifold of N.*

Proof Let $P = f(M)$ have the relative topology. A proper map is closed. Since f viewed as a map $M \to P$ is bijective and closed, it is an open map and thus a homeomorphism. Due to Proposition 1.4.5, P is an embedded submanifold of N. The properness of the inclusion $P \to N$ clearly follows from that of f. □

1.4.7 Exercise Give an example of an embedded submanifold of a smooth manifold which is not properly embedded. ■

1.4.8 Exercise Decide whether a closed embedded submanifold of a smooth manifold is necessarily properly embedded. ■

Exercise 1.1.5 dealt with a situation in which a smooth map $f : M \to N$ factors through an immersed submanifold (P, g) of N (namely, $f(M) \subset g(P)$) and the induced map $f_0 : M \to P$ (namely, $g \circ f_0 = f$) is discontinuous.

1.4.9 Proposition *Suppose that $f : M \to N$ is smooth and (P, g) is an immersed submanifold of N such that $f(M) \subset g(P)$. Consider the induced map $f_0 : M \to P$ that satisfies $g \circ f_0 = f$.*

a. If g is an embedding, then f_0 is continuous.

b. If f_0 is continuous, then it is smooth.

Proof (a) In this case g is a homeomorphism onto $g(P)$ with the relative topology. If $V \subset P$ is open, then $g(V) = W \cap g(P)$ for some open subset $W \subset N$. By continuity of f, we have that $f_0^{-1}(V) = f_0^{-1}(g^{-1}(W)) = f^{-1}(W)$ is open in M, hence also f_0 is continuous.

(b) Let $p \in M$ and $q = f_0(p) \in P$. By Proposition 1.4.4, there exists a neighborhood U of q and a local chart (V, ψ) of N^n, with $g(U) \subset V$, adapted to $g(U)$. In particular, there exists a projection π from $\mathbb{R}^n$ onto a subspace obtained by setting some coordinates equal to 0 such that $\tau = \pi \circ \psi \circ g$ is a local chart of P around q. Note that $f_0^{-1}(U)$ is a neighborhood of p in M. Now

$$\tau \circ f_0|_{f_0^{-1}(U)} = \pi \circ \psi \circ g \circ f_0|_{f_0^{-1}(U)} = \pi \circ \psi \circ f|_{f_0^{-1}(U)},$$

and the latter is smooth. □

An immersed submanifold (P, g) of N with the property that $f_0 : M \to P$ is smooth for every smooth map $f : M \to N$ with $f(M) \subset g(P)$ will be called an *initial submanifold*.

1.4.10 Exercise Use Exercise 1.3.9 and Propositions 1.4.5 and 1.4.9 to deduce that an embedding $f : M \to N$ induces a diffeomorphism from M onto an embedded submanifold P of N. In particular, viewing $T_q P$ as subspace of $T_q N$ for $q \in P$ (cf. Exercise 1.4.1), we have $T_{f(p)} P = df_p(T_p M)$ for all $p \in M$. ■

1.4.11 Exercise For an immersed submanifold (M, f) of N, show that there is a natural structure of smooth manifold on $f(M)$ and that $(f(M), \iota)$ is an immersed submanifold of N, where $\iota : f(M) \to N$ denotes the inclusion. ■

Submersions

Let $f : M \to N$ be a smooth map between smooth manifolds. The map f is called a *submersion* at $p \in M$ if $df_p : T_p M \to T_{f(p)} N$ is surjective. If f is a submersion everywhere, it is simply called a *submersion*. A point $q \in N$ is called a *regular value* of f if f is a submersion at all points in $f^{-1}(q)$; otherwise q is called a *singular value* or a *critical value* of f. A point $p \in M$ is called a *critical point* of f if f is not a submersion at p.

1.4.12 Lemma (Local Form of a Submersion) *Let M and N be smooth manifolds of dimensions $n + k$ and k, respectively, and suppose that $f : M \to N$ is a submersion at $p \in M$. Then there exist local charts of M and N such that the local expression of f at p is the standard projection of $\mathbb{R}^{n+k}$ onto $\mathbb{R}^k$.*

Proof Let (U, φ) and (V, ψ) be local charts of M and N around p and $q = f(p)$, respectively, and set $\alpha = \psi \circ f \circ \varphi^{-1}$. Then $d\alpha_{\varphi(p)} : \mathbb{R}^{n+k} \to \mathbb{R}^k$ is surjective, so, up to rearranging indices, we can assume that $d(\alpha \circ \iota_2)_{\varphi(p)} = d\alpha_{\varphi(p)} \circ \iota_2 : \mathbb{R}^k \to \mathbb{R}^k$ is an isomorphism, where $\iota_2 : \mathbb{R}^k \to \mathbb{R}^{n+k} = \mathbb{R}^n \times \mathbb{R}^k$ is the standard inclusion. Define

$\tilde{\alpha} : \varphi(U) \subset \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n \times \mathbb{R}^k$ by $\tilde{\alpha}(x, y) = (x, \alpha(x, y))$. Since $d\alpha_{\varphi(p)} \circ \iota_2$ is an isomorphism, it is clear that $d\tilde{\alpha}_{\varphi(p)} : \mathbb{R}^n \oplus \mathbb{R}^k \to \mathbb{R}^n \oplus \mathbb{R}^k$ is an isomorphism. By the inverse mapping theorem, there exists an open neighborhood U_0 of $\varphi(p)$ contained in $\varphi(U)$ such that $\tilde{\alpha}$ is a diffeomorphism from U_0 onto its image V_0; let $\tilde{\beta} : V_0 \to U_0$ be its smooth inverse. We put $\tilde{\varphi} = \tilde{\alpha} \circ \varphi$. Then $(\varphi^{-1}(U_0), \tilde{\varphi})$ is a local chart of M around p and

$$\begin{aligned}\psi \circ f \circ \tilde{\varphi}^{-1}(x, y) &= \psi \circ f \circ \varphi^{-1} \circ \tilde{\beta}(x, y) = \alpha \circ \tilde{\beta}(x, y) \\ &= \pi_2 \circ \tilde{\alpha} \circ \tilde{\beta}(x, y) = y.\end{aligned}$$

□

1.4.13 Proposition *Let $f : M \to N$ be a smooth map, and let $q \in N$ be a regular value of f such that $f^{-1}(q) \neq \varnothing$. Then $P = f^{-1}(q)$ is an embedded submanifold of M of dimension* $\dim M - \dim N$. *Moreover, for $p \in P$ we have $T_p P = \ker df_p$.*

Proof To show the first assertion, it is enough to construct local charts of M that are adapted to P and whose domains cover P. So suppose $\dim M = n + k$, $\dim N = k$, let $p \in P$ and consider local charts $(W := \varphi^{-1}(U_0), \tilde{\varphi})$ and (V, ψ) as in Theorem 1.4.12 such that $p \in U$ and $q \in V$. We can assume that $\psi(q) = 0$. Now

$$\pi_2 \circ \tilde{\varphi}(W \cap P) = \alpha \circ \varphi(W \cap P) = \psi \circ f(W \cap P) = \{0\},$$

so $\tilde{\varphi}(W \cap P) = \tilde{\varphi}(W) \cap \mathbb{R}^n$ and thus φ is an adapted chart around p. Finally, the local representation of f at p is the projection $\mathbb{R}^{n+k} \to \mathbb{R}^k$. This is a linear map with kernel $\mathbb{R}^n$. It follows that $\ker df_p = (d\tilde{\varphi}^{-1})_{\varphi(p)}(\mathbb{R}^n) = T_p P$. □

1.4.14 Examples (i) Let A be a non-singular real symmetric matrix of order $n + 1$ and define $f : \mathbb{R}^{n+1} \to \mathbb{R}$ by $f(p) = \langle Ap, p\rangle$ where $\langle , \rangle$ is the standard scalar product. Then $df_p : \mathbb{R}^{n+1} \to \mathbb{R}$ is given by $df_p(v) = 2\langle Ap, v\rangle$, so it is surjective if $p \neq 0$. It follows that f is a submersion on $\mathbb{R}^{n+1} \setminus \{0\}$, and then $f^{-1}(r)$ for $r \neq 0$ is an embedded submanifold of $\mathbb{R}^{n+1}$ of dimension n if it is non-empty. In particular, by taking A to be the identity matrix, we get a manifold structure for S^n which coincides with the one previously constructed.

(ii) Denote by $Sym(n, \mathbb{R})$ the vector space of real symmetric matrices of order n, and define $f : M(n, \mathbb{R}) \to Sym(n, \mathbb{R})$ by $f(A) = AA^t$. This is a map between vector spaces whose local representation has quadratic polynomials as components. It follows that f is smooth and df_A can be viewed as a map $M(n, \mathbb{R}) \to Sym(n, \mathbb{R})$ for all $A \in M(n, \mathbb{R})$. We claim that I is a regular value of f. For the purpose of checking that, we first compute for $A \in f^{-1}(I)$ and $B \in M(n, \mathbb{R})$ that

$$\begin{aligned}df_A(B) &= \lim_{h\to 0} \frac{(A + hB)(A + hB)^t - I}{h} \\ &= \lim_{h\to 0} \frac{h(AB^t + BA^t) + h^2 BB^t}{h}\end{aligned}$$

$$= AB^t + BA^t.$$

Now, given $C \in Sym(n, \mathbb{R})$, we have $df_A(\frac{1}{2}CA) = C$, and this proves that f is a submersion at A, as desired. Hence $f^{-1}(I) = \{ A \in M(n, \mathbb{R}) \mid AA^t = I \}$ is an embedded submanifold of $M(n, \mathbb{R})$ of dimension

$$\dim M(n, \mathbb{R}) - \dim Sym(n, \mathbb{R}) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

Note that $f^{-1}(I)$ is a group with respect to the multiplication of matrices; it is called the *orthogonal group* of order n and is usually denoted by $O(n)$. It is obvious that $O(n) \subset GL(n, \mathbb{R})$.

■

1.4.15 Remark *Sard's theorem* is the statement that the set of critical values of a smooth map has measure zero (cf. Problem 22 for the notion of measure zero and [GP10, App. 1] for a proof of the theorem). In particular the set of regular values is dense. It follows from Sard's theorem that there always exists a regular value of a given smooth map.

We close this section by mentioning a generalization of Proposition 1.4.13. Let $f : M \to N$ be a smooth map and let Q be an embedded submanifold of N. We say that f is *transverse* to Q, in symbols $f \pitchfork Q$, if

$$df_p(T_pM) + T_{f(p)}Q = T_{f(p)}N$$

for every $p \in f^{-1}(Q)$. Transversality is a formalization of the idea of "generic intersection."

1.4.16 Exercise Let $f : M \to N$ be a smooth map and let $q \in N$. Prove that $f \pitchfork \{q\}$ if and only if q is a regular value of f. ■

For an immersed submanifold (M, f) of a smooth manifold N, its *codimension* is the number $\dim N - \dim M$.

1.4.17 Proposition *If $f : M \to N$ is a smooth map which is transverse to an embedded submanifold Q of N of codimension k and $P = f^{-1}(Q)$ is non-empty, then P is an embedded submanifold of M of codimension k. Moreover $T_pP = (df_p)^{-1}(T_{f(p)}Q)$ for every $p \in P$.*

Proof For the first assertion, it suffices to check that P is an embedded submanifold of M in a neighborhood of a point $p \in P$. Let (V, ψ) be a local chart of N adapted to Q around $q := f(p)$. Then $\psi : V \to \mathbb{R}^{n+k}$ and $\psi(V \cap Q) = \psi(V) \cap \mathbb{R}^n$, where $n = \dim Q$. Let $\pi_2 : \mathbb{R}^{n+k} = \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^k$ be the standard projection and put $g = \pi_2 \circ \psi$. Then $g : V \to \mathbb{R}^k$ is a submersion and $g^{-1}(0) = V \cap Q$. Moreover

$$\begin{aligned} d(g \circ f)_p(T_pM) &= dg_q \circ df_p(T_pM) \\ &= dg_q(T_qN) \\ &= \mathbb{R}^k, \end{aligned}$$

where, in view of $\ker dg_q = T_qQ$, the second equality follows from the assumption $f \pitchfork Q$. Now $h := g \circ f : f^{-1}(V) \to \mathbb{R}^k$ is a submersion at p and $h^{-1}(0) = f^{-1}(V \cap Q) = f^{-1}(V) \cap P$, and $f^{-1}(V)$ is an open neighborhood of p in M, so we can apply Proposition 1.4.13. All assertions follow. □

As a most important special case, two embedded submanifolds M, P of N are called *transverse*, denoted $M \pitchfork P$, if the inclusion map $\iota : M \to N$ is transverse to P. It is easy to see that this is a symmetric relation.

1.4.18 Corollary *If M and P are transverse embedded submanifolds of N, then $M \cap P$ is an embedded submanifold of N and*

$$\operatorname{codim}(M \cap P) = \operatorname{codim}(M) + \operatorname{codim}(P).$$

A very good source to read more about transversality is [GP10].

1.5 Partitions of Unity

Many important constructions for smooth manifolds rely on the existence of smooth partitions of unity. This technique allows for a much greater flexibility of smooth manifolds as compared, for instance, with real analytic or complex manifolds.

Bump Functions

We start with the remark that the function

$$f(t) = \begin{cases} e^{-1/t}, & \text{if } t > 0 \\ 0, & \text{if } t \leq 0 \end{cases}$$

is smooth everywhere. Therefore the function

$$g(t) = \frac{f(t)}{f(t) + f(1-t)}$$

is smooth, flat and equal to 0 on $(-\infty, 0]$, and flat and equal to 1 on $[1, +\infty)$. Finally,

$$h(t) = g(t+2)g(2-t)$$

is smooth, flat and equal to 1 on $[-1, 1]$, and vanishes outside $(-2, 2)$; h is called a *bump function*. We can also make an n-dimensional version of a bump function by setting

$$k(x_1, \ldots, x_n) = h(\sqrt{x_1^2 + \cdots + x_n^2}),$$

and we can rescale k by precomposing with $x \mapsto r^{-1}x$ to have a smooth function on $\mathbb{R}^n$ which is flat and equal to 1 on a closed ball of radius r and with support contained in an open ball of radius $2r$.

Bump functions are very useful. As one application, note that, for a given smooth manifold M, so far we do not know whether the algebra $C^\infty(M)$ of smooth functions on M contains functions other than the constants (of course, the components of local charts are smooth, but these are not *globally* defined on M). We claim that $C^\infty(M)$ is indeed, in general, huge. In fact, let (U, φ) be a local chart of M and take a bump function $k : \mathbb{R}^n \to \mathbb{R}$ whose support lies in $\varphi(U)$. Then

$$f(x) := \begin{cases} k \circ \varphi(x) & \text{if } x \in U, \\ 0 & \text{if } x \in M \setminus U \end{cases}$$

is a smooth function on M: this is clear for a point $p \in U$; if $p \notin U$, then we can find a neighborhood V of p which does not meet the compact set $\varphi^{-1}(\text{supp}(k))$, so $f|_V = 0$ and thus f is smooth at p.

Partitions of Unity

Let M be a smooth manifold. A *partition of unity* on M is a collection $\{\rho_i\}_{i\in I}$ of smooth functions on M, where I is an index set, such that:

(i) $\rho_i(p) \geq 0$ for all $p \in M$ and all $i \in I$;
(ii) the collection of supports $\{\text{supp}(\rho)\}_{i\in I}$ is locally finite (i.e., every point of M admits a neighborhood meeting $\text{supp}(\rho_i)$ for only finitely many indices i);
(iii) $\sum_{i\in I} \rho_i(p) = 1$ for all $p \in M$ (the sum is finite in view of (ii)).

Let $\{U_\alpha\}_{\alpha\in A}$ be a cover of M by open sets. We say that a partition of unity $\{\rho_i\}_{i\in I}$ is *subordinate* to $\{U_\alpha\}_{\alpha\in A}$ if, for every $i \in I$, there is $\alpha \in A$ such that $\text{supp}(\rho_i) \subset U_\alpha$, and we say $\{\rho_i\}_{i\in I}$ is *strictly subordinate* to $\{U_\alpha\}_{\alpha\in A}$ if $I = A$ and $\text{supp}(\rho_\alpha) \subset U_\alpha$ for every $\alpha \in A$.

Partitions of unity are used to piece together global objects out of local ones and, conversely, to decompose global objects as locally finite sums of locally defined ones. For instance, suppose $\{U_\alpha\}_{\alpha\in A}$ is an open cover of M and $\{\rho_\alpha\}_{\alpha\in A}$ is a partition of unity strictly subordinate to $\{U_\alpha\}$. If we are given $f_\alpha \in C^\infty(U_\alpha)$ for all $\alpha \in A$, then $f = \sum_{\alpha\in A} \rho_\alpha f_\alpha$ is a smooth function on M. Indeed for $p \in M$ and $\alpha \in A$, it is true that either $p \in U_\alpha$ and then f_α is defined at p, or $p \notin U_\alpha$ and then $\rho_\alpha(p) = 0$. Moreover, since the sum is locally finite, f is locally the sum of finitely many smooth functions and hence smooth. Conversely, if we start with $f \in C^\infty(M)$, then $f = \sum_{\alpha\in A} f_\alpha$ for smooth functions f_α with $\text{supp}(f_\alpha) \subset U_\alpha$, namely $f_\alpha := \rho_\alpha f$.

If M is compact, it is a lot easier to prove the existence of a partition of unity strictly subordinate to any given open cover $\{U_\alpha\}$ of M. In fact, for each $x \in M$, we choose α such that $x \in U_\alpha$ and construct as above a bump function λ_x which is flat and equal to 1 on a neighborhood V_x of x and whose (compact) support lies in U_α. Now $\{V_x\}$ is an open cover of M and, owing to compactness, we can extract a finite subcover $\{V_{x_i}\}_{i=1}^a$ with associated non-negative smooth functions $\lambda_i := \lambda_{x_i}$ such that λ_i is 1 on V_{x_i}. In particular, their sum is positive, so

$$\rho_i := \frac{\lambda_i}{\sum_{i=1}^a \lambda_i}$$

for $i = 1, \ldots, a$ yields the desired partition of unity.

1.5.1 Theorem (Easy Whitney Embedding Theorem) *Let M be a compact smooth manifold. Then there exists an embedding of M into $\mathbb{R}^m$ for sufficiently large m.*

Proof Let $n = \dim M$. Since M is compact, there exists a finite atlas $\{(U_i, \varphi_i)\}_{i=1}^a$ such that $\varphi_i(U_i)$ is the open ball of radius 2 centered at the origin of $\mathbb{R}^n$ and M is covered by $V_1, \ldots, V_a$, where V_i is the preimage of the open ball of radius 1 under φ_i. Now we can find for each i a bump function $\rho_i \in C^\infty(M)$ such that $0 \le \rho_i \le 1$, $\rho_i|_{\bar{V}_i} = 1$ and $\operatorname{supp}(\rho_i) \subset U_i$. Put

$$f_i(x) = \begin{cases} \rho_i(x)\varphi_i(x), & \text{if } x \in U_i, \\ 0, & \text{if } x \in M \setminus U_i. \end{cases}$$

Then $f_i : M \to \mathbb{R}^n$ is smooth. Define also smooth functions

$$g_i = (f_i, \rho_i) : M \to \mathbb{R}^{n+1} \quad \text{and} \quad g = (g_1, \ldots, g_a) : M \to \mathbb{R}^{a(n+1)}.$$

It is enough to check that g is an injective immersion. In fact, on the open set V_i, we have that $g_i = (\varphi_i, 1)$ is an immersion, so g is an immersion. Further, if $g(x) = g(y)$ for $x, y \in M$, then $\rho_i(x) = \rho_i(y)$ and $f_i(x) = f_i(y)$ for all i. Take an index j such that $\rho_j(x) = \rho_j(y) \neq 0$. Then $x, y \in U_j$ and $\varphi_j(x) = \varphi_j(y)$. Due to the injectivity of φ_j, we must have $x = y$. Hence g is injective. □

1.5.2 Remark In the noncompact case, one can still construct partitions of unity and modify the proof of Theorem 1.5.1 to prove that M properly embeds into $\mathbb{R}^m$ for some m. Then a standard trick involving Sard's theorem (cf. Remark 1.4.15) and projections into lower dimensional subspaces of $\mathbb{R}^m$ allows one to find the bound $m \le 2n + 1$, where $n = \dim M$ [GP10, Chap. 1, §8]. A more difficult result, the *strong Whitney embedding theorem*, asserts that in fact $m \le 2n$ [Whi44].

In general, a reasonable substitute for compactness is paracompactness. A topological space is called *paracompact* if every open cover admits an open locally finite

refinement. It turns out that every locally compact, second-countable, Hausdorff space is paracompact. Hence manifolds are paracompact. Now the above argument can be extended to give the following theorem, for whose proof we refer the reader to [War83, Theorem 1.11].

1.5.3 Theorem (Existence of Partitions of Unity) *Let M be a smooth manifold and let $\{U_\alpha\}_{\alpha\in A}$ be an open cover of M. Then there exists a countable partition of unity $\{\rho_i : i = 1, 2, 3, \ldots\}$ subordinate to $\{U_\alpha\}$ with* $\operatorname{supp}(\rho_i)$ *compact for each i. If one does not require compact supports, then there is a partition of unity $\{\lambda_\alpha\}_{\alpha\in A}$ strictly subordinate to $\{U_\alpha\}$ with at most countably many of the ρ_α not zero.*

1.5.4 Exercise Let C be closed in M and let U be open in M with $C \subset U$. Prove that there exists a smooth function $\lambda \in C^\infty(M)$ such that $0 \le \lambda \le 1$, $\lambda|_C = 1$ and $\operatorname{supp}(\lambda) \subset U$. (Hint: Apply Theorem 1.5.3 to a suitable cover of M.) ■

1.6 Vector Fields

Let M be a smooth manifold of dimension n. A *vector field* on M is an assignment of a tangent vector $X(p)$ in T_pM for all $p \in M$. Sometimes, we also write X_p instead of $X(p)$. So a vector field is a map $X : M \to TM$, where $TM = \dot\cup_{p\in M} T_pM$ (disjoint union), and

$$\pi \circ X = \mathrm{id}, \tag{1.6.1}$$

where $\pi : TM \to M$ is the natural projection $\pi(v) = p$ for $v \in T_pM$. In account of property (1.6.1), we say that X is a *section* of TM.

We shall need to talk about continuity and differentiability of vector fields, so we next explain that TM carries a canonical manifold structure induced from that of M.

The Tangent Bundle

Let M be a smooth manifold and consider the disjoint union

$$TM = \dot\bigcup_{p\in M} T_pM.$$

We can view the elements of TM as equivalence classes of triples (p, a, φ), where $p \in M$, $a \in \mathbb{R}^n$ and (U, φ) is a local chart of M such that $p \in U$, and

$$(p, a, \varphi) \sim (q, b, \psi) \quad \text{if and only if } p = q \text{ and } d(\psi \circ \varphi^{-1})_{\varphi(p)}(a) = b.$$

There is a natural projection $\pi : TM \to M$ given by $\pi[p, a, \varphi] = p$, and then $\pi^{-1}(p) = T_pM$.

Note that we have n degrees of freedom for a point p in M and n degrees of freedom for a vector $v \in T_pM$, so we expect TM to be $2n$-dimensional. We next use Proposition 1.2.10 to simultaneously introduce a topology and a smooth structure on TM; the resulting manifold is called the *tangent bundle* of M. Let $\{(U_\alpha, \varphi_\alpha)\}$ be the maximal atlas defining the smooth structure of M. For each α, $\varphi_\alpha : U_\alpha \to \varphi_\alpha(U_\alpha)$ is a diffeomorphism and, for each $p \in U_\alpha$, $d(\varphi_\alpha)_p : T_pU_\alpha = T_pM \to \mathbb{R}^n$ is the isomorphism mapping $[p, a, \varphi]$ to a. Set

$$\tilde{\varphi}_\alpha : \pi^{-1}(U_\alpha) \to \varphi_\alpha(U_\alpha) \times \mathbb{R}^n, \qquad [p, a, \varphi] \to (\varphi_\alpha(p), a).$$

Then $\tilde{\varphi}_\alpha$ is a bijection and $\tilde{\varphi}_\alpha(\pi^{-1}(U_\alpha))$ is an open subset of $\mathbb{R}^{2n}$. Moreover, the maps

$$\tilde{\varphi}_\beta \circ \tilde{\varphi}_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \times \mathbb{R}^n \to \varphi_\beta(U_\alpha \cap U_\beta) \times \mathbb{R}^n$$

are defined on open subsets of $\mathbb{R}^{2n}$ and are given by

$$(x, a) \mapsto (\varphi_\beta \circ \varphi_\alpha^{-1}(x)\,,\ d(\varphi_\beta \circ \varphi_\alpha^{-1})_x(a)).$$

Since $\varphi_\beta \circ \varphi_\alpha^{-1}$ is a smooth diffeomorphism, we have that $d(\varphi_\beta \circ \varphi_\alpha^{-1})_x$ is a linear isomorphism and $d(\varphi_\beta \circ \varphi_\alpha^{-1})_x(a)$ is also smooth on x. Countably many elements of $\{\pi^{-1}(U_\alpha)\}$ cover TM because countably many elements of $\{U_\alpha\}$ cover M (recall that every second-countable space is Lindelöf). It follows that $\{(\pi^{-1}(U_\alpha), \tilde{\varphi}_\alpha)\}$ defines a second-countable topology and a smooth structure for M. It remains only to check the Hausdorff condition. Namely, let $v, w \in TM$ with $v \neq w$. If $v, w \in TM$ and $\pi(v) \neq \pi(w)$, we can use the Hausdorff property of M to separate v and w from each other with open sets of TM. On the other hand, if $v, w \in T_pM$, they lie in the domain of the same local chart of TM and the result also follows.

The local representation of $\pi : TM \to M$ with respect to the charts $(\pi^{-1}U_\alpha, \tilde{\varphi}_\alpha)$ and $(U_\alpha, \varphi_\alpha)$ is the projection onto the first factor $\varphi_\alpha(U_\alpha) \times \mathbb{R}^n \to \varphi_\alpha(U_\alpha)$. It follows that π is a smooth submersion (and an open map).

If $f \in C^\infty(M, N)$, then we define the *differential of* f to be the map

$$df : TM \to TN$$

that restricts to $df_p : T_pM \to T_{f(p)}N$ for each $p \in M$. Using the above atlases for TM and TN, we immediately see that $df \in C^\infty(TM, TN)$.

1.6.2 Exercise Check the last sentence. ■

1.6.3 Remark The mapping that associates to each manifold M its tangent bundle TM and associates to each smooth map $f : M \to N$ its tangent map $df : TM \to TN$ can be thought of a functor **DIFF** $\to$ **VB** from the category of smooth manifolds to the category of smooth vector bundles. In fact, $d(\mathrm{id}_M) = \mathrm{id}_{TM}$, and $d(g \circ f) = dg \circ df$ for a sequence of smooth maps $M \xrightarrow{f} N \xrightarrow{g} P$.

Smooth Vector Fields

A vector field X on M is called *smooth* (resp. *continuous*) if the map $X : M \to TM$ is smooth (resp. continuous), where TM carries the structure of smooth manifold defined above.

More generally, let $f : M \to N$ be a smooth mapping. Then a (smooth, continuous) *vector field along* f is a (smooth, continuous) map $X : M \to TN$ such that $X(p) \in T_{f(p)}N$ for all $p \in M$, or $\pi \circ X = f$. The most important case is that in which f is a smooth curve $\gamma : [a, b] \to N$. A vector field along γ is a map $X : [a, b] \to TN$ such that $X(t) \in T_{\gamma(t)}N$ for all $t \in [a, b]$. A typical example is the tangent vector field $\dot{\gamma}$.

For practical purposes, we reformulate the notion of smoothness as follows. Let X be a vector field on M. Given a smooth function $f \in C^\infty(U)$ where U is an open subset of M, the directional derivative $X(f) : U \to \mathbb{R}$ is defined to be the function $p \in U \mapsto X_p(f)$. Further, if $(x_1, \ldots, x_n)$ is a coordinate system on U, we have already seen that $\{\frac{\partial}{\partial x_1}|_p, \ldots, \frac{\partial}{\partial x_n}|_p\}$ is a basis of T_pM for $p \in U$. It follows that there are functions $a_i : U \to \mathbb{R}$ such that

$$X|_U = \sum_{i=1}^{n} a_i \frac{\partial}{\partial x_i}. \tag{1.6.4}$$

Note that we have also shown above that $(x_1, \ldots, x_n)$ induces a system of local coordinates $(x_1 \circ \pi, \ldots, x_n \circ \pi, dx_1, \ldots, dx_n)$ on $\pi^{-1}U =: TM|_U$.

1.6.5 Proposition *Let X be a vector field on M. Then the following assertions are equivalent:*

a. X is smooth.

b. For every coordinate system $(U, x_1, \ldots, x_n)$ of M, the functions a_i defined by (1.6.4) are smooth.

c. For every open set V of M and $f \in C^\infty(V)$, the function $X(f) \in C^\infty(V)$.

Proof Suppose X is smooth and let $(x_1, \ldots, x_n)$ be a coordinate system on U. Then $X|_U$ is smooth and $a_i = dx_i \circ X|_U$ is also smooth.

Next, assume (b) and let $f \in C^\infty(V)$. Take a coordinate system

$$(U, x_1, \ldots, x_n)$$

with $U \subset V$. Then, by using (b) and the fact that $\frac{\partial f}{\partial x_i}$ is smooth,

$$X(f)|_U = \sum_{i=1}^{n} a_i \frac{\partial f}{\partial x_i} \in C^\infty(U).$$

Since V can be covered by such U, this proves (c).

Finally, assume (c). For every coordinate system $(U, x_1, \ldots, x_n)$ of M, we have a corresponding coordinate system

$$(\pi^{-1}(U), x_1 \circ \pi, \ldots, x_n \circ \pi, dx_1, \ldots, dx_n)$$

of TM. Then

$$(x_i \circ \pi) \circ X|_U = x_i \quad \text{and} \quad dx_i \circ X|_U = X(x_i)$$

are smooth. This proves that X is smooth. □

In particular, the proposition shows that the coordinate vector fields $\frac{\partial}{\partial x_i}$ associated with a local chart are smooth. Since $a_i = X(x_i)$ in (1.6.4), we have

1.6.6 Scholium *If X is a smooth vector field on M and $X(f) = 0$ for every smooth function, then $X = 0$.*

1.6.7 Remark Part (c) of Proposition 1.6.5 in fact says that every smooth vector field on M defines a *derivation* of the algebra $C^\infty(M)$, namely an $\mathbb{R}$-linear operator on $C^\infty(M)$ that satisfies the Leibniz identity $X(fg) = X(f)g + fX(g)$. Moreover, because of the local expression (1.6.4), one says that X defines a *differential operator* on $C^\infty(M)$.

Flow of a Vector Field

We have now come to the integration of vector fields. Let $\varphi_t : M \to M$ be a diffeomorphism for $t \in (-\epsilon, \epsilon)$ such that $\varphi_0 = \mathrm{id}_M$ and the curve $t \mapsto \varphi_t(p)$ is smooth for each p. Then $X_p := \frac{d}{dt}\big|_{t=0}\varphi_t(p)$ defines a vector field on M. Going the other way, one can integrate smooth vector fields to obtain families of local diffeomorphisms. Actually, this is the extension of ODE theory to smooth manifolds that we discuss below.

An *integral curve* of a vector field X on M is a smooth curve $\gamma : I \to M$, where I is an open interval, such that

$$\dot{\gamma}(t) = X(\gamma(t))$$

for all $t \in I$. We write this equation in local coordinates. Suppose $X|_U$ has the form (1.6.4), where $(U, x_1, \ldots, x_n)$ is a coordinate system on M. Write $\gamma_i = x_i \circ \gamma$ and $\tilde{a}_i = a_i \circ \varphi^{-1}$. Then γ is an integral curve of X in $\gamma^{-1}(U)$ if and only if

$$\frac{d\gamma_i}{dr}\bigg|_t = \tilde{a}_i(\gamma_1(t), \ldots, \gamma_n(t)) \tag{1.6.8}$$

for $i = 1, \ldots, n$ and $t \in \gamma^{-1}(U)$. Equation (1.6.8) is a system of first order ordinary differential equations for which existence and uniqueness theorems are known. These, translated into manifold terminology, yield local existence and uniqueness of integral curves for smooth vector fields, which we state in the following proposition.

1.6.9 Proposition *Let X be a smooth vector field on M. For each $p \in M$, there exists a (possibly infinite) interval $(a(p), b(p)) \subset \mathbb{R}$ and a smooth curve $\gamma_p : (a(p), b(p)) \to M$ such that:*

a. $0 \in (a(p), b(p))$ *and* $\gamma_p(0) = p$.

b. γ_p *is an integral curve of* X.

c. γ_p *is maximal in the sense that, if* $\mu : (c, d) \to M$ *is a smooth curve satisfying (a) and (b), then* $(c, d) \subset (a(p), b(p))$ *and* $\mu = \gamma_p|_{(c,d)}$.

Proof Consider the set $\mathcal{I}$ of integral curves of X passing through p at instant 0. By the local existence theorem [Hur70, Theorem 4, p. 28] applied to Eq. (1.6.8), this set is non-empty. Note that, given elements $\gamma : I \to M$, $\eta : J \to M$ of $\mathcal{I}$, the subset of points of $I \cap J$ where they agree is open by the uniqueness theorem [Hur70, Theorem 3, p. 28] and closed by continuity, hence it coincides with $I \cap J$. It follows that we can define an integral curve of X on the union of domains of all elements of $\mathcal{I}$, proving assertions (a), (b), and (c). □

Even more interesting is to reverse the rôles of p and t as follows. Put

$$\mathcal{D}_t = \{p \in M \mid t \in (a(p), b(p))\}$$

and define $\varphi_t : \mathcal{D}_t \to M$ by setting

$$\varphi_t(p) := \gamma_p(t).$$

The collection $\{\varphi_t\}$ is called the *flow* of X. Next we refer to the smooth dependence of solutions of ODE on the initial conditions to study flows from the global point of view.

1.6.10 Proposition *a. For each $p \in M$, there exists an open neighborhood V of p and $\epsilon > 0$ such that the map*

$$\Phi : (-\epsilon, \epsilon) \times V \to M, \quad (t, q) \mapsto \varphi_t(q)$$

is well-defined and smooth.

b. The domain $\mathrm{dom}(\varphi_s \circ \varphi_t) \subset \mathcal{D}_{s+t}$ *and* $\varphi_{s+t}|_{\mathrm{dom}(\varphi_s \circ \varphi_t)} = \varphi_s \circ \varphi_t$. *Further,* $\mathrm{dom}(\varphi_s \circ \varphi_t) = \mathcal{D}_{s+t}$ *if* $st > 0$.

c. $\mathcal{D}_t$ *is open for all* t, $\cup_{t>0} \mathcal{D}_t = M$ *and* $\varphi_t : \mathcal{D}_t \to \mathcal{D}_{-t}$ *is a diffeomorphism with inverse* φ_{-t}.

Proof The existence assertion in part (a) follows from [Hur70, Theorem 7, p. 29], and the smoothness assertion follows from [Hur70, Theorem 9, p. 29] on the smooth dependence of solutions of ODE on their initial conditions. We prove part (b). First, we remark the obvious fact that, if $p \in \mathcal{D}_t$, then $s \mapsto \gamma_p(s + t)$ is an integral curve of X with initial condition $\gamma_p(t)$ and maximal domain $(a(p) - t, b(p) - t)$; therefore $(a(p) - t, b(p) - t) = (a(\gamma_p(t)), b(\gamma_p(t)))$. Next, let $p \in \mathrm{dom}(\varphi_s \circ \varphi_t)$. This means that $p \in \mathrm{dom}(\varphi_t) = \mathcal{D}_t$ and $\gamma_p(t) = \varphi_t(p) \in \mathrm{dom}(\varphi_s) = \mathcal{D}_s$. Then $s \in (a(\gamma_p(t)), b(\gamma_p(t)))$, so $s + t \in (a(\gamma_p(t))+t, b(\gamma_p(t))+t) = (a(p), b(p))$, that is, $p \in \mathcal{D}_{s+t}$. Further, $\varphi_{s+t}(p) = \gamma_p(s+t) =$

$\gamma_{\gamma_p(t)}(s) = \varphi_s(\varphi_t(p))$, and we have already proved the first two assertions. Next, assume that $s, t > 0$ (the case $s, t < 0$ is similar); we need to show that $\mathcal{D}_{s+t} \subset \mathrm{dom}(\varphi_s \circ \varphi_t)$. But this follows from reversing the argument above, as $p \in \mathcal{D}_{s+t}$ implies that $s + t \in (a(p), b(p))$, and this implies that $t \in (a(p), b(p))$ and $s = (s + t) - t \in (a(p) - t, b(p) - t) = (a(\gamma_p(t)), b(\gamma_p(t)))$. Finally, we prove part (c). The statement about the union follows from part (a). Note that $\mathcal{D}_0 = M$ and φ_0 is the identity. Fix $t > 0$ and $p \in \mathcal{D}_t$; we prove that p is an interior point of $\mathcal{D}_t$ and φ_t is smooth on a neighborhood of p (the case $t < 0$ is analogous). Indeed, since $\gamma_p([0, t])$ is compact, part (a) yields an open neighborhood W_0 of this set and $\epsilon > 0$ such that $(s, q) \in (-\epsilon, \epsilon) \times W_0 \mapsto \varphi_s(q) \in M$ is well-defined and smooth. Take an integer $n > 0$ such that $t/n < \epsilon$ and put $\alpha_1 = \varphi_{\frac{t}{n}}|_{W_0}$ and $W_1 = \alpha_1^{-1}(W_0)$. Then, define inductively $\alpha_i = \varphi_{\frac{t}{n}}|_{W_{i-1}}$ and $W_i = \alpha_i^{-1}(W_{i-1}) \subset W_{i-1}$ for $i = 2, \ldots, n$. It is clear that α_i is smooth and W_i is an open neighborhood of $\gamma_p(\frac{n-i}{n}t)$ for all i. In particular, W_n is an open neighborhood of p in W. Moreover,

$$\alpha_1 \circ \alpha_2 \circ \cdots \circ \alpha_n|_{W_n} = (\varphi_{\frac{t}{n}})^n|_{W_n} = \varphi_t|_{W_n}$$

by the last assertion of part (b), so φ_t is smooth on W_n. Now $\mathcal{D}_t$ is open and φ_t is smooth on $\mathcal{D}_t$. It is obvious that the image of φ_t is $\mathcal{D}_{-t}$. Since φ_{-t} is also smooth on $\mathcal{D}_{-t}$, it follows from part (b) that φ_t and φ_{-t} are inverses one of the other and this completes the proof of the theorem. □

Owing to Proposition 1.6.10(b), the flow of X is also called the *local one-parameter group* of X, and X is called the *infinitesimal generator* of $\{\varphi_t\}$. The vector field X is called *complete* if all of its maximal integral curves are defined on $\mathbb{R}$. This is equivalent to requiring that $\mathcal{D}_t = M$ for all $t \in \mathbb{R}$. In this case, we refer to $\{\varphi_t\}$ as the *one-parameter group* of diffeomorphisms of M generated by X.

1.6.11 Proposition *Every smooth vector field X defined on a compact smooth manifold M is complete.*

Proof If M is compact, using Proposition 1.6.10(a), we can find a finite open covering $\{V_i\}$ of M and $\epsilon_i > 0$ such that $(-\epsilon_i, \epsilon_i) \times V_i \to M$, $(t, p) \mapsto \varphi_t(p)$ is well-defined and smooth for all i. Let $\epsilon = \min_i\{\epsilon_i\}$. Now this map is defined on $(-\epsilon, \epsilon) \times M \to M$. This means that any integral curve of X, starting at any point of M, is defined at least on the interval $(-\epsilon, \epsilon)$. The argument using the uniqueness of solutions of ODE as above and piecing together integral curves of X shows that any integral curve of X is defined on $(-k\epsilon, k\epsilon)$ for all positive integers k, hence it is defined on $\mathbb{R}$. □

1.6.12 Examples (i) Take $M = \mathbb{R}^2$ and $X = \frac{\partial}{\partial x_1}$. Then X is complete and $\varphi_t(x_1, x_2) = (x_1 + t, x_2)$ for $(x_1, x_2) \in \mathbb{R}^2$. Note that if we replace $\mathbb{R}^2$ by the punctured plane $\mathbb{R}^2 \setminus \{(0, 0)\}$, the domains of φ_t become proper subsets of M.

(ii) Consider the smooth vector field on $\mathbb{R}^{2n}$ defined by

$$X(x_1, \ldots, x_{2n}) = -x_2 \frac{\partial}{\partial x_1} + x_1 \frac{\partial}{\partial x_2} + \cdots - x_{2n} \frac{\partial}{\partial x_{2n-1}} + x_{2n-1} \frac{\partial}{\partial x_{2n}}.$$

The flow of X is given by the linear map

$$\varphi_t \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{2n-1} \\ x_{2n} \end{pmatrix} = \begin{pmatrix} R_t & & \\ & \ddots & \\ & & R_t \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{2n-1} \\ x_{2n} \end{pmatrix}$$

where R_t is the 2×2 block

$$\begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}.$$

It is true that X restricts to a smooth vector field $\bar{X}$ on S^{2n-1} (cf. Problem 32). The flow of $\bar{X}$ is of course the restriction of φ_t to S^{2n-1}. X and $\bar{X}$ are complete vector fields.

(iii) Take $M = \mathbb{R}$ and $X(x) = x^2 \frac{\partial}{\partial x}$. Solving the ODE we find $\varphi_t(x) = \frac{x}{1-tx}$. It follows that the domain of φ_t is $(-\infty, \frac{1}{t})$ if $t > 0$ and $(\frac{1}{t}, +\infty)$ if $t < 0$. ■

Lie Bracket

If X is a smooth vector field on M and $f : M \to \mathbb{R}$ is a smooth function, the directional derivative $X(f) : M \to \mathbb{R}$ is also smooth and so it makes sense to derivate it again as in $Y(X(f))$, where Y is another smooth vector field on M. For instance, in a local chart $(U, \varphi = (x_1, \ldots, x_n))$, we have the first order partial derivative

$$\frac{\partial}{\partial x_i}\Big|_p (f) = \frac{\partial f}{\partial x_i}\Big|_p$$

and the second order partial derivative

$$\left(\frac{\partial}{\partial x_j}\right)_p \left(\frac{\partial}{\partial x_i}(f)\right) = \frac{\partial^2 f}{\partial x_j \partial x_i}\Big|_p,$$

and it follows from Schwarz's theorem on the commutativity of mixed partial derivatives of smooth functions on $\mathbb{R}^n$ that

$$\frac{\partial^2 f}{\partial x_j \partial x_i}\Big|_p = \frac{\partial^2 (f \circ \varphi^{-1})}{\partial r_j \partial r_i}\Big|_{\varphi(p)} = \frac{\partial^2 (f \circ \varphi^{-1})}{\partial r_i \partial r_j}\Big|_{\varphi(p)} = \frac{\partial^2 f}{\partial x_i \partial x_j}\Big|_p, \tag{1.6.13}$$

where $\mathrm{id} = (r_1, \ldots, r_n)$ denotes the canonical coordinates on $\mathbb{R}^n$.

On the other hand, for general smooth vector fields X, Y on M, the second derivative depends on the order of the vector fields and the failure of commutativity is measured by the *commutator* or *Lie bracket*, given by

$$[X, Y](f) = X(Y(f)) - Y(X(f)) \tag{1.6.14}$$

for every smooth function $f : M \to \mathbb{R}$. We say that X, Y *commute* if $[X, Y] = 0$. It turns out that formula (1.6.14) defines a smooth vector field on M! Indeed, Scholium 1.6.6 says that such a vector field is unique, if it exists. In order to prove its existence, consider a coordinate system $(U, x_1, \ldots, x_n)$. Then we can write

$$X|_U = \sum_{i=1}^{n} a_i \frac{\partial}{\partial x_i} \quad \text{and} \quad Y|_U = \sum_{j=1}^{n} b_j \frac{\partial}{\partial x_j}$$

for $a_i, b_j \in C^\infty(U)$. If $[X, Y]$ exists, we must have

$$[X, Y]|_U = \sum_{i,j=1}^{n} \left(a_i \frac{\partial b_j}{\partial x_i} - b_i \frac{\partial a_j}{\partial x_i} \right) \frac{\partial}{\partial x_j}, \tag{1.6.15}$$

because the coefficients of $[X, Y]|_U$ in the local frame $\{\frac{\partial}{\partial x_j}\}_{j=1}^n$ must be given by $[X, Y](x_j) = X(Y(x_j)) - Y(X(x_j))$. We can use formula (1.6.15) as the definition of a vector field on U; note that such a vector field is smooth and satisfies property (1.6.14) for functions in $C^\infty(U)$. We finally define $[X, Y]$ globally by covering M with domains of local charts: on the overlap of two charts, the different definitions coming from the two charts must agree by the above uniqueness result; it follows that $[X, Y]$ is well-defined.

1.6.16 Exercise Check that the vector field $[X, Y]$ just defined satisfies property (1.6.14) for $f \in C^\infty(M)$. (Hint: Cover M by domains of local charts, $M = \cup_\alpha U_\alpha$, write $f = \sum_\alpha f_\alpha$ where $f_\alpha \in C^\infty(M)$ and $\operatorname{supp}(f_\alpha) \subset U_\alpha$, and check the property for each f_α.) ■

1.6.17 Examples (i) Schwarz's theorem (1.6.13) now means $[\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j}] = 0$ for coordinate vector fields associated with a local chart.

(ii) Let $X = \frac{\partial}{\partial x} - \frac{y}{2}\frac{\partial}{\partial z}$, $Y = \frac{\partial}{\partial y} + \frac{x}{2}\frac{\partial}{\partial z}$, $Z = \frac{\partial}{\partial z}$ be smooth vector fields on $\mathbb{R}^3$. Then $[X, Y] = Z, [Z, X] = [Z, Y] = 0$.

■

1.6.18 Proposition *Let X, Y, and Z be smooth vector fields on M. Then*

a. $[Y, X] = -[X, Y]$.

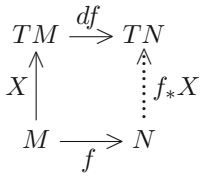
b. If $f, g \in C^\infty(M)$, then

$$[fX, gY] = fg[X, Y] + f(Xg)Y - g(Yf)X.$$

c. $[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0$ *(Jacobi identity).*

1.6.19 Exercise Prove Proposition 1.6.18. (Hint: Use (1.6.14).) ■

Let $f : M \to N$ be a diffeomorphism. For every smooth vector field X on M, the formula $df \circ X \circ f^{-1}$ defines a smooth vector field on N, called the *push-forward* of X under f, which we denote by f_*X, as in the following commutative diagram:



If the flow of X is $\{\varphi_t\}$, then the flow of f_*X is $f \circ \varphi_t \circ f^{-1}$, as

$$\frac{d}{dt} f(\varphi_t(f^{-1}(p))) = df\left(\frac{d}{dt}\varphi_t(f^{-1}(p))\right) = df(X_{f^{-1}(p)}).$$

More generally, if $f : M \to N$ is a smooth map which needs not be a diffeomorphism, smooth vector fields X on M and Y on N are called *f-related* if $df \circ X = Y \circ f$.

1.6.20 Proposition *Let $f : M \to M'$ be smooth. Let X, Y be smooth vector fields on M, and let X', Y' be smooth vector fields on M'. If X and X' are f-related and Y and Y' are f-related, then also $[X, Y]$ and $[X', Y']$ are f-related.*

Proof Let $h \in C^\infty(M')$ and $q \in M$. Note first that

$$\begin{aligned} X_q(h \circ f) &= d(h \circ f)(X_q) \\ &= dh(df(X_q)) \\ &= (df \circ X)_q(h) \\ &= X'_{f(q)}(h), \end{aligned}$$

namely

$$X(h \circ f) = X'(h) \circ f. \tag{1.6.21}$$

Similarly, $Y(h \circ f) = Y'(h) \circ f$.

We now prove $df \circ [X, Y] = [X', Y'] \circ f$. Let $g \in C^\infty(M')$ and $p \in M$. We use (1.6.14) and (1.6.21) a few times to write

$$\begin{aligned} df([X, Y]_p)(g) &= [X, Y]_p(g \circ f) \\ &= X_p(Y(g \circ f)) - Y_p(X(g \circ f)) \\ &= X_p(Y'(g) \circ f) - Y_p(X'(g) \circ f) \\ &= X'_{f(p)}(Y'(g)) - Y'_{f(p)}(X'(g)) \\ &= [X', Y']_{f(p)}(g), \end{aligned}$$

as we wished. □

What is the relation between flows and Lie brackets? In order to discuss that, let X, Y be smooth vector fields on M. Denote the flow of X by $\{\varphi_t\}$ and let f be a smooth function on M. Then

$$\frac{d}{dt}(f \circ \varphi_t) = X(f) \circ \varphi_t,$$

and

$$((\varphi_{-t})_* Y)(f \circ \varphi_t) = Y(f) \circ \varphi_t \tag{1.6.22}$$

as $(\varphi_{-t})_* Y$ and Y are φ_t-related (cf. (1.6.21)).

1.6.23 Exercise Let Z_t be a smooth curve in T_pM and let $h_t(x) = H(t, x)$, where $H \in C^\infty(\mathbb{R} \times M)$. Prove that

$$\frac{d}{dt}\Big|_{t=0} Z_t(h_t) = \left(\frac{d}{dt}\Big|_{t=0} Z_t\right)(h_0) + Z_0\left(\frac{d}{dt}\Big|_{t=0} h_t\right).$$

(Hint: Here $\frac{d}{dt}|_{t=0} h_t(x)$ means $\frac{\partial H}{\partial t}(0, x)$. Consider $\Gamma \in C^\infty(\mathbb{R} \times \mathbb{R}, M)$ such that $\Gamma(t, 0) = p$ and $\frac{\partial}{\partial s}|_{s=0}\Gamma(t, s) = Z_t$ for all t, and use the chain rule.) ■

Differentiate identity (1.6.22) at $t = 0$ to get

$$\frac{d}{dt}\Big|_{t=0} (\varphi_{-t})_* Y(f) + Y(X(f)) = X(Y(f)).$$

Note that $t \mapsto ((\varphi_{-t})_* Y)_p$ is a smooth curve in T_pM; indeed, we can write $((\varphi_{-t})_* Y)_p = d\Phi(0_{-t}, Y_{\Phi(t,p)})$ where Φ is the flow viewed as a function of two arguments, as in Proposition 1.6.10(a). Its tangent vector at $t = 0$ is called the *Lie derivative* of Y with respect to X at p, denoted by $(L_X Y)_p$, and this defines the Lie derivative $L_X Y$ as a smooth vector field on M. The above calculation shows that

$$L_X Y = [X, Y]. \tag{1.6.24}$$

1.6.25 Proposition *X and Y commute if and only if the corresponding flows $\{\varphi_t\}$, $\{\psi_s\}$ commute.*

Proof For each $p \in M$, $t \mapsto ((\varphi_{-t})_* Y)_p$ is a curve in T_pM defined on an open interval. Since $\{\varphi_t\}$ is a one-parameter group,

$$\frac{d}{dt}\Big|_{t=t_0} (\varphi_{-t})_* Y = \frac{d}{dh}\Big|_{h=0} (\varphi_{-(t_0+h)})_* Y$$

$$= d(\varphi_{-t_0})\left(\frac{d}{dh}\Big|_{h=0} (\varphi_{-h})_* Y \circ \varphi_{t_0}\right). \tag{1.6.26}$$

By (1.6.24), $[X, Y] = 0$ if and only if $0 = \frac{d}{dt}\big|_{t=0}(\varphi_{-t})_* Y$ and, due to (1.6.26), this is equivalent to $(\varphi_{-t})_* Y = (\varphi_0)_* Y = Y$ for all t. However, the flow of $(\varphi_{-t})_* Y$ is $\{\varphi_{-t}\psi_s\varphi_t\}$, so this means $\varphi_{-t}\psi_s\varphi_t = \psi_s$. □

We know that, for a local chart (U, φ), the set of coordinate vector fields $\{\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n}\}$ is linearly independent at every point of U and the $\frac{\partial}{\partial x_i}$ pairwise commute. It turns out these two conditions locally characterize coordinate vector fields. Namely, we call a set $\{X_1, \ldots, X_k\}$ of smooth vector fields defined on an open set V of M a *local k-frame* if it is linearly independent at every point of V; if $k = \dim M$, we simply say *local frame*.

1.6.27 Proposition *Let $\{X_1, \ldots, X_k\}$ be a local k-frame on V such that $[X_i, X_j] = 0$ for all $i, j = 1, \ldots, k$. Then, for every $p \in V$, there exists an open neighborhood U of p in V and a local chart (U, φ) whose first k coordinate vector fields are exactly the X_i.*

Proof Complete $\{X_1, \ldots, X_k\}$ to a local frame $\{X_1, \ldots, X_n\}$ on a smaller neighborhood $\tilde{V} \subset V$ of p. (One can do that by first completing

$$\{X_1(p), \ldots, X_k(p)\}$$

to a basis

$$\{X_1(p), \ldots, X_k(p), v_{k+1}, \ldots, v_n\}$$

of T_pM and then declaring $X_{k+1}, \ldots, X_n$ to be the vector fields defined on the domain of a system of local coordinates $(W, y_1, \ldots, y_n)$ around p, $W \subset V$, with constant coefficients in $\{\frac{\partial}{\partial y_1}, \ldots, \frac{\partial}{\partial y_n}\}$, that extend $v_{k+1}, \ldots, v_n$. By continuity, $\{X_1, \ldots, X_k\}$ will be a local frame on a neighborhood $\tilde{V} \subset W$ of p.) Let $\{\varphi_t^i\}$ be the flow of X_i and put $F(t_1, \ldots, t_n) := \varphi_{t_1}^1 \circ \cdots \circ \varphi_{t_n}^n(p)$, smooth map defined on a neighborhood of 0 in $\mathbb{R}^n$. Then $dF_0(e_i) = X_i(p)$ for all i, so F is a local diffeomorphism at 0 by the inverse mapping theorem. The local inverse F^{-1} defines a local chart $(U, x_1, \ldots, x_n)$ around p. Finally, for $q = F(t_1, \ldots, t_n)$ and $1 \le i \le k$,

$$\begin{aligned}
\frac{\partial}{\partial x_i}\Big|_q &= dF_{(t_1,\ldots,t_n)}(e_i) \\
&= \frac{d}{dh}\Big|_{h=0} \varphi_{t_i+h}^i \varphi_{t_1}^1 \cdots \hat{\varphi_{t_i}^i} \cdots \varphi_{t_n}^n(p) \\
&= X_i\left(\varphi_{t_i}^i \varphi_{t_1}^1 \cdots \hat{\varphi_{t_i}^i} \cdots \varphi_{t_n}^n(p)\right) \\
&= X_i\left(\varphi_{t_1}^1 \cdots \varphi_{t_n}^n(p)\right) \\
&= X_i(q),
\end{aligned}$$

where we have used Proposition 1.6.25 twice. □

1.7 Distributions and Foliations

We seek to generalize the theory of the previous section to higher dimensions, so let us rephrase it in the following terms. Let X be a smooth vector field on M which is *nowhere zero*. On the one hand, the $\mathbb{R}$-span of X_p defines a family $\mathcal{D}$ of one-dimensional subspaces $\mathcal{D}_p$ of T_pM for each $p \in M$. On the other hand, the maximal integral curves of X define a partition $\mathcal{F}$ of M into regular parameterized curves, or 1-dimensional immersed submanifolds of M. The relation between $\mathcal{D}$ and $\mathcal{F}$ is that $T_pL = \mathcal{D}_p$ for every $L \in \mathcal{F}$ and every $p \in L$.

In view of the above, we give the following definition. Suppose $\dim M = n$. A *rank k (smooth) distribution* $\mathcal{D}$ on M, $0 \leq k \leq n$, is an assignment of a k-dimensional subspace $\mathcal{D}_p$ of T_pM to each $p \in M$, where any $p \in M$ admits an open neighborhood U with the property that there exist smooth vector fields $X_1, \ldots, X_k$ on U such that the span of $X_1(q), \ldots, X_k(q)$ coincides with $\mathcal{D}_q$ for all $q \in U$.

Before continuing, we make explicit a consequence of Proposition 1.6.27. Loosely speaking, it says that the flow of a non-vanishing vector field can be locally "rectified":

1.7.1 Proposition *Let X be a smooth vector field on M such that $X_p \neq 0$ for some $p \in M$. Then there exists a system of local coordinates $(U, x_1, \ldots, x_n)$ around p such that $X|_U = \frac{\partial}{\partial x_1}$. It follows that the flow of X is given by*

$$\varphi_t(x_1(q), \ldots, x_n(q)) = (x_1(q) + t, x_2(q), \ldots, x_n(q))$$

for all $q \in U$, and the integral curves of X in U are of the form $x_2 = c_2, \ldots, x_n = c_n$ for some $c_2, \ldots, c_n \in \mathbb{R}$.

Based on Proposition 1.7.1, we make the following definition. A *k-dimensional foliation* of M, $0 \leq k \leq n$, is a partition $\mathcal{F}$ of M into piecewise smooth path-connected subsets, where any $p \in M$ admits a coordinate neighborhood $(U, x_1, \ldots, x_n)$ such that U is connected and, for every $L \in \mathcal{F}$, the piecewise smooth path-connected components of $L \cap U$ coincide with the "slices"

$$x_{k+1} = c_{k+1}, \ \ldots, \ x_n = c_n$$

for some $c_{k+1}, \ldots, c_n \in \mathbb{R}$. The elements of $\mathcal{F}$ are called *leaves*. A coordinate system $(U, x_1, \ldots, x_n)$ as above will be called *distinguished*. If $L \in \mathcal{F}$, the piecewise smooth path-components of $L \cap U$ are called *plaques*.

1.7.2 Examples (i) The connected components of the levels sets of a submersion $M \to N$ form a foliation of dimension $\dim M - \dim N$, by the local form of a submersion, where the leaves are embedded submanifolds. Indeed, this is the local model of a general foliation, as follows from the definition.

(ii) Recall the skew-line in the torus in Example 1.4.2. The images of the immersions

$$F_s : \mathbb{R} \to \mathbb{R}^4, \qquad F(t) = (\cos at, \sin at, \cos(bt + s), \sin(bt + s)),$$

where $a, b \in \mathbb{R}$, $a \neq 0$, for $s \in [0, 2\pi)$, form a foliation of dimension 1 of T^2. If b/a is an irrational number, the leaves are dense in T^2. ■

Each leaf $L \in \mathcal{F}$ has a canonical structure of immersed submanifold of M of dimension k. In fact, we can use Proposition 1.2.10. For any distinguished chart (U, φ), $\varphi|_P$ is a bijective map from a plaque (path-component) P of $L \cap U$ onto an open subset of $\mathbb{R}^k$. In this way, if we start with a countable collection $\{(U_m, \varphi_m)\}_{m\in\mathbb{N}}$ of distinguished charts of M whose domains cover L, we construct a collection $\{(P_\alpha, \varphi_\alpha)\}_{\alpha\in A}$, where P_α is a plaque of $L \cap U_m$ for some m and φ_α is the restriction of φ_m to P_α. It is clear that this collection satisfies conditions (b), (c), and (d) of Proposition 1.2.10, but (a) remains to be checked. We shall indeed show that the index set A is countable. For that purpose, it suffices to see that $U_m \cap L$ has countably many path-components, for every m. Fix a plaque P_0 of L in $\{U_m\}$. Since L is path-connected, for any other plaque P, there exists a sequence $P_1, \ldots, P_\ell = P$ of plaques such that $P_{i-1} \cap P_i \neq \varnothing$ for all $i = 1, \ldots, \ell$. So any plaque of L in $\{U_m\}$ can be reached by a finite path of plaques that originates at P_0. It suffices to show that the collection of such paths is countable. In order to do that, it is enough to prove that a given plaque P' of L in $\{U_m\}$ can meet only countably many other plaques of L in $\{U_m\}$. For fixed m_0, $P' \cap (L \cap U_{m_0}) = P' \cap U_{m_0}$ is an open subset of the locally Euclidean space P' and thus has countably many path-components, each such component being contained in a plaque of $L \cap U_{m_0}$. It follows that P' can meet at most countably many path-components of $L \cap U_{m_0}$, as we wished. In this way, we have a structure of smooth manifold on L such that each plaque of L is an open submanifold of L. The underlying topology in L can be much finer than the induced topology. In any case, the Hausdorff condition follows because the inclusion map $L \to M$ is continuous and M is Hausdorff. In addition (recall Proposition 1.4.9):

1.7.3 Proposition *Every leaf L of a foliation of a smooth manifold N is an initial submanifold.*

Proof Let $f : M \to N$ be a smooth map such that $f(M) \subset L$ and consider the induced map $f_0 : M \to L$ such that $\iota \circ f_0 = f$, where $\iota : L \to N$ is the inclusion. We need to show that f_0 is continuous. We will prove that $f_0^{-1}(U)$ is open in M for any given open subset U of L. We may assume $f_0^{-1}(U) \neq \varnothing$, so let $p \in f_0^{-1}(U)$ and $q = f_0(p) \in U$. It suffices to show that p is an interior point of $f_0^{-1}(U)$. Let $(V, y_1, \ldots, y_n)$ be a distinguished chart of N around q, so that the plaques of L in V are of the form

$$y_i = \text{constant} \qquad \text{for } i = k + 1, \ldots, n \tag{1.7.4}$$

and the plaque containing q is

$$y_{k+1} = \cdots = y_n = 0. \tag{1.7.5}$$

By shrinking V, we may assume that (1.7.5) is an open set $\tilde{U} \subset U$. Note that $f^{-1}(V)$ is an open neighborhood of p in M; let W be its path-connected component containing p. Of course, W is open. It is enough to show that $f_0(W) \subset \tilde{U}$, or what amounts to the same, $f(W)$ is contained in (1.7.5). Since $f(W)$ is path-connected, it is contained in a plaque of $V \cap L$; since $f(W)$ meets q, it must be (1.7.5). □

The Frobenius Theorem

Let M be a smooth manifold. It is clear that every foliation of M gives rise to a distribution simply by taking the tangent spaces to the leaves at each point; locally, for a distinguished chart $(U, x_1, \ldots, x_n)$, the vector fields $\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_k}$ span the distribution on U. What about the converse? If we start with a distribution, can we produce an "integral" foliation? Well, in case $k = 1$, locally we can find a smooth vector field X that spans the line distribution and we have seen how to construct a local foliation by integral curves of X; in fact, the global problem can also be solved by passing to a double covering of M. It turns out that in case $k = 1$ there are no obstructions to the local integrability of distributions, and this is in line with the fact that there are no obstructions to the local integrability of ordinary differential equations. On the other hand, the situation is different when we pass to distributions of rank $k > 1$, what amounts to considering certain kinds of partial differential equations.

Let $\mathcal{D}$ be a distribution on M. We say that $\mathcal{D}$ is *integrable* if there exists a foliation $\mathcal{F}$ such that $T_pL_p = \mathcal{D}_p$ for every $p \in M$, where $L_p \in \mathcal{F}$ denotes the leaf through p. Such an $\mathcal{F}$ is called an *integral foliation* of $\mathcal{D}$ and each $L \in \mathcal{F}$ is called an *integral manifold* of $\mathcal{D}$.

1.7.6 Proposition *If $\mathcal{D}$ is an integrable distribution on M, then the integral foliation $\mathcal{F}$ is unique.*

Proof Each leaf of $\mathcal{F}$ is an initial submanifold of M by Proposition 1.7.3, so the structure of smooth manifold on the leaf is unique up to equivalence, as in Problem 19(d). Let us characterize the leaves of $\mathcal{F}$ as subsets of M in terms of $\mathcal{D}$. Define an equivalence relation on M by declaring two points equivalent if and only if they can be joined by a piecewise smooth curve whose smooth segments are tangent to $\mathcal{D}$. For $p \in M$, denote by L_p the leaf of $\mathcal{F}$ through p. Since L_p is path-connected, it is a union of equivalence classes. Now the existence of distinguished charts implies that each such equivalence class is open in L_p, so L_p coincides with the equivalence class of p, as wished. □

More generally, an *integral manifold* of a distribution $\mathcal{D}$ on M is a submanifold (N, f) of M, where N is connected, such that $df_p(T_pN) = \mathcal{D}_{f(p)}$ for all $p \in N$.

1.7.7 Corollary *If (N, f) is an integral manifold of an integrable distribution $\mathcal{D}$ on M, then $f(N)$ is an open submanifold of L for some $L \in \mathcal{F}$, where $\mathcal{F}$ is the integral foliation of $\mathcal{D}$.*

Proof Let $p \in f(N)$ and let L be the leaf of $\mathcal{F}$ passing through p. For any $q \in f(N)$, there is a smooth curve $\gamma : [0, 1] \to N$ joining $f^{-1}(p)$ to $f^{-1}(q)$. Now $f \circ \gamma$ is a smooth curve on M which is everywhere tangent to $\mathcal{D}$, hence is entirely contained in L by the discussion in Proposition 1.7.6. In particular, $q \in L$. We have shown that $f(N) \subset L$. The induced map $f_0 : N \to L$ is smooth by Proposition 1.7.3 and thus an injective immersion. Since N and L have the same dimension, $f(N)$ is an open submanifold of L. □

We say that a vector field X on M *lies* in $\mathcal{D}$ if $X(p) \in \mathcal{D}_p$ for all $p \in M$; in this case, we write $X \in \mathcal{D}$. We say that $\mathcal{D}$ is *involutive* if for all locally defined vectors fields $X, Y \in \mathcal{D}$ we have $[X, Y] \in \mathcal{D}$, namely $\mathcal{D}$ is closed under Lie brackets. Involutivity is a necessary condition for a distribution to be integrable.

1.7.8 Proposition *Every integrable distribution is involutive.*

Proof Let $\mathcal{D}$ be an integrable distribution on a smooth manifold M. Given smooth vector fields $X, Y \in \mathcal{D}$ defined on an open set $V \subset M$ and $p \in V$, we need to show that $[X, Y]_p \in \mathcal{D}_p$. By assumption, there exists a distinguished coordinate system $(U, x_1, \ldots, x_n)$ around p such that $U \subset V$ and the vector fields $\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_k}$ span the distribution $\mathcal{D}$ on U. Now $X|_U, Y|_U$ are linear combinations of $\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_k}$ with $C^\infty(U)$-coefficients, and so is their bracket, as we wished. □

It so happens that involutivity is also a sufficient condition for a distribution to be integrable. This is the contents of the celebrated Frobenius theorem. Despite being named after Frobenius, the theorem seems to be proved first by Clebsch and Deahna. The merit of Frobenius in his 1877 Crelle's paper was to apply the theorem to Pfaffian systems, or systems of partial differential equations that are usually formulated, from the point of view of their underlying geometric and algebraic structure, in terms of a system of differential forms of degree one. The proof below is accredited to Lundell [Lun92] who found inspiration in Chern and Wolfson (referenced therein).

We first prove an elementary, general lemma.

1.7.9 Lemma *Let $\mathcal{D}$ be a rank k distribution on a smooth manifold M. Then there exists a system of local coordinates $(U, x_1, \ldots, x_n)$ around any given point p in M such that $\mathcal{D}$ is spanned by the k vector fields*

$$X_j = \frac{\partial}{\partial x_j} + \sum_{i=k+1}^{n} a_{ij} \frac{\partial}{\partial x_i} \qquad \text{for } j = 1, \ldots, k$$

at all points in U, where $a_{ij} \in C^\infty(U)$.

Proof Let $(V, x_1, \ldots, x_n)$ be any system of local coordinates around p. Let $Y_1, \ldots, Y_k$ be arbitrary smooth vector fields spanning $\mathcal{D}$ on an open neighborhood $\tilde{U}$ of p contained in V. Then $Y_j = \sum_{i=1}^{n} b_{ij} \frac{\partial}{\partial x_i}$ for $j = 1, \ldots, k$ and $b_{ij} \in C^\infty(\tilde{U})$. Since $Y_1, \ldots, Y_k$ is linearly independent at every point of $\tilde{U}$, the matrix $B(q) = (b_{ij}(q))$ has rank k for all $q \in \tilde{U}$. By

relabeling the x_i, we may assume that the $1 \le i, j \le k$-block B' is non-singular in an open neighborhood $U \subset \tilde{U}$ of p. Now the $1 \le i, j \le k$-block of $B(B')^{-1}$ is the identity, namely $X_j = \sum_{i=1}^{k} \hat{b}_{ij} Y_i$ has the desired form, where $(B')^{-1} = (\hat{b}_{ij})$. □

1.7.10 Theorem (Frobenius) *Every involutive distribution is integrable.*

Proof Let $\mathcal{D}$ be an involutive distribution on a smooth manifold M. We first prove the local integrability, namely the existence around any given point $p \in M$ of a system of local coordinates $(V, y_1, \ldots, y_n)$ such that $\mathcal{D}_q$ is spanned by $\frac{\partial}{\partial y_1}|_q, \ldots, \frac{\partial}{\partial y_k}|_q$ for every $q \in V$. Indeed let $(U, x_1, \ldots, x_n)$ and $X_1, \ldots, X_k$ be as in Lemma 1.7.9. Note that

$$[X_i, X_j] \in \operatorname{span}\left\{\frac{\partial}{\partial x_{k+1}}, \ldots, \frac{\partial}{\partial x_n}\right\},$$

so the involutivity of $\mathcal{D}$ implies that $[X_i, X_j] = 0$ for $i, j = 1, \ldots, k$. The desired result now follows from Proposition 1.6.27.

Finally, we construct the integral foliation. In view of Proposition 1.7.6, the leaf L_p through a given point $p \in M$ must be the set of points $q \in M$ that can be reached from p by a piecewise smooth curve whose smooth segments are tangent to $\mathcal{D}$. This defines a partition $\mathcal{F}$ of M into piecewise smooth path-connected subsets. Given $q \in L_p$, let $(V, y_1, \ldots, y_n)$ be a system of local coordinates around q such that $\mathcal{D}$ is spanned by $\frac{\partial}{\partial y_1}, \ldots, \frac{\partial}{\partial y_k}$ at all points in V. It is clear that the path-connected components of $L_p \cap V$ are all of the form

$$y_{k+1} = \text{constant}, \ \ldots, \ y_n = \text{constant}.$$

This proves that $\mathcal{F}$ is a foliation. □

Frobenius' work on the problem of Pfaff was an important source of inspiration for Élie Cartan to create the exterior calculus of differential forms, or the theory of *exterior differential systems*, as they are now called, which is used to investigate systems of PDE in a geometric and coordinate-free manner, particularly well-suited to problems arising in local differential geometry; see, e.g., [IL03].

1.8 Problems

► *Section 1.2*

1

a. Use stereographic projection $\varphi_N : U_N = S^2 \setminus \{(0, 0, 1)\} \to \mathbb{R}^2$ to define a local chart on S^2 and write a formula for φ_N in terms of the coordinates of $\mathbb{R}^3$. Do the same for $\varphi_S : U_S = S^2 \setminus \{(0, 0, -1)\} \to \mathbb{R}^2$.

b. Show that $\{(U_N, \varphi_N), (U_S, \varphi_S)\}$ is a smooth atlas for S^2. Compare the smooth structure defined by this atlas with that defined in Example 1.2.7(v) (viewing S^2 as a union of graphs of smooth maps).

2 Let M be the set of all (affine) lines in $\mathbb{R}^2$. Construct a natural structure of smooth manifold on M. What is the dimension of M? (Hint: Parameterize lines in terms of the coefficients of their equations.)

3 Let M, N, P be smooth manifolds and denote by $\pi_1 : M \times N \to M, \pi_2 : M \times N \to N$ the canonical projections. For fixed $p \in M$ and $q \in N$, define maps $\iota_1 : M \to M \times N$ and $\iota_2 : N \to M \times N$ by $\iota_1(x) = (x, q)$ and $\iota_2(y) = (p, y)$.

a. Show that $\pi_1, \pi_2, \iota_1, \iota_2$ are smooth maps.

b. Show that $f : P \to M \times N$ is smooth if and only if $\pi_1 \circ f$ and $\pi_2 \circ f$ are smooth.

4 (Real Grassmann Manifolds) In this problem, we aim to generalize Example 1.2.9. Let $Gr_k(\mathbb{R}^n)$ denote the set of all k-dimensional subspaces of $\mathbb{R}^n$, where $0 \leq k \leq n$. For each subset I of $\{1, \ldots, n\}$ of cardinality k, consider the linear projection $\pi_I : M(k \times n, \mathbb{R}) \to M(k \times k, \mathbb{R})$ obtained by selecting the k columns indexed by the elements of I, and denote the subset complementary to I in $\{1, \ldots, n\}$ by I'. Let also $\mathcal{U}$ denote the open subset of $M(k \times n, \mathbb{R})$ consisting of matrices of maximal rank.

a. Check that the map $\mathcal{S} : \mathcal{U} \to Gr_k(\mathbb{R}^n)$ that takes a $k \times n$ matrix x of rank k to the span of its lines in $\mathbb{R}^n$ is surjective.

b. Let $U_I = \{\sigma \in Gr_k(\mathbb{R}^n) : \det \pi_I(x) \neq 0 \quad \text{for some } x \in \mathcal{S}^{-1}(\sigma)\}$. Check that $Gr_k(\mathbb{R}^n) = \cup_I U_I$.

c. Show that $\varphi_I : U_I \to M(k \times (n-k), \mathbb{R})$ given by $\varphi_I(\sigma) = \pi_I(x)^{-1}\pi_{I'}(x)$ for some $x \in \mathcal{S}^{-1}(\sigma)$ is well-defined and bijective.

d. For $I, J \subset \{1, \ldots, n\}$, prove that $\varphi_I(U_I \cap U_J)$ is open and $\varphi_J\varphi_I^{-1} : \varphi_I(U_I \cap U_J) \to \varphi_J(U_I \cap U_J)$ is smooth.

e. Deduce that this construction defines on $Gr_k(\mathbb{R}^n)$ a structure of smooth manifold of dimension $k(n-k)$ such that the map $\mathcal{S}$ is a submersion. This is called the *Grassmann manifold* of k-planes in $\mathbb{R}^n$. (Hint: Use Proposition 1.2.10.)

f. Show that $Gr_k(\mathbb{R}^n)$ is compact. (Hint: Every $\sigma \in Gr_k(\mathbb{R}^n)$ is of the form $\sigma = \mathcal{S}(x)$, where the lines of $x \in M(k \times n, \mathbb{R})$ form an orthonormal set in $\mathbb{R}^n$, by the Gram–Schmidt orthonormalization process.)

g. For $\sigma \in Gr_k(\mathbb{R}^n)$, exhibit a canonical identification $T_\sigma Gr_k(\mathbb{R}^n) \cong \mathrm{Hom}(\sigma, \sigma^\perp)$ (the space of linear maps from σ to its orthogonal complement in $\mathbb{R}^n$). (Hint: For $A \in \mathrm{Hom}(\sigma, \sigma^\perp)$, consider the curve in $Gr_k(\mathbb{R}^n)$ given by the graph of tA, for $|t| < \epsilon$.)

5 Let $p : \tilde{M} \to M$ be a topological covering of a smooth manifold M. Check that $\tilde{M}$ is necessarily Hausdorff, second-countable (here you need to know that the fundamental group $\pi_1(M)$ is at most countable) and locally Euclidean. Prove also that there exists a unique smooth structure on $\tilde{M}$ which makes π smooth and a local diffeomorphism (compare Appendix A).

▶ *Section 1.4*

6

a. Prove that the composition and the product of immersions are immersions.

b. In case $\dim M = \dim N$, check that the immersions $M \to N$ coincide with the local diffeomorphisms.

7 Show that every smooth real function on a compact manifold of positive dimension has at least two critical points.

8 Prove that every submersion is an open map.

9

a. Prove that if M is compact and N is connected then every submersion $M \to N$ is surjective.

b. Show that there are no submersions of compact manifolds into Euclidean spaces.

10 Let $p(z) = z^m + a_{m-1}z^{m-1} + \cdots + a_0$ be a polynomial with complex coefficients and consider the associated polynomial map $\mathbb{C} \to \mathbb{C}$. Show that this map is a submersion out of finitely many points.

11 (Generalized Inverse Mapping Theorem) Let $f : M \to N$ be a smooth map which is injective on a compact embedded submanifold P of M. Assume that $df_p : T_pM \to T_{f(p)}N$ is an isomorphism for every $p \in P$.

a. Note that $f(P)$ is an embedded submanifold of N and that f restricts to a diffeomorphism from P onto $f(P)$.

b. Prove that indeed f maps some open neighborhood of P in M diffeomorphically onto an open neighborhood of $f(P)$ in N. (Hint: It suffices to show that f is injective on some neighborhood of P; if this is not the case, there exist sequences $\{p_i\}$, $\{q_i\}$ in M both converging to a point $p \in P$, with $p_i \neq q_i$ but $f(p_i) = f(q_i)$ for all i, and this contradicts the non-singularity of df_p; you may want to use the metrizability of M to define suitable neighborhoods of P.)

12 Let p be a homogeneous polynomial of degree m in n variables $t_1, \ldots, t_n$. Show that $p^{-1}(a)$ is a submanifold of codimension one of $\mathbb{R}^n$ if $a \neq 0$. Show that the submanifolds obtained with $a > 0$ are all diffeomorphic, as well as those with $a < 0$. (Hint: Use Euler's identity

$$\sum_{i=1}^{n} t_i \frac{\partial p}{\partial t_i} = mp.)$$

13 The $n \times n$ real matrices with determinant 1 form a group denoted by $SL(n, \mathbb{R})$. Prove that $SL(n, \mathbb{R})$ is a submanifold of $GL(n, \mathbb{R})$. (Hint: Use Problem 12.)

14 Consider the submanifolds $GL(n, \mathbb{R})$, $O(n)$, and $SL(n, \mathbb{R})$ of the vector space $M(n, \mathbb{R})$ (see Examples 1.2.7(ix) and 1.4.14(ii), and Problem 13, respectively).

a. Check that the tangent space of $GL(n, \mathbb{R})$ at the identity is canonically isomorphic to $M(n, \mathbb{R})$.

b. Check that the tangent space of $SL(n, \mathbb{R})$ at the identity is canonically isomorphic to the subspace of $M(n, \mathbb{R})$ consisting of the matrices of trace zero.

c. Check that the tangent space of $O(n)$ at the identity is canonically isomorphic to the subspace of $M(n, \mathbb{R})$ consisting of the skew-symmetric matrices.

15 Denote by $M(m \times n, \mathbb{R})$ the vector space of $m \times n$ real matrices.

a. Show that the subset of $M(m \times n, \mathbb{R})$ consisting of matrices of rank at least k ($0 \leq k \leq \min\{m, n\}$) is a smooth manifold.

b. Show that the subset of $M(m \times n, \mathbb{R})$ consisting of matrices of rank equal to k ($0 \leq k \leq \min\{m, n\}$) is a smooth manifold. What is its dimension? (Hint: We may work on a neighborhood of a matrix

$$g = \begin{matrix} \\ m-k \end{matrix} \overset{\begin{matrix} k & n-k \end{matrix}}{\left(\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right)},$$

where A is non-singular and right multiply by

$$\left(\begin{array}{c|c} I & -A^{-1}B \\ \hline 0 & I \end{array}\right)$$

to check that g has rank k if and only if $D - CA^{-1}B = 0$.)

16 Let $M \xrightarrow{f} N \xrightarrow{g} P$ be a sequence of smooth maps between smooth manifolds. Assume that $g \pitchfork Q$ for a submanifold Q of P. Prove that $f \pitchfork g^{-1}(Q)$ if and only if $g \circ f \pitchfork Q$.

17 Let $f : M \to \mathbb{R}^k$ be a smooth map and let N be an embedded submanifold of $\mathbb{R}^k$. Prove that there is a dense subset U of $\mathbb{R}^k$ such that $f + a \pitchfork N$ for all $a \in U$. (Hint: Apply Sard's theorem (see Remark 1.4.15) to the restriction of the projection $M \times \mathbb{R}^k \to \mathbb{R}^k$ to $P = F^{-1}(N)$, where $F : M \times \mathbb{R}^k \to \mathbb{R}^k$ is given by $F(p, a) = f(p) + a$.)

18 Let $G \subset \mathbb{R}^2$ be the graph of $g : \mathbb{R} \to \mathbb{R}$, $g(x) = |x|^{1/3}$. Show that G admits a smooth structure so that the inclusion $G \to \mathbb{R}^2$ is smooth. Is it an immersion? (Hint: consider the map $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(t) = \begin{cases} te^{-1/t} & \text{if } t > 0, \\ 0 & \text{if } t = 0, \\ te^{1/t} & \text{if } t < 0.) \end{cases}$$

19 Define immersed submanifolds (M_1, f_1), (M_2, f_2) of N to be *equivalent* if there exists a diffeomorphism $g : M_1 \to M_2$ such that $f_2 \circ g = f_1$.

a. Show that this is indeed an equivalence relation.

b. Show that each equivalence class of submanifolds of N contains a unique representative of the form (M, ι), where M is a subset of N with a manifold structure such that the inclusion $\iota : M \to N$ is a smooth immersion.

c. Let N be a smooth manifold and let M be a subset of N equipped with a given topology. Prove that there exists at most one smooth structure on M, up to equivalence, which makes (M, ι) into an immersed submanifold of N, where $\iota : M \to N$ is the inclusion.

(Hint: Use Proposition 1.4.9 with P being M equipped with the same topology but an arbitrary smooth structure.)

d. Let N be a smooth manifold and let M be a subset of N. Prove that there exists at most one structure of smooth manifold on M, up to equivalence, which makes (M, ι) into an initial submanifold of N, where $\iota : M \to N$ is the inclusion. (Hint: Similar to part (c), use Proposition 1.4.9.)

20 Let N be a smooth manifold of dimension $n+k$. For a point $q \in N$ and a subset $A \subset N$, denote by $C_q(A)$ the set of all points of A that can be joined to q by a smooth curve in N whose image lies in A.

a. Prove that if (P, g) is an initial submanifold of dimension n of N, then, for every $p \in P$, there exists a local chart (V, ψ) of N around $g(p)$ such that

$$\psi(C_{g(p)}(V \cap g(P))) = \psi(V) \cap (\mathbb{R}^n \times \{0\}).$$

(Hint: Use Proposition 1.4.3.)

b. Conversely, assume P is a subset of N with the property that around any point $p \in P$ there exists a local chart (V, ψ) of N such that

$$\psi(C_p(V \cap P)) = \psi(V) \cap (\mathbb{R}^n \times \{0\}).$$

Prove that there exists a topology on P that makes each connected component of P into an initial submanifold of dimension n of N with respect to the inclusion. (Hint: Apply Proposition 1.2.10 to the restrictions $\psi|_{C_p(V\cap P)}$. Proving second-countability of the topology so obtained requires the following facts: for locally Euclidean Hausdorff spaces, paracompactness is equivalent to the property that each connected component is second-countable; every metric space is paracompact; the topology on P is metrizable since it is compatible with the Riemannian distance for the Riemannian metric induced from a given Riemannian metric on N; Riemannian metrics can be constructed on N using partitions of unity.)

21 Show that the product of any number of spheres can be embedded in some Euclidean space with codimension one.

22 We say a subset $A \subset \mathbb{R}^n$ has *measure zero* in $\mathbb{R}^n$ if given $\epsilon > 0$ there exists a countable collection of open balls $\{B_i\}$ that cover A and has total volume $\sum_i \mathrm{vol}(B_i) < \epsilon$.

a. Prove that $\mathbb{R}^{n-1}$ has measure zero in $\mathbb{R}^n$.

b. Let $f : U \to \mathbb{R}^n$ be a smooth map, where U is an open subset of $\mathbb{R}^n$. Prove that if $A \subset U$ has measure zero in $\mathbb{R}^n$, then $f(A)$ has measure zero in $\mathbb{R}^n$. (Hint: Use the local Lipschitz property of smooth maps).

c. Let $f : U \to \mathbb{R}^n$ be a smooth map, where U is an open subset of $\mathbb{R}^m$. Prove that the image $f(U)$ has measure zero in $\mathbb{R}^n$ if $m < n$. (Hint: Combine parts (a) and (b).)

A subset A of a smooth manifold M^n has *measure zero* if for every local chart (U, φ) of M we have that $\varphi(A \cap U)$ has measure zero in $\mathbb{R}^n$.

d. (*Proto-Sard*) Let $f : M \to N$ be a smooth map where $\dim M < \dim N$. Prove that the image of f has measure zero in N. (Hint: Use part (c) and the second-countability of smooth manifolds.)

▶ *Section 1.5*

23 Let M be a smooth manifold of dimension n. Given $p \in M$, construct a local chart (U, φ) of M around p such that φ is the restriction of a smooth map $M \to \mathbb{R}^n$.

24 Let $f : M \to N$ be a map. Prove that $f \in C^\infty(M, N)$ if and only if $g \circ f \in C^\infty(M)$ for all $g \in C^\infty(N)$.

25 Let M be a closed embedded submanifold of N. Prove that the restriction map $C^\infty(N) \to C^\infty(M)$ is well-defined and surjective. Show that the result ceases to be true if: (i) M is not closed in N; or (ii) M is closed in N but merely assumed to be an immersed submanifold.

26 Prove that $C^\infty(M)$ is infinite-dimensional as a real vector space for any smooth manifold M of positive dimension. (Hint: Given $p_1, \dots p_n \in M$, choose mutually disjoint neighborhoods of those points and construct suitable bump functions.)

27 Prove that on any smooth manifold M there exists a proper smooth map $f : M \to \mathbb{R}$. (Hint: Use σ-compactness of manifolds and partitions of unity.)

▶ *Section 1.6*

28 Determine the vector field on $\mathbb{R}^2$ with flow $\varphi_t(x, y) = (xe^{2t}, ye^{-3t})$.

29 Determine the flow of the vector field X on $\mathbb{R}^2$ when:

a. $X = y\frac{\partial}{\partial x} - x\frac{\partial}{\partial y}$.

b. $X = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}$.

30 Given the following vector fields in $\mathbb{R}^3$,

$$X = y\frac{\partial}{\partial x} - x\frac{\partial}{\partial y}, \quad Y = z\frac{\partial}{\partial y} - y\frac{\partial}{\partial z}, \quad Z = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z},$$

compute their Lie brackets.

31 Let (M, f) be an immersed submanifold of N and let X be a smooth vector field on N. Suppose there exists a vector field $\bar{X}$ on M such that $\bar{X}$ and X are f-related. Prove that $\bar{X}$ is smooth. (Hint: Use Propositions 1.4.4 and 1.6.5.)

32 Show that the restriction of the vector field defined on $\mathbb{R}^{2n}$

$$X = -x_2 \frac{\partial}{\partial x_1} + x_1 \frac{\partial}{\partial x_2} + \cdots - x_{2n} \frac{\partial}{\partial x_{2n-1}} + x_{2n-1} \frac{\partial}{\partial x_{2n}}$$

to the unit sphere S^{2n-1} defines a nowhere vanishing smooth vector field.

33 Let X and Y be smooth vector fields on M and N with flows $\{\varphi_t\}$ and $\{\psi_t\}$, respectively, and let $f : M \to N$ be smooth. Show that X and Y are f-related if and only if $f \circ \varphi_t = \psi_t \circ f$ for all t.

34 Let M be a properly embedded submanifold of N. Prove that every smooth vector field on M can be smoothly extended to a vector field on N.

35 Construct a natural diffeomorphism $TS^1 \approx S^1 \times \mathbb{R}$ which restricts to a linear isomorphism $T_pS^1 \to \{p\} \times \mathbb{R}$ for every $p \in S^1$—we say that such a diffeomorphism maps fibers to fibers and is linear on the fibers. (Hint: Problem 32.)

36 Construct a natural diffeomorphism $T(M \times N) \approx TM \times TN$ that maps fibers to fibers and is linear on the fibers.

37 Construct a natural diffeomorphism $T\mathbb{R}^n \approx \mathbb{R}^n \times \mathbb{R}^n$ that maps fibers to fibers and is linear on the fibers.

38 Prove that TS^n is an embedded submanifold of $T\mathbb{R}^{n+1}$.

39 Show that $TS^n \times \mathbb{R}$ is diffeomorphic to $S^n \times \mathbb{R}^{n+1}$. (Hint: There are natural isomorphisms $T_pS^n \oplus \mathbb{R} \cong \mathbb{R}^{n+1}$.)

40 A smooth manifold M of dimension n is called *parallelizable* if $TM \approx M \times \mathbb{R}^n$ by a diffeomorphism that maps fibers to fibers and is linear on the fibers. Prove that M is parallelizable if and only if there exists a globally defined n-frame $\{X_1, \ldots, X_n\}$ on M.

▶ *Section 1.7*

41 Is there a non-constant smooth function f defined on a connected open subset of $\mathbb{R}^3$ such that

$$\frac{\partial f}{\partial x} - y\frac{\partial f}{\partial z} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} + x\frac{\partial f}{\partial z} = 0?$$

(Hint: Consider the Lie bracket of suitable vector fields.)

42 Consider the first order system of partial differential equations

$$\frac{\partial z}{\partial x} = \alpha(x, y, z), \quad \frac{\partial z}{\partial y} = \beta(x, y, z),$$

where α, β are smooth functions defined on an open subset of $\mathbb{R}^3$.

a. Show that if f is a solution, then the smooth vector fields $X = \frac{\partial}{\partial x} + \alpha \frac{\partial}{\partial z}$ e $Y = \frac{\partial}{\partial y} + \beta \frac{\partial}{\partial z}$ span the tangent space to the graph of f at all points.

b. Prove that the system admits local solutions if and only if

$$\frac{\partial \beta}{\partial x} + \alpha \frac{\partial \beta}{\partial z} = \frac{\partial \alpha}{\partial y} + \beta \frac{\partial \alpha}{\partial z}.$$

43 Prove that there exists a smooth function f defined on a neighborhood of $(0, 0)$ in $\mathbb{R}^2$ such that $f(0, 0) = 0$ and $\frac{\partial f}{\partial x} = ye^{-(x+y)} - f$, $\frac{\partial f}{\partial y} = xe^{-(x+y)} - f$.

Tensor Fields and Differential Forms

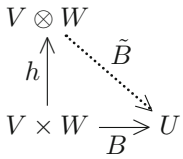
Multilinear algebra is an extension of the methods of Linear Algebra to spaces of multivectors. On a smooth manifold, tensor fields and differential forms are fields of smoothly varying multivectors of various types. In this chapter, we introduce these objects and the exterior derivative of differential forms. This operator will be used in ▶ Chap. 4 to define de Rham cohomology and prove some results in Differential Topology.

2.1 Multilinear Algebra

Let V be a real vector space. In this section, we construct the tensor algebra $T(V)$ and the exterior algebra $\Lambda(V)$ over V. Elements of $T(V)$ are called tensors on V. Later we will apply these constructions to the tangent space T_pM of a manifold M and let p vary in M, similarly to the definition of the tangent bundle.

Tensor Algebra

All vector spaces are real and finite-dimensional. Let V and W be vector spaces. It is less important what the tensor product of V and W *is* than what it *does*. Namely, a tensor product of V and W is a vector space $V \otimes W$ together with a bilinear map $h : V \times W \to V \otimes W$ such that the following *universal property* holds: for every vector space U and every bilinear map $B : V \times W \to U$, there exists a unique linear map $\tilde{B} : V \otimes W \to U$ such that $\tilde{B} \circ h = B$:



There are different ways to construct $V \otimes W$. It does not actually matter which one we choose, in view of the following exercise.

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2.1.1 Exercise Prove that the tensor product of V and W is uniquely defined by the universal property. In other words, if $(V \otimes_1 W, h_1)$, $(V \otimes_2 W, h_2)$ are two tensor products, then there exists an isomorphism $\ell : V \otimes_1 W \to V \otimes_2 W$ such that $\ell \circ h_1 = h_2$. ■

We proceed as follows. Start with the canonical isomorphism $V^{**} \cong V$ between V and its bidual. It says that we can view an element v in V as the linear map on V^* given by $f \mapsto f(v)$. Well, we can extend this idea and consider the space $\mathrm{Bil}(V \times W)$ of bilinear forms on $V \times W$. Then there is a natural map $h : V \times W \to \mathrm{Bil}(V \times W)^*$ given by $h(v, w)(b) = b(v, w)$ for $b \in \mathrm{Bil}(V \times W)$. We claim that $(\mathrm{Bil}(V \times W)^*, h)$ satisfies the universal property: given a bilinear map $B : V \times W \to U$, there is an associated map $U^* \to \mathrm{Bil}(V \times W)$, $f \mapsto f \circ B$; let $\tilde{B} : \mathrm{Bil}(V \times W)^* \to U^{**} = U$ be its transpose.

2.1.2 Exercise Check that $\tilde{B} \circ h = B$. ■

2.1.3 Exercise Let $\{e_i\}$, $\{f_j\}$ be bases of V, W, respectively. Define $b_{ij} \in \mathrm{Bil}(V \times W)$ to be the bilinear form whose value on (e_k, f_ℓ) is 1 if $(k, \ell) = (i, j)$ and 0 otherwise. Prove that $\{b_{ij}\}$ is a basis of $\mathrm{Bil}(V \times W)$. Prove also that $\{h(e_i, f_j)\}$ is the dual basis of $\mathrm{Bil}(V \times W)^*$. Deduce that the image of h spans $\mathrm{Bil}(V \times W)^*$ and hence $\tilde{B}$ as in Exercise 2.1.2 is uniquely defined. ■

Now that $V \otimes W$ is constructed, we can forget about its definition and keep in mind its properties only (in the same way as when we work with real numbers and we do not need to know that they are equivalence classes of Cauchy sequences), namely the universal property and those listed in the sequel. The elements of $V \otimes W$ are called *tensors*, and those in the image of h are called *pure tensors*. Henceforth, we write $v \otimes w = h(v, w)$ for $v \in V$ and $w \in W$. As a consequence of Exercise 2.1.3, note that $V \otimes W$ is spanned by the subset of pure tensors and $\dim V \otimes W = (\dim V)(\dim W)$.

2.1.4 Proposition *Let V and W be vector spaces. Then:*

a. $(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w$;
b. $v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2$;
c. $av \otimes w = v \otimes aw = a(v \otimes w)$;

for all $v, v_1, v_2 \in V$; $w, w_1, w_2 \in W$; $a \in \mathbb{R}$.

2.1.5 Proposition *Let U, V, and W be vector spaces. Then there are* canonical *isomorphisms (isomorphisms that are independent of choices of bases):*

a. $\mathbb{R} \otimes V \cong V$;
b. $V \otimes W \cong W \otimes V$;
c. $(V \otimes W) \otimes U \cong V \otimes (W \otimes U)$;
d. $V^* \otimes W \cong \mathrm{Hom}(V, W)$;

where $\mathrm{Hom}(V, W)$ *denotes the space of homomorphisms (or linear maps) $V \to W$.*

2.1.6 Exercise Prove Propositions 2.1.4 and 2.1.5. ■

2.1.7 Exercise Let $\{e_1, \ldots, e_m\}$ and $\{f_1, \ldots, f_n\}$ be bases for V and W, respectively. Prove that $\{e_i \otimes f_j \ : \ i = 1, \ldots, m \text{ and } j = 1, \ldots, n\}$ is a basis for $V \otimes W$. ■

2.1.8 Exercise Let $A = (a_{ij})$ be a $m \times n$ real matrix, viewed as an element of $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$. Use the standard scalar product in $\mathbb{R}^n$ to identify $(\mathbb{R}^n)^* \cong \mathbb{R}^n$. What is the element of $\mathbb{R}^n \otimes \mathbb{R}^m$ that corresponds to A? ■

In view of Proposition 2.1.5, henceforth we abolish the use of parentheses in iterated tensor products. Taking $V = W$ we can now inductively form the nth tensor power $\otimes^n V = \otimes^{n-1} V \otimes V$ for $n \geq 1$, where we adopt the convention that $\otimes^0 V = \mathbb{R}$. The *tensor algebra* $T(V)$ over V is the direct sum

$$T(V) = \bigoplus_{r,s \geq 0} V^{r,s}$$

where

$$V^{r,s} = \left(\otimes^r V\right) \otimes \left(\otimes^s V^*\right)$$

is called the *tensor space of type* (r, s). The elements of $T(V)$ are called *tensors*, and those of $V^{r,s}$ are called *homogeneous tensors of type* (r, s). The multiplication $\otimes$ on $T(V)$, read "tensor," is defined to be the $\mathbb{R}$-linear extension of

$$\begin{aligned}&(u_1 \otimes \cdots \otimes u_{r_1} \otimes u_1^* \otimes \cdots \otimes u_{s_1}^*) \otimes (v_1 \otimes \cdots \otimes v_{r_2} \otimes v_1^* \otimes \cdots \otimes v_{s_2}^*) \\ &= u_1 \otimes \cdots \otimes u_{r_1} \otimes v_1 \otimes \cdots \otimes v_{r_2} \otimes u_1^* \otimes \cdots \otimes u_{s_1}^* \otimes v_1^* \otimes \cdots \otimes v_{s_2}^*.\end{aligned}$$

By construction, we have that $\otimes : V^{r_1,s_1} \times V^{r_2,s_2} \to V^{r_1+r_2,s_1+s_2}$ equals the composition

$$V^{r_1,s_1} \times V^{r_2,s_2} \to V^{r_1,s_1} \otimes V^{r_2,s_2} \to V^{r_1+r_2,s_1+s_2},$$

where the first map is the canonical one and the second map is the canonical isomorphism coming from Proposition 2.1.5, parts (b) and (c). This turns $T(V)$ into a non-commutative, associative *graded* algebra, in the sense that tensor multiplication is compatible with the natural grading:

$$V^{r_1,s_1} \otimes V^{r_2,s_2} \subset V^{r_1+r_2,s_1+s_2}.$$

Note that $V^{0,0} = \mathbb{R}$, $V^{1,0} = V$, $V^{0,1} = V^*$, so real numbers, vectors, and linear forms are examples of tensors.

Exterior Algebra

Even more important to us will be a certain quotient of the subalgebra $T^+(V) = \bigoplus_{k \geq 0} V^{k,0}$ of $T(V)$. Let $\mathfrak{I}$ be the two-sided ideal of $T^+(V)$ generated by the set of

elements of the form

$$v \otimes v \tag{2.1.9}$$

for $v \in V$.

2.1.10 Exercise Prove that another set of generators for $\mathfrak{I}$ is given by the elements of the form $u \otimes v + v \otimes u$ for $u, v \in V$. ■

The *exterior algebra* over V is the quotient

$$\Lambda(V) = T^+(V)/\mathfrak{I}.$$

The induced multiplication is denoted by $\wedge$, and read "wedge" or "exterior product". In particular, the class of $v_1 \otimes \cdots \otimes v_k$ modulo $\mathfrak{I}$ is denoted $v_1 \wedge \cdots \wedge v_k$. This is also a graded algebra, as $\mathfrak{I}$ is generated by homogeneous elements, and the space of elements of degree k is

$$\Lambda^k(V) = V^{k,0}/\mathfrak{I} \cap V^{k,0}.$$

Since $\mathfrak{I}$ is generated by elements of degree 2, we immediately get

$$\Lambda^0(V) = \mathbb{R} \quad \text{and} \quad \Lambda^1(V) = V.$$

$\Lambda(V)$ is not commutative, but we have

2.1.11 Proposition *$\alpha \wedge \beta = (-1)^{k\ell}\beta \wedge \alpha$ for $\alpha \in \Lambda^k(V)$, $\beta \in \Lambda^\ell(V)$.*

Proof Since $v \otimes v \in \mathfrak{I}$ for all $v \in V$, we have $v \wedge v = 0$. Since $\mathbb{R}$ is not a field of characteristic two, this relation is equivalent to $v_1 \wedge v_2 = -v_2 \wedge v_1$ for all $v_1, v_2 \in V$.

By linearity, we may assume that $\alpha = u_1 \wedge \cdots \wedge u_k$, $\beta = v_1 \wedge \cdots \wedge v_\ell$. Now

$$\begin{aligned}
\alpha \wedge \beta &= u_1 \wedge \cdots \wedge u_k \wedge v_1 \wedge \cdots \wedge v_\ell \\
&= -u_1 \wedge \cdots \wedge u_{k-1} \wedge v_1 \wedge u_k \wedge v_2 \cdots \wedge v_\ell \\
&= u_1 \wedge \cdots \wedge u_{k-1} \wedge v_1 \wedge v_2 \wedge u_k \wedge v_3 \cdots \wedge v_\ell \\
&= \cdots \\
&= (-1)^\ell u_1 \wedge \cdots \wedge u_{k-1} \wedge v_1 \wedge \cdots \wedge v_\ell \wedge u_k \\
&= (-1)^{2\ell} u_1 \wedge \cdots \wedge u_{k-2} \wedge v_1 \wedge \cdots \wedge v_\ell \wedge u_{k-1} \wedge u_k \\
&= \cdots \\
&= (-1)^{k\ell}\beta \wedge \alpha,
\end{aligned}$$

as we wished. □

2.1.12 Lemma *If* $\dim V = n$, *then* $\dim \Lambda^n(V) = 1$ *and* $\Lambda^k(V) = 0$ *for* $k > n$.

Proof Let $\{e_1, \dots, e_n\}$ be a basis of V. Since

$$\{e_{i_1} \otimes \cdots \otimes e_{i_k} \ : \ i_1, \dots, i_k \in \{1, \dots, n\}\} \tag{2.1.13}$$

is a basis of $V^{k,0}$ (see Exercise 2.1.7), the image of this set under the projection $V^{k,0} \to \Lambda^k(V)$ is a set of generators of $\Lambda^k(V)$. Taking into account Proposition 2.1.11 yields $\Lambda^k(V) = 0$ for $k > n$ and that $\Lambda^n(V)$ is generated by $e_1 \wedge \cdots \wedge e_n$, so we need only to show that this element is not zero.

Suppose, on the contrary, that $e_1 \otimes \cdots \otimes e_n \in \mathfrak{I}$. Then $e_1 \otimes \cdots \otimes e_n$ is a linear combination of elements of the form $\alpha \otimes v \otimes v \otimes \beta$ where $v \in V, \alpha \in V^{k,0}, \beta \in V^{\ell,0}$, and $k + \ell + 2 = n$. Writing α (resp. β) in terms of the basis (2.1.13), we may assume that the only appearing basis elements are of the form $e_1 \otimes \cdots \otimes e_k$ (resp. $e_{n-\ell+1} \otimes \cdots \otimes e_n$). It follows that we can write

$$e_1 \otimes \cdots \otimes e_n = \sum_{k=0}^{n-2} c_k \, e_1 \otimes \cdots \otimes e_k \otimes v_k \otimes v_k \otimes e_{k+3} \otimes \cdots \otimes e_n, \tag{2.1.14}$$

where $c_k \in \mathbb{R}$ and $v_k \in V$ for all k. Finally, write $v_k = \sum_{i=1}^n a_{ik} e_i$ for $a_k \in \mathbb{R}$. For $m = 0, \dots, n-2$, the coefficient of

$$e_1 \otimes \cdots \otimes e_m \otimes e_{m+2} \otimes e_{m+1} \otimes e_{m+3} \otimes \cdots \otimes e_n$$

on the right-hand side of (2.1.14) is

$$c_m \, a_{m+2,m} a_{m+1,m};$$

by comparison with the left-hand side, it must be zero. However, the coefficient of $e_1 \otimes \cdots \otimes e_n$ on the right-hand side is

$$\sum_{k=0}^{n-2} c_k \, a_{k+1,k} a_{k+2,k};$$

in view of the above, it is also zero, a contradiction. □

2.1.15 Proposition *If* $\{e_1, \dots, e_n\}$ *is a basis of* V, *then*

$$\{e_{i_1} \wedge \cdots \wedge e_{i_k} \ : \ i_1 < \cdots < i_k\}$$

is a basis of $\Lambda^k(V)$ *for all* $0 \le k \le n$; *in particular,* $\dim \Lambda^k(V) = \binom{n}{k}$.

2

Proof Fix $k \in \{0, \dots, n\}$. The above set is clearly a set of generators of $\Lambda^k(V)$ and we need only to show linear independence. Suppose

$$\sum a_{i_1 \cdots i_k} e_{i_1} \wedge \cdots \wedge e_{i_k} = 0,$$

which we write as

$$\sum a_I e_I = 0,$$

where each I denotes an increasing k-multi-index, and $e_\varnothing = 1$. Multiply through this equation by e_J, where J is an increasing $(n-k)$-multi-index, and note that $e_I \wedge e_J = 0$ unless I is the multi-index J^c complementary to J, in which case $e_{J^c} \wedge e_J = \pm e_1 \wedge \cdots \wedge e_n$. Since $e_1 \wedge \cdots \wedge e_n \neq 0$ by Lemma 2.1.12, this shows that $a_I = 0$ for all I. □

2.2 Tensor Bundles

Cotangent Bundle

In the same way as the fibers of the tangent bundle of M are the tangent spaces T_pM for $p \in M$, the fibers of the cotangent bundle of M will be the dual spaces T_pM^*. Indeed, form the disjoint union

$$T^*M = \dot{\bigcup}_{p \in M} T_pM^*.$$

There is a natural projection $\pi^* : T^*M \to M$ given by $\pi(\tau) = p$ if $\tau \in T_pM^*$. Recall that every local chart (U, φ) of M induces a local chart $\tilde{\varphi} : \pi^{-1}(U) \to \mathbb{R}^n \times \mathbb{R}^n = \mathbb{R}^{2n}$ of TM, where $\tilde{\varphi}(v) = (\varphi(\pi(v)), d\varphi(v))$, and thus a map $\tilde{\varphi}^* : (\pi^*)^{-1}(U) \to \mathbb{R}^n \times (\mathbb{R}^n)^* = \mathbb{R}^{2n}$, $\tilde{\varphi}^*(\tau) = (\varphi(\pi^*(\tau)), ((d\varphi_{\pi^*(\tau)})^*)^{-1}(\tau))$, where $(d\varphi_{\pi^*(\tau)})^*$ denotes the transpose map of $d\varphi_{\pi^*(\tau)}$ and we have identified $\mathbb{R}^n = \mathbb{R}^{n*}$ using the standard scalar product. The collection

$$\{((\pi^*)^{-1}(U), \tilde{\varphi}^*) \mid (U, \varphi) \in \mathcal{A}\}, \tag{2.2.1}$$

for the maximal atlas $\mathcal{A}$ of M, satisfies the conditions of Proposition 1.2.10 and defines a Hausdorff, second-countable topology and a smooth structure on T^*M such that $\pi^* : T^*M \to M$ is smooth. The resulting manifold T^*M is called the *cotangent bundle* of M.

A section of T^*M is a map $\omega : M \to T^*M$ such that $\pi^* \circ \omega = \mathrm{id}_M$. A smooth section of T^*M is also called a *differential form of degree 1* or *differential 1-form*. For instance, if $f : M \to \mathbb{R}$ is a smooth function, then $df_p : T_pM \to \mathbb{R}$ is an element of T_pM^* for all $p \in M$ and hence defines a differential 1-form df on M.

If $(U, x_1, \dots, x_n)$ is a system of local coordinates on M, the differentials $dx_1, \dots, dx_n$ yield local smooth sections of T^*M that form the dual basis to $\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}$ at each point (recall (1.3.7)). Therefore any section ω of T^*M can be

locally written as $\omega|_U = \sum_{i=1}^n a_i dx_i$, and one proves similarly to Proposition 1.6.5 that ω is smooth if and only if the a_i are smooth functions on U, for every coordinate system $(U, x_1, \ldots, x_n)$.

2.2.2 Exercise Prove that the differential of a smooth function on M indeed gives a smooth section of T^*M by using the atlas (2.2.1). ■

Tensor Bundles

We now generalize the construction of the tangent and cotangent bundles using the notion of tensor algebra. Let M be a smooth manifold. Set:

$$\begin{aligned} T^{r,s}(M) &= \dot{\bigcup}_{p\in M}(T_pM)^{r,s} \quad \textit{tensor bundle of type } (r,s) \textit{ over } M;\\ \Lambda^k(M^*) &= \dot{\bigcup}_{p\in M}\Lambda^k(T_pM^*) \qquad \textit{exterior } k\textit{-bundle over } M;\\ \Lambda(M^*) &= \dot{\bigcup}_{p\in M}\Lambda(T_pM^*) \qquad \textit{exterior algebra bundle over } M. \end{aligned}$$

Then $T^{r,s}(M)$, $\Lambda^k(M^*)$, and $\Lambda(M^*)$ admit natural structures of smooth manifolds such that the projections onto M are smooth. Indeed, similar to the case of T^*M, we construct atlases for $T^{r,s}(M)$, $\Lambda^k(M^*)$, and $\Lambda(M^*)$ by starting with the maximal atlas of M. For instance, given a local chart (U, φ) of M, we construct a local chart of $\Lambda^k(M^*)$ by inducing a map (isomorphism) $\Lambda^k((d\varphi_p^*)^{-1}) : \Lambda^k(T_pM^*) \to \Lambda^k(\mathbb{R}^{n*}) \cong \mathbb{R}^{\binom{n}{k}}$ from the isomorphism $d\varphi_p : T_pM \to \mathbb{R}^n$, for each $p \in U$ (compare Problem 6(a)).

2.2.3 Exercise Check that $T^{1,0}(M) = TM$, $T^{0,1}(M) = T^*M = \Lambda^1(M^*)$ and $\Lambda^0(M^*) = M \times \mathbb{R}$. ■

A *section* of $T^{r,s}(M)$, $\Lambda^k(M^*)$, or $\Lambda(M^*)$ is a map from M into $T^{r,s}(M)$, $\Lambda^k(M^*)$, or $\Lambda(M^*)$, respectively, whose composition with the canonical projection onto M is the identity map. The smooth sections of $T^{r,s}(M)$, $\Lambda^k(M^*)$, $\Lambda(M^*)$ are, respectively, called *tensor fields of type* (r, s), *differential k-forms*, *differential forms* on M.

We will denote the space of differential k-forms on M by $\Omega^k(M)$ and the space of all differential forms on M by $\Omega^\bullet(M)$. Note that $\Omega^\bullet(M)$ is a graded algebra over $\mathbb{R}$ with wedge multiplication, and a module over the ring $C^\infty(M)$; those operations are defined pointwise. We will also denote by $\mathcal{T}^{r,s}(M)$ the $C^\infty(M)$-module of tensor fields of type (r, s) on M, and by $\mathcal{T}(M)$ the $C^\infty(M)$-module of finite sums of tensor fields of certain types.

If $(U, \varphi = (x_1, \ldots, x_n))$ is a coordinate system on M, then the bases $\{\frac{\partial}{\partial x_i}|_p\}_{i=1}^n$ of T_pM and $\{dx_i|_p\}_{i=1}^n$ of T_pM^*, for $p \in U$, define bases of $(T_pM)^{r,s}$, $\Lambda^k(T_pM^*)$, and $\Lambda(T_pM^*)$. For instance, a section ω of $\Lambda^k(M^*)$ can be locally written as

$$\omega|_U = \sum_{i_1<\cdots<i_k} a_{i_1\cdots i_k} dx_{i_1} \wedge \cdots \wedge dx_{i_k}, \tag{2.2.4}$$

where the $a_{i_1\cdots i_k}$ are functions on U. Note that $\Lambda^k((d\varphi_p^*)^{-1})$ maps $dx_{i_1}\wedge\cdots\wedge dx_{i_k}$ to $e^*_{i_1}\wedge\cdots\wedge e^*_{i_k}$, where $\{e_i^*\}_{i=1}^n$ is the dual basis of the canonical basis of $\mathbb{R}^n$; it follows that $(a_{i_1,\ldots,i_k}(p))$ are exactly the coordinates of ω_p with respect to the local chart of $\Lambda^k(M^*)$ induced from (U,φ). Hence a section ω of $\Lambda^k(M)$ is a differential k-form if and only if the functions $a_{i_1\ldots i_k}$, in all of its local representations (2.2.4), are smooth. Similarly, a (resp. smooth) section S of $T^{r,s}(M)$ can be locally written as

$$S|_U = \sum_{\substack{i_1,\ldots,i_r\\ j_1,\ldots,j_s}} a^{i_1\cdots i_r}_{j_1\cdots j_s}\frac{\partial}{\partial x_{i_1}}\otimes\cdots\otimes\frac{\partial}{\partial x_{i_r}}\otimes dx_{j_1}\otimes\cdots\otimes dx_{j_s},$$

where the $a^{i_1\cdots i_r}_{j_1\cdots j_s}$ are (resp. smooth) functions on U.

2.2.5 Exercise Let $\omega\in\Omega^k(M)$ and $\eta\in\Omega^\ell(M)$. Use local representations as in (2.2.4) to check that $\omega\wedge\eta$ is a *smooth* section of $\Lambda^{k+\ell}(M^*)$, and hence an element of $\Omega^{k+\ell}(M)$. ■

Let ω be a section of $\Lambda^k(M^*)$. It follows from Problems 4 and 7(a) that, for each $p\in M$, we can view ω_p as an alternating k-multilinear map on T_pM, namely

$$\omega_p\in\Lambda^k(T_pM^*)\cong\Lambda^k(T_pM)^*\cong A_k(T_pM)^{**}\cong A_k(T_pM). \tag{2.2.6}$$

In this sense, the coefficients on the left-hand side of (2.2.4) are given by

$$a_{i_1\ldots i_k}(p)=\omega_p\Big(\frac{\partial}{\partial x_{i_1}}\Big|_p,\ldots,\frac{\partial}{\partial x_{i_k}}\Big|_p\Big),$$

for $p\in U$.

The next proposition presents a different perspective on the concept of a differential k-form. We first prove a lemma. Hereafter, it is convenient to denote the $C^\infty(M)$-module of smooth vector fields on M by $\mathfrak{X}(M)$.

2.2.7 Lemma *Let*

$$\omega:\underbrace{\mathfrak{X}(M)\times\cdots\times\mathfrak{X}(M)}_{k\ \text{factors}}\to C^\infty(M)$$

be a $C^\infty(M)$-multilinear map. Then the value of $\omega(X_1,\ldots,X_k)$ at any given point p depends only on the values of $X_1,\ldots,X_k$ at p.

Proof For simplicity of notation, let us do the proof for $k=1$; the case $k>1$ is similar. We first claim that, if $X|_U=X'|_U$ for some open subset U of M, then $\omega(X)|_U=\omega(X')|_U$. Indeed let $p\in U$ be arbitrary, take an open neighborhood V of p such that $\bar V\subset U$ and a smooth function $\lambda\in C^\infty(M)$ with $\lambda|_{\bar V}=1$ and $\operatorname{supp}(\lambda)\subset U$ (Exercise 1.5.4). Then

$$\omega(X)(p)=\lambda(p)\omega(X)(p)$$

$$
\begin{aligned}
&= (\lambda(\omega(X)))(p)\\
&= \omega(\lambda X)(p)\\
&= \omega(\lambda X')(p)\\
&= (\lambda(\omega(X')))(p)\\
&= \lambda(p)\omega(X')(p)\\
&= \omega(X')(p),
\end{aligned}
$$

where in the third and fifth equalities we have used $C^\infty(M)$-linearity of ω, and in the fourth equality we have used that $\lambda X = \lambda X'$ as vector fields on M. This proves the claim above.

Finally, we prove that $\omega(X)(p)$ depends only on $X(p)$. By linearity, it suffices to prove that $X(p) = 0$ implies $\omega(X)(p) = 0$. Let $(W, x_1, \ldots, x_n)$ be a coordinate system around p and write $X|_W = \sum_{i=1}^n a_i \frac{\partial}{\partial x_i}$ for $a_i \in C^\infty(W)$. By assumption, $a_i(p) = 0$ for all i. Let λ be a smooth function on M equal to 1 on an open neighborhood U of p and with support contained in W. Define also

$$
\tilde{X}_i = \begin{cases} \lambda \frac{\partial}{\partial x_i} & \text{on } W \\ 0 & \text{on } M \setminus \operatorname{supp}(\lambda) \end{cases} \qquad \text{and} \qquad \tilde{a}_i = \begin{cases} \lambda a_i & \text{on } W \\ 0 & \text{on } M \setminus \operatorname{supp}(\lambda). \end{cases}
$$

Then $\tilde{X} := \sum_{i=1}^n \tilde{a}_i \tilde{X}_i$ is a globally defined smooth vector field on M such that $\tilde{X}|_U = X|_U$ and we can apply the result in the previous paragraph to write

$$
\begin{aligned}
\omega(X)(p) &= \omega(\tilde{X})(p)\\
&= \left(\sum_{i=1}^n \tilde{a}_i \omega(\tilde{X}_i)\right)(p)\\
&= \sum_{i=1}^n \tilde{a}_i(p)\omega(\tilde{X}_i)(p)\\
&= 0
\end{aligned}
$$

because $\tilde{a}_i(p) = a_i(p) = 0$ for all i. □

2.2.8 Proposition *$\Omega^k(M)$ is canonically isomorphic as a $C^\infty(M)$-module to the $C^\infty(M)$-module of alternating $C^\infty(M)$-multilinear maps*

$$
\underbrace{\mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M)}_{k \text{ factors}} \to C^\infty(M). \tag{2.2.9}
$$

2

Proof Let $\omega \in \Omega^k(M)$. Then $\omega_p \in A_k(T_pM)$ for every $p \in M$, as in (2.2.6); namely, ω_p can be considered to be an alternating k-multilinear form on T_pM. Therefore, for smooth vector fields $X_1, \dots, X_k$ on M,

$$\tilde{\omega}(X_1, \dots, X_k)(p) := \omega_p(X_1(p), \dots, X_k(p))$$

defines a smooth function on M; here $\tilde{\omega}(X_1, \dots, X_k)$ is $C^\infty(M)$-linear in each argument X_i, thus $\tilde{\omega}$ is an alternating $C^\infty(M)$-multilinear map as in (2.2.9).

Conversely, let $\tilde{\omega}$ be a $C^\infty(M)$-multilinear map as in (2.2.9). Due to Lemma 2.2.7, we can construct a section ω of $\Lambda^k(M^*)$ as follows. For each $p \in M$, we define $\omega_p \in A_k(T_pM) \cong \Lambda^k(T_pM^*)$: given $v_1, \dots, v_k \in T_pM$, choose $X_1, \dots, X_k \in \mathfrak{X}(M)$ such that $X_i(p) = v_i$ for all i and put

$$\omega_p(v_1, \dots, v_k) := \tilde{\omega}(X_1, \dots, X_k)(p);$$

owing to Lemma 2.2.7, the value on the right-hand-side is independent on the extensions X_i chosen. The multilinear and alternating properties of ω_p are obvious. In order to check smoothness of the section ω around a point $p \in M$, fix a coordinate system $(U, x_1, \dots, x_n)$ around p and write $\omega|_U = \sum_{i_1<\dots<i_k} a_{i_1\dots i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}$. Take a neighborhood V of p in U and construct $\tilde{X}_j \in \mathfrak{X}(M)$ such that $\tilde{X}_j|_V = \frac{\partial}{\partial x_j}$ for $j = 1, \dots, n$. Then

$$\begin{aligned} a_{i_1\dots i_k}(q) &= \omega_q\left(\tfrac{\partial}{\partial x_{i_1}}\big|_q, \dots, \tfrac{\partial}{\partial x_{i_k}}\big|_q\right) \\ &= \omega_q(\tilde{X}_{i_1}(q), \dots, \tilde{X}_{i_k}(q)) \\ &= \tilde{\omega}(\tilde{X}_{i_1}, \dots, \tilde{X}_{i_k})(q) \end{aligned}$$

for all $q \in V$, and thus $a_{i_1\dots i_k} \in C^\infty(V)$. It follows that ω is a differential k-form on M. □

Henceforth we will not distinguish between differential k-forms and alternating multilinear maps (2.2.9). Similarly to Proposition 2.2.8:

2.2.10 Proposition *The $C^\infty(M)$-module of tensor fields of type (r, s) on M is canonically isomorphic to the $C^\infty(M)$-module of $C^\infty(M)$-multilinear maps*

$$\underbrace{\Omega^1(M) \times \dots \times \Omega^1(M)}_{r \text{ factors}} \times \underbrace{\mathfrak{X}(M) \times \dots \times \mathfrak{X}(M)}_{s \text{ factors}} \to C^\infty(M).$$

2.3 The Exterior Derivative

Recall that $\Lambda^0(M^*) = M \times \mathbb{R}$, so a smooth section of this bundle is a map $M \to M \times \mathbb{R}$ of the form $p \mapsto (p, f(p))$ where $f \in C^\infty(M)$. This shows that $\Omega^0(M) \cong C^\infty(M)$. Furthermore, we have seen that the differential of $f \in C^\infty(M)$ can be viewed as a differential 1-form $df \in \Omega^1(M)$, so we have an operator $C^\infty(M) \to \Omega^1(M)$, $f \mapsto df$.

In this section, we extend this operator to an operator $d : \Omega^\bullet(M) \to \Omega^\bullet(M)$, called *exterior derivative*, mapping $\Omega^k(M)$ to $\Omega^{k+1}(M)$ for all $k \geq 0$. It so happens that d plays an *extremely* important rôle in the theory of smooth manifolds.

2.3.1 Theorem *There exists a unique $\mathbb{R}$-linear operator $d : \Omega^\bullet(M) \to \Omega^\bullet(M)$ with the following properties:*

a. $d\left(\Omega^k(M)\right) \subset \Omega^{k+1}(M)$ *for all* $k \geq 0$ *(d has* degree $+1$*);*

b. $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ *for every* $\omega \in \Omega^k(M)$, $\eta \in \Omega^\ell(M)$ *(d is an* anti-derivation*);*

c. $d^2 = 0$*;*

d. df *is the differential of* f *for every* $f \in C^\infty(M) \cong \Omega^0(M)$.

Proof We start with uniqueness, so let d be as in the statement. The first case is when M is a coordinate neighborhood $(U, x_1, \ldots, x_n)$. Then any $\omega \in \Omega^k(U)$ can be written as $\omega = \sum_I a_I dx_I$, where I runs over increasing multi-indices $(i_1, \ldots, i_k)$ and $a_I \in C^\infty(U)$, and we get

$$\begin{aligned} d\omega &= \sum_I d(a_I\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}) \quad \text{(by } \mathbb{R}\text{-linearity)} \\ &= \sum_I d(a_I) \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k} \\ &\quad + \sum_{r=1}^{k} (-1)^{r-1}\, a_I\, dx_{i_1} \wedge \cdots \wedge d(dx_{i_r}) \wedge \cdots \wedge dx_{i_k} \quad \text{(by (b))} \qquad (2.3.2) \\ &= \sum_I \sum_{r=1}^{n} \frac{\partial a_I}{\partial x_r} dx_r \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k} \quad \text{(by (c) and (d).)} \end{aligned}$$

Next we go to the case of a general manifold M and show that d is a *local operator*, in the sense that $(d\omega)|_U = 0$ whenever $\omega|_U = 0$ and U is an open subset of M. So assume $\omega|_U = 0$, take an arbitrary point $p \in U$, and choose $\lambda \in C^\infty(M)$ such that $0 \leq \lambda \leq 1$, λ is flat equal to 1 on $M \setminus U$ and has support disjoint from $\bar{V}$, where V is a neighborhood of p with $\bar{V} \subset U$. Then $\omega = \lambda\omega$ on the entire M, so using (b) we get

$$(d\omega)_p = d(\lambda\omega)_p = d\lambda_p \wedge \underbrace{\omega_p}_{=0} + \underbrace{\lambda(p)}_{=0} d\omega_p = 0,$$

as wished.

To continue, we verify that d induces an operator d_U on $\Omega^\bullet(U)$ satisfying (a)–(d) for every open subset U of M. So given $\omega \in \Omega^k(U)$ and $p \in U$, construct $\tilde{\omega} \in \Omega^k(M)$ which coincides with ω on a neighborhood V of p with $\bar{V} \subset U$, as usual by means of a bump function, and define $(d_U\omega)_p := (d\tilde{\omega})_p$. The definition is independent of the chosen extension, as d is a local operator. It is easy to check that d_U indeed satisfies (a)–(d); for instance, for (b), note that $\tilde{\omega} \wedge \tilde{\eta}$ is an extension of $\omega \wedge \eta$ and hence $d_U(\omega \wedge \eta)_p = (d(\tilde{\omega} \wedge \tilde{\eta}))_p = (d\tilde{\omega})_p \wedge \tilde{\eta}_p + (-1)^{\deg\tilde{\omega}} \tilde{\omega}_p \wedge (d\tilde{\eta})_p = (d_U\omega)_p \wedge \eta_p + (-1)^{\deg\omega} \omega_p \wedge (d_U\eta)_p$.

Note also that the collection $\{d_U\}$ is *natural with respect to restrictions*, in the sense that if $U \subset V$ are open subsets of M, then $d_V|_U = d_U$.

Finally, for $\omega \in \Omega^\bullet(M)$ and a coordinate neighborhood $(U, x_1, \ldots, x_n)$, on the one hand $d_U(\omega|_U)$ is uniquely defined by formula (2.3.2). On the other hand, ω itself is an extension of $\omega|_U$, and hence $(d\omega)_p = (d_U(\omega|_U))_p$ for every $p \in U$. This proves that $d\omega$ is uniquely defined.

To prove existence, we first use formula (2.3.2) to define an $\mathbb{R}$-linear operator d_U on $\Omega^k(U)$ for every coordinate neighborhood U of M. It is clear that d_U satisfies (a) and (d); let us prove that it also satisfies (b) and (c). So let $\omega = \sum_I a_I dx_I \in \Omega^k(U)$. Then $d_U\omega = \sum_I da_I \wedge dx_I$ and

$$\begin{aligned} d_U^2\omega &= \sum_{I,r} d_U\left(\frac{\partial a_I}{\partial x_r} dx_r \wedge dx_I\right) \\ &= \sum_{I,r,s} \frac{\partial^2 a_I}{\partial x_s \partial x_r} dx_s \wedge dx_r \wedge dx_I \\ &= 0, \end{aligned}$$

since $\dfrac{\partial^2 a_I}{\partial x_s \partial x_r}$ is symmetric and $dx_s \wedge dx_r$ is skew-symmetric in r, s. Let also $\eta = \sum_J b_J dx_J$. Then $\omega \wedge \eta = \sum_{I,J} a_I b_J dx_I \wedge dx_J$ and

$$\begin{aligned} d_U(\omega \wedge \eta) &= \sum_{I,J} d_U(a_I b_J dx_I \wedge dx_J) \\ &= \sum_{I,J,r} \frac{\partial a_I}{\partial x_r} b_J dx_r \wedge dx_I \wedge dx_J + \sum_{I,J,s} a_I \frac{\partial b_J}{\partial x_s} dx_s \wedge dx_I \wedge dx_J \\ &= \left(\sum_{I,r} \frac{\partial a_I}{\partial x_r} dx_r \wedge dx_I\right) \wedge \left(\sum_J b_J dx_J\right) \\ &\qquad + (-1)^{|I|}\left(\sum_I a_I dx_I\right) \wedge \left(\sum_{J,s} \frac{\partial b_J}{\partial x_s} dx_s \wedge dx_J\right) \\ &= d_U\omega \wedge \eta + (-1)^{\deg\omega} \omega \wedge d_U\eta, \end{aligned}$$

where we have used Proposition 2.1.11 in the third equality to write $dx_s \wedge dx_I = (-1)^{|I|} dx_I \wedge dx_s$.

We finish by noting that the operators d_U for each coordinate system U of M can be pieced together to define a global operator d. Indeed for two coordinate systems U and V, the operators d_U and d_V induce two operators on $\Omega^\bullet(U \cap V)$ satisfying (a)–(d) by the remarks above which must coincide by the uniqueness part. Note also that the resulting d satisfies (a)–(d) since it locally coincides with some d_U. □

2.3.3 Remark We have constructed the exterior derivative d as an operator between sections of vector bundles which, locally, is such that the local coordinates of $d\omega$ are linear combinations of partial derivatives of the local coordinates of ω (cf. 2.3.2). For this reason, d is called a *differential operator* (compare Remark 1.6.7).

Pull-Back

A nice feature of differential forms is that they can always be pulled-back under a smooth map. In contrast, the push-forward of a vector field under a smooth map need not exist if the map is not a diffeomorphism.

Let $f : M \to N$ be a smooth map. The differential $df_p : T_pM \to T_{f(p)}N$ at a point p in M has a transpose map $(df_p)^* : T_{f(p)}N^* \to T_pM^*$ and there is an induced algebra homomorphism $\delta f_p := \Lambda((df_p)^*) : \Lambda(T_{f(p)}N^*) \to \Lambda(T_pM^*)$ (cf. Problem 6). A differential form ω on N is a section of $\Lambda(N^*)$, and the composite map $\omega \circ f : M \to \Lambda(N^*)$ is a *differential form along* f. The *pull-back of* ω *under* f is the section of $\Lambda(M^*)$ given by

$$(f^*\omega)_p = \delta f_p(\omega_{f(p)}) \tag{2.3.4}$$

for all $p \in M$; we prove below that $f^*\omega$ is smooth, so that it is in fact a differential form on M. In particular, if ω is a k-form, then $(f^*\omega)_p \in \Lambda^k(T_pM^*) = \Lambda^k(T_pM)^* = A_k(T_pM)$ and

$$(f^*\omega)_p(v_1, \ldots, v_k) = \omega_{f(p)}(df_p(v_1), \ldots, df_p(v_k)) \tag{2.3.5}$$

for all $v_1, \ldots, v_k \in T_pM$.

2.3.6 Exercise Let $f : M \to N$ be a smooth map.

a. In the case of 0-forms, that is, smooth functions, check that $f^*(g) = g \circ f$ for all $g \in \Omega^0(N) = C^\infty(N)$.

b. In the case $\omega = dg \in \Omega^1(N)$ for some $g \in C^\infty(N)$, check that $f^*(dg) = d(g \circ f)$.

■

2.3.7 Proposition *Let $f : M \to N$ be a smooth map. Then:*

a. $f^* : \Omega^\bullet(N) \to \Omega^\bullet(M)$ *is a homomorphism of algebras;*

b. $d \circ f^* = f^* \circ d$*;*

c. $(f^*\omega)(X_1, \ldots, X_k)(p) = \omega_{f(p)}(df(X_1(p)), \ldots, df(X_k(p)))$ *for all* $\omega \in \Omega^k(N)$ *and all* $X_1, \ldots, X_k \in \mathfrak{X}(M)$.

2

Proof Result (c) follows from (2.3.5). The fact that f^* is compatible with vector space operations and the wedge product can be checked pointwise; then one can refer to Eq. (2.3.4) and note that $\delta f_p = \Lambda(df_p^*)$ is an algebra homomorphism according to Problem 6(b). For (a), it only remains to prove that $f^*\omega$ is a *smooth* section of $\Lambda(M^*)$, for a differential form $\omega \in \Omega^\bullet(N)$. So let $p \in M$, choose a coordinate system $(V, y_1, \ldots, y_n)$ of N around $f(p)$ and a neighborhood U of p in M with $f(U) \subset V$. Since f^* is linear, we may assume that ω is a k-form. As ω is smooth, we can write

$$\omega|_V = \sum_I a_I dy_{i_1} \wedge \cdots \wedge dy_{i_k}$$

where $a_I \in C^\infty(V)$. It follows from Exercise 2.3.6 and part (a) that

$$f^*\omega|_U = \sum_I (a_I \circ f)\, d(y_{i_1} \circ f) \wedge \cdots \wedge d(y_{i_k} \circ f), \tag{2.3.8}$$

which indeed is a smooth form on U. Finally, (b) is proved using (2.3.8):

$$\begin{aligned} d(f^*\omega)_p &= d\left(\sum_I (a_I \circ f)\, d(y_{i_1} \circ f) \wedge \cdots \wedge d(y_{i_k} \circ f)\right)\Big|_p \\ &= \sum_I \big(d(a_I \circ f) \wedge d(y_{i_1} \circ f) \wedge \cdots \wedge d(y_{i_k} \circ f)\big)\,|_p \\ &= f^*\left(\sum_I da_I \wedge dy_{i_1} \wedge \cdots \wedge dy_{i_k}\right)\Big|_p \\ &= f^*(d\omega)_p, \end{aligned}$$

as desired. □

2.4 The Lie Derivative of Tensors

In ▶ Sect. 1.6, we defined the Lie derivative of a smooth vector field Y on M with respect to another smooth vector field X by using the flow $\{\varphi_t\}$ of X to identify different tangent spaces of M along an integral curve of X. The same idea can be used to define the Lie derivative of a differential form ω or a tensor field S with respect to X. The main point is to understand the action of $\{\varphi_t\}$ on the spaces of differential forms and tensor fields.

So let $\{\varphi_t\}$ denote the flow of a vector field X on M and let ω be a differential form on M. Then the pull-back $\varphi_t^*\omega$ is a differential form and $t \mapsto (\varphi_t^*\omega)_p$ is a smooth curve in $\Lambda(T_pM^*)$, for all $p \in M$ (compare argument for vector fields on page 32). The *Lie derivative* of ω with respect to X is the section $L_X\omega$ of $\Lambda(M^*)$ given by

$$(L_X\omega)_p = \frac{d}{dt}\Big|_{t=0} (\varphi_t^*\omega)_p. \tag{2.4.1}$$

We prove below that $L_X\omega$ is smooth, so it indeed yields a differential form on M. In view of (2.3.5), it is clear that the Lie derivative preserves the degree of a differential form.

We extend the definition of Lie derivative to an arbitrary tensor field S of type (r, s) as follows. Suppose

$$S_{\varphi_t(p)} = v_1 \otimes \cdots \otimes v_r \otimes v_1^* \otimes \cdots \otimes v_s^*.$$

We define $(\varphi_t^* S)_p \in (T_pM)^{r,s}$ to be

$$d\varphi_{-t}(v_1) \otimes \cdots \otimes d\varphi_{-t}(v_r) \otimes d\varphi_t^*(v_1^*) \otimes \cdots \otimes d\varphi_t^*(v_s^*)$$

and put

$$(L_X S)_p = \frac{d}{dt}\Big|_{t=0} (\varphi_t^* S)_p. \tag{2.4.2}$$

One can view Definition 2.4.1 as the operator in the quotient obtained from Definition 2.4.2 in the sense that the exterior algebra is a subquotient of the tensor algebra.

Before stating properties of the Lie derivative, it is convenient to introduce two more operators. For $X \in \mathfrak{X}(M)$ and $\omega \in \Omega^{k+1}(M)$ with $k \geq 0$, the *interior multiplication* $\iota_X\omega \in \Omega^k(M)$ is the k-differential form given by

$$\iota_X\omega(X_1, \ldots, X_k) = \omega(X, X_1, \ldots, X_k)$$

for $X_1, \ldots, X_k \in \mathfrak{X}(M)$; we also define ι_X to be zero on 0-forms.

2.4.3 Exercise Prove that $\iota_X\omega$ is indeed a *smooth* section of $\Lambda^{k-1}(M^*)$ for $\omega \in \Omega^k(M)$. Prove also that ι_X is an *anti-derivation*, in the sense that

$$\iota_X(\omega \wedge \eta) = \iota_X\omega \wedge \eta + (-1)^k \omega \wedge \iota_X\eta$$

for $\omega \in \Omega^k(M)$ and $\eta \in \Omega^\ell(M)$. (Hint: For the last assertion, it suffices to check the identity at one point.) ■

Let V be a vector space. The *contraction* $c_{i,j} : V^{r,s} \to V^{r-1,s-1}$ is the linear map that operates on pure tensors as

$$\begin{aligned} &v_1 \otimes \cdots \otimes v_r \otimes v_1^* \otimes \cdots \otimes v_s^* \\ &\qquad \mapsto v_j^*(v_i)\; v_1 \otimes \cdots \otimes \hat{v_i} \otimes \cdots \otimes v_r \otimes v_1^* \otimes \cdots \hat{v_j^*} \otimes \cdots \otimes v_s^*. \end{aligned}$$

It is easy to see that $c_{i,j}$ extends to a map $\mathcal{T}^{r,s}(M) \to \mathcal{T}^{r-1,s-1}(M)$.

2.4.4 Exercise Let V be a vector space. Recall the canonical isomorphism $V^{1,1} \cong \mathrm{End}(V)$ from Proposition 2.1.5, where $\mathrm{End}(V)$ denotes the space of endomorphisms of V. Check that $c_{1,1} : V^{1,1} \to V^{0,0}$ is the trace map $\mathrm{tr} : \mathrm{End}(V) \to \mathbb{R}$. ■

2.4.5 Proposition *Let X be a smooth vector field on M. Then:*

a. $L_X f = X(f)$ *for all* $f \in C^\infty(M)$.

b. $L_X Y = [X, Y]$ *for all* $Y \in \mathfrak{X}(M)$.

c. L_X *is a type-preserving* $\mathbb{R}$*-linear operator on the space* $\mathcal{T}(M)$ *of tensor fields on* M.

d. $L_X : \mathcal{T}(M) \to \mathcal{T}(M)$ *is a derivation, in the sense that*

$$L_X(S \otimes S') = (L_X S) \otimes S' + S \otimes (L_X S').$$

e. $L_X : \mathcal{T}(M) \to \mathcal{T}(M)$ *commutes with contractions:*

$$L_X(c(S)) = c(L_X S)$$

for any contraction $c : \mathcal{T}^{r,s}(M) \to \mathcal{T}^{r-1,s-1}(M)$.

f. L_X *is a degree-preserving* $\mathbb{R}$*-linear operator on the space of differential forms* $\Omega^\bullet(M)$ *which is a derivation and commutes with exterior differentiation.*

g. $L_X = \iota_X \circ d + d \circ \iota_X$ *on* $\Omega^\bullet(M)$ *(*Cartan's magical formula*).*

Moreover:

h. For $\omega \in \Omega^k(M)$ *and* $X_0, \ldots, X_k \in \mathfrak{X}(M)$*, we have*

$$L_{X_0}\omega(X_1, \ldots, X_k) = X_0(\omega(X_1, \ldots, X_k)) - \sum_{i=1}^{k} \omega(X_1, \ldots, X_{i-1}, [X_0, X_i], X_{i+1}, \ldots, X_k).$$

i. Same assumption as in (h), we have

$$d\omega(X_0, \ldots, X_k) = \sum_{i=0}^{k} (-1)^i X_i \omega(X_0, \ldots, \hat{X}_i, \ldots, X_k) + \sum_{i<j} (-1)^{i+j} \omega([X_i, X_j], X_0, \ldots, \hat{X}_i, \ldots, \hat{X}_j, \ldots, X_k).$$

Proof (a) follows from differentiation of $(\varphi_t^* f)_p = f(\varphi_t(p))$ at $t = 0$. (b) was proved in ▶ Sect. 1.6. The type-preserving part of (c) is clear from the definition. For (d), differentiate the obvious formula $\varphi_t^*(S \otimes S')|_p = (\varphi_t^* S)_p \otimes (\varphi_t^* S')_p$ at $t = 0$; the derivation property follows using the fact that tensor multiplication $\otimes : (T_pM)^{r,s} \times (T_pM)^{r',s'} \to (T_pM)^{r+r',s+s'}$ is $\mathbb{R}$-bilinear. Smoothness of $L_X S$ as a section of $T^{r,s}(M)$ is proved noting that L_X is a local operator and expressing $L_X S$ in a system of local coordinates, see below for the analogous argument in the case of differential forms. This covers (c) and (d).

(e) follows from the easily checked fact that φ_t^* commutes with contractions. As a consequence, which we will use below, if $\omega \in \Omega^1(M)$ and $Y \in \mathfrak{X}(M)$, then $\omega(Y) = c(Y \otimes \omega)$ so

$$\begin{aligned} X(\omega(Y)) &= L_X(c(Y \otimes \omega)) \qquad \text{(using (a))} \\ &= c(L_X(Y \otimes \omega)) \\ &= c(L_X Y \otimes \omega + Y \otimes L_X \omega) \qquad \text{(using (d))} \\ &= \omega([X, Y]) + L_X \omega(Y) \qquad \text{(using (b));} \end{aligned}$$

in other words,

$$L_X \omega(Y) = X(\omega(Y)) - \omega([X, Y]). \tag{2.4.6}$$

For (f), we first remark that L_X is a derivation as a map from $\Omega^\bullet(M)$ to non-necessarily smooth sections of $\Lambda(M^*)$: this is a pointwise check, and follows from the fact that φ_t^* defines an automorphism of the algebra $\Omega^\bullet(M)$. Next, check that L_X commutes with d on functions, using (2.4.6):

$$\begin{aligned} L_X(df)(Y) &= X(df(Y)) - df([X, Y]) \\ &= X(Y(f)) - [X, Y](f) \\ &= Y(X(f)) \\ &= d(X(f))(Y) \\ &= d(L_X f)(Y) \end{aligned}$$

for all $f \in C^\infty(M)$ and $Y \in \mathfrak{X}(M)$. To continue, note that L_X is a local operator: formula (2.4.1) shows that $L_X \omega|_U$ depends only on $\omega|_U$, for any open subset U of M. Finally, to see that $L_X \omega$ is smooth for any $\omega \in \Omega^\bullet(M)$, we may assume that ω has degree k and work on a coordinate system $(U, x_1, \ldots, x_n)$, where ω has a local representation as in (2.2.4). Using the above collected facts:

$$\begin{aligned} L_X \omega|_U = &\sum_{i_1 < \cdots < i_k} X(a_{i_1 \ldots i_k})\, dx_{i_1} \wedge \cdots \wedge dx_{i_k} \\ &+ \sum_{i_1 < \cdots < i_k} \sum_{j=1}^{k} a_{i_1 \ldots i_k} dx_{i_1} \wedge \cdots \wedge d(X(x_{i_j})) \wedge \cdots \wedge dx_{i_k}, \end{aligned} \tag{2.4.7}$$

which is indeed a smooth form on U. This formula can also be used to show that L_X commutes with d, in general.

To prove (g), let $P_X = d \circ \iota_X + \iota_X \circ d$. Then P_X and L_X are local operators, derivations of $\Omega^\bullet(M)$, coincide on functions and commute with d. Since any differential form is locally a sum of wedge products of functions and differentials of functions, it follows that $L_X = P_X$.

The case $k = 1$ in (h) is formula (2.4.6). The proof for $k > 1$ is analogous. One can also work locally; then the formula in (h) is equivalent to (2.4.7).

Finally, (i) is proved by induction on k. The initial case $k = 0$ is immediate. Assuming (i) holds for $k - 1$, one proves it holds for k by starting with (h) and using (g) and the induction hypothesis. □

2.4.8 Exercise Carry out the calculations to prove (h) and (i) in Proposition 2.4.5. (Hint: See hints in the proof of the proposition.) ■

2.5 Vector Bundles

The tangent, cotangent, and all tensor bundles we have constructed so far are smooth manifolds of a special kind in that they have a fibered structure over another manifold, each fiber being a vector space. For instance, TM fibers over M so that the fiber over any point p in M is the tangent space T_pM. Moreover, there is some control on how the fibers vary with the point. In case of TM, this is reflected on the way a chart $(\pi^{-1}(U), \tilde{\varphi})$ is constructed from a given chart (U, φ) of M. Recall that $\tilde{\varphi} : \pi^{-1}(U) \to \mathbb{R}^n \times \mathbb{R}^n$ where $\tilde{\varphi}(v) = (\varphi(\pi(v)), d\varphi(v))$. So $\tilde{\varphi}$ induces a diffeomorphism $\cup_{p\in U} T_pM \to \varphi(U) \times \mathbb{R}^n$ such that each fiber T_pM is mapped linearly and isomorphically onto $\{\varphi(p)\} \times \mathbb{R}^n$. We could also compose this map with $\varphi^{-1} \times \mathrm{id}$ to get a diffeomorphism

$$TM|_U := \cup_{p\in U} T_pM \to \varphi(U) \times \mathbb{R}^n \to U \times \mathbb{R}^n.$$

Of course each T_pM is abstractly isomorphic to $\mathbb{R}^n$, where $n = \dim M$, but here we are saying that the part of TM consisting of fibers lying over points in U is diffeomorphic to a product $U \times \mathbb{R}^n$ in such a way that T_pM corresponds to $\{p\} \times \mathbb{R}^n$. This is the idea of a vector bundle.

2.5.1 Definition A (smooth) *vector bundle* of rank k over a smooth manifold M is a smooth manifold E, called the *total space*, together with a smooth projection $\pi : E \to M$, such that:

a. $E_p := \pi^{-1}(p)$ is a vector space of dimension k for all $p \in M$;

b. M can be covered by open sets U such that there exists a diffeomorphism $E|_U = \pi^{-1}(U) \to U \times \mathbb{R}^k$ mapping E_p linearly and isomorphically onto $\{p\} \times \mathbb{R}^k$ for all $p \in U$.

□

The *trivial vector bundle* of rank k over M is the direct product $M \times \mathbb{R}^k$ with the projection onto the first factor. A vector bundle of rank $k = 1$ is also called a *line bundle*.

An equivalent definition of vector bundle, more similar in spirit to the definition of smooth manifold, is as follows.

2.5.2 Definition A (smooth) *vector bundle* of rank k over a smooth manifold M is a set E, called the *total space*, together with a projection $\pi : E \to M$, with the following properties:

a. M admits a covering by open sets U such that there exists a bijection $\varphi_U : E|_U = \pi^{-1}(U) \to U \times \mathbb{R}^k$ satisfying $\pi = \pi_1 \circ \varphi_U$, where $\pi_1 : U \times \mathbb{R}^k \to U$ is the projection onto the first factor. Such a φ_U is called a *local trivialization*.

b. Given local trivializations φ_U, φ_V with $U \cap V \neq \varnothing$, the change of local trivialization or transition function

$$\varphi_U \circ \varphi_V^{-1} : (U \cap V) \times \mathbb{R}^k \to (U \cap V) \times \mathbb{R}^k$$

has the form

$$(x, a) \mapsto (x, g_{UV}(x)a),$$

where

$$g_{UV} : U \cap V \to GL(k, \mathbb{R})$$

is smooth.

□

2.5.3 Exercise Prove that the family of transition functions $\{g_{UV}\}$ in Definition 2.5.2 satisfies the *cocycle conditions*:

$$g_{UU}(x) = \mathrm{id} \qquad (x \in U),$$
$$g_{UV}(x)g_{VW}(x)g_{WU}(x) = \mathrm{id} \qquad (x \in U \cap V \cap W).$$

■

2.5.4 Exercise Let M be a smooth manifold.

a. Prove that, for a vector bundle $\pi : E \to M$ as in Definition 2.5.2, the total space E has a natural structure of smooth manifold such that π is smooth and the local trivializations are diffeomorphisms.

b. Prove that Definitions 2.5.1 and 2.5.2 are equivalent.

■

2.5.5 Example In this example, we construct a very important example of vector bundle which is not a tensor bundle, called the *tautological (line) bundle* over $\mathbb{R}P^n$. Recall that a point p in real projective space $M = \mathbb{R}P^n$ is a 1-dimensional subspace of $\mathbb{R}^{n+1}$ (Example 1.2.9). Set $E = \dot{\cup}_{p \in M} E_p$ where E_p is the subspace of $\mathbb{R}^{n+1}$ corresponding to p, namely, E_p consists of vectors $v \in \mathbb{R}^{n+1}$ such that $v \in p$. Let $\pi : E \to M$ map E_p to p. We will prove that this is a smooth vector bundle by constructing local trivializations and using Definition 2.5.2. Recall the atlas $\{\varphi_i\}_{i=1}^{n+1}$ of Example 1.2.9. Set

$$\tilde{\varphi}_i : \pi^{-1}(U_i) \to U_i \times \mathbb{R} \qquad v \mapsto (\pi(v), x_i(v)).$$

This is a bijection and the cocycle

$$g_{ij}(x_1, \ldots, x_{n+1}) = x_i/x_j \in GL(1, \mathbb{R}) = \mathbb{R} \setminus \{0\}$$

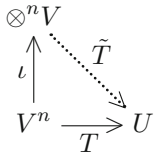
is smooth on $U_i \cap U_j$, as wished. ■

In general, vector bundles are used to describe certain objects or encode additional structures on the base manifolds. Characteristic classes and topological K-theory are powerful tools coming from vector bundles with many applications in Algebraic Topology. To learn more about vector bundles, we refer the reader to [MS74] or [Hat17].

2.6 Problems

► *Section 2.1*

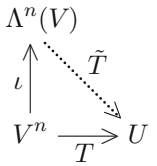
1 Let V be a vector space and let $\iota : V^n \to \otimes^n V$ be defined as $\iota(v_1, \ldots, v_n) = v_1 \otimes \cdots \otimes v_n$, where $V^n = V \times \cdots \times V$ (n factors on the right-hand side). Prove that $\otimes^n(V)$ satisfies the following universal property: for every vector space U and every n-multilinear map $T : V^n \to U$, there exists a unique linear map $\tilde{T} : \otimes^n V \to U$ such that $\tilde{T} \circ \iota = T$.



2 Prove that $\otimes^n V$ is canonically isomorphic to the dual space of the space of n-multilinear forms on V^n. (Hint: Use Problem 1.)

3 Let V be a vector space. An n-multilinear map $T : V^n \to U$ is called *alternating* if $T(v_{\sigma(1)}, \ldots, v_{\sigma(n)}) = (\operatorname{sgn}\sigma) T(v_1, \ldots, v_n)$ for every $v_1, \ldots, v_n \in V$ and every permutation σ of $\{1, \ldots, n\}$, where sgn denotes the sign ± 1 of the permutation.

Let $\iota : V^n \to \Lambda^n(V)$ be defined as $\iota(v_1, \ldots, v_n) = v_1 \wedge \cdots \wedge v_n$. Note that ι is alternating. Prove that $\Lambda^n(V)$ satisfies the following universal property: for every vector space U and every alternating n-multilinear map $T : V^n \to U$, there exists a unique linear map $\tilde{T} : \Lambda^n(V) \to U$ such that $\tilde{T} \circ \iota = T$.



4 Denote the vector space of all alternating multilinear forms $V^n \to \mathbb{R}$ by $A_n(V)$. Prove that $\Lambda^n(V)$ is canonically isomorphic to $A_n(V)^*$. (Hint: Use Problem 3.)

5 Prove that $v_1, \ldots, v_k \in V$ are linearly independent if and only if $v_1 \wedge \cdots \wedge v_k \neq 0$.

6 Let V and W be vector spaces and let $T : V \to W$ be a linear map.

a. Show that T naturally induces a linear map $\Lambda^k(T) : \Lambda^k(V) \to \Lambda^k(W)$ for all $k \geq 0$. (Hint: Use Problem 3.)

b. Show that the maps $\Lambda^k(V)$ for various k induce an algebra homomorphism $\Lambda(T) : \Lambda(V) \to \Lambda(W)$.

Let now $V = W$ and $n = \dim V$. The operator $\Lambda^n(T)$ is multiplication by a scalar, as $\dim \Lambda^n(V) = 1$; define the *determinant* of T to be this scalar.

c. Any $n \times n$ real matrix $A = (a_{ij})$ can be viewed as a linear operator on $\mathbb{R}^n$, simply by left-multiplying column vectors. Prove that

$$\det A = \sum_{\sigma} (\operatorname{sgn} \sigma)\, a_{1,\sigma(i)} \cdots a_{n,\sigma(n)},$$

where $\operatorname{sgn} \sigma$ is the sign of the permutation σ, and σ runs over the set of all permutations of the set $\{1, \ldots, n\}$. Use this formula to check that a square matrix and its transpose have the same determinant.

d. Prove that the determinant of the product of two square matrices is the product of their determinants.

e. Prove that the determinant defines an alternate n-multilinear function on the rows (resp. columns) of a $n \times n$-matrix.

7 Let V be a vector space.

a. Prove that there is a canonical isomorphism

$$\Lambda^k(V^*) \to \Lambda^k(V)^*$$

given by

$$v_1^* \wedge \cdots \wedge v_k^* \mapsto \left(u_1 \wedge \cdots \wedge u_k \mapsto \det(v_i^*(u_j)) \right).$$

b. Use part (a) to show that $\Lambda^k(T)^* = \Lambda^k(T^*)$ for a linear map $T : V \to W$.

c. Let $\alpha, \beta \in V^* \cong \Lambda^1(V^*) \cong A_1(V)$. Show that $\alpha \wedge \beta \in \Lambda^2(V^*)$, viewed as an element of $\Lambda^2(V^*) \cong A_2(V)$, is given by

$$\alpha \wedge \beta\,(u, v) = \alpha(u)\beta(v) - \alpha(v)\beta(u)$$

for all $u, v \in V$.

8 Let V be an Euclidean vector space, that is, a vector space equipped with a positive-definite inner product $\langle , \rangle$. Prove that there is an induced positive-definite inner product on $\Lambda^k(V)$ given by

$$\langle u_1 \wedge \cdots \wedge u_k, v_1 \wedge \cdots \wedge v_k \rangle \mapsto \det\left(\langle v_i, u_j \rangle\right).$$

9 Let V be a vector space.

a. In analogy with the exterior algebra, construct the *symmetric algebra* $Sym(V)$, a commutative graded algebra, as a quotient of $T(V)$.

b. Determine a basis of the homogeneous subspace $Sym^n(V)$.

c. State and prove that $Sym^n(V)$ satisfies a certain universal property.

d. Show that the $Sym^n(V)$ is canonically isomorphic to the dual of the space $S_n(V)$ of symmetric n-multilinear forms $V^n \to \mathbb{R}$.

In view of (d), $Sym(V^*)$ is usually defined to be the space $\mathcal{P}(V)$ of *polynomials* on V.

10 An element of $\Lambda^n(V)$ is called *decomposable* if it lies in the subset $\Lambda^1(V) \wedge \cdots \wedge \Lambda^1(V)$ (n factors).

a. Show that, in general, not every element of $\Lambda^n(V)$ is decomposable.

b. Show that, for $\dim V \leq 3$, every homogeneous element in $\Lambda(V)$ is decomposable.

c. Let ω be a differential form. Is $\omega \wedge \omega = 0$?

11 (Hodge Star) Let V be an oriented vector space equipped with a non-degenerate symmetric bilinear form (we do not require positive-definiteness from the outset). Let $\dim V = n$.

a. Prove there exists an element $\omega \in \Lambda^n(V)$ such that

$$\omega = e_1 \wedge \cdots \wedge e_n$$

for every positively oriented orthonormal basis $\{e_1, \ldots, e_n\}$ of V (here *orthonormal* means that $\langle e_i, e_j \rangle = \pm\delta_{ij}$ (Kronecker delta)).

b. Check that the bilinear form on V induces an isomorphism $V \to V^*$, which induces an isomorphism $\Lambda^k(V) \to \Lambda^k(V^*)$ via Problem 6(a).

c. Show that the bilinear map

$$\Lambda^k(V) \times \Lambda^{n-k}(V) \to \Lambda^n(V), \qquad (\alpha, \beta) \mapsto \alpha \wedge \beta$$

together with the isomorphism

$$\mathbb{R} \to \Lambda^n(V), \qquad a \mapsto a\,\omega$$

define a canonical isomorphism

$$(\Lambda^k(V))^* \to \Lambda^{n-k}(V).$$

d. Combine the isomorphisms of (b) and (c) with that in Problem 7(a) to get a linear isomorphism

$$* : \Lambda^k(V) \to \Lambda^{n-k}(V)$$

for $0 \leq k \leq n$, called the *Hodge star*.

e. Show that

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle \, \omega$$

for all $\alpha, \beta \in \Lambda^k(V)$, where we use the inner product of Problem 8 extended to the case of non-necessarily positive-definite inner products.

f. Prove that $*T = T*$ for every linear isometry $T : V \to T$. (Hint: Use (e).)

g. Assume the inner product is positive definite and let $\{e_1, \ldots, e_n\}$ be a positively oriented orthonormal basis of V. Show that

$$*1 = e_1 \wedge \cdots \wedge e_n, \quad *(e_1 \wedge \cdots \wedge e_n) = 1,$$

and

$$*(e_1 \wedge \cdots \wedge e_k) = e_{k+1} \wedge \cdots \wedge e_n.$$

Show also that

$$** = (-1)^{k(n-k)}$$

on $\Lambda^k(V)$.

► *Section 2.2*

12 (Riemannian Manifolds) Let M be a smooth manifold. A *Riemannian metric* g on M is an assignment of a positive-definite inner product g_p on each tangent space T_pM which is smooth, in the sense that $g(X, Y)(p) = g_p(X(p), Y(p))$ defines a smooth function for every $X, Y \in \mathfrak{X}(M)$. A *Riemannian manifold* is a smooth manifold equipped with a Riemannian metric.

a. Show that a Riemannian metric g on M is the same as a tensor field $\tilde{g}$ of type $(0, 2)$ which is *symmetric*, in the sense that $\tilde{g}(Y, X) = \tilde{g}(X, Y)$ for every $X, Y \in \mathfrak{X}(M)$, with the additional property of positive-definiteness at each point.

b. Fix a local coordinate system $(U, x_1, \ldots, x_n)$ on M.

(i) Let g be a Riemannian metric on M. Show that

$$g|_U = \sum_{i,j} g_{ij} dx_i \otimes dx_j,$$

where $g_{ij} = g(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j}) \in C^\infty(U)$, $g_{ij} = g_{ji}$ and the matrix (g_{ij}) is everywhere positive definite.

(ii) Conversely, given functions $g_{ij} = g_{ji} \in C^\infty(U)$ such that the matrix (g_{ij}) is positive definite everywhere in M, show how to define a Riemannian metric on U.

c. Use part (b)(ii) and a partition of unity to prove that every smooth manifold can be equipped with a Riemannian metric.

d. On a Riemannian manifold M there exists a natural diffeomorphism $TM \approx T^*M$ that maps fibers to fibers and is linear on the fibers. (Hint: There exists a linear isomorphism $v \in T_pM \mapsto g_p(v, \cdot) \in T_pM^*$.)

▶ *Section 2.3*

13 Consider $\mathbb{R}^3$ with coordinates (x, y, z). In each case, decide whether $d\omega = 0$ or there exists η such that $d\eta = \omega$.

a. $\omega = yzdx + xzdy + xydz$.

b. $\omega = xdx + x^2y^2dy + yzdz$.

c. $\omega = 2xy^2dx \wedge dy + zdy \wedge dz$.

14 (The Operator d on $\mathbb{R}^3$) Identify 1- and 2-forms on $\mathbb{R}^3$ with vector fields on $\mathbb{R}^3$, and 0- and 3-forms on $\mathbb{R}^3$ with smooth functions on $\mathbb{R}^3$, and check that:

d on 0-forms is the gradient;
d on 1-forms is the curl;
d on 2-forms is the divergent.

Also, interpret $d^2 = 0$ in those terms.

▶ *Section 2.4*

15 Let M and N be smooth manifolds, where M is connected, and consider the projection $\pi : M \times N \to N$ onto the second factor. Prove that a k-form ω on $M \times N$ is of the form $\pi^*\eta$ for some k-form η on N if and only if $\iota_X\omega = L_X\omega = 0$ for every $X \in \mathfrak{X}(M \times N)$ satisfying $d\pi \circ X = 0$.

16 Let M be a smooth manifold.

a. Prove that $\iota_X\iota_X = 0$ for every $X \in \mathfrak{X}(M)$.

b. Prove that $\iota_{[X,Y]}\omega = L_X\iota_Y\omega - \iota_Y L_X\omega$ for every $X, Y \in \mathfrak{X}(M)$ and $\omega \in \Omega^k(M)$.

▶ *Section 2.5*

17 The *Whitney sum* $E_1 \oplus E_2$ of two vector bundles $\pi_1 : E_1 \to M$, $\pi_2 : E_2 \to M$ is the vector bundle $\pi : E = E_1 \oplus E_2 \to M$ where $E_p = (E_1)_p \oplus (E_2)_p$ for all $p \in M$.

a. Show that $E_1 \oplus E_2$ is indeed a vector bundle by expressing its local trivializations in terms of those of E_1 and E_2 and checking the conditions of Definition 2.5.2.

b. Similarly, construct the tensor product bundle $E_1 \otimes E_2$ and the dual bundle E^*.

Lie Groups



Lie groups are amongst the most important examples of smooth manifolds. A Lie group is a smooth manifold with an additional, compatible structure of group. Here compatibility refers to the fact that the group operations are smooth (another point of view is to regard a Lie group as a group with an additional structure of manifold...). The reader can keep in mind the matrix group $GL(n, \mathbb{R})$ of non-singular $n \times n$ real matrices (Examples 1.2.7(ix) and 3.1.1(vii)) in which the n^2 matrix coefficients form a global coordinate system. The conjunction of the smooth and the group structures allows one to give a more explicit description of the differential invariants attached to a manifold. For this reason, Lie groups form a class of manifolds suitable for testing general hypotheses and conjectures. The same remarks apply to homogeneous spaces, which are certain quotients of Lie groups. In this chapter, we introduce the fundamental notions about Lie groups, including a discussion of Lie's fundamental theorems in modern formulation based on Frobenius' theorem.

3.1 Basic Definitions and Examples

A *Lie group* G is a smooth manifold endowed with a group structure such that the group operations are smooth. More concretely, the multiplication map $\mu : G \times G \to G$ and the inversion map $\iota : G \to G$ are required to be smooth.

3.1.1 Examples

(i) The Euclidean space $\mathbb{R}^n$ with its additive vector space structure is a Lie group. Since the multiplication is commutative, this is an example of an *Abelian* (or *commutative*) Lie group.

(ii) The multiplicative group of non-zero complex numbers $\mathbb{C}^\times$ is a Lie group. The subgroup of unit complex numbers is also a Lie group and, as a smooth manifold, it is diffeomorphic to the circle S^1. These are also Abelian Lie groups.

(iii) If G and H are Lie groups, the direct product group structure turns the product manifold $G \times H$ into a Lie group.

(iv) It follows from (ii) and (iii) that the *n-torus* $T^n = S^1 \times \cdots \times S^1$ (n times) is a Lie group. Of course, T^n is a compact connected Abelian Lie group. Conversely, we will see in Theorem 3.5.3 that every compact connected Abelian Lie group is an n-torus.

(v) If G is a Lie group, the connected component of the identity of G, denoted by G°, is also a Lie group. Indeed, G° is open in G, so it inherits a smooth structure from G just by restricting the local charts. Since $\mu(G^\circ \times G^\circ)$ is connected and $\mu(1, 1) = 1$,

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we must have $\mu(G^\circ \times G^\circ) \subset G^\circ$. Similarly, $\iota(G^\circ) \subset G^\circ$. Since $G^\circ \subset G$ is an open submanifold, it follows that the group operations restricted to G° are smooth.

(vi) Any finite or countable group endowed with the discrete topology becomes a 0-dimensional Lie group. Such examples are called *discrete Lie groups*.

(vii) We now turn to some of the classical matrix groups. The general linear group $GL(n, \mathbb{R})$ is a Lie group since each entry of the product of two matrices is a quadratic polynomial on the entries of the two matrices, and each entry of the inverse of a non-singular matrix is a rational function on the entries of the matrix.

Similarly, one defines the *complex general linear group of order n*, which is denoted by $GL(n, \mathbb{C})$, as the group consisting of all non-singular $n \times n$ complex matrices, and checks that it is a Lie group. Note that $\dim GL(n, \mathbb{C}) = 2n^2$ and $GL(1, \mathbb{C}) = \mathbb{C}^\times$.

We have already encountered the orthogonal group $O(n)$ as a closed embedded submanifold of $GL(n, \mathbb{R})$ in 1.4.14. Since $O(n)$ is an embedded submanifold, it follows from Theorem 1.4.9 that the group operations of $O(n)$ are smooth, and hence $O(n)$ is a Lie group.

Similarly to $O(n)$, one checks that

$$SL(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) \mid \det(A) = 1\} \quad (\textit{real special linear group}),$$

$$SL(n, \mathbb{C}) = \{A \in GL(n, \mathbb{C}) \mid \det(A) = 1\} \quad (\textit{complex special linear group}),$$

$$U(n) = \{A \in GL(n, \mathbb{C}) \mid AA^* = I\} \quad (\textit{unitary group}),$$

$$SO(n) = \{A \in O(n) \mid \det(A) = 1\} \quad (\textit{special orthogonal group}),$$

$$SU(n) = \{A \in U(n) \mid \det(A) = 1\} \quad (\textit{special unitary group})$$

are Lie groups, where A^* denotes the complex conjugate transpose matrix of A. It is easy to see that $O(n)$, $U(n)$, $SO(n)$, and $SU(n)$ are compact, as they are closed and bounded in $M(n, \mathbb{R})$. Note also that $U(1) = S^1$.

(viii) Let V be an n-dimensional real or complex vector space and consider the group $GL(V)$ of invertible operators on V. A choice of basis of V determines a diffeomorphism of $GL(V)$ with $GL(n, \mathbb{R})$ (resp. $GL(n, \mathbb{C})$). It follows that $GL(V)$ inherits a Lie group structure.

■

Lie Algebras

For an arbitrary smooth manifold M, the space $\mathfrak{X}(M)$ of smooth vector fields on M is an infinite-dimensional vector space over $\mathbb{R}$. In addition, we have already encountered the Lie bracket, a bilinear map $[\cdot, \cdot] : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$ satisfying:

a. $[Y, X] = -[X, Y]$;

b. $[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0$ (Jacobi identity);

for every $X, Y \in \mathfrak{X}(M)$. In general, a vector space with a bilinear operation satisfying (a) and (b) above is called a *Lie algebra*. So $\mathfrak{X}(M)$ is a Lie algebra over $\mathbb{R}$.

It turns out in case of a Lie group G, we can single out a finite-dimensional subalgebra of $\mathfrak{X}(M)$. For that purpose, let us first introduce translations in G. The *left translation* defined by $g \in G$ is the map $L_g : G \to G$, $L_g(x) = gx$. It is a smooth map and indeed a diffeomorphism of G, its inverse being given by $L_{g^{-1}}$. Similarly, the *right translation* defined by $g \in G$ is the map $R_g : G \to G$, $R_g(x) = xg$. It is also a diffeomorphism of G and its inverse is given by $R_{g^{-1}}$.

The translations in G define canonical identifications between the tangent spaces to G at different points. For instance, $dL_g : T_hG \to T_{gh}G$ is an isomorphism for every g, $h \in G$. This allows us to consider invariant tensors, the most important case being that of vector fields. A vector field X on G is called *left-invariant* if $d(L_g)_x(X_x) = X_{gx}$ for every $g, x \in G$. This condition is simply $dL_g \circ X = X \circ L_g$ for every $g \in G$; equivalently, X is L_g-related to itself, or yet $L_{g*}X = X$ (since L_g is a diffeomorphism), for all $g \in G$. We can similarly define *right-invariant* vector fields, but most often we will be considering the left-invariant variety. Note that left-invariance and right-invariance are the same property in case of an Abelian group.

3.1.2 Lemma *Every left invariant vector field X in G is smooth.*

Proof Let f be a smooth function defined on a neighborhood of g in G, and let $\gamma : (-\epsilon, \epsilon) \to G$ be a smooth curve with $\gamma(0) = 1$ and $\gamma'(0) = X_1$. Then the value of X on f is given by

$$X_g(f) = dL_g(X_1)(f) = X_1(f \circ L_g) = \frac{d}{dt}\Big|_{t=0} f(g\gamma(t)) = \frac{d}{dt}\Big|_{t=0} f \circ \mu(g, \gamma(t)),$$

and hence, it is a smooth function of g. □

Let $\mathfrak{g}$ denote the set of left-invariant vector fields on G. It follows that $\mathfrak{g}$ is a vector subspace of $\mathfrak{X}(M)$. Further, $\mathfrak{g}$ is a subalgebra of $\mathfrak{X}(M)$ for, given $X, Y \in \mathfrak{g}$, we have by Proposition 1.6.20 that

$$L_{g*}[X, Y] = [L_{g*}X, L_{g*}Y] = [X, Y],$$

for every $g \in G$. Finally, we explain why $\mathfrak{g}$ is finite-dimensional: the map $X \in \mathfrak{g} \mapsto X_1$ defines a linear isomorphism between $\mathfrak{g}$ and the tangent space to G at the identity, T_1G, since any left-invariant vector field is completely defined by its value at the identity.

The discussion above shows that with any Lie group G is naturally associated a (real) finite-dimensional Lie algebra $\mathfrak{g}$ of the same dimension as G, consisting of the left-invariant vector fields on G. This Lie algebra is an infinitesimal object associated with G and, as we shall see, completely determines its local structure. Also, it is often convenient to view $\mathfrak{g}$ as T_1G equipped with the Lie algebra structure such that the

evaluation map $\mathfrak{g} \to T_1 G$ is a Lie algebra isomorphism, and at times we shall follow this practice without further comment.

3.1.3 Examples (The Lie Algebras of Some Known Lie Groups). (i) The left-invariant vector fields on $\mathbb{R}^n$ are precisely the constant vector fields, namely linear combinations of the coordinate vector fields (in the canonical coordinate system) with constant coefficients. The bracket of two constant vector fields on $\mathbb{R}^n$ is zero. It follows that the Lie algebra of $\mathbb{R}^n$ is $\mathbb{R}^n$ itself with the null bracket. In general, a vector space equipped with the null bracket is called an *Abelian* Lie algebra.

(ii) The Lie algebra of the direct product $G \times H$ is the direct sum of Lie algebras $\mathfrak{g} \oplus \mathfrak{h}$, where the bracket is taken componentwise.

(iii) Owing to the skew-symmetry of the Lie bracket, every one-dimensional Lie algebra is Abelian. In particular, the Lie algebra of S^1 is Abelian. It follows from (ii) that also the Lie algebra of T^n is Abelian.

(iv) G and G° have canonically isomorphic Lie algebras. In fact the differential of the inclusion $\iota : G^\circ \to G$ defines a linear isomorphism $T_1 G^\circ \cong T_1 G$, and the corresponding left-invariant vector fields on G° and G are ι-related.

(v) The Lie algebra of a discrete group is $\{0\}$.

3.1.4 Examples (Some Abstract Lie Algebras). (i) Let A be any real associative algebra and set $[a, b] = ab - ba$ for $a, b \in A$. It is easy to see that A becomes a Lie algebra.

(ii) The cross-product $\times$ on $\mathbb{R}^3$ is easily seen to define a Lie algebra structure.

(iii) If V is a two-dimensional vector space and $X, Y \in V$ are linearly independent, the conditions $[X, X] = [Y, Y] = 0$, $[X, Y] = X$ define a Lie algebra structure on V.

(iv) If V is a three-dimensional vector space spanned by X, Y, Z, the conditions $[X, Y] = Z$, $[Z, X] = [Z, Y] = 0$ define a Lie algebra structure on V, called the *(3-dimensional) Heisenberg algebra*. It can be realized as a Lie algebra of smooth vector fields on $\mathbb{R}^3$ as in Examples 1.6.17(ii).

3.1.5 Exercise Check the assertions of Examples 3.1.3 and 3.1.4. ■

3.2 The Exponential Map

For a Lie group G, we have constructed its most basic invariant, its Lie algebra $\mathfrak{g}$. Our next step will be to present the fundamental map that relates G and $\mathfrak{g}$, namely the exponential map $\exp : \mathfrak{g} \to G$.

Matrix Exponential

Recall that the exponential of a matrix $A \in M(n, \mathbb{R})$ (or $M(n, \mathbb{C})$) is given by the formula:

$$e^A = I + A + \frac{1}{2}A^2 + \frac{1}{3!}A^3 + \cdots$$
$$= \sum_{k=0}^{\infty} \frac{1}{k!}A^k.$$

Since $||\frac{1}{k!}A^k|| \leq \frac{1}{k!}||A||^k$ and $\sum_{k=0}^{\infty} \frac{1}{k!}||A||^k = e^{||A||}$ for all $k \geq 0$, the series is absolutely convergent on the entire $M(n, \mathbb{R})$ and uniformly convergent on compact subsets; here $||\cdot||$ denotes the operator norm in $M(n, \mathbb{R}) \cong \mathrm{End}(\mathbb{R}^n)$. In case $n = 1$, we recover the usual exponential map on the line. In general, note that:

a. $e^0 = I$;
b. $e^{A+B} = e^A e^B$ if A and B commute.

Indeed, to check (b) notice that one can compute the product of e^A and e^B by multiplying the individual terms and rearranging, by absolute convergence. In particular:

c. $e^{(s+t)A} = e^{sA}e^{tA}$ for all $s, t \in \mathbb{R}$;
d. e^A is invertible and $(e^A)^{-1} = e^{-A}$.

View $t \in \mathbb{R} \mapsto e^{tA}$ as a curve in $M(n, \mathbb{R})$. The last property worth mentioning is

e. $\frac{d}{dt}\big|_{t=t_0} e^{tA} = A\, e^{t_0 A} = e^{t_0 A}\, A$.

This is proved via differentiation term by term, again using absolute convergence.

Flow of Left-Invariant Vector Fields

Let G be a Lie group and let $\mathfrak{g}$ denote its Lie algebra.

3.2.1 Proposition *Every left-invariant vector field is complete.*

Proof Given $X \in \mathfrak{g}$, there exists a maximal integral curve $\gamma_X : (a, b) \to G$ of X with $0 \in (a, b)$ $(a, b \in [-\infty, \infty])$ and $\gamma_X(0) = 1$; namely, $\gamma_X'(t) = X_{\gamma_X(t)}$. Since

$$\frac{d}{dt}\Big|_{t=t_0} L_g(\gamma_X(t)) = d(L_g)(X_{\gamma_X(t_0)}) = X_{L_g(\gamma_X(t_0))},$$

we have that $L_g \circ \gamma_X$ is an integral curve of X starting at g. In particular, if $b < \infty$, by taking $g = \gamma(s)$ with s very close to b, this shows that γ_X can be extended beyond b, leading to a contradiction. Similarly, one sees that $a = -\infty$. Hence X is complete. □

Now the integral curve γ_X of any $X \in \mathfrak{g}$ starting at the identity is defined on $\mathbb{R}$. The *exponential map* of G is the map $\exp : \mathfrak{g} \to G$ defined by $\exp X = \gamma_X(1)$.

Note that $\frac{d}{ds}\big|_{s=s_0} \gamma_X(ts) = t\gamma_X'(ts_0) = tX(\gamma_X(ts_0))$. This implies $\gamma_X(ts) = \gamma_{tX}(s)$ for all $s, t \in \mathbb{R}$ and therefore

$$\gamma_X(t) = \gamma_{tX}(1)$$

$$= \exp(tX), \tag{3.2.2}$$

namely every integral curve of a left-invariant vector field through the identity factors over the exponential map.

3.2.3 Exercise Check that the flow $\{\varphi_t\}$ of a left-invariant vector field X is given by $\varphi_t = R_{\exp tX}$ (recall that R_g denotes a right translation). What is the corresponding result for right-invariant vector fields? ■

Moreover, we state

3.2.4 Proposition *The exponential map* $\exp : \mathfrak{g} \to G$ *is smooth and it is a local diffeomorphism at* 0.

Proof Recall that $\exp X$, for $X \in \mathfrak{g}$, is the value at time 1 of the integral curve of X starting at the identity of G. One can view the functions $\tilde{a}_i$ in (1.6.8) as smoothly depending on $X \in \mathfrak{g}$, and then smoothness of exp follows from the smooth dependence on parameters of solutions of ODE's [Hur70, Theorem 10, p. 31]. Alternatively, consider the vector field $V_{(g,X)} = (X_g, 0)$ on $G \times \mathfrak{g}$. It is clearly left-invariant, hence smooth and complete. According to Exercise 3.2.3, its flow is given by $\Xi(t, g, X) = (g \exp tX, X)$. In particular, $\Xi(1, 1, X) = (\exp X, X)$ is well-defined and smooth on $X \in \mathfrak{g}$, and so is $\exp X$.

Moreover, $d \exp_0 : T_0\mathfrak{g} \cong \mathfrak{g} \to T_1 G \cong \mathfrak{g}$ is the identity, since

$$d \exp_0(X) = \frac{d}{dt}\Big|_{t=0} \exp(tX) = \gamma_X'(0) = X.$$

Thus, exp is a diffeomorphism from a neighborhood of 0 in $\mathfrak{g}$ onto a neighborhood of 1 in G by the inverse mapping theorem (1.3.8). □

Recall that the identity component G° is an open subgroup of G.

3.2.5 Proposition *G° is generated as a group by any neighborhood U of* 1 *in G°, namely*

$$G^\circ = \bigcup_{n \geq 1} U^n,$$

where U^n denotes the set of n-fold products of elements in U. In particular, G° is generated by $\exp[\mathfrak{g}]$.

Proof By replacing U by $U \cap U^{-1}$, we may assume that $U = U^{-1}$. Define $V = \cup_{n \geq 1} U^n$ (note that V is a subgroup of G°) and consider the relation in G° given by $g \sim g'$ if and only if $g^{-1}g' \in V$. Note that this is an equivalence relation and the equivalence classes (which

coincide with the left cosets of V in G°) are open, as $g' \sim g$ implies $g'U \sim g$, where $g'U$ is an open neighborhood of g'. Hence $V = G^\circ$. □

The Case of $GL(n, \mathbb{R})$

Recall that $G = GL(n, \mathbb{R})$ inherits its manifold structure as an open subset of the Euclidean space $M(n, \mathbb{R})$. In particular, the tangent space at the identity $T_I G = M(n, \mathbb{R})$. Let $A \in M(n, \mathbb{R})$ and denote by $\tilde{A} \in \mathfrak{g}$ the corresponding left-invariant vector field on G. For any $g \in G$, we have $\tilde{A}_g = (dL_g)(A) = gA$ (matrix multiplication on the right-hand side).

Using property (e) of the matrix exponential,

$$\frac{d}{dt}\Big|_{t=t_0} e^{tA} = e^{t_0 A} A = \tilde{A}_{e^{t_0 A}},$$

shows that $t \mapsto e^{tA}$ is the integral curve of $\tilde{A}$ through the identity, namely

$$\exp \tilde{A} = e^A$$

for all $A \in M(n, \mathbb{R})$.

Finally, to determine the Lie bracket in $\mathfrak{g}$, we resort to (1.6.24). Let $A, B \in M(n, \mathbb{R})$, denote by $\tilde{A}$, $\tilde{B}$ the corresponding left-invariant vector fields on G and let $\{\varphi_t = R_{e^{tA}}\}$ be the flow of $\tilde{A}$ (cf. Exercise 3.2.3):

$$\begin{aligned}
[A, B] &= [\tilde{A}, \tilde{B}]_I \\
&= (L_{\tilde{A}} \tilde{B})_I \\
&= \frac{d}{dt}\Big|_{t=0} d\varphi_{-t}(\tilde{B}_{\varphi_t(I)}) \\
&= \frac{d}{dt}\Big|_{t=0} e^{tA} B e^{-tA} \\
&= AB - BA.
\end{aligned}$$

Note that the Lie algebra structure in $M(n, \mathbb{R})$ is induced from its associative algebra structure as in Examples 3.1.4(i). The space $M(n, \mathbb{R})$ with this Lie algebra structure will be denoted by $\mathfrak{gl}(n, \mathbb{R})$.

The case of $GL(n, \mathbb{C})$ is completely analogous. In the case of the Lie group $GL(V)$, where V is a finite-dimensional real or complex vector space, the Lie algebra consists of the space $\mathrm{End}(V)$ of all linear operators or endomorphisms of V with the Lie bracket $[L_1, L_2] = L_1 \circ L_2 - L_2 \circ L_1$. The space $\mathrm{End}(V)$ with this Lie algebra structure will be denoted by $\mathfrak{gl}(V)$.

3.3 Homomorphisms and Lie Subgroups

3

A *(Lie group) homomorphism* between Lie groups G and H is map $\varphi : G \to H$ which is both a group homomorphism and a smooth map. φ is called an *isomorphism* if, in addition, it is a diffeomorphism. An *automorphism* of a Lie group is an isomorphism of the Lie group with itself. In the cases $H = GL(n, \mathbb{R})$ or $GL(n, \mathbb{C})$, or $H = GL(V)$ for some vector space V, a homomorphism $\varphi : G \to H$ is called a *representation* of G. A *(Lie algebra) homomorphism* between Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$ is a linear map $\Phi : \mathfrak{g} \to \mathfrak{h}$ which preserves brackets. Φ is called an *isomorphism* if, in addition, it is bijective. An *automorphism* of a Lie algebra is an isomorphism of the Lie algebra with itself. In the cases $\mathfrak{h} = \mathfrak{gl}(n, \mathbb{R})$ or $\mathfrak{gl}(n, \mathbb{C})$, or $\mathfrak{h} = \mathfrak{gl}(V)$ for some vector space, a homomorphism $\Phi : \mathfrak{g} \to \mathfrak{h}$ is called a *representation* of $\mathfrak{g}$.

3.3.1 Exercise For a homomorphism $\varphi : G \to H$, check that $L_{\varphi(g)} \circ \varphi = \varphi \circ L_g$ for all $g \in G$. ■

A homomorphism $\varphi : G \to H$ between Lie groups induces a linear map $d\varphi_1 : T_1G \to T_1H$ and hence a linear map $d\varphi : \mathfrak{g} \to \mathfrak{h}$. Indeed, if X is a left invariant vector field on G, let X' be the unique left-invariant vector field on H such that $X'_1 = d\varphi_1(X_1)$ and put $d\varphi(X) = X'$.

3.3.2 Proposition *If $\varphi : G \to H$ is a homomorphism between Lie groups, then $d\varphi : \mathfrak{g} \to \mathfrak{h}$ is a homomorphism between the corresponding Lie algebras.*

Proof Let $X \in \mathfrak{g}$. We first claim that X and $X' := d\varphi(X)$ are φ-related. In fact,

$$X'_{\varphi(g)} = d(L_{\varphi(g)})_1(X'_1) = d(L_{\varphi(g)} \circ \varphi)_1(X_1) = d(\varphi \circ L_g)_1(X_1) = d\varphi_g(X_g),$$

proving the claim. Now, if $Y \in \mathfrak{g}$, then Y and $d\varphi(Y)$ are φ-related. Therefore $[X, Y]$ and $[d\varphi(X), d\varphi(Y)]$ are φ-related and thus

$$d\varphi([X, Y]_1) = [d\varphi(X), d\varphi(Y)]_{\varphi(1)},$$

or

$$d\varphi([X, Y]) = [d\varphi(X), d\varphi(Y)].$$

This shows that $d\varphi$ is a Lie algebra homomorphism. □

Let G be a Lie group. A *Lie subgroup* of G is an immersed submanifold (H, φ) of G such that H is a Lie group and $\varphi : H \to G$ is a homomorphism.

3.3.3 Remark Similarly to the case of immersed submanifolds (Problem 19 in ▶ Chap. 1), we consider two Lie subgroups (H_1, φ_1) and (H_2, φ_2) of G to be equivalent if there exists a

Lie group isomorphism $\alpha : H_1 \to H_2$ such that $\varphi_1 = \varphi_2 \circ \alpha$. This is an equivalence relation in the class of Lie subgroups of G and each equivalence class contains a unique representative of the form (A, ι), where A is a subset of G (an actual subgroup) and $\iota : A \to G$ is the inclusion. So we lose no generality in assuming that a Lie subgroup of G is an abstract subgroup H of G which is an immersed submanifold of G and a Lie group with respect to the operations induced from G; namely, the multiplication and inversion in G must restrict to smooth maps $H \times H \to H$ and $H \to H$, respectively.

3.3.4 Example The skew-line $(\mathbb{R}, f)$ in T^2 (Example 1.4.2) is an example of a Lie subgroup of T^2 which is not closed. ■

If $\mathfrak{g}$ is a Lie algebra, a subspace $\mathfrak{h}$ of $\mathfrak{g}$ is called a *Lie subalgebra* if $\mathfrak{h}$ is closed under the bracket of $\mathfrak{g}$.

Let H be a Lie subgroup of G, say, $\iota : H \to G$ is the inclusion map. Since ι is an immersion, $d\iota : \mathfrak{h} \to \mathfrak{g}$ is an injective homomorphism of Lie algebras, and we will view $\mathfrak{h}$ as a Lie subalgebra of $\mathfrak{g}$. Conversely, as our most important application of Frobenius' theorem, we have

3.3.5 Theorem (Lie) *Let G be a Lie group and let $\mathfrak{g}$ denote its Lie algebra. If $\mathfrak{h}$ is a Lie subalgebra of $\mathfrak{g}$, then there exists a unique connected Lie subgroup H of G such that the Lie algebra of H is $\mathfrak{h}$.*

Proof We have that $\mathfrak{h}$ is a subspace of $\mathfrak{g}$ and so defines a subspace $\mathfrak{h}(1) := \{X(1) \mid X \in \mathfrak{h}\}$ of $T_1 G$. Let $\mathcal{D}$ be the left-invariant distribution on G defined by $\mathfrak{h}(1)$, namely $\mathcal{D}_g = dL_g(\mathfrak{h}(1))$ for all $g \in G$. Then $\mathcal{D}$ is a smooth distribution, as it is globally generated by left-invariant vector fields $X_1, \ldots, X_k$ in $\mathfrak{h}$. The fact that $\mathcal{D}$ is involutive follows from (and is equivalent to) $\mathfrak{h}$ being a Lie subalgebra of $\mathfrak{g}$. In fact, suppose X and Y lie in $\mathcal{D}$ over the open subset U of G. Write $X = \sum_i a_i X_i$, $Y = \sum_j b_j X_j$ for $a_i, b_j \in C^\infty(U)$. Then

$$[X, Y] = \sum_{i,j} a_i b_j [X_i, X_j] + a_i X_i(b_j) X_j - b_j X_j(a_i) X_i$$

also lies in $\mathcal{D}$, as $[X_i, X_j] \in \mathfrak{h}$.

By Frobenius' theorem (1.7.10), there is an integral foliation $\mathcal{F}$ of $\mathcal{D}$. Denote by H the leaf of $\mathcal{F}$ through 1. Since $\mathcal{D}$ is left-invariant, for every $h \in H$, $L_{h^{-1}}(H) = h^{-1}H$ is also leaf of $\mathcal{F}$, and it passes through $h^{-1}h = 1$. This implies $h^{-1}H = H$, by uniqueness. It follows that H is a subgroup of G. The operations induced by G on H are smooth because H is an initial submanifold, due to Proposition 1.7.3. This proves that H is a Lie group. Its Lie algebra is $\mathfrak{h}$ because $\mathfrak{h}$ consists precisely of the elements of $\mathfrak{g}$ whose value at 1 lies in $\mathcal{D}_1 = T_1 H$, and these are exactly the elements of the Lie algebra of H.

Suppose now H' is another connected Lie subgroup of G with Lie algebra $\mathfrak{h}$. Then H' must be an integral manifold of $\mathcal{D}$ through 1. By Corollary 1.7.7, H' is an open submanifold of H and contains a neighborhood of 1 in H. Owing to Proposition 3.2.5, $H' = H$. □

3.3.6 Corollary *There is a bijective correspondence between connected Lie subgroups of a Lie group and subalgebras of its Lie algebra.*

3.3.7 Example Let G be a Lie group. A subgroup H of G which is an embedded submanifold of G is a Lie subgroup of G by Proposition 1.4.9. It follows from Examples 1.4.14(ii) that $O(n)$ is a (closed) Lie subgroup of $GL(n,\mathbb{R})$. Similarly, the other matrix groups listed in Examples 3.1.1(vii) are (closed) embedded submanifolds and hence Lie subgroups of $GL(n,\mathbb{R})$ or $GL(n,\mathbb{C})$. In particular, the Lie bracket in those subgroups is given by $[A,B] = AB - BA$. ■

3.3.8 Exercise Show that the Lie algebras of the matrix groups listed in Examples 3.1.1(vii) are, respectively, as follows:

$$\begin{aligned}
\mathfrak{o}(n) &= \{A \in \mathfrak{gl}(n,\mathbb{R}) \mid A + A^t = 0\},\\
\mathfrak{sl}(n,\mathbb{R}) &= \{A \in \mathfrak{gl}(n,\mathbb{R}) \mid \operatorname{trace}(A) = 0\},\\
\mathfrak{sl}(n,\mathbb{C}) &= \{A \in \mathfrak{gl}(n,\mathbb{C}) \mid \operatorname{trace}(A) = 0\},\\
\mathfrak{u}(n) &= \{A \in \mathfrak{gl}(n,\mathbb{C}) \mid A + A^* = 0\},\\
\mathfrak{so}(n) &= \mathfrak{o}(n),\\
\mathfrak{su}(n) &= \{A \in \mathfrak{u}(n) \mid \operatorname{trace}(A) = 0\}.
\end{aligned}$$

■

A Lie group homomorphism $\gamma : \mathbb{R} \to G$ is called a *(smooth) one-parameter subgroup*. Note that such a γ is the integral curve of $X := d\gamma(1) \in \mathfrak{g}$, and we have seen in (3.2.2) that

$$\gamma(t) = \exp(tX) \tag{3.3.9}$$

for all $t \in \mathbb{R}$.

More generally, let $\varphi : G \to H$ be a homomorphism between Lie groups. Then, for a left-invariant vector field X on G, $t \mapsto \varphi(\exp^G(tX))$ is a one-parameter subgroup of H with $\frac{d}{dt}\big|_{t=0}\varphi(\exp^G tX) = d\varphi(X_1)$. In view of fact (3.3.9),

$$\varphi \circ \exp^G X = \exp^H \circ d\varphi(X), \tag{3.3.10}$$

for every $X \in \mathfrak{g}$. In particular, if K is a Lie subgroup of G, then the inclusion map $\iota : K \to G$ is a Lie group homomorphism, so that the exponential map of G restricts to the exponential map of K, and the connected component of K is generated by $\exp^G[\mathfrak{k}]$, where $\mathfrak{k}$ is the Lie algebra of K. It also follows that

$$d\iota[\mathfrak{k}] = \{X \in \mathfrak{g} : \exp^G(tX) \in \iota(K), \text{ for all } t \in \mathbb{R}\}. \tag{3.3.11}$$

Indeed, let $X \in \mathfrak{g}$ with $\exp^G(tX) \in \iota(K)$ for all $t \in \mathbb{R}$. Since K is an integral manifold of an involutive distribution (compare Theorem 3.3.5), $t \mapsto \exp^G(tX)$ defines a smooth map $\varphi : \mathbb{R} \to K$, and thus a one-parameter subgroup of K, such that $\exp^G(tX) = \iota \circ \varphi(t)$ for all $t \in \mathbb{R}$. Owing to (3.3.9), $\varphi(t) = \exp^K(tX')$ for some $X' \in \mathfrak{k}$. Therefore $\exp^G(tX) = \iota \circ \exp^K(tX') = \exp^G(td\iota(X'))$, and hence $X = di(X')$.

3.4 Covering Lie Groups

Let G be a connected Lie group. Consider the universal covering $\pi : \tilde{G} \to G$. By Problem 5 in ▶ Chap. 1 or the results in Appendix A, $\tilde{G}$ has a unique smooth structure for which π is a local diffeomorphism.

3.4.1 Theorem *Every connected Lie group G has a simply connected covering $\pi : \tilde{G} \to G$ such that $\tilde{G}$ is a Lie group and π is a Lie group homomorphism.*

Proof Consider the smooth map $\alpha : \tilde{G} \times \tilde{G} \to G$ given by $\alpha(\tilde{g}, \tilde{h}) = \pi(\tilde{g})\pi(\tilde{h})^{-1}$. Choose $\tilde{1} \in \pi^{-1}(1)$. As $\tilde{G}$ is simply connected, so is $\tilde{G} \times \tilde{G}$. By the lifting criterion A.3.1, there exists a unique smooth map $\tilde{\alpha} : \tilde{G} \times \tilde{G} \to \tilde{G}$ such that $\pi \circ \tilde{\alpha} = \alpha$ and $\tilde{\alpha}(\tilde{1}, \tilde{1}) = \tilde{1}$. Put

$$\tilde{g}^{-1} := \tilde{\alpha}(\tilde{1}, \tilde{g}) \quad \text{and} \quad \tilde{g}\tilde{h} := \tilde{\alpha}(\tilde{g}, \tilde{h}^{-1}),$$

for $\tilde{g}, \tilde{h} \in \tilde{G}$. These operations are shown to make $\tilde{G}$ into a group by use of the uniqueness part in the lifting criterion. As an example,

$$\pi(\tilde{g}\tilde{1}) = \pi\tilde{\alpha}(\tilde{g}, \tilde{1}^{-1}) = \alpha(\tilde{g}, \tilde{1}^{-1}) = \pi(\tilde{g})\pi(\tilde{1}^{-1})^{-1} = \pi(\tilde{g}) \tag{3.4.2}$$

since $\tilde{1}^{-1} = \tilde{\alpha}(\tilde{1}, \tilde{1}) = \tilde{1}$ and $\pi(\tilde{1}) = 1$. Identity (3.4.2) shows that $\tilde{g} \mapsto \tilde{g}\tilde{1}$ is a lifting of $\tilde{g} \mapsto \pi(\tilde{g})$, $\tilde{G} \to G$, to a map $\tilde{G} \to \tilde{G}$ which takes $\tilde{1}$ to $\tilde{1} \cdot \tilde{1} = \tilde{\alpha}(\tilde{1}, \tilde{1}^{-1}) = \tilde{\alpha}(\tilde{1}, \tilde{1}) = \tilde{1}$. However, the identity map of $\tilde{G}$ is also a lifting of $\tilde{g} \mapsto \pi(\tilde{g})$ which takes $\tilde{1}$ to $\tilde{1}$. By uniqueness, both liftings coincide and $\tilde{g}\tilde{1} = \tilde{g}$ for all $\tilde{g} \in \tilde{G}$.

Now $\tilde{G}$ is a group. Since $\tilde{\alpha}$ is smooth, $\tilde{G}$ is a Lie group. Finally,

$$\pi(\tilde{g}^{-1}) = \pi\tilde{\alpha}(\tilde{1}, \tilde{g}) = \alpha(\tilde{1}, \tilde{g}) = \pi(\tilde{1})\pi(\tilde{g})^{-1} = \pi(\tilde{g})^{-1}$$

and

$$\pi(\tilde{g}\tilde{h}) = \pi\tilde{\alpha}(\tilde{g}, \tilde{h}^{-1}) = \alpha(\tilde{g}, \tilde{h}^{-1}) = \pi(\tilde{g})\pi(\tilde{h}^{-1})^{-1} = \pi(\tilde{g})\pi(\tilde{h}).$$

Hence, $\pi : \tilde{G} \to G$ is a Lie group homomorphism. □

3.4.3 Remark It follows from Lemma 3.4.4(c) and Theorem 3.4.5 below that the structure of Lie group on the universal covering $\tilde{G}$ of G is unique, up to isomorphism.

3.4.4 Lemma *Let $\varphi : G \to H$ be a homomorphism between Lie groups. Consider the induced homomorphism between the corresponding Lie algebras $d\varphi : \mathfrak{g} \to \mathfrak{h}$. Then:*

a. $d\varphi$ is injective if and only if the kernel of φ is discrete.

b. $d\varphi$ is surjective if and only if $\varphi(G^\circ) = H^\circ$.

c. $d\varphi$ is bijective if and only if φ is a smooth covering (here we assume G and H connected). In this case, the group of deck transformations is isomorphic to $\ker\varphi$.

Proof (a) If $d\varphi : \mathfrak{g} \to \mathfrak{h}$ is injective, then φ is an immersion at 1 and thus everywhere by Exercise 3.3.1. Therefore φ is locally injective and hence $\ker\varphi$ is discrete. Conversely, if $d\varphi : \mathfrak{g} \to \mathfrak{h}$ is not injective, $\ker d\varphi_g$ is of constant, positive dimension for all $g \in G$ and thus defines a smooth distribution $\mathcal{D}$. Note that X lies in $\mathcal{D}$ if and only if X is φ-related to the null vector field on H. It follows that $\mathcal{D}$ is involutive. The leaf of the integral foliation of $\mathcal{D}$ through the identity is collapsed to a point under φ, implying that $\ker\varphi$ is not discrete.

(b) Since $\varphi \circ \exp = \exp \circ d\varphi$ and G° is generated by $\exp[\mathfrak{g}]$, we have that $\varphi(G^\circ)$ is the subgroup of H° generated by $\exp[d\varphi(\mathfrak{g})]$; thus $\varphi(G^\circ) = H^\circ$ if $d\varphi$ is surjective. On the other hand, if $d\varphi$ is not surjective, $d\varphi(\mathfrak{g})$ is a proper subalgebra of $\mathfrak{h}$ to which there corresponds a connected, proper subgroup K of H°, and $\exp[d\varphi(\mathfrak{g})]$ generates K.

(c) Assume G, H connected. If φ is a covering, then $\ker d\varphi$ is discrete and φ is surjective, so $d\varphi$ is an isomorphism by (a) and (b). Conversely, suppose that $d\varphi : \mathfrak{g} \to \mathfrak{h}$ is an isomorphism. Then φ is surjective by (b). Let U be a connected neighborhood of 1 in G such that $\varphi : U \to \varphi(U) =: V$ is a diffeomorphism. We can choose U so that $U \cap \ker d\varphi = \{1\}$, by (a). Then $\varphi^{-1}(V) = \cup_{n\in\ker\varphi} nU$, and this is a disjoint union, for $ng = n'g'$ with $n, n' \in \ker\varphi$ and $g, g' \in U$ implies $gg'^{-1} = n^{-1}n' \in \ker\varphi$, and so $\varphi(g) = \varphi(g')$, hence $g = g'$ and $n = n'$. Since $\varphi \circ L_n = \varphi$ for $n \in \ker\varphi$, we also have that $\varphi|nU$ is a diffeomorphism onto V. This shows that V is an evenly covered neighborhood of 1. Now hV is an evenly covered neighborhood of any given $h \in H$, and thus φ is a covering.

For the last assertion, note that the assignment $n \mapsto R_{n^{-1}}$ defines an injective homomorphism from $\ker\varphi$ to the group of deck transformations, which is onto because there is an element in the image mapping $1 \in G$ to any given point in the fiber $\pi^{-1}(1) = \ker\varphi$.

□

3.4.5 Theorem (Lie) *Let G_1, G_2 be Lie groups and assume that G_1 is connected and simply connected. Then, given a homomorphism $\Phi : \mathfrak{g}_1 \to \mathfrak{g}_2$ between the corresponding Lie algebras, there exists a unique homomorphism $\varphi : G_1 \to G_2$ such that $d\varphi = \Phi$.*

Proof The graph of Φ, $\mathfrak{h} = \{(X, \Phi(X)) : X \in \mathfrak{g}_1\}$, is a subalgebra of $\mathfrak{g}_1 \oplus \mathfrak{g}_2$. Let H be the subgroup of $G_1 \times G_2$ defined by $\mathfrak{h}$ (Theorem 3.3.5). Consider the projections

$$\Phi_i : \mathfrak{g}_1 \oplus \mathfrak{g}_2 \to \mathfrak{g}_i, \qquad \varphi_i : G_1 \times G_2 \to G_i,$$

for $i = 1, 2$. Since $\Phi_1|_\mathfrak{h} : \mathfrak{h} \to \mathfrak{g}_1$ is an isomorphism, we have that $\Phi = \Phi_2 \circ (\Phi_1|_\mathfrak{h})^{-1}$ and $\varphi_1|_H : H \to G_1$ is a covering. Since G_1 is simply connected, $\varphi_1|_H : H \to G_1$ is

an isomorphism of Lie groups, and we can thus define $\varphi = \varphi_2 \circ (\varphi_1|_H)^{-1}$. This proves the existence part. The uniqueness part comes from the fact that $d\varphi = \Phi$ specifies φ in a neighborhood of 1 (by using the exponential map as in (3.3.10)) and G_1 is generated by this neighborhood. □

3.5 The Adjoint Representation

Let G be a Lie group and denote its Lie algebra by $\mathfrak{g}$. The noncommutativity of G is organized by the adjoint representation. In order to introduce it, let $g \in G$, and define a map $\mathrm{Inn}_g : G \to G$ by $\mathrm{Inn}_g(x) = gxg^{-1}$. Then Inn_g is an automorphism of G, which is called the *inner automorphism* defined by g. The differential $d(\mathrm{Inn}_g) : \mathfrak{g} \to \mathfrak{g}$ defines an automorphism of $\mathfrak{g}$, which we denote by Ad_g. Then

$$(\mathrm{Ad}_g X)_1 = \frac{d}{dt}\Big|_{t=0} \mathrm{Inn}(g)(\exp tX_1) = \frac{d}{dt}\Big|_{t=0} g \exp tX_1 g^{-1}$$

or $\mathrm{Ad}_g X = \frac{d}{dt}\big|_{t=0} g \exp tXg^{-1}$. This defines a homomorphism

$$\mathrm{Ad} : g \in G \to \mathrm{Ad}_g \in GL(\mathfrak{g}),$$

which is called the *adjoint representation* of G on $\mathfrak{g}$.

3.5.1 Example In case $G = GL(n, \mathbb{R})$, Inn_g is the restriction of the linear map $M(n, \mathbb{R}) \to M(n, \mathbb{R})$, $X \mapsto gXg^{-1}$, so $\mathrm{Ad}_g X = d(\mathrm{Inn}_g)(X) = gXg^{-1}$. ■

We have

$$\begin{aligned}(\mathrm{Ad}_g X)_1 &= (dL_g)(dR_{g^{-1}})X_1 \\ &= (dR_{g^{-1}})(dL_g)X_1 \\ &= (dR_{g^{-1}})(X_g) \\ &= (dR_g^{-1} \circ X \circ R_g)(1) \\ &= \left((R_{g^{-1}})_* X\right)_1\end{aligned}$$

and $R_{g^{-1}*}X$ is left-invariant, so $\mathrm{Ad}_g X = R_{g^{-1}*}X$ for all $g \in G$, $X \in \mathfrak{g}$.

Recall that $GL(\mathfrak{g})$ is itself a Lie group isomorphic to $GL(n, \mathbb{R})$, where $n = \dim \mathfrak{g}$. Its Lie algebra consists of all linear endomorphisms of $\mathfrak{g}$ with the bracket $[A, B] = AB - BA$ and is denoted by $\mathfrak{gl}(\mathfrak{g})$. Note that $\mathrm{Ad}_g = (d_2 F)_{(g,1)}$, where $F : G \times G \to G$ is the smooth function $F(g, x) = gxg^{-1}$, so the linear endomorphism Ad_g of $\mathfrak{g}$ depends smoothly on g. Now $\mathrm{Ad} : g \in G \to \mathrm{Ad}_g \in GL(\mathfrak{g})$ is a homomorphism of Lie groups and its differential $d(\mathrm{Ad})$ defines the *adjoint representation* of $\mathfrak{g}$ on $\mathfrak{g}$:

$$\mathrm{ad} : X \in \mathfrak{g} \to \mathrm{ad}_X = \frac{d}{dt}\Big|_{t=0} \mathrm{Ad}_{\exp tX} \in \mathfrak{gl}(\mathfrak{g}).$$

Since $\varphi_t = R_{\exp tX}$ is the flow of X, we get

$$\mathrm{ad}_X Y = \frac{d}{dt}\Big|_{t=0} \mathrm{Ad}_{\exp tX} Y = \frac{d}{dt}\Big|_{t=0} (R_{\exp(-tX)})_* Y = L_X Y = [X, Y].$$

As an important special case of (3.3.10), we have

$$\begin{aligned} \mathrm{Ad}_{\exp X} &= e^{\mathrm{ad}_X} \\ &= I + \mathrm{ad}_X + \frac{1}{2}\mathrm{ad}_X^2 + \frac{1}{3!}\mathrm{ad}_X^3 + \cdots \end{aligned}$$

for all $X \in \mathfrak{g}$.

3.5.2 Lemma *For given $X, Y \in \mathfrak{g}$, we have that $[X, Y] = 0$ if and only if $\exp X \exp Y = \exp Y \exp X$. In that case, $\exp t(X + Y) = \exp tX \exp tY$ for all $t \in \mathbb{R}$. It follows that a connected Lie group is Abelian if and only if its Lie algebra is Abelian.*

Proof The first assertion is a special case of Proposition 1.6.25 using that $\varphi_t = R_{\exp tX}$ is the flow of X and $\psi_s = R_{\exp sY}$ is the flow of Y. The second one follows from noting that both $t \mapsto \exp(t(X+Y))$ and $t \mapsto \exp tX \exp tY$ are one-parameter groups with initial speed $X + Y$. Finally, we have just seen that $\mathfrak{g}$ is Abelian if and only if any two elements in $\exp[\mathfrak{g}]$ commute, but the latter generates G°. □

3.5.3 Theorem *Every connected Abelian Lie group G is isomorphic to $\mathbb{R}^{n-k} \times T^k$. In particular, a connected and simply connected Abelian Lie group is isomorphic to $\mathbb{R}^n$ and a compact connected Abelian Lie group is isomorphic to T^n.*

Proof It follows from Lemma 3.5.2 that the Lie algebra $\mathfrak{g}$ of G is Abelian and $\exp : \mathfrak{g} \to G$ is a homomorphism, where $\mathfrak{g} \cong \mathbb{R}^n$ as a Lie group; thus exp is a smooth covering by Lemma 3.4.4(c). Hence G is isomorphic to the quotient of $\mathbb{R}^n$ by the discrete group $\ker \exp$, but any discrete subgroup of a vector space is generated by (finitely many) linearly independent vectors [Arn89, Chap. 10, §49]. □

3.6 Homogeneous Manifolds

Let G be a Lie group and let H be a closed subgroup. Consider the set G/H of left cosets of H in G equipped with the quotient topology with respect to the projection $\pi : G \to G/H$. Note also that left multiplication in G induces a map $\lambda : G \times G/H \to G/H$, namely $\lambda(g, xH) = (gx)H$, and that

$$\pi \circ L_g = \lambda_g \circ \pi \tag{3.6.1}$$

for all $g \in G$, where $\lambda_g(p) = \lambda(g, p)$ for $p \in G/H$.

3.6.2 Lemma *A closed Lie subgroup H of a Lie group G must have the induced topology.*

Proof We need to prove that the inclusion map $\iota : H \to G$ is an embedding. Since ι commutes with left translations, it suffices to find an open subset V of H such that the restriction $\iota|_V$ is an embedding into G. By Theorem 3.3.5 and the discussion in ▶ Sect. 1.7, there exists a distinguished chart $(U, \varphi = (x_1, \ldots, x_n))$ of G around 1 such that $H \cap U$ consists of at most countably many plaques, each plaque being a slice of the form

$$x_{k+1} = c_{k+1}, \ \ldots, \ x_n = c_n$$

for some $c_{k+1}, \ldots, c_n \in \mathbb{R}$, where $k = \dim H$. Denote by $\tau : \mathbb{R}^n = \mathbb{R}^k \times \mathbb{R}^{n-k} \to \mathbb{R}^{n-k}$ the projection. Let $\tilde{U}$ be a compact neighborhood of 1 contained in U. Now $H \cap \tilde{U}$ is compact, so $\tau(H \cap \tilde{U})$ is a non-empty closed countable subset of $\mathbb{R}^{n-k}$, which by the Baire category theorem [Mun00, Theorem 48.2] must have an isolated point. This point specifies an isolated plaque V of H in U along which ι is an open mapping and hence a homeomorphism onto its image, as desired. □

3.6.3 Theorem *If G is a Lie group and H is a closed subgroup of G, then there is a unique smooth structure on the topological quotient G/H such that $\lambda : G \times G/H \to G/H$ is smooth. Moreover, $\pi : G \to G/H$ is a surjective submersion and* $\dim G/H = \dim G - \dim H$.

Proof For an open set V of G/H we have that $\pi^{-1}(\pi(V)) = \bigcup_{g \in G} gV$ is a union of open sets and thus open. This shows that π is an open map and hence the projection of a countable basis of open sets of G yields a countable basis of open sets of G/H. To prove that G/H is Hausdorff, we use closedness of H. Indeed it implies that the equivalence relation $\mathcal{R} \subset G \times G$, defined by specifying that $g \sim g'$ if and only if $g^{-1}g' \in H$, is a closed subset of $G \times G$. Now if $gH \neq g'H$ in G/H then $(g, g') \notin \mathcal{R}$ and there exist open neighborhoods W of g and W' of g' in G such that $(W \times W') \cap \mathcal{R} = \varnothing$. It follows that $\pi(W)$ and $\pi(W')$ are disjoint neighborhoods of g and g' in G/H, respectively.

We first construct a local chart of G/H around $p_0 = \pi(1) = 1H$. Recall from Proposition 3.2.4 and (3.3.10) that the exponential map $\exp = \exp^G$ gives a parameterization of G around the identity element and restricts to the exponential map of H. Denote the Lie algebras of G and H by $\mathfrak{g}$ and $\mathfrak{h}$, respectively, and choose a complementary subspace $\mathfrak{m}$ to $\mathfrak{h}$ in $\mathfrak{g}$. We can choose a product neighborhood of 0 in $\mathfrak{g}$ of the form $U_0 \times V_0$, where U_0 is a neighborhood of 0 in $\mathfrak{h}$ and V_0 is a neighborhood of 0 in $\mathfrak{m}$, such that the map

$$f : U_0 \times V_0 \to G, \qquad f(X, Y) = \exp Y \exp X$$

is a diffeomorphism onto its image (by the inverse mapping theorem 1.3.8). Owing to Lemma 3.6.2, H has the topology induced from G, so we may choose a neighborhood W of 1 in G such that $W \cap H = \exp(U_0)$. We also shrink V_0 so that $(\exp V_0)^{-1} \exp V_0 \subset W$. Now we claim that $\pi \circ \exp|_{V_0}$ is injective. Indeed, if $\pi(\exp Y) = \pi(\exp Y')$ for some Y,

$Y' \in V_0$, then $(\exp Y)^{-1} \exp Y' \in H \cap W = \exp(U_0)$, so $\exp Y' = \exp Y \exp X$ for some $X \in U_0$. Since f is injective on $U_0 \times V_0$, this implies that $X = 0$ and $Y' = Y$ and proves the claim. Note $\exp V_0 \exp U_0$ is open in G, so the image $\pi(\exp V_0) = \pi(\exp V_0 \exp U_0)$ is open in G/H. We have shown that $\pi \circ \exp$ defines a homeomorphism from V_0 onto the open neighborhood $V = \pi(\exp V_0)$ of p in G/H, whose inverse can then be used to define a local chart (V, ψ) of G/H around p_0.

Now the collection $\{(V^g, \psi^g)\}_{g \in G}$ defines an atlas of G/H, where $V^g = \lambda_g(V)$ and $\psi^g = \psi \circ \lambda_{g^{-1}}$, and we need to check its smoothness. Suppose $g, g' \in G$ are such that $V^g \cap V^{g'} \neq \varnothing$, and let $p = (g \exp Y)H = (g' \exp Y')H$ be an element there, namely $\psi^g(p) = Y$ and $\psi^{g'}(p) = Y'$. Then $\exp Y' = (g')^{-1} g \exp Y h \in \exp V_0$ for some $h \in H$, so there exists a neighborhood $\tilde{V}_0$ of Y in V_0 such that $(g')^{-1} g (\exp \tilde{V}_0) h \subset \exp V_0$, and thus $\psi^{g'} \circ (\psi^g)^{-1}|_{\tilde{V}_0}$ can be written as the composite map

$$\tau \circ f^{-1} \circ R_h \circ L_{(g')^{-1} g} \circ \exp,$$

where f^{-1} denotes the inverse map of $f : U_0 \times V_0 \to f(U_0 \times V_0)$ and $\tau : \mathfrak{g} \to \mathfrak{m}$ denotes the projection along $\mathfrak{h}$. Hence the change of charts $\psi^{g'} \circ (\psi^g)^{-1}$ is smooth.

The local representation of π in the above charts is τ, namely there is a commutative diagram

$$\begin{array}{ccc} g \exp V_0 \exp U_0 & \xrightarrow{\pi} & V^g \\ {\scriptstyle f^{-1} \circ L_{g^{-1}}} \downarrow & & \uparrow {\scriptstyle \lambda_g \circ \pi \circ \exp} \\ U_0 \times V_0 & \xrightarrow[\tau]{} & V_0 \end{array}$$

which shows that π is a submersion. Similarly, the commutative diagram

$$\begin{array}{ccc} G \times g \exp V_0 & \xrightarrow{\mu} & G \\ {\scriptstyle \mathrm{id} \times \pi} \downarrow & & \downarrow {\scriptstyle \pi} \\ G \times V^g & \xrightarrow[\lambda|_{G \times V^g}]{} & G/H \end{array}$$

(where μ is the multiplication map of G) proves that λ is smooth. The uniqueness of the smooth structure follows from Proposition 3.6.4 below. □

Let M be a smooth manifold and let G be a Lie group. An *action* of G on M is a smooth map $\mu : G \times M \to M$ such that $\mu(1, p) = p$ and $\mu(g, \mu(g', p)) = \mu(gg', p)$ for all $p \in M$ and $g, g' \in G$. For brevity of notation, in case μ is fixed and clear from the context, we will simply write $\mu(g, p) = gp$.

An action of G on M is called *transitive* if for every $p, q \in M$ there exists $g \in G$ such that $gp = q$. In this case, M is called *homogeneous under* G, G*-homogeneous*, or

simply a *homogeneous manifold*. Examples of homogeneous manifolds are given by the quotients G/H, where H is a closed Lie subgroup of G, according to Theorem 3.6.3. Conversely, the next proposition proves that every homogeneous manifold is of this form. For an action of G on M and $p \in M$, the *isotropy group* at p is the subgroup G_p of G consisting of the elements that fix p, namely $G_p = \{ g \in G \mid gp = p \}$. Plainly, G_p is a closed subgroup of G, and so a Lie subgroup of G, owing to Theorem 3.7.1 below.

3.6.4 Proposition *Let $\mu : G \times M \to M$ be a transitive action of a Lie group G on a smooth manifold M. Fix $p_0 \in M$ and let $H = G_{p_0}$ be the isotropy group at p_0. Define a map*

$$f : G/H \to M, \qquad f(gH) = \mu(g, p_0).$$

Then f is well-defined and a diffeomorphism.

Proof It is easy to see that f is well-defined, bijective, and smooth. We can write $f \circ \pi = \omega$, where $\omega : G \to M$ is the "orbit map" $\omega(g) = gp_0$. For $X \in \mathfrak{g}$, we have

$$d\omega_1(X) = \frac{d}{ds}\Big|_{s=0} (\exp sX)p_0 = d(\exp(-sX))\frac{d}{dt}\Big|_{t=s} (\exp tX)p_0,$$

so $X \in \ker d\omega_1$ if and only if $\exp tX \in H$ for all $t \in \mathbb{R}$ if and only if X belongs to the Lie algebra $\mathfrak{h}$ of H, due to (3.3.11). Since $df_{1H} \circ d\pi_1 = d\omega_1$ and $\ker d\pi_1 = \mathfrak{h}$, this implies that f is an immersion at $1H$, and thus an immersion everywhere by the equivariance property $f \circ \lambda_g = \mu_g \circ f$ for all $g \in G$.

This already implies that $\dim G/H \leq \dim M$ and $(G/H, f)$ is a submanifold of M, but the strict inequality cannot hold as f is bijective and the image of a smooth map from a smooth manifold into a smooth manifold of strictly larger dimension has measure zero (see Problem 22). It follows that f is a local diffeomorphism and hence a diffeomorphism. □

3.6.5 Examples (i) Let $\{e_1, \dots, e_n\}$ be the canonical basis of $\mathbb{R}^n$ and view elements of $\mathbb{R}^n$ as column-vectors ($n \times 1$ matrices). Then $GL(n, \mathbb{R})$ acts on $\mathbb{R}^n$ by left-multiplication:

$$GL(n, \mathbb{R}) \times \mathbb{R}^n \to \mathbb{R}^n. \tag{3.6.6}$$

The basis $\{e_i\}$ is orthonormal with respect to the standard scalar product in $\mathbb{R}^n$. The orthogonal group $O(n)$ consists precisely of those elements of $GL(n, \mathbb{R})$ whose action on $\mathbb{R}^n$ preserves the lengths of vectors. In particular, the action (3.6.6) restricts to an action

$$O(n) \times S^{n-1} \to S^{n-1} \tag{3.6.7}$$

which is smooth, since S^{n-1} is an embedded submanifold of $\mathbb{R}^n$. The action (3.6.7) is transitive due to the facts that any unit vector can be completed to an orthonormal basis of $\mathbb{R}^n$ and any two orthonormal bases of $\mathbb{R}^n$ differ by an orthogonal transformation. The isotropy group of (3.6.7) at e_1 consists of transformations that leave the orthogonal

complement $e_1^\perp$ invariant, and indeed any orthogonal transformation of $e_1^\perp \cong \mathbb{R}^{n-1}$ can occur. It follows that the isotropy group is isomorphic to $O(n-1)$ and hence

$$S^{n-1} = O(n)/O(n-1)$$

presents the unit sphere as a homogeneous space, where the diffeomorphism is given by $gO(n-1) \mapsto g(e_1)$. If we use only orientation-preserving transformations on $\mathbb{R}^n$, also the elements of the isotropy group will act by orientation-preserving transformations and hence

$$S^{n-1} = SO(n)/SO(n-1). \tag{3.6.8}$$

(ii) The group $SO(n)$ also acts transitively on the set of lines through the origin in $\mathbb{R}^n$. Besides the orthogonal transformations of $e_1^\perp$, the isotropy group at the line $\mathbb{R}e_1$ now also contains transformations that map e_1 to $-e_1$. It follows that

$$\mathbb{R}P^n = SO(n)/O(n-1),$$

where $O(n-1)$ is identified with the subgroup of $SO(n)$ consisting of matrices of the form

$$\begin{pmatrix} \det A & 0 \\ 0 & A \end{pmatrix},$$

where $A \in O(n-1)$.

(iii) Let $\{e_1, \ldots, e_n\}$ be the canonical basis of $\mathbb{C}^n$. It is a unitary basis with respect to the standard Hermitian inner product in $\mathbb{C}^n$. Similarly to (a), one shows that $U(n)$ and $SU(n)$ act transitively on the set of unit vectors of $\mathbb{C}^n$, namely the sphere S^{2n-1}. More interesting is to consider the set $\mathbb{C}P^{n-1}$ of one-dimensional complex subspaces of $\mathbb{C}^n$. This set is homogeneous under $SU(n)$ and the isotropy group at the line $\mathbb{C}e_1$ consists of matrices of the form

$$\begin{pmatrix} (\det A)^{-1} & 0 \\ 0 & A \end{pmatrix},$$

where $A \in U(n-1)$. It follows from Theorems 3.6.3 and 3.6.4 that $\mathbb{C}P^{n-1}$ is a smooth manifold, called *complex projective space*, and

$$\mathbb{C}P^{n-1} = SU(n)/U(n-1)$$

as a homogeneous manifold.

(iv) Let $\{e_1, \ldots, e_n\}$ be the canonical basis of $\mathbb{R}^n$, and let $V_k(\mathbb{R}^n)$ be the set of orthonormal k-frames in $\mathbb{R}^n$, that is, ordered k-tuples of orthonormal vectors in $\mathbb{R}^n$. There is an action

$$O(n) \times V_k(\mathbb{R}^n) \to V_k(\mathbb{R}^n), \quad g \cdot (v_1, \ldots, v_k) = (gv_1, \ldots, gv_k)$$

which is clearly transitive. The isotropy group at $(e_1, \ldots, e_k)$ is the subgroup of $O(n)$ consisting of matrices of the form

$$\begin{pmatrix} I & 0 \\ 0 & A \end{pmatrix}, \tag{3.6.9}$$

where $A \in O(n-k)$. The resulting homogeneous space

$$V_k(\mathbb{R}^n) = O(n)/O(n-k)$$

is called the *Stiefel manifold* of k-frames in $\mathbb{R}^n$. Note that the restricted action of $SO(n)$ on $V_k(\mathbb{R}^n)$ is also transitive and

$$V_k(\mathbb{R}^n) = SO(n)/SO(n-k).$$

■

More generally, for an action of G on M and $p \in M$, the *orbit* through p is the subset $G(p) = \{ gp \in M \mid g \in M \}$. The orbit map $G \to M$, $g \mapsto gp$, is an immersion and induces a bijection $G/G_p \to G(p)$, so that each orbit can be viewed as a homogeneous manifold, and the set of orbits of the action determines a partition of M into immersed submanifolds.

3.6.10 Proposition *Let G be a Lie group and let H be a closed subgroup of G.*

a. If the homogeneous manifold G/H is connected, then G/G° is isomorphic to a quotient group of H/H°. In particular, G is connected if H has that property.

b. Assume G is connected. If the homogeneous manifold G/H is simply connected, then the fundamental group $\pi_1(G, 1)$ is isomorphic to a quotient group of $\pi_1(H, 1)$. In particular, G is simply connected if H has that property.

Proof (a) The projection $G \to G/H$ is a submersion, and so is its restriction to G°. It follows that its image, which coincides with the G°-orbit through $1H \in G/H$, is open. Since G° is normal in G, also the other orbits $G^\circ \cdot (gH) = (gG^\circ)H \subset G/H$ $(g \in G)$ are open. Now we use the connectedness of G/H to deduce that there is one orbit only, that is, G° is transitive on G/H.

The inclusion $H \to G$ induces a homomorphism $\varphi : H/H^\circ \to G/G^\circ$; we finish this part by showing that it is onto. Let $gG^\circ \in G/G^\circ$. By the result in the previous paragraph, there is $g' \in G^\circ$ mapping $1H$ to $g^{-1}H$, that is, $g'H = g^{-1}H$. Therefore $gg' \in H$, namely $g = hg'^{-1}$ for some $h \in H$. It follows that $gG^\circ = hg'^{-1}G^\circ = hG^\circ$, which shows that gG° belongs to the image of φ, as desired.

(b) Note that H is connected, because $G/H^\circ \to G/H$ is a covering but the base is simply connected, so $H = H^\circ$.

Let $\pi : \tilde{G} \to G$ be the universal covering Lie group. Put $H' = \pi^{-1}(H)$. Then H' is a closed subgroup of $\tilde{G}$, and hence a Lie subgroup (Theorem 3.7.1 below). The

map π induces a continuous bijection $\bar{\pi} : \tilde{G}/H' \to G/H$, which is also open as the composition $\tilde{G} \to G \to G/H$ is open. It follows that $\bar{\pi}$ is a homeomorphism and hence $\tilde{G}/H'$ is simply connected. We deduce that H' is connected. Let $\pi' : \tilde{H} \to H'$ be the universal covering Lie group. Note that $\pi_1(G, 1) \cong \ker \pi \subset H'$ and $\pi_1(H, 1) \cong \ker(\pi \circ \pi') \subset \tilde{H}$. It suffices to show that $\pi' : \ker(\pi \circ \pi') \to \ker \pi$ is onto. But this is obvious: given $h' \in \ker \pi$, take $\tilde{h} \in \tilde{H}$ such that $\pi'(\tilde{h}) = h'$; of course, $\pi \circ \pi'(\tilde{h}) = \pi(h') = 1$, that is, $\tilde{h} \in \ker(\pi \circ \pi')$.

□

3.6.11 Remark The results in Proposition 3.6.10 are part of a more general construction called the *long exact homotopy sequence of a fibration* [Hat02, p. 375].

3.7 Additional Results

In this section, we state some important, additional results about basic Lie theory. We omit the proofs, but give references and add some remarks.

Closed Subgroups

3.7.1 Theorem *Let G be a Lie group and let A be a closed (abstract) subgroup of G. Then A admits a unique manifold structure which makes it into a Lie group; moreover, the topology in this manifold structure must be the relative topology.*

Proof See Lemma 3.6.2 and [War83, 3.42]. □

3.7.2 Corollary *Let $\varphi : G \to H$ be a homomorphism of Lie groups. Then $A = \ker \varphi$ is a closed Lie subgroup of G with Lie algebra $\mathfrak{a} = \ker d\varphi$.*

Proof A is a closed subgroup and hence a Lie subgroup of G by Theorem 3.7.1. The rest follows from (3.3.10) and (3.3.11). □

Continuous Homomorphisms

3.7.3 Theorem *Let $\varphi : G \to H$ be a* continuous *homomorphism between Lie groups. Then φ is smooth.*

Proof See [War83, 3.38]. □

3.7.4 Definition A *topological group* is an abstract group equipped with a topology such that the group operations are continuous maps. □

3.7.5 Corollary *A Hausdorff and second-countable locally Euclidean topological group G can have at most one smooth structure making it into a Lie group.*

Proof Let $[\mathcal{A}]$ and $[\mathcal{B}]$ two such smooth structures on G. The identity map $(G, [\mathcal{A}]) \to (G, [\mathcal{B}])$ is a homomorphism and a homeomorphism, and hence a diffeomorphism by Theorem 3.7.3. □

Hilbert's fifth problem is the fifth mathematical problem posed by David Hilbert in his famous address to the International Congress of Mathematicians in 1900. One (restricted) interpretation of the problem in modern language asks whether a connected Hausdorff second-countable locally Euclidean topological group admits a smooth structure which makes it into a Lie group. In 1952, A. Gleason proved that a connected locally compact topological group satisfying the "no-small subgroups" (NSS) condition is a Lie group, and then immediately afterwards Montgomery and Zippin used Gleason's result to prove inductively that locally Euclidean groups of any dimension satisfy NSS. The two papers appeared together in the same issue of the Annals of Mathematics. Here one says that a topological group satisfies *NSS* if there exists a neighborhood of the identity which contains no subgroups other than the trivial group; compare Problem 17. (Actually, the above is not quite the full story: Gleason assumed a weak form of finite-dimensionality in his original argument that NSS implies Lie, but shortly thereafter Yamabe showed that finite-dimensionality was not needed in the proof.)

Theorem of Ado

Recall that a representation of a Lie algebra $\mathfrak{g}$ is a homomorphism φ from $\mathfrak{g}$ into $\mathfrak{gl}(V)$, where V is a real or complex vector space; if, in addition, φ is injective, it is called a *faithful* representation.

A faithful representation of a Lie algebra $\mathfrak{g}$ can be thought of as a "linear picture" of $\mathfrak{g}$ and allows one to view $\mathfrak{g}$ as a Lie algebra of matrices.

3.7.6 Theorem (Ado) *Every Lie algebra admits a faithful representation.*

Proof See [Var84, 3.17.7]. □

In contrast, not every Lie group admits a faithful representation; e.g. the universal covering Lie group of $SL(2, \mathbb{R})$ [Var84, Chap. 2, Exercise 15].

3.7.7 Theorem (Lie) *There is a bijective correspondence between isomorphism classes of Lie algebras and isomorphism classes of simply connected Lie groups.*

Proof If $\mathfrak{g}$ is a Lie algebra, then $\mathfrak{g}$ is isomorphic to a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{R})$ by Theorem 3.7.6. Owing to Theorem 3.3.5, there is a connected Lie subgroup of $GL(n, \mathbb{R})$ with Lie algebra $\mathfrak{g}$. Due to Theorem 3.4.1 and Lemma 3.4.4(c), there is also a simply connected Lie group with Lie algebra $\mathfrak{g}$. Two simply connected Lie groups with isomorphic Lie algebras are isomorphic in view of Theorem 3.4.5. □

Theorem of Yamabe

3.7.8 Theorem (Yamabe) *An arcwise connected subgroup of a Lie group is a Lie subgroup.*

Proof See [KN96, App. 4] for a proof in case the group is connected by piecewise differentiable curves of class C^1. □

3.7.9 Corollary *Let G be a connected Lie group and let A and B be connected Lie subgroups. Then the subgroup (A, B) generated by the commutators*

$$S = \{aba^{-1}b^{-1} \ : \ a \in A, \ b \in B\}$$

is a Lie subgroup of G. In particular, the commutator of G, (G, G), is a Lie subgroup of G.

Proof As a continuous image of $A \times B$, S is arcwise connected, and so is $T = S \cup S^{-1}$, since $S \cap S^{-1} \ni 1$. As a continuous image of $T \times \cdots \times T$ (n factors), also T^n is arcwise connected and hence so is $(A, B) = \cup_{n\geq 1} T^n$, since it is an increasing union of arcwise connected subsets. The result follows from Yamabe's theorem 3.7.8. □

3.7.10 Example In general, the subgroup (A, B) does not have to be closed for closed connected subgroups A and B of G, even if G is simply connected. Indeed, take G to be the simply connected covering of $SL(4, \mathbb{R})$, and let $\mathfrak{a}$ and $\mathfrak{b}$ be one-dimensional and, respectively, spanned by

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 \end{pmatrix}.$$

Then A and B are closed one-dimensional subgroups isomorphic to $\mathbb{R}$ but their commutator is a dense line in a 2-torus. ■

Where does Lie theory go from this point? We suggest the Killing–Cartan classification of compact Lie groups, the Cartan–Weyl theory of representations of compact Lie groups, and the Peter–Weyl theorem. To learn more about Lie theory, see e.g. [BtD95] or [Kna02]. In a more geometric vein, see the book [Arv03] about homogeneous spaces.

3.8 Problems

► *Section 3.1*

1 Let $\alpha, \beta : (-\epsilon, \epsilon) \to G$ be smooth curves in a Lie group G such that $\alpha(0) = \beta(0) = 1$ and consider $\gamma(t) = \alpha(t)\beta(t)$. Prove that $\dot{\gamma}(0) = \dot{\alpha}(0) + \dot{\beta}(0)$. (Hint: consider the multiplication map $\mu : G \times G \to G$ and show that $d\mu(v, w) = d\mu((v, 0) + (0, w)) = v + w$ for $v, w \in T_1 G$.)

2

a. Show that

$$SO(2) = \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \,:\, a,\ b \in \mathbb{R},\ a^2 + b^2 = 1 \right\}.$$

Deduce that $SO(2)$ is diffeomorphic to S^1.

b. Show that

$$SU(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} \,:\, \alpha,\ \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}.$$

Deduce that $SU(2)$ is diffeomorphic to S^3.

3 Let

$$H^3 = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : \quad x, y, z \in \mathbb{R} \right\}.$$

a. Prove that H^3 is closed under matrix multiplication and it has the structure of a Lie group (the so-called *Heisenberg group*).

b. Show that $A = \frac{\partial}{\partial x}$, $B = \frac{\partial}{\partial y} + x \frac{\partial}{\partial z}$, $C = \frac{\partial}{\partial z}$ are left-invariant vector fields. Compute their Lie brackets.

c. Identify the Lie algebra of H^3 as a space of matrices with the commutator as Lie bracket.

4 In this problem, we classify all real Lie algebras of dimension two and three.

a. Show that a non-Abelian two-dimensional Lie algebra contains two vectors X, Y such that $[X, Y] = X$.

b. For an arbitrary Lie algebra $\mathfrak{g}$, denote by $[\mathfrak{g}, \mathfrak{g}]$ the subspace spanned by all elements of the form $[X, Y]$ for $X, Y \in \mathfrak{g}$. Show that $[\mathfrak{g}, \mathfrak{g}]$ is a subalgebra of $\mathfrak{g}$ (this is called the *derived subalgebra* of $\mathfrak{g}$).

Throughout the remainder of this problem, we let $\mathfrak{g}$ be a three-dimensional Lie algebra. Put $n = \dim[\mathfrak{g}, \mathfrak{g}]$. Note that $n = 0$ if and only if $\mathfrak{g}$ is Abelian.

c. Assume $n = 3$. Fix a non-zero $\omega \in \Lambda^3 \mathfrak{g}$ and show that

$$X \wedge Y \wedge Z = \langle [X, Y], Z \rangle\, \omega$$

for all X, Y, $Z \in \mathfrak{g}$ defines a nondegenerate symmetric bilinear form $\langle , \rangle$ on $\mathfrak{g}$ such that ad_X is skew-symmetric for all $X \in \mathfrak{g}$. Deduce that $\mathfrak{g}$ is isomorphic to $\mathfrak{so}(3)$ or $\mathfrak{sl}(2, \mathbb{R})$.

d. Now consider $n = 2$. Choose linearly independent $X, Y \in [\mathfrak{g}, \mathfrak{g}]$. Use (a) and the Jacobi identity to show that $[X, Y] = 0$. Show also that it is possible to chose a non-zero $Z \notin [\mathfrak{g}, \mathfrak{g}]$ such that one of the following holds:

(i) $[Z, X] = X, [Z, Y] = X + Y$.
(ii) $[Z, X] = X, [Z, Y] = \lambda Y, \lambda \in \mathbb{R} \setminus \{0\}$.
(iii) $[Z, X] = aX + Y, [Z, Y] = -X + aY, a \in \mathbb{R}$.

(Hint: Consider the Jordan canonical form of $\mathrm{ad}_Z : [\mathfrak{g}, \mathfrak{g}] \to [\mathfrak{g}, \mathfrak{g}]$.)

e. Finally, show that $\mathfrak{g}$ is either the Heisenberg algebra or a certain product algebra in case $n = 1$.

f. Conclude that the above reasoning classifies real Lie algebras in dimension 3 (Bianchi 1898). Can you find corresponding Lie groups?

5 Let $G = O(n)$.

a. Show that $G^\circ \subset SO(n)$.

b. Prove that any element in $SO(n)$ is conjugate in G to a matrix of the form

$$\begin{pmatrix} R_{t_1} & & & & & \\ & \ddots & & & & \\ & & R_{t_p} & & & \\ & & & 1 & & \\ & & & & \ddots & \\ & & & & & 1 \end{pmatrix},$$

where R_t is the 2×2 block

$$\begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

and $t_1, \ldots, t_p \in \mathbb{R}$ (that is, for every $A \in SO(n)$ there is $B \in G$ such that BAB^{-1} has the above form).

c. Deduce from the above that $SO(n)$ is connected. Conclude that $O(n)$ has two connected components and $SO(n)$ is the identity component.

d. Use a similar idea to show that $U(n)$ and $SU(n)$ are connected.

6 Prove that Lie groups are parallelizable manifolds (cf. Problem 40 in ▶ Chap. 1).

7 (Quaternions) Hamilton's *quaternions* $\mathbb{H}$ consist of a (non-commutative) associative division algebra over $\mathbb{R}$ with basis $1, i, j, k$ and relations

$$i^2 = j^2 = k^2 = -1,\ ij = -ji = k,\ jk = -kj = i,\ ki = -ik = j.$$

Let $q = a + bi + cj + dk \in \mathbb{H}$, where $a, b, c, d' \in \mathbb{R}$, and define: the *conjugate* of q as $\bar{q} = a - bi - cj - dk$; the *real part* of q as $\Re q = \frac{1}{2}(q + \bar{q}) = a$; and the *imaginary part* of q as $\Im q = q - \Re q$. We also say that q is *imaginary* if $\Re q = 0$.

a. Check that $\overline{qq'} = \bar{q}'\bar{q}$ for all $q, q' \in \mathbb{H}$.

b. Verify that: $\langle q, q' \rangle = \Re(\bar{q}q')$ yields the standard scalar product on $\mathbb{R}^4$ with respect to the basis $1, i, j, k$; and the associated norm $|\cdot|$ is multiplicative in the sense that $|qq'| = |q||q'|$ for all $q, q' \in \mathbb{H}$.

c. View $\mathbb{H}^n$ as a right vector space over $\mathbb{H}$ and check that the space $M(n, \mathbb{H})$ of $n \times n$ matrices with quaternionic entries can be identified with the quaternionic linear transformations of $\mathbb{H}^n$ via $A \mapsto L_A$, where vectors of $\mathbb{H}^n$ are viewed as column vectors and L_A is left multiplication by A.

d. Show that the group $GL(n, \mathbb{H})$ of invertible matrices in $M(n, \mathbb{H})$ is a Lie group of dimension $4n^2$. Note that $GL(1, \mathbb{H})$ is the multiplicative group $\mathbb{H}^\times$, and the subgroup of unit quaternions of $\mathbb{H}^\times$ is a closed Lie subgroup which is diffeomorphic to S^3.

8 (Symplectic Groups) We refer to Problem 7 for information on $\mathbb{H}$.

a. Define the *conjugate transpose* A^* of $A \in M(n, \mathbb{H})$ as the matrix obtained by transposition and conjugation of the entries of A. Prove that

$$Sp(n) = \{ A \in GL(n, \mathbb{H}) \mid AA^* = I \}$$

is a compact Lie group of dimension $2n^2 + n$; it is called the *symplectic group*.

b. Show that $\langle q, q' \rangle = \sum_{i=1}^n \Re(\bar{q}_i q'_i)$ for $q = (q_1, \ldots, q_n), q' = (q'_1, \ldots, q'_n) \in \mathbb{H}^n$ defines a positive-definite inner product on $\mathbb{H}^n$, and $Sp(n)$ corresponds to the quaternionic linear transformations of $\mathbb{H}^n$ that preserve this inner product.

c. Check that $Sp(1)$ is isomorphic to S^3.

► *Section 3.2*

9 Show that

$$\exp \begin{pmatrix} 0 & -t \\ t & 0 \end{pmatrix} = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

for $t \in \mathbb{R}$.

10 Give examples of matrices $A, B \in \mathfrak{gl}(2, \mathbb{R})$ such that $e^{A+B} \neq e^A e^B$.

11 In this problem, we show that the exponential map in a Lie group does not have to be surjective.

a. Show that every element g in the image of $\exp : \mathfrak{g} \to G$ has a square root, namely there is $h \in G$ such that $h^2 = g$.

b. Prove that trace $A^2 \geq -2$ for any $A \in SL(2, \mathbb{R})$. (Hint: A satisfies its characteristic polynomial equation $X^2 - 2(\text{trace } X)X + (\det X)I = 0$.)

c. Deduce from the above that $\begin{pmatrix} -2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$ does not lie in the image of $\exp : \mathfrak{sl}(2, \mathbb{R}) \to SL(2, \mathbb{R})$.

12 Let $X \in \mathfrak{sl}(2, \mathbb{R})$. Show that

$$e^X = \begin{cases} \cosh(-\det X)^{1/2} I + \frac{\sinh(-\det X)^{1/2}}{(-\det X)^{1/2}} X & \text{if } \det X < 0, \\ \cos(\det X)^{1/2} I + \frac{\sin(\det X)^{1/2}}{(\det X)^{1/2}} X & \text{if } \det X > 0, \\ I + X & \text{if } \det X = 0. \end{cases}$$

13 Prove that the exponential map $\exp : \mathfrak{gl}(n, \mathbb{C}) \to GL(n, \mathbb{C})$ is surjective. Deduce that $GL(n, \mathbb{C})$ and $SL(n, \mathbb{C})$ are connected. (Hint: We may assume the matrix is in Jordan canonical form. Write an elementary Jordan block as the product of a diagonal matrix S and an upper triangular matrix N with 1's on the diagonal. Observe that N is *unipotent*, that is, sufficiently high powers of $N - I$ vanish. Deduce that

$$\log N = -\sum_{n=1}^{\infty} \frac{(I - N)^n}{n}$$

is well-defined and satisfies $e^{\log N} = N$.)

14 Let S denote the vector space of $n \times n$ real symmetric matrices and let S^+ denote the open subset (positive cone) of S consisting of positive definite matrices.

a. Prove that the exponential map of matrices sets up a bijection from S onto S^+. (Hint: Prove it first for diagonal matrices.)

b. Show that $\langle X, Y \rangle = \operatorname{trace}(XY)$ for $X, Y \in S$ defines a positive-definite symmetric bilinear form on S.

c. Check that $\operatorname{ad}_Z X \in S$ for all $Z \in \mathfrak{so}(n)$ and $X \in S$, and $\langle \operatorname{ad}_Z X, Y \rangle + \langle X, \operatorname{ad}_Z Y \rangle = 0$ for all $Z \in \mathfrak{so}(n)$ and $X, Y \in S$.

d. For $X \in S$, show that there is an orthogonal decomposition $S = T_X \oplus N_X$, where $T_X = \{[Z, X] \in S \mid Z \in \mathfrak{so}(n)\}$ and $N_X = \{Y \in S \mid [Y, X] = 0\}$.

e. For $X \in S$, check that $e^{-X}(d \exp)_X : T_X S \cong S \to T_I S^+ \cong S$ is given by

$$e^{-X}(d \exp)_X = \frac{I - e^{-\operatorname{ad}_X}}{\operatorname{ad}_X}.$$

(Hint: Check the formula separately on N_X and T_X.)

f. Deduce from the above that $\exp : S \to S^+$ is a global diffeomorphism.

15 (Polar Decomposition of Matrices)

a. Prove that any $g \in GL(n, \mathbb{R})$ can be uniquely written as $g = hk$ where $h \in O(n)$ and k is a positive-definite symmetric matrix.

b. Deduce from part (a) and Problem 14 that $GL(n, \mathbb{R})$ is diffeomorphic to $O(n) \times \mathbb{R}^{\frac{n(n+1)}{2}}$ and $SL(n, \mathbb{R})$ is diffeomorphic to $SO(n) \times \mathbb{R}^{\frac{n(n+1)}{2}-1}$.

c. Deduce from part (b) and Problem 5 that $SL(n, \mathbb{R})$ is connected, but $GL(n, \mathbb{R})$ has two connected components and $SL(n, \mathbb{R})$ is its identity component.

16

a. Prove that any $g \in GL(n, \mathbb{C})$ can be uniquely written as $g = hk$ where $h \in U(n)$ and k is a positive-definite *Hermitian* matrix (that is, $k = k^*$, where k^* is the conjugate transpose matrix of k).

b. Prove that $GL(n, \mathbb{C})$ is diffeomorphic to $U(n) \times \mathbb{R}^{n^2}$ and $SL(n, \mathbb{C})$ is diffeomorphic to $SU(n) \times \mathbb{R}^{n^2-1}$. This gives another way to see that $SL(n, \mathbb{C})$ and $GL(n, \mathbb{C})$ are connected (compare Problems 5 and 13).

17 Let G be a Lie group. Prove that it does not have small subgroups; i.e. prove the existence of an open neighborhood U of 1 such that $\{1\}$ is the only subgroup of G entirely contained in U.

18 For a connected Lie group, prove that the second-countability of its topology is a consequence of the other conditions in the definition of a Lie group. (Hint: Use Proposition 3.2.5).

▶ *Section 3.3*

19 Check that

$$A + iB \in GL(n, \mathbb{C}) \mapsto \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \in GL(2n, \mathbb{R})$$

defines an injective homomorphism φ of $GL(n, \mathbb{C})$ onto a closed subgroup of $GL(2n, \mathbb{R})$. Check also that φ restricts to an injective homomorphism of $U(n)$ onto a closed subgroup of $SO(2n)$.

20 Check that

$$A + jB \in GL(n, \mathbb{H}) \mapsto \begin{pmatrix} A & -\bar{B} \\ B & \bar{A} \end{pmatrix} \in GL(2n, \mathbb{C})$$

defines an injective homomorphism φ of $GL(n, \mathbb{H})$ onto a closed subgroup of $GL(2n, \mathbb{C})$ (here A and B are complex matrices). Check also that φ restricts to an injective homomorphism of $Sp(n)$ onto a closed subgroup of $SU(2n)$.

21 Determine the center of $SU(n)$.

22 Construct a diffeomorphism between $U(n)$ and $S^1 \times SU(n)$. Is it an isomorphism of Lie groups?

▶ *Section 3.4*

23 Prove that a discrete normal subgroup of a connected Lie group is central. Deduce that the fundamental group of a Lie group is always Abelian.

24 View S^3 as the unit quaternions and $\mathbb{R}^3$ as the imaginary quaternions $\Im\mathbb{H}$. Prove that the map $\pi : S^3 \to SO(3)$ given by $\pi(q)x = qxq^{-1}$ for $q \in S^3$ and $x \in \mathbb{R}^3$ is well-defined, a smooth homomorphism, and a double covering.

25 View S^3 as the unit quaternions and $\mathbb{R}^4$ as the quaternions $\mathbb{H}$. Prove that the map $\pi : S^3 \times S^3 \to SO(4)$ given by $\pi(q_1, q_2)x = q_1 x q_2^{-1}$ for $q_1, q_2 \in S^3$ and $x \in \mathbb{R}^4$ is well-defined, a smooth homomorphism, and a double covering.

▶ *Section 3.5*

26 Consider $G = SU(2)$ and its Lie algebra $\mathfrak{g} = \mathfrak{su}(2)$.

a. Check that

$$\mathfrak{g} = \left\{ \begin{pmatrix} ix & y+iz \\ -y+iz & -ix \end{pmatrix} : x,\ y,\ z \in \mathbb{R} \right\}.$$

b. Identify $\mathfrak{g}$ with $\mathbb{R}^3$ and check that $\det : \mathfrak{g} \to \mathbb{R}$ corresponds to the usual quadratic form on $\mathbb{R}^3$. Check also that Ad_g preserves this quadratic form for all $g \in G$.

c. Deduce from the above that there is a smooth homomorphism

$$SU(2) \to SO(3),$$

which is the universal covering of $SO(3)$. What are the adjoint orbits of $SU(2)$?

27 Prove that the kernel of the adjoint representation of a connected Lie group coincides with its center.

28 Let A be a connected subgroup of a connected Lie group G. Prove that A is a normal subgroup of G if and only if the Lie algebra $\mathfrak{a}$ of A is an ideal of the Lie algebra $\mathfrak{g}$ of G.

▶ *Section 3.6*

29

a. Prove that the Grassmann manifold of k-planes in $\mathbb{R}^n$ (cf. Problem 4 in ▶ Chap. 1) is a homogeneous manifold:

$$Gr_k(\mathbb{R}^n) = SO(n)/S(O(k) \times O(n-k)).$$

b. Consider now the set $Gr_k^+(\mathbb{R}^n)$ of *oriented* k-dimensional subspaces of $\mathbb{R}^n$ and prove that

$$Gr_k^+(\mathbb{R}^n) = SO(n)/(SO(k) \times SO(n-k)).$$

c. Define the Grassmann manifold $Gr_k(\mathbb{C}^n)$ of k-planes in $\mathbb{C}^n$ and prove that

$$\begin{aligned} Gr_k(\mathbb{C}^n) &= U(n)/\left(U(k) \times U(n-k)\right) \\ &= SU(n)/S(U(k) \times U(n-k)). \end{aligned}$$

30

a. Check that $\pi_1(SO(2)) \cong \mathbb{Z}$ and $\pi_1(SO(3)) \cong \pi_1(SO(4)) \cong \mathbb{Z}_2$. (Hint: Use Problems 2, 24 and 25.)

b. Use Proposition 3.6.10(b) to prove that $\pi_1(SO(n))$ for $n \geq 5$ has order at most two. (Hint: Consider the homogeneous space (3.6.8) and proceed by induction on n.)

It can be shown that $\pi_1(SO(n)) \cong \mathbb{Z}_2$ for all $n \geq 3$; hence the universal covering group of $SO(n)$ for $n \geq 3$ is a double covering, called the *spin group*, and denoted by $Spin(n)$ (see, e.g., [BtD95, Chap. 1, §6]).

31

a. Use Problem 2 to check that $SU(2)$ is connected and simply-connected.

b. Use Proposition 3.6.10 to prove that $SU(n)$ is connected and simply connected for all n. (Hint: View $S^{2n-1} = SU(n)/SU(n-1)$.)

c. Deduce from (b) and Problem 22 that $U(n)$ is connected an $\pi_1(U(n)) \cong \mathbb{Z}$ for all n.

Integration



The integration of differential forms yields de Rham cohomology, the single most important differential invariant of a smooth manifold. It is a cohomology theory particularly suited to computation and the concrete representation of cohomology classes. De Rham cohomology measures the failure to globally solve certain types of PDE, by reason of topology, and in this way expresses some basic topological information about smooth manifolds. In this chapter, we prove a version of Stokes' theorem for manifolds with boundary and introduce de Rham cohomology. Then we go on to derive some results about the differential topology of manifolds.

4.1 Orientation

Recall the formula for change of variables in a multiple integral [Spi65, Theorem 3-13]:

$$\int_{\varphi(D)} f(y_1, \ldots, y_n) dy_1 \cdots dy_n$$

$$= \int_D f(\varphi(x_1, \ldots, x_n)) |J\varphi(x_1, \ldots, x_n)| dx_1 \cdots dx_n. \tag{4.1.1}$$

Here $(x_1, \ldots, x_n)$ and $(y_1, \ldots, y_n)$ are two sets of coordinates on $\mathbb{R}^n$ related by a diffeomorphism $\varphi : U \to V$ between open subsets of $\mathbb{R}^n$, D is a bounded domain of integration in U, f is a bounded real continuous function on $\varphi(D)$,

$$J\varphi = \det\left(\frac{\partial(y_i \circ \varphi)}{\partial x_j}\right)$$

is the Jacobian determinant of φ, and $\int$ refers to the Riemann integral. Let us interpret (4.1.1) in terms of differential forms. We have

$$d\varphi\left(\frac{\partial}{\partial x_j}\Big|_p\right) = \sum_i \frac{\partial(y_i \circ \varphi)}{\partial x_j}\Big|_p \frac{\partial}{\partial y_i}\Big|_{\varphi(p)}$$

and

$$(d\varphi)^*(dy_i|_{\varphi(p)}) = \sum_j \frac{\partial(y_i \circ \varphi)}{\partial x_j}\Big|_p dx_j|_p,$$

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so, in view of Problem 6 in ▶ Chap. 2,

$$\varphi^*(dy_1 \wedge \cdots \wedge dy_n) = (J\varphi)\, dx_1 \wedge \cdots \wedge dx_n.$$

If we define, as we do, the left-hand side of (4.1.1) as the integral of the n-form $\omega = f\, dy_1 \wedge \cdots \wedge dy_n$ over $\varphi(D)$, that formula says that

$$\int_{\varphi(D)} \omega = \pm \int_D \varphi^*\omega, \tag{4.1.2}$$

where the sign is positive or negative according to whether the sign of the Jacobian determinant is positive or negative throughout D. In general, a diffeomorphism between open subsets of $\mathbb{R}^n$ is called *orientation-preserving* if its Jacobian determinant is everywhere positive. The above discussion shows that integration of n-forms on bounded domains is not invariant under diffeomorphisms in general, but only under those that preserve orientation. This suggests that if we want to transfer these ideas to smooth manifolds via local charts, and define integration of n-forms in a manner independent of local coordinates, we should try to sort out a consistent sign for the transition maps.

Let M be a smooth manifold. In the following, we only consider atlases in the smooth structure of M. A smooth atlas for M is called *oriented* if all the transition maps are orientation-preserving, and M is called *orientable* if it admits an oriented atlas. If M is orientable, two oriented atlases are said to define the same orientation if their union is an oriented atlas; this defines an equivalence relation on the set of oriented atlases, and a choice of equivalence class is called an *orientation* for M. A manifold with an orientation is called an *oriented* manifold.

If M is orientable, an oriented atlas for M defines an orientation on each tangent space, which is induced from the canonical orientation of $\mathbb{R}^n$ via the local charts. For this reason, an orientation on M can also be viewed as a coherent choice of orientations on the tangent spaces to M.

4.1.3 Exercise Recall that an *orientation* on a vector space V of positive dimension is an equivalence class of (ordered) bases, where two bases are said to be equivalent if the matrix of change of basis has positive determinant. Clearly, a vector space admits exactly two orientations. Show that for any non-zero element $\omega \in \Lambda^n(V^*)$ ($n = \dim V$) and any basis $(e_1, \ldots, e_n)$ of V, the number $\omega(e_1, \ldots, e_n)$ is not zero and its sign is constant in each equivalence class of bases. Deduce that the components of $\Lambda^n(V^*) \setminus \{0\} \cong \mathbb{R} \setminus \{0\}$ naturally correspond to the orientations in V. ■

4.1.4 Remark In case $V = \{0\}$, we make the convention that an orientation is a choice of sign ± 1.

4.1.5 Proposition *A smooth manifold M of dimension n is orientable if and only if it has a nowhere vanishing n-form.*

Proof Let $\omega_0 = dx_1 \wedge \cdots \wedge dx_n$ be the canonical n-form on $\mathbb{R}^n$. The basic fact we need is that a diffeomorphism τ between open subsets of $\mathbb{R}^n$ is orientation-preserving if and only if $\tau^*\omega_0 = f\,\omega_0$ for an everywhere positive smooth function f.

Assume first ω is a nowhere vanishing n-form on M. Let $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ be a smooth atlas for M where each U_α is connected. For all α, $\varphi_\alpha^*\omega_0 = f_\alpha\omega$ where f_α is a nowhere zero smooth function on U_α. Thus f_α is everywhere positive or everywhere negative on U_α; in the latter case, we replace φ_α by $\psi_\alpha = \tau \circ \varphi_\alpha$ where $\tau(x_1, \ldots, x_n) = (-x_1, \ldots, x_n)$. Since $\psi_\alpha^*\omega_0 = \varphi_\alpha^*\tau^*\omega_0 = -\varphi_\alpha^*\omega_0 = -f_\alpha\omega$, this shows that, by replacing $\mathcal{A}$ with an equivalent atlas, we may assume that $f_\alpha > 0$ for all α. Now $(\varphi_\beta\varphi_\alpha^{-1})^*\omega_0 = (f_\beta \circ \varphi_\alpha^{-1})/(f_\alpha \circ \varphi_\alpha^{-1})\omega_0$ with $f_\beta/f_\alpha > 0$ for all α, β, which proves that $\mathcal{A}$ is oriented.

Conversely, assume $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ is an oriented atlas for M. Define $\omega_\alpha = \varphi_\alpha^*\omega_0$. Then ω_α is a nowhere vanishing n-form on U_α, and ω_α, ω_β are positive multiples of one another on $U_\alpha \cap U_\beta$. It follows that $\omega := \sum_\alpha \rho_\alpha\omega_\alpha$ is a well-defined, nowhere vanishing n-form on M, where $\{\rho_\alpha\}$ is a partition of unity strictly subordinate to $\{U_\alpha\}$. □

In view of the proof of Proposition 4.1.5, on an orientable manifold M of dimension n, there exists a bijection between equivalence classes of oriented atlases and equivalence classes of nowhere vanishing n-forms, where two nowhere vanishing n-forms on M are said to be equivalent if they differ by a positive smooth function. On a connected manifold, the sign of a nowhere zero continuous function cannot change, so on a connected orientable manifold there are exactly two possible orientations.

A local diffeomorphism $f : M \to N$ between oriented smooth manifolds is called *orientation-preserving* if $df_p : T_pM \to T_{f(p)}N$ is orientation-preserving for all $p \in M$.

4.1.6 Exercise Let M and N be smooth manifolds of the same dimension n, respectively oriented by nowhere vanishing n-forms ω and η. Show that a local diffeomorphism $f : M \to N$ is orientation-preserving if and only if $f^*\eta = g\omega$ for some $g \in C^\infty(M)$ with $g > 0$. ■

4.1.7 Example Let M be the preimage of a regular value of a smooth map $f : \mathbb{R}^{n+1} \to \mathbb{R}$. Then M is an (embedded) submanifold of $\mathbb{R}^{n+1}$ and we show in the following that M is orientable by constructing a nowhere vanishing n-form on M. Let $U_i = \{p \in M \mid \frac{\partial f}{\partial x_i}(p) \neq 0\}$ for $i = 1, \ldots, n+1$. Then $\{U_i\}$ is an open cover of M and we can take $(x_1, \ldots, \hat{x}_i, \ldots, x_{n+1})$ as local coordinates on each component of U_i. Define a nowhere vanishing n-form on U_i by

$$\omega_i = (-1)^i \left(\frac{\partial f}{\partial x_i}\right)^{-1} dx_1 \wedge \cdots \wedge \hat{dx_i} \wedge \cdots \wedge dx_{n+1}.$$

Since f is constant on M, $\sum_k \frac{\partial f}{\partial x_k} dx_k = 0$, so we have on U_j that

$$dx_j = -\left(\frac{\partial f}{\partial x_j}\right)^{-1} \sum_{k \neq j} \frac{\partial f}{\partial x_k} dx_k.$$

Now one easily checks that

$$\omega_i|_{U_i\cap U_j} = \omega_j|_{U_i\cap U_j}$$

and hence the ω_i can be pieced together to yield a global n-form on M. ■

4

Let M be an orientable smooth manifold and fix an orientation for M, say given by an oriented atlas $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$. We want to define the integral of a compactly supported n-form ω on M. For that purpose, consider first the special case in which the support of ω is contained in the domain of some local chart, say, $(U_\alpha, \varphi_\alpha) \in \mathcal{A}$. Then we set

$$\int_M \omega = \int_{U_\alpha} \omega = \int_{\varphi_\alpha(U_\alpha)} (\varphi_\alpha^{-1})^*\omega.$$

Note that choosing another local chart in $\mathcal{A}$ whose domain contains the support of ω yields the same result due to (4.1.2). In the general case, we choose a partition of unity $\{\rho_i\}$ subordinate to $\{U_\alpha\}$, $\operatorname{supp}(\rho_i) \subset U_{\alpha(i)}$, and put

$$\int_M \omega = \sum_i \int_{U_{\alpha(i)}} \rho_i\omega.$$

Note that only finitely many terms in this sum are non-zero as $\operatorname{supp}(\omega)$ is compact and $\{\operatorname{supp}(\rho_i)\}$ is locally finite. Let us check that this definition is independent of the choices made. Namely, let $\{(V_\beta, \psi_\beta)\}$ be another oriented atlas defining the same orientation, and let $\{\lambda_j\}$ be a partition of unity subordinate to $\{V_j\}$, that is, $\operatorname{supp}(\lambda_j) \subset V_{\beta(j)}$. Note that $\rho_i\lambda_j\omega$ has support contained in $U_{\alpha(i)} \cap V_{\beta(j)}$, so, by the special case,

$$\int_{U_{\alpha(i)}} \rho_i\lambda_j\omega = \int_{V_{\beta(j)}} \rho_i\lambda_j\omega.$$

It follows that

$$\begin{aligned}\sum_i \int_{U_{\alpha(i)}} \rho_i\omega &= \sum_{i,j} \int_{U_{\alpha(i)}} \rho_i\lambda_j\omega \\ &= \sum_{i,j} \int_{V_{\beta(j)}} \rho_i\lambda_j\omega \\ &= \sum_j \int_{V_{\beta(j)}} \lambda_j\omega,\end{aligned}$$

as we wished, where we have used that $\sum_i \rho_i = \sum_j \lambda_j = 1$.

4.1.8 Exercise Let $f : M \to N$ be a diffeomorphism between connected oriented smooth manifolds of dimension n and let ω be a compactly supported n-form on N. Prove that

$$\int_M f^*\omega = \pm \int_N \omega$$

where the sign is "+" if f is orientation-preserving and "−" if f is orientation-reversing. (Hint: Use (4.1.2).) ■

4.1.9 Exercise Let M be a connected orientable smooth manifold of dimension n and denote by $-M$ the same manifold with the opposite orientation. Show that

$$\int_{-M} \omega = -\int_M \omega$$

for every compactly supported n-form ω on M. ■

4.2 Stokes' Theorem

Stokes' theorem for manifolds is the exact generalization of the classical theorems of Green, Gauss, and Stokes of Vector Calculus. In order to proceed, we need to develop a notion of boundary.

Manifolds with Boundary

In the same way as manifolds are modeled on $\mathbb{R}^n$, manifolds with boundary are modeled on the *upper half-space*

$$\mathbb{R}^n_+ = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}.$$

An n-dimensional *smooth manifold with boundary* is given by a smooth atlas $\{(U_\alpha, \varphi_\alpha)\}$ where φ_α maps U_α homeomorphically onto an open subset of $\mathbb{R}^n_+$ and the transition maps are diffeomorphisms between open subsets of $\mathbb{R}^n_+$. Recall a function f from an arbitrary subset A of $\mathbb{R}^n$ is called *smooth* if it admits a smooth extension $\tilde{f}$ to an open neighborhood of A. In case A is an open subset of $\mathbb{R}^n_+$, by continuity all partial derivatives of $\tilde{f}$ at points in the boundary $\partial\mathbb{R}^n_+$ (see below) are determined by the values of f on $\mathbb{R}^n_+$, and therefore in particular are independent of the choice of extension.

Of course, $\mathbb{R}^n_+$ is itself a manifold with boundary. There is a natural decomposition of $\mathbb{R}^n_+$ into the *boundary*

$$\partial\mathbb{R}^n_+ = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_n = 0\}$$

and its complement, the *interior*, and both are smooth manifolds in the previous (restricted) sense, with a natural diffeomorphism $\partial\mathbb{R}^n_+ \approx \mathbb{R}^{n-1}$. For an open subset U of $\mathbb{R}^n_+$, we also put $\partial U = U \cap \partial\mathbb{R}^n_+$.

4.2.1 Lemma *Let $\tau : U \to V$ be a diffeomorphism between open subsets of $\mathbb{R}^n_+$ with everywhere positive Jacobian determinant. Then τ restricts to a diffeomorphism $\partial\tau : \partial U \to \partial V$ with everywhere positive Jacobian determinant.*

Proof By assumption, τ is smooth, bijective and τ^{-1} is smooth; moreover, its Jacobian determinant is nowhere zero. Due to the inverse mapping theorem, a point in $U \setminus \partial U$ admits an open neighborhood in $\mathbb{R}^n$ which is mapped diffeomorphically onto an open subset of $\mathbb{R}^n$ entirely contained in V. It follows that $\tau(U \setminus \partial U) \subset V \setminus \partial V$; applying the same argument to τ^{-1} (note that τ^{-1} also has non-zero Jacobian determinant at $q \in V$, given by the inverse of the Jacobian determinant of τ at $\tau^{-1}(q)$), we get equality and hence $\tau(\partial U) = \partial V$.

Write $x' = (x_1, \ldots, x_{n-1}) \in \mathbb{R}^{n-1}$. By assumption, the Jacobian matrix of $\tau = (\tau_1, \ldots, \tau_n)$ at $(x', 0) \in \partial U$ has positive determinant and block form

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where

$$C = \left(\frac{\partial \tau_n}{\partial x_1}(x', 0), \ldots, \frac{\partial \tau_n}{\partial x_{n-1}}(x', 0) \right) = (0, \ldots, 0)$$

since $\tau_n(x', 0) = 0$ for all x', and

$$D = \frac{\partial \tau_n}{\partial x_n}(x', 0) > 0$$

since τ maps the upper half-space into itself. It follows that A, which is the Jacobian of $\partial\tau$ at $(x', 0)$, also has positive determinant, as desired. □

Let M be a smooth manifold with boundary. It follows from Lemma 4.2.1 that the *boundary* of M, namely, the subset ∂M consisting of points of M mapped to $\partial\mathbb{R}^n_+$ under a coordinate chart, is well-defined. Moreover, it is a smooth manifold of dimension $(n-1)$, and an (oriented) atlas for M induces an (oriented) atlas for ∂M by restricting the coordinate charts. Note also that $M \setminus \partial M$ is a smooth manifold of dimension n.

4.2.2 Examples (i) The closed unit ball $\bar{B}^n$ in $\mathbb{R}^n$ is a smooth manifold with boundary S^{n-1}.

(ii) The Möbius band is smooth manifold with boundary a circle S^1. ■

Let M be a smooth manifold of dimension n with boundary. The tangent space to M at a point p is an n-dimensional vector space defined in the same way as in the case of a smooth manifold (even in case $p \in \partial M$). The definition of the tangent bundle also works, and TM is itself a manifold with boundary $\partial(TM) = \dot{\cup}_{p \in \partial M} T_p M$. More generally, tensor bundles and differential forms are also defined. If M is in addition oriented, the integral of compactly supported n-forms is defined similarly to above.

In general, for an oriented smooth manifold with boundary, we will always use the so-called *induced orientation* on its boundary. Namely, if on $\mathbb{R}^n_+$ we use the standard orientation given by $dx_1 \wedge \cdots \wedge dx_n$, then the induced orientation on $\partial\mathbb{R}^n_+$ is specified by $(-1)^n\, dx_1 \wedge \cdots \wedge dx_{n-1}$ (the sign is required to make the statement of Stokes' theorem right); in particular, the induced orientation on $\partial\mathbb{R}^1_+$ is -1. On an oriented smooth manifold with boundary M, for any local chart (U, φ) in an oriented atlas of M, we declare the restriction of φ to $\partial U \to \partial\mathbb{R}^n_+$ to be orientation-preserving.

A 0-manifold M is just a countable discrete collection of points. In this case, an orientation for M is an assignment of sign $\sigma(p) = \pm 1$ for each $p \in M$ and $\int_M f = \sum_{p\in M} \sigma(p) f(p)$ for any 0-form $f \in C^\infty(M)$ with compact support.

As it is, a closed interval $[a, b] \subset \mathbb{R}$ $(a < b)$ admits an orientation given by the nowhere vanishing 1-form dx_1, but no oriented atlas consisting of local charts with values on $\mathbb{R}^1_+$! (Note that in the proof of Proposition 4.1.5 we used the fact that, if $(x_1, \ldots, x_n)$ are local coordinates on our manifold, then so are $(-x_1, \ldots, x_n)$.) To remedy this situation, we introduce a slight modification in the definition of manifold with boundary in case $n = 1$ and also allow local charts with values on the *left-line* $\mathbb{R}^1_-$. Accordingly, for the standard orientation dx_1 of $\mathbb{R}^1_-$, the induced orientation on $\partial\mathbb{R}^1_-$ is specified by $+1$. With such conventions, the induced orientation at a is -1 and that at b is $+1$.

4.2.3 Exercise Let M be an oriented n-manifold with boundary and give ∂M the induced orientation. Fix $p \in \partial M$. A vector $v \in T_pM \setminus T_p(\partial M)$ is called *inward pointing* if $v = \dot\gamma(0)$ for some smooth curve $\gamma : [0, \epsilon) \to M$ with $\gamma(0) = p$; and it is called *outward-pointing* if $-v$ is inward pointing. Show that $(v_1, \ldots, v_{n-1})$ is a basis that defines the orientation of $T_p(\partial M)$ if and only if $(v, v_1, \ldots, v_{n-1})$ is a basis that defines the orientation of T_pM, where $v \in T_pM$ is outward-pointing. ■

4.2.4 Remark A smooth manifold M in the old sense is a smooth manifold with boundary with $\partial M = \varnothing$. Indeed, we can always find an atlas for M whose local charts have images in $\mathbb{R}^n_+ \setminus \partial\mathbb{R}^n_+$.

Statement and Proof of the Theorem

4.2.5 Theorem (Stokes) *Let ω be an $(n-1)$-form with compact support on an oriented smooth n-manifold M with boundary and give ∂M the induced orientation. Then*

$$\int_M d\omega = \int_{\partial M} \omega.$$

On the right-hand side of Stokes' theorem, ω is viewed as $\iota^*\omega$, where $\iota : \partial M \to M$ is the inclusion, and the integral vanishes if $\partial M = \varnothing$. In the case $n = 1$, the integral on the right-hand side is a finite sum and the result specializes to the fundamental theorem of Calculus.

Proof of Theorem 4.2.5 We first consider two special cases.

Case 1 M is an open subset U of $\mathbb{R}^n$. View ω as an $(n-1)$-form on $\mathbb{R}^n$ with compact support contained in U. Write $\omega = \sum_i a_i\, dx_1 \wedge \cdots \wedge \hat{dx_i} \wedge \cdots \wedge dx_n$. Then $d\omega = \sum_i (-1)^{i-1} \frac{\partial a_i}{\partial x_i}\, dx_1 \wedge \cdots \wedge dx_n$. By Fubini's theorem,

$$\begin{aligned}\int_U d\omega &= \int_{\mathbb{R}^n} d\omega \\ &= \sum_i (-1)^{i-1} \int_{\mathbb{R}^{n-1}} \left(\int_{-\infty}^{\infty} \frac{\partial a_i}{\partial x_i}\, dx_i \right) dx_1 \cdots \hat{dx_i} \cdots dx_n \\ &= 0\end{aligned}$$

because

$$\begin{aligned}&\int_{-\infty}^{\infty} \frac{\partial a_i}{\partial x_i}\, dx_i \\ &= a_i(\ldots, x_{i-1}, \infty, x_{i+1}, \ldots) - a_i(\ldots, x_{i-1}, -\infty, x_{i+1}, \ldots) \\ &= 0,\end{aligned}$$

as a_i has compact support. Since M has no boundary, this case is settled.

Case 2 M is an open subset U of $\mathbb{R}^n_+$. View ω as an $(n-1)$-form on $\mathbb{R}^n_+$ with compact support contained in U. Write $\omega = \sum_i a_i\, dx_1 \wedge \cdots \wedge \hat{dx_i} \wedge \cdots \wedge dx_n$ as before, but note that the a_i are smooth on an open subset of $\mathbb{R}^n_+$, whereas the linear forms dx_i are defined on the entire $\mathbb{R}^n$. Since a_i has compact support, $\int_{-\infty}^{\infty} \frac{\partial a_i}{\partial x_i}\, dx_i = 0$ for $i < n$, so by Fubini's theorem

$$\begin{aligned}\int_U d\omega &= \int_{\mathbb{R}^n_+} d\omega \\ &= (-1)^{n-1} \int_{\mathbb{R}^{n-1}} \left(\int_0^{\infty} \frac{\partial a_n}{\partial x_n}\, dx_n \right) dx_1 \cdots dx_{n-1} \\ &= (-1)^{n-1} \int_{\mathbb{R}^{n-1}} -a_n(x_1, \ldots, x_{n-1}, 0)\, dx_1 \cdots dx_{n-1} \\ &= \int_{\partial \mathbb{R}^n_+} \omega \\ &= \int_{\partial U} \omega,\end{aligned}$$

where we have used that $dx_n = 0$ on $\mathbb{R}^{n-1} \times \{0\}$, as well as the induced orientation of $\partial \mathbb{R}^n_+$ as defined above, finishing this case.

General Case M is an arbitrary manifold with boundary of dimension n. Let $\{(U_\alpha, \varphi_\alpha)\}$ be an oriented atlas for M and let $\{\rho_\alpha\}$ be a partition of unity strictly subordinate to $\{U_\alpha\}$. Then $\omega = \sum_\alpha \rho_\alpha \omega$ where each term has compact support. By linearity, it suffices to prove Stokes'

formula for $\rho_\alpha\omega$, which has support contained in U_α. Since U_α is diffeomorhic to an open set in $\mathbb{R}^n$ or $\mathbb{R}^n_+$, cases 1 and 2 imply that the formula holds on U_α, so

$$\int_M d\rho_\alpha\omega = \int_{U_\alpha} d\rho_\alpha\omega = \int_{\partial U_\alpha} \rho_\alpha\omega = \int_{\partial M} \rho_\alpha\omega,$$

which finishes the proof of the theorem.

□

Manifolds with Corners and Stokes' Theorem

The theory of smooth manifolds with corners is not a completely standardized theory and we just sketch a group of definitions which leads us quickly to a version of Stokes' theorem.

Define the *n-quadrant* Q_n to be the subset of $\mathbb{R}^n$ defined by

$$x_1 \geq 0, \ldots, x_n \geq 0.$$

A point $p \in Q_n$ is called a *corner of index* k ($0 \leq k \leq n$) if exactly k coordinates of p vanish. The basic result that we need is contained in the following lemma.

4.2.6 Lemma *Let $\tau : U \to V$ be a diffeomorphism between open subsets of Q_n. Then $p \in Q_n$ is a corner of index k if and only if $\tau(p)$ is a corner of index k.*

Proof Denote by U_k (resp. V_k) the subset of U (resp. V) consisting of points of index k. First, we use the inverse mapping theorem as in the proof of Lemma 4.2.1 to show that $\tau(U_0) = V_0$. Next, we take up the case $k = 1$. Let $p \in U_1$ and denote the hyperplane of $\mathbb{R}^n$ defined by $x_j = 0$ by H_j. Then $p \in H_i$ for exactly one index $i \in \{1, \ldots, n\}$. Suppose $q = \tau(p) \in H_{i'}$. Note that $\tau_{i'}$ has a local minimum at p, and that p lies in the interior of $U \cap H_i$, therefore $\partial\tau_{i'}/\partial x_j$ vanishes at p for all $j \neq i$. Since $d\tau_p$ has rank n, we deduce that it maps H_i onto $H_{i'}$. In particular, q cannot lie in H_j for $j \neq i'$, that is, $q \in V_1$.

It is easy to see that U_k for $k \geq 2$ can be characterized as consisting of those points in $U \setminus U_0$ admitting a sufficiently small connected open neighborhood that meets exactly k connected components of U_1 (and similarly for V_k). Hence $\tau(U_k) = V_k$ for all k. □

An n-dimensional *smooth manifold with corners* M is given by a smooth atlas where the local charts are homeomorphisms onto open subsets of Q_n and the transition maps are diffeomorphisms between open subsets of Q_n. Tensor bundles, differential forms, orientation and integration of forms are defined in the usual way.

A smooth n-manifold with corners M is stratified by locally closed subsets M_k consisting of corners of index k, that is, points corresponding under a chart to corners of index k in Q_n (this is a good definition due to Lemma 4.2.6). In general, M_k is a smooth manifold without boundary of dimension $n - k$, and the locus $M_{\leq k}$ of corners of index at most k is an open subset of M and a smooth manifold with corners. In particular, $M_{\leq 1}$

is a smooth manifold with boundary. For instance, if $M = [0, 1]^2$ is the unit square, then $M_{\leq 1}$ is M with the four vertices removed.

We define the *boundary* ∂M to be the locus $M_{\geq 1}$ of corners of index at least 1. Since the locus $M_{\geq 2}$ of corners of index at least 2 is a closed subset of measure zero in ∂M, it is immaterial for the calculation of integrals. Note that $(Q_n)_1 = \cup_{i=1}^n \tilde{H}_i$ where $\tilde{H}_i$ is defined by $x_i = 0$ and $x_j > 0$ for $j \neq i$. Each $\tilde{H}_i$ is oriented by the $(n-1)$-form $(-1)^i dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n$. For an oriented smooth manifold with corners M, the induced orientation at a point in M_1 by definition corresponds under a local chart to the orientation at the corresponding point of $\tilde{H}_i$ for some i.

If ω is a compactly supported $(n-1)$-form on a smooth n-manifold with corners M, then Stokes' theorem $\int_M d\omega = \int_{\partial M} \omega$ holds, where the right-hand side is interpreted as the integral of the pull-back of ω to M_1 (since, in principle, ∂M is not a manifold with corners). Indeed, for an oriented atlas $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ of M with partition of unity $\{\rho_i\}$ subordinate to $\{U_\alpha\}$, $\mathrm{supp}(\rho_i) \subset U_{\alpha(i)}$, we have $\int_{M_1} \omega = \sum_i \int_{M_1 \cap U_{\alpha(i)}} \rho_i \omega$, where the sum on the right-hand side is finite. Stokes' formula in this case follows as in Theorem 4.2.5 by using the following result.

4.2.7 Exercise Let ω be a compactly supported $(n-1)$-form on an open subset U of Q_n. Then $\int_U d\omega = \int_{\partial U} \omega$. ■

4.3 De Rham Cohomology

De Rham theory, named after Georges de Rham, is a cohomology theory in the realm of smooth manifolds and "constitutes in some sense the most perfect example of a cohomology theory" (Bott and Tu). The de Rham complex of a smooth manifold is defined as a differential invariant, but turns out to be a topological invariant. This follows from de Rham's isomorphism theorem, which states that the de Rham cohomology of a smooth manifold is isomorphic to its singular cohomology [Mas91, App. A]. We will not prove this, but in the next section we shall prove that it is an invariant of the *smooth* homotopy type.

The most basic invariant of a topological space X is perhaps its number of connected components. In terms of continuous functions, a component is characterized by the property that on it every locally constant continuous function is globally constant. If we define $H^0(X)$ to be the vector space of real valued locally constant continuous functions on X, then $\dim H^0(X)$ is the number of connected components of X. Of course, in case $X = M$ is a smooth manifold and we define $H^0(M)$ to be the vector space of real valued locally constant *smooth* functions on M, again $\dim H^0(X)$ is the number of connected components of M.

In seeking to define $H^k(M)$ for $k > 0$, assume for simplicity M is an open subset of $\mathbb{R}^n$ with coordinates $(x_1, \ldots, x_n)$. In this case, the locally constant smooth functions f on M are exactly those satisfying

$$df = \sum_i \frac{\partial f}{\partial x_i}\, dx_i = 0.$$

Therefore $H^0(M)$ appears as the space of solutions of a differential equation. In case $k > 0$, points and functions are replaced by k-dimensional submanifolds and k-forms, respectively. For instance, if $k = 1$, a 1-form $\omega = \sum_i a_i\, dx_i$ defines a function on smooth paths

$$\gamma \mapsto \int_\gamma \omega$$

and we look for locally constant functions, namely, those left unchanged under a small perturbation of γ keeping the endpoints fixed. In general, if we homotope γ to a nearby curve with endpoints fixed, the difference between the line integrals is given by the integral of $d\omega$ along the spanned surface, owing to Stokes' theorem. Therefore the condition of local constancy is here $d\omega = 0$ or, equivalently, the system of partial differential equations

$$\frac{\partial a_i}{\partial x_j} - \frac{\partial a_j}{\partial x_i} = 0 \tag{4.3.1}$$

for all i, j. On the other hand, $\int_\gamma df = f(q) - f(p)$ where p, q are the endpoints of γ, so 1-forms of type df yield trivial solutions of (4.3.1). This suggests that $H^1(M)$ be defined as the vector space of locally constant line integrals modulo the trivially constant ones, and similarly for bigger k.

4.3.2 Definition Let M be a smooth manifold. A k-form ω on M is called *closed* if $d\omega = 0$, and it is called *exact* if $\omega = d\eta$ for some $(k-1)$-form η on M. These conditions define subspaces of the real vector space of k-forms on M. Since $d^2 = 0$, every exact form is closed. The *k-th de Rham cohomology space* of M is the quotient vector space

$$H^k(M) = \{\text{closed } k\text{-forms}\}/\{\text{exact } k\text{-forms}\}.$$

□

4.3.3 Examples (i) For any smooth manifold M of dimension n, there are no exact 0-forms, and all n-forms are closed. Moreover $H^0(M) = \mathbb{R}^p$ where p is the number of connected components of M, and $H^k(M) = 0$ for $k > n$ since in this case there are no non-zero k-forms.

(ii) Let $\omega = f(x)dx$ be a 1-form on $\mathbb{R}$. Then $\omega = dg$ where $g(x) = \int_0^x f(t)\, dt$. Therefore every 1-form on $\mathbb{R}$ is exact and hence $H^1(\mathbb{R}) = 0$. It follows from Poincaré lemma 4.4.6 to be proved in the next section that $H^k(\mathbb{R}^n) = 0$ for all $k > 0$ and all n.

(iii) Owing to Stokes' theorem, an n-form ω on an n-dimensional orientable smooth manifold M (without boundary) can be of the form $d\eta$ for a *compactly supported*

$(n-1)$-form η only if $\int_M \omega = 0$; in particular, if M is compact, ω can be exact only if $\int_M \omega = 0$. On the other hand, if M is compact and oriented, let $(U, x_1, \ldots, x_n)$ be a positively oriented local coordinate system and let f be a non-zero, non-negative smooth function with compact support contained in U. Then $\omega = f\, dx_1 \wedge \cdots \wedge dx_n$ on U extends to an n-form on M with $\int_M \omega > 0$ and hence $H^n(M) \neq 0$. We will see later (Proposition 4.5.3 and Corollary 4.5.4) that "integration over M" defines an isomorphism $H^n(M) \cong \mathbb{R}$ for a compact connected orientable smooth n-manifold M.

(iv) The 1-form

$$\omega = \frac{-y\, dx + x\, dy}{x^2 + y^2}$$

on $M = \mathbb{R}^2 \setminus \{(0,0)\}$ is easily checked to be closed by a direct calculation. Let $\iota : S^1 \to M$ be the unit circle. If ω is exact, $\omega = df$ for some $f \in C^\infty(M)$, then $d(\iota^* f) = \iota^* df = \iota^* \omega$, so also $\iota^*\omega$ is exact. Since $\int_{S^1} \iota^*\omega = 2\pi \neq 0$, the discussion in part (iii) says that ω cannot be exact. It follows that $H^1(M) \neq 0$.

(v) Consider $M = S^1$. The polar coordinate function θ on S^1 is defined only locally, but any two determinations of the angle differ by a constant multiple of 2π, so its differential is a well-defined 1-form called the "angular form" and usually denoted by $d\theta$, although it is not globally exact (be careful!). It is easily seen that $\iota^*\omega = d\theta$, where ω is as in (iv), and so $H^1(S^1) \neq 0$. We next show that $\int_{S^1} : \Omega^1(S^1) \to \mathbb{R}$ induces an isomorphism $H^1(S^1) \to \mathbb{R}$. Every 1-form is closed, so we need only to identify its kernel with the exact 1-forms. Since $d\theta$ never vanishes, any 1-form α on S^1 can be written as $\alpha = f\, d\theta$ where $f \in C^\infty(S^1)$. Now $\int_{S^1} \alpha = 0$ says that $\int_0^{2\pi} f(e^{it})\, dt = 0$, so

$$\tilde{g}(t) = \int_0^t f(e^{is})\, ds$$

is a smooth, 2π-periodic function on $\mathbb{R}$ which induces $g \in C^\infty(S^1)$ such that $g(e^{it}) = \tilde{g}(t)$ for all $t \in \mathbb{R}$. It is clear that $dg = \alpha$, completing the argument.

■

4.3.4 Exercise Prove that the restriction of ω from Examples 4.3.3(iv) to the cut plane $\mathbb{R}^2 \setminus \{(x,0) \mid x \leq 0\}$ is exact. ■

Induced Maps in Cohomology

Let $f : M \to N$ be smooth. Since $d(f^*\omega) = f^*(d\omega)$ for any $\omega \in \Omega^\bullet(N)$, $f^*\omega$ is closed if ω is closed, and it is exact if ω is exact. Thus there is an induced homomorphism

$$f^* : H^k(N) \to H^k(M)$$

for each $k \geq 0$. In addition, if $g : N \to P$ is smooth, then

$$(g \circ f)^* = f^* \circ g^*.$$

Of course, the identity map $\mathrm{id} : M \to M$ induces the identity map in cohomology. Such properties show that de Rham cohomology defines a family of contravariant functors and, in particular, a diffeomorphism $f : M \to N$ induces an isomorphism between each one of the corresponding cohomology spaces of M and N. Thus de Rham cohomology is a differential invariant of smooth manifolds. We prove in the next section that it is a (smooth) homotopy invariant.

4.4 Homotopy Invariance of Cohomology

Let $f, g : M \to N$ be smooth maps between smooth manifolds. A (smooth) *homotopy* between f and g is a smooth map $F : M \times [0, 1] \to N$ such that

$$\begin{cases} F(p, 0) = f(p) \\ F(p, 1) = g(p) \end{cases}$$

for all $p \in M$. If there exists a homotopy between f and g, we say that they are *homotopic* (in the smooth sense).

4.4.1 Proposition *Let f, g be homotopic maps. Then the induced maps in de Rham cohomology*

$$f^*, g^* : H^k(N) \to H^k(M)$$

are equal.

The proof of this proposition is given below. First, we need to make some remarks. For $t \in [0, 1]$, consider the inclusions $i_t : M \to M \times [0, 1]$ given by

$$i_t(p) = (p, t)$$

for $p \in M$, and consider the natural projection $\pi : M \times [0, 1] \to M$ given by $\pi(p, t) = p$. Then, obviously,

$$\pi \circ i_t = \mathrm{id}_M$$

implying that

$$i_t^* \pi^* = \mathrm{id} \quad \text{on } \Omega^k(M) \text{ and } H^k(M).$$

We also consider the projection $t : M \times [0, 1] \to [0, 1]$. Then there exist a "vertical" vector field $\frac{\partial}{\partial t}$ and a 1-form dt on $M \times [0, 1]$. Note that $\ker d\pi$ is spanned by $\frac{\partial}{\partial t}$.

4.4.2 Lemma *Let $\omega \in \Omega^k(M \times [0,1])$. Then we can write*

$$\omega = \zeta + dt \wedge \eta, \tag{4.4.3}$$

where $\zeta \in \Omega^k(M \times [0,1])$ has the property that it vanishes if some of its arguments belong to $\ker d\pi$, and $\eta \in \Omega^{k-1}(M \times [0,1])$ has the same property.

Proof Set $\eta = \iota_{\frac{\partial}{\partial t}} \omega$ and $\zeta = \omega - dt \wedge \eta$. Since

$$\iota_{\frac{\partial}{\partial t}} \eta = \iota_{\frac{\partial}{\partial t}} \iota_{\frac{\partial}{\partial t}} \omega = 0,$$

it is clear that η has the claimed property. Similarly,

$$\begin{aligned} \iota_{\frac{\partial}{\partial t}} \zeta &= \iota_{\frac{\partial}{\partial t}} \omega - \iota_{\frac{\partial}{\partial t}} (dt \wedge \eta) \\ &= \eta - \iota_{\frac{\partial}{\partial t}} dt \wedge \eta + dt \wedge \iota_{\frac{\partial}{\partial t}} \eta \\ &= \eta - \eta + 0 \\ &= 0, \end{aligned}$$

as desired, where we have used that interior multiplication is an anti-derivation. □

We define the homotopy operator

$$H_k : \Omega^k(M \times [0,1]) \to \Omega^{k-1}(M)$$

by the formula

$$(H_k \omega)_p(v_1, \ldots, v_{k-1}) = \int_0^1 \eta_{(p,t)}(di_t(v_1), \ldots, di_t(v_{k-1}))\, dt,$$

where ω is decomposed as in (4.4.3) and $p \in M$, $v_1, \ldots, v_{k-1} \in T_p M$. Note that H_k is "integration along the fiber of π". For simplicity, we henceforth drop the subscript and write just H for the homotopy operator.

Proof of Proposition 4.4.1 Let $\omega \in \Omega^k(M \times [0,1])$. We first claim that

$$dH\omega + Hd\omega = i_1^* \omega - i_0^* \omega. \tag{4.4.4}$$

The proof is by direct computation: since this is a pointwise identity, we can work on a coordinate system. Let $(U, x_1, \ldots, x_n)$ be a coordinate system in M. Then $(U \times [0,1], x_1 \circ \pi, \ldots, x_n \circ \pi, t)$ is a coordinate system in $M \times [0,1]$ and we can write

$$\omega|_{U \times [0,1]} = \sum_I a_I dx_I + dt \wedge \sum_J b_J dx_J,$$

where a_I, b_J are smooth functions on $U \times [0,1]$ and I, J are increasing multi-indices. On $U \times [0,1]$, we have

$$H\omega = \sum_J \left(\int_0^1 b_J dt \right) dx_J,$$

$$dH\omega = \sum_{J,i} \left(\int_0^1 \frac{\partial b_J}{\partial x_i} dt \right) dx_i \wedge dx_J,$$

$$d\omega = \sum_{I,i} \frac{\partial a_I}{\partial x_i} dx_i \wedge dx_I + \sum_I \frac{\partial a_I}{\partial t} dt \wedge dx_I - dt \wedge \sum_{J,i} \frac{\partial b_J}{\partial x_i} dx_i \wedge dx_J,$$

$$Hd\omega = \sum_I \left(\int_0^1 \frac{\partial a_I}{\partial t} dt \right) dx_I - \sum_{J,i} \left(\int_0^1 \frac{\partial b_J}{\partial x_i} dt \right) dx_i \wedge dx_J.$$

It follows that

$$\begin{aligned} dH\omega + Hd\omega|_p &= \sum_I \left(\int_0^1 \frac{\partial a_I}{\partial t}(p,t)\, dt \right) dx_I \\ &= \sum_I (a_I(p,1) - a_I(p,0)) dx_I \\ &= i_1^*\omega - i_0^*\omega|_p, \end{aligned}$$

as claimed.

Suppose now that $F : M \times [0,1] \to N$ is a homotopy between f and g. Let α be a closed k-form in N representing the cohomology class $[\alpha] \in H^k(N)$. Applying identity (4.4.4) to $\omega = F^*\alpha$ yields

$$dHF^*\alpha + HF^*d\alpha = i_1^*F^*\alpha - i_0^*F^*\alpha.$$

Since $d\alpha = 0$ and $F \circ i_0 = f$, $F \circ i_1 = g$, we get

$$d\left(HF^*\alpha\right) = g^*\alpha - f^*\alpha.$$

Hence $g^*\alpha$ and $f^*\alpha$ are cohomologous. □

Two smooth manifolds M and N are said to have the same *homotopy type* (in the smooth sense) and are called *homotopy equivalent* (in the smooth sense) if there exist smooth maps $f : M \to N$ and $g : N \to M$ such that $g \circ f$ and $f \circ g$ are (smoothly) homotopic to the identity maps on M and N, respectively. Each of the maps f and g is then called a *homotopy equivalence*, and f and g are called *inverses up to homotopy* or *homotopy inverses*. A manifold homotopy equivalent to a point is called *contractible*.

4.4.5 Corollary *A homotopy equivalence between smooth manifolds induces an isomorphism in de Rham cohomology.*

4.4.6 Corollary (Poincaré Lemma) *The de Rham cohomology of $\mathbb{R}^n$ (or a star-shaped open subset of $\mathbb{R}^n$) is $\mathbb{R}$ in dimension zero and zero otherwise:*

$$H^k(\mathbb{R}^n) = \begin{cases} \mathbb{R} \text{ if } k = 0, \\ 0 \text{ if } k > 0. \end{cases}$$

Consider an inclusion $\iota : A \to M$. A (smooth) map $r : M \to A$ satisfying $r \circ \iota = \mathrm{id}_A$ is called a *retraction*. A special case of homotopy equivalence is the case in which $\iota \circ r : M \to M$ is homotopic to id_M; if that happens, r is called a *deformation retraction* of M onto A and A is called a *deformation retract* of M.

4.4.7 Exercise Check that $r : \mathbb{R}^2 \setminus \{0\} \to S^1$ given by $r(x) = \frac{x}{||x||}$ is a deformation retraction. Compare with Examples 4.3.3(iv) and (v). ■

4.4.8 Lemma *There exists no smooth retraction $r : \bar{B}^n \to \partial \bar{B}^n$ from the closed unit n-ball onto its boundary.*

Proof The case $n = 1$ is easy as a retraction is surjective, the closed interval $\bar{B}^1$ is connected and its boundary is disconnected. Assume $n \geq 2$ and suppose, to the contrary, that such a retraction r exists. From $r \circ \iota = \mathrm{id}_{\partial \bar{B}^n}$ we deduce that $\iota^* r^* = \mathrm{id}$ and thus that $r^* : H^{n-1}(\partial \bar{B}^n) \to H^{n-1}(\bar{B}^n)$ is injective. However, $\partial \bar{B}^n = S^{n-1}$ and $H^{n-1}(S^{n-1}) \neq 0$ (Examples 4.3.3(iii)), whereas $H^{n-1}(\bar{B}^n) = 0$ (Corollary 4.4.6), which is a contradiction. □

4.4.9 Theorem (Smooth Brouwer's Fixed Point Theorem) *Let $f : \bar{B}^n \to \bar{B}^n$ be a smooth map. Then there exists $p \in \bar{B}^n$ such that $f(p) = p$. In other words, every smooth self-map of the closed unit n-ball admits a fixed point.*

Proof Suppose, on the contrary, that $f(x) \neq x$ for all $x \in \bar{B}^n$. The half-line originating at $f(x)$ and going through x meets $\partial \bar{B}^n$ at a unique point; call it $r(x)$. It is easy to see that this defines a smooth retraction $r : \bar{B}^n \to \partial \bar{B}^n$, which is prohibited by Lemma 4.4.8. □

4.4.10 Remark The theorem is not true in the case of the open n-ball, as is easily seen.

For the next result, consider the unit sphere $\iota : S^n \to \mathbb{R}^{n+1}$. It is useful to have an explicit expression for a non-zero element in $H^n(S^n)$ (Example 4.1.7):

$$\omega = (-1)^i \frac{1}{x_i} \, dx_1 \wedge \cdots \wedge \hat{dx_i} \wedge \cdots \wedge dx_{n+1} \tag{4.4.11}$$

on $x_i \neq 0$, for $i = 1, \ldots, n+1$ (see also Problem 3).

4.4.12 Theorem (Smooth Hairy Ball Theorem) *Let X be a smooth vector field on S^{2m} $(m > 0)$. Then there exists $p \in S^{2m}$ such that $X_p = 0$. In other words, every smooth vector field on an even-dimensional sphere has a zero.*

Proof Suppose, on the contrary, that X never vanishes. By rescaling, we may assume that X is a unit vector field with respect to the metric induced from Euclidean space. Set

$$F_t : S^{2m} \to S^{2m}, \quad F_t(p) = \cos t\, p + \sin t\, X(p).$$

It is clear that F_t defines a homotopy between the identity map and the antipodal map of S^{2m}:

$$F_0 = \mathrm{id}_{S^{2m}} \quad \text{and} \quad F_\pi = -\mathrm{id}_{S^{2m}}.$$

Note that

$$F_\pi^*(x_i \circ \iota) = -x_i \circ \iota.$$

It follows that

$$F_\pi^* \omega = (-1)^{2m+1} \omega = -\omega,$$

where ω is as in (4.4.11). On the other hand,

$$F_0^* \omega = \omega,$$

and by Proposition 4.4.1, $F_0^*\omega$ and $F_\pi^*\omega$ are cohomologous, which contradicts the fact that ω is not cohomologous to zero. □

4.4.13 Corollary *An even-dimensional sphere (of positive dimension) cannot admit a structure of Lie group compatible with its standard topology and smooth structure (compare Remark 4.4.18 below).*

Proof It follows from the theorem together with Problem 40 in ▶ Chap. 1 that even-dimensional spheres are never parallelizable, but Lie groups are always parallelizable according to Problem 6 in ▶ Chap. 3. □

4.4.14 Remark Regarding the proof of Corollary 4.4.13, it is known that the only parallelizable spheres are S^0, S^1, S^3, and S^7 [Ker58, BM58]; the proof requires some algebraic topology.

4.4.15 Remark Theorems 4.4.9 and 4.4.12 can be extended to the continuous category by using appropriate approximation results.

We close this section computing the de Rham cohomology of the n-sphere. The argument is a nice presentation of the "Mayer–Vietoris principle" in a very special case.

4.4.16 Proposition *The de Rham cohomology of S^n vanishes except in dimensions 0 and n.*

Proof We may assume $n > 1$. We prove first that $H^1(S^n) = 0$. Let ω be a closed 1-form on S^n. We must show that ω is exact. Decompose S^n into the union of two open sets U and V, where U is a neighborhood of the northern hemisphere diffeomorphic to an open n-ball, V is a neighborhood of the southern hemisphere diffeomorphic to an open n-ball, and $U \cap V$ is a neighborhood of the equator which is diffeomorphic to $S^{n-1} \times (-1, 1)$. Since U and V are contractible, $\omega|_U = df$ for a smooth function f on U and $\omega|_V = dg$ for a smooth function g on V. In general on $U \cap V$, f and g do not agree, but the difference $h := f|_{U\cap V} - g|_{U\cap V}$ has $dh = \omega|_{U\cap V} - \omega|_{U\cap V} = 0$. Since $n > 1$, S^{n-1} is connected and thus h is a constant. Setting

$$k := \begin{cases} f & \text{on } U \\ g + h & \text{on } V \end{cases}$$

defines a smooth function on S^n such that $dk = \omega$, as we wished.

We proceed by induction on n. Let ω be a closed k-form on S^n for $1 < k < n$. We shall prove that ω is exact using the same decomposition $S^n = U \cup V$ as above and the induction hypothesis. As above, $\omega|_U = d\alpha$ for a $(k-1)$-form α on U an $\omega|_V = d\beta$ for a $(k-1)$-form β on V. Let $\gamma = \alpha|_{U\cap V} - \beta|_{U\cap V}$. Then $d\gamma = 0$. Since γ is a closed $(k-1)$-form ξ on $U \cap V$ and $U \cap V$ is homotopy equivalent to S^{n-1}, by the induction hypothesis, $\gamma = d\xi$ for a $(k-2)$-form ξ on $U \cap V$. Let $\{\rho_U, \rho_V\}$ be a partition of unity subordinate to $\{U, V\}$. Setting

$$\eta := \begin{cases} \alpha - d(\rho_V \xi) & \text{on } U \\ \beta + d(\rho_U \xi) & \text{on } V \end{cases}$$

defines a $(k-1)$-form on S^n such that $d\eta = \omega$. This completes the induction step and the proof of the theorem. □

4.4.17 Remark The "Mayer–Vietoris principle" indeed yields a long exact sequence in cohomology. One nice application is to show that the de Rham cohomology spaces of a compact manifold are always finite-dimensional [BT82, §5].

4.4.18 Remark *The only spheres that admit a structure of Lie group compatible with their standard topology and smooth structure are S^0, S^1, and S^3.* This result will not be needed elsewhere in the text, but for the sake of the interested reader we sketch an argument. Note that S^0 is simply the group with two elements, and S^1 and S^3 are, respectively, diffeomorphic to $SO(2)$ and $SU(2)$ (cf. Problem 2 in ▶ Chap. 3). Another way is to see S^0, S^1, S^3 as subgroups of the multiplicative groups $\mathbb{R}^\times$, $\mathbb{C}^\times$, $\mathbb{H}^\times$, respectively (for $\mathbb{H}$, see Problems 7 and 8 in ▶ Chap. 3). Conversely, suppose a compact connected Lie group G is diffeomorphic to S^n. If G is Abelian, then its universal covering Lie group $\tilde{G}$ is also Abelian (Lemma 3.5.2) and hence is $\mathbb{R}^n$ (Theorem 3.5.3). It follows that $n = 1$. On the other hand, if G is not Abelian, we claim that $H^3(G) \neq 0$ and hence Proposition 4.4.16 yields $n = 3$. To check the claim, let $\langle , \rangle$ be an Ad-invariant inner product on $\mathfrak{g}$ as in Problem 9 and put $\omega(X, Y, Z) = \langle [X, Y], Z\rangle$

for $X, Y, Z \in \mathfrak{g}$. This defines a bi-invariant 3-form ω on G which is thus closed (Problem 21), so it will define a non-trivial element of $H^3(G)$ as long as we show it is not exact. Suppose, to the contrary, that $\omega = d\eta$ for a 2-form η on G. Fix a bi-invariant integral on $G \times G$ with total volume 1 (Problem 8). Then $\bar{\eta} = \int_{G\times G} L_g^* R_{h^{-1}}^* \eta$ defines a bi-invariant form (where the integrand is viewed as a function of $(g, h) \in G \times G$) and

$$\begin{aligned} d\bar{\eta} &= d\left(\int_{G\times G} L_g^* R_{h^{-1}}^* \eta\right) \\ &= \int_{G\times G} d(L_g^* R_{h^{-1}}^* \eta) \\ &= \int_{G\times G} L_g^* R_{h^{-1}}^* d\eta \\ &= \int_{G\times G} L_g^* R_{h^{-1}}^* \omega \\ &= \int_{G\times G} \omega \\ &= \omega. \end{aligned}$$

(To see differentiation under the integral sign in the second equality is correct, one can write all integrals in local coordinates.) But $\bar{\eta}$ is bi-invariant by construction, so $\omega = d\bar{\eta} = 0$ (Problem 21), a contradiction to the assumption that G is non-Abelian.

4.5 Degree Theory

Our first aim is to prove that the top dimensional de Rham cohomology of a compact connected orientable smooth manifold is one-dimensional. We start with a lemma in Calculus.

4.5.1 Lemma *Let f be a smooth function on $\mathbb{R}^n$ with support in the open cube $C^n = (-1, 1)^n$ and*

$$\int_{\mathbb{R}^n} f\, dx_1 \cdots dx_n = 0.$$

Then there exist smooth functions $f_1, \ldots, f_n$ on $\mathbb{R}^n$ with support in C^n such that

$$f = \sum_{i=1}^{n} \frac{\partial f_i}{\partial x_i}.$$

Proof By induction on n. If $n = 1$, we simply define $f_1(x_1) = \int_{-\infty}^{x_1} f(t)\,dt$. If $n \geq 2$, define a smooth function g on $\mathbb{R}^{n-1}$ by

$$g(x_1, \ldots, x_{n-1}) = \int_{-\infty}^{+\infty} f(x_1, \ldots, x_{n-1}, t)\,dt.$$

Then g has total integral zero by Fubini's theorem, and clearly support contained in C^{n-1}, so by the induction hypothesis we can write

$$g = \sum_{i=1}^{n-1} \frac{\partial g_i}{\partial x_i}$$

for smooth functions g_i on $\mathbb{R}^{n-1}$ with support in C^{n-1}. Now choose a smooth function ρ on $\mathbb{R}$ with support in $(-1, 1)$ and total integral 1, and define $f_j : \mathbb{R}^n \to \mathbb{R}$ by

$$f_j(x_1, \ldots, x_{n-1}, x_n) = g_j(x_1, \ldots, x_{n-1})\rho(x_n)$$

for $j = 1, \ldots, n-1$. Clearly the f_j have support in C^n. Set

$$h = f - \sum_{i=1}^{n-1} \frac{\partial f_j}{\partial x_j}$$

and

$$f_n(x_1, \ldots, x_{n-1}, x_n) = \int_{-\infty}^{x_n} h(x_1, \ldots, x_{n-1}, t)\,dt.$$

Clearly h has support in C^n, and an easy calculation shows that

$$\int_{-\infty}^{+\infty} h(x_1, \ldots, x_{n-1}, t)\,dt = 0,$$

so $\operatorname{supp}(f_n) \subset C^n$ and we are done. □

4.5.2 Lemma *Let ω be an n-form on $\mathbb{R}^n$ with support contained in the open cube C^n such that $\int_{\mathbb{R}^n} \omega = 0$. Then there exists an $(n-1)$-form η on $\mathbb{R}^n$ with support contained in C^n such that $d\eta = \omega$.*

Proof Poincaré's lemma (4.4.6) yields η with $d\eta = \omega$ but does not give information about the support of η. Instead, write $\omega = f\,dx_1 \wedge \cdots \wedge dx_n$ for $f \in C^\infty(\mathbb{R}^n)$. Then $\operatorname{supp}(f) \subset C^n$ and $\int_{\mathbb{R}^n} f\,dx_1 \cdots dx_n = 0$, so $f = \sum_i \frac{\partial f_i}{\partial x_i}$ as in Lemma 4.5.1, and thus $\omega = d\eta$ where $\eta = \sum_i (-1)^{i-1} f_i\,dx_1 \wedge \cdots \wedge \hat{dx_i} \wedge \cdots \wedge dx_n$. □

4.5.3 Proposition *If M is a compact connected orientable smooth manifold of dimension n, then $H^n(M) = \mathbb{R}$.*

Proof By compactness, there is a finite cover $\{U_1, \ldots, U_m\}$ by coordinate neighborhoods diffeomorphic to the open cube C^n. Let ω_0 be a bump n-form as in Examples 4.3.3(iii) with support contained in U_1 and total integral 1. Then ω_0 defines a non-zero cohomology class in $H^n(M)$. We shall prove that any n-form ω on M is cohomologous to a multiple of ω_0, namely, $\omega = c\,\omega_0 + d\eta$ for some $c \in \mathbb{R}$ and some $(n-1)$-form η. Using a partition of unity $\{\rho_i\}$ strictly subordinate to $\{U_i\}$, we can write $\omega = \sum_{i=1}^m \rho_i \omega$ where $\rho_i \omega$ is an n-form with support in U_i. By linearity, it suffices to prove the result for $\rho_i \omega$, so we may assume from the outset that the support of ω is contained in U_k, for some $k = 1, \ldots, m$.

Owing to the connectedness of M, we can find a chain $U_{i_1}, \ldots, U_{i_r}$ such that $U_{i_1} = U_1$, $U_{i_r} = U_k$, and $U_{i_j} \cap U_{i_{j+1}} \neq \varnothing$ for all $j = 1, \ldots, r-1$. For all $j = 1, \ldots, r-1$, choose an n-form ω_j with support in $U_{i_j} \cap U_{i_{j+1}}$ and total integral 1. Now $\omega_0 - \omega_1$ has support in $U_{i_1} = U_1$ and total integral zero, so, by Lemma 4.5.2, there exists η_1 with support in U_1 such that

$$\omega_0 - \omega_1 = d\eta_1.$$

Next, $\omega_1 - \omega_2$ has support in U_{i_2} and total integral zero, so the lemma yields η_2 with support in U_{i_2} such that

$$\omega_1 - \omega_2 = d\eta_2.$$

Continuing, we find η_j with support in U_{i_j} such that

$$\omega_{j-1} - \omega_j = d\eta_j$$

for all $j = 1, \ldots, r-1$. Adding up, we get

$$\omega_0 - \omega_{r-1} = d\eta,$$

where $\eta = \sum_{j=1}^{r-1} \eta_j$. Now $U_{i_r} = U_k$ contains the support of ω and ω_{r-1}, and $\omega - c\omega_{r-1}$ has total integral zero, where $c = \int_M \omega$. By applying the lemma again,

$$\omega - c\omega_{r-1} = d\zeta$$

and hence

$$\omega = c\omega_0 + d(\zeta - c\eta),$$

as required. □

4.5.4 Corollary *Let M be a compact connected oriented smooth manifold of dimension n. Then "integration over M"*

$$\int_M : H^n(M) \to \mathbb{R}$$

is a well-defined linear isomorphism which is positive precisely on the cohomology classes defined by nowhere vanishing n-forms belonging to the orientation of M.

Proof By Stokes' formula, the integral of an exact form is zero, so the integral of an n-form depends only on its cohomology class and thus the map is well-defined. By the proposition, $H^n(M)$ is one-dimensional and there exist bump n-forms with non-zero integral, so the map is an isomorphism.

Let ω be a nowhere vanishing n-form belonging to the orientation of M, choose an oriented atlas $\{(U_\alpha, \varphi_\alpha = (x_1^\alpha, \dots, x_n^\alpha))\}$ and a partition of unity $\{\rho_\alpha\}$ strictly subordinate to $\{U_\alpha\}$. Then $\omega = \sum_\alpha \rho_\alpha \omega$, where $\rho_\alpha \omega$ has support in U_α and on which its local representation is of the form $f_\alpha \, dx_1^\alpha \wedge \cdots \wedge dx_n^\alpha$ for a non-zero, non-negative smooth function f_α on U_α. It follows that

$$\int_M \omega = \sum_\alpha \int_{\varphi_\alpha(U_\alpha)} (f_\alpha \circ \varphi_\alpha^{-1}) \, dx_1 \cdots dx_n > 0$$

since $f_\alpha \geq 0$ and it is positive somewhere. Conversely, if ω' is an n-form with $\int_M \omega' > 0$, then ω' is cohomologous to $c\omega$, where $c = \int_M \omega' / \int_M \omega > 0$, and $c\omega$ and ω are nowhere vanishing n-forms defining the same orientation on M. □

Let $f : M \to N$ be a smooth map between compact connected oriented manifolds of the same dimension n. Let ω_M, ω_N be n-forms on M, N, respectively, with total integral one. Then $f^* : H^n(N) \to H^n(M)$ carries $[\omega_N]$ to a multiple of $[\omega_M]$; this number is called the *degree* of f, denoted $\deg f$. It follows from Proposition 4.4.1 that homotopic maps have the same degree.

4.5.5 Remark In case $N = S^n$, Hopf's degree theorem [GP10, Chap. 3, §6] asserts that non-homotopic maps have different degrees. For the case $n = 1$, see Problem 29.

4.5.6 Proposition *Let $f : M \to N$ be a smooth map.*

a. The degree of f is an integer.

b. For all $\omega \in \Omega^n(N)$,

$$\int_M f^*\omega = (\deg f) \int_N \omega.$$

c. If $q \in N$ is a regular value of f, then

$$\deg f = \sum_{p \in f^{-1}(q)} \operatorname{sgn}(\det df_p) \qquad \textit{(finite sum)}.$$

Proof (b) follows from the commutativity of the diagram
where the vertical arrows are isomorphisms, and (a) follows from (c). Let us prove (c).

Consider first the case in which q is a point outside the image of f. Since $f(M)$ is compact, we can find a bump n-form α on N with total integral one and support disjoint

$$\begin{array}{ccc} H^n(N) & \xrightarrow{f^*} & H^n(M) \\ \int_N \downarrow & & \downarrow \int_M \\ \mathbb{R} & \xrightarrow[\deg f]{} & \mathbb{R} \end{array}$$

from $f(M)$. It follows from (b) that $\deg f = \int_M f^*\alpha = 0$. Since $f^{-1}(q) = \varnothing$, (c) is proved in this case.

Suppose now q lies in the image of f. Since q is a regular value and $\dim M = \dim N$, f is a local diffeomorphism at each $p \in f^{-1}(q)$. In particular, $f^{-1}(q)$ is discrete and thus finite, due to the compactness of M. Write $f^{-1}(q) = \{p_1, \ldots, p_m\}$ and choose open neighborhoods $\tilde{U}_i$ of p_i and V_i of q such that $f : \tilde{U}_i \to V_i$ is a diffeomorphism for all $i = 1, \ldots, m$. Setting $V = \cap_{i=1}^m V_i$ and $U_i = \tilde{U}_i \cap f^{-1}(V)$, now $f : U_i \to V$ is a diffeomorphism for all i. Moreover, $f(M \setminus \cup_{i=1}^m \tilde{U}_i)$ is a compact subset of N disjoint from q, so by further shrinking V we can ensure that $f^{-1}(V) = \cup_{i=1}^m U_i$.

Let α be an n-form on N with total integral one and support contained in V. Then $f^*\alpha$ is an n-form on M with support in $\cup_{i=1}^m U_i$. In view of Exercise 4.1.8,

$$\int_{U_i} f^*\alpha = \operatorname{sgn}(\det df_{p_i}) \int_V \alpha = \operatorname{sgn}(\det df_{p_i}),$$

where we consider the determinant of the Jacobian matrix of f at p_i relative to orientation-preserving local charts around p_i and q, so its sign is $+1$ if $df_{p_i} : T_{p_i}M \to T_qN$ preserves orientation and -1 if it reverses orientation. It follows that

$$\deg f = \int_M f^*\alpha = \sum_{i=1}^p \int_{U_i} f^*\alpha = \sum_{i=1}^p \operatorname{sgn}(\det df_{p_i}),$$

as desired. □

4.5.7 Corollary *The degree of a non-surjective map is zero.*

4.5.8 Remark It follows from Sard's theorem (cf. Remark 1.4.15) that in Proposition 4.5.6 there always exists a regular value of f.

More generally, in case M has finitely many connected components $M_1, \ldots, M_r$, the degree of $f : M \to N$ can still be defined as the sum of the degrees of the restrictions $f : M_i \to N$, and Proposition 4.5.6 remains true.

4.5.9 Example Consider S^1 as the set of unit complex numbers. Then $f : S^1 \to S^1$ given by $f(z) = z^n$ is smooth and has degree n, which we can show as follows. Recall that the angular form $d\theta = -y\,dx + x\,dy|_{S^1}$ generates $H^1(S^1)$. Removal of one point does not change the integral below on the left-hand side, and $h : (0, 2\pi) \to S^1 \setminus \{1\}$, $h(t) = e^{it}$ is an orientation-preserving diffeomorphism, so

$$\int_{S^1} f^* d\theta = \int_0^{2\pi} h^* f^* d\theta = \int_0^{2\pi} (f \circ h)^* d\theta,$$

where $(f \circ h)^* d\theta$ is exact on $(0, 2\pi)$ and in fact equal to $n\,dt$ by direct computation. Therefore

$$\int_{S^1} f^* d\theta = \int_0^{2\pi} n\,dt = 2\pi n = n \cdot \int_{S^1} d\theta,$$

as we wished. ■

4.5.10 Example Let $f : S^1 \to \mathbb{R}^2$ be a smooth map. Its image is a closed curve in the plane. Fix a point q not in the curve. The *winding number* $W(f, q)$ of f with respect to q is the degree of the map $u : S^1 \to S^1$ given by

$$u(x) = \frac{f(x) - q}{||f(x) - q||}.$$

Note that $W(f, q_1) = W(f, q_2)$ if q_1 and q_2 lie in the same connected component of the complement of the image of f.

Introducing the complex variable $z = x + iy$ we have

$$\frac{-y\,dx + x\,dy}{x^2 + y^2} = \Im\left\{\frac{1}{z} dz\right\}$$

(compare Examples 4.3.3(iv)). Using this formula, it is easy to arrive at the complex integral for the winding number,

$$W(f, q) = \frac{1}{2\pi i} \int_C \frac{dz}{z - q}, \tag{4.5.11}$$

where C is the image of f (Cauchy 1825 [Cau25]).

■

4.5.12 Example Let $f, g : S^1 \to \mathbb{R}^3$ be two smooth maps. Their images yield two closed curves in $\mathbb{R}^3$ which we assume to be disjoint. The *linking number* $\mathrm{Lk}(f, g)$ is the degree of the map $F : S^1 \times S^1 \to S^2$ given by

$$F(x, y) = \frac{f(x) - g(y)}{||f(x) - g(y)||}.$$

If $f_t, g_t : S^1 \to \mathbb{R}^3$ are homotopies of f, g such that f_t and g_t have disjoint images for all t, then $\mathrm{Lk}(f_t, g_t)$ is independent of t.

In case $f, g : S^1 \to S^3$, one chooses $q \in S^3$ not in the image of those maps and performs stereographic projection $S^3 \setminus \{q\} \to \mathbb{R}^3$ to define their linking number. Moving q continuously yields homotopies of the associated maps $S^1 \to \mathbb{R}^3$, so, since the union of the images of f and g does not disconnect S^3, this definition does not depend on the choice of q.

According to Problems 7 and 15, the volume form of S^2, normalized to have total integral 1, is

$$dA = \frac{1}{4\pi}\left(x_1\, dx_2 \wedge dx_3 + x_2\, dx_3 \wedge dx_1 + x_3\, dx_1 \wedge dx_2\right).$$

Since

$$F^* dA = \frac{1}{4\pi} F \cdot \frac{\partial F}{\partial x} \times \frac{\partial F}{\partial y}\, dx \wedge dy,$$

an easy calculation yields the formula for the linking number (Gauss 1833 [RN11])

$$\text{Lk}(f, g) = -\frac{1}{4\pi} \int_{S^1} \int_{S^1} \frac{f(x) - g(y)}{||f(x) - g(y)||^3} \cdot \frac{df}{dx} \times \frac{dg}{dy}\, dx dy. \tag{4.5.13}$$

■

4.5.14 Example We can generalize Example 4.5.10 as follows. Let $f : M \to \mathbb{R}^{n+1}$ be a smooth map from a compact, connected oriented manifold M of dimension n. If $q \in \mathbb{R}^{n+1}$ does not lie in the image of f, the winding number $W(f, q)$ of f with respect to q is the degree of the map $u : M \to S^n$ given by

$$u(x) = \frac{f(x) - q}{||f(x) - q||}.$$

It records how f "wraps" around q. ■

4.5.15 Exercise Check formulae (4.5.11) and (4.5.13). ■

4.6 The Borsuk–Ulam Theorem

The Borsuk–Ulam theorem is one of the theorems in topology with most applications in practice. It was conjectured by Ulam at the Scottish Cafe in Lvov. The theorem proven in one form by Borsuk in 1933 has several other equivalent formulations and many different proofs. One well-known of these was first given by Lyusternik and Shnirel´man in 1930. A host of extensions and generalizations, and numerous interesting applications to areas that include combinatorics, differential equations, and even economics add to its importance.

4.6.1 Lemma *Let $F : \bar{B}^n \to \mathbb{R}^n$ be a smooth map (recall $\bar{B}^n$ denotes the closed unit n-ball). Denote the restriction of F to the boundary $\partial \bar{B}^n$ by f and let $q \in \mathbb{R}^n$ be a regular value of F that does not lie in the image of f. Then the winding number $W(f, q)$ equals the number of preimages of q under F counted with signs according to whether F preserves or reverses orientation at the point, as in Proposition 4.5.6.*

Proof Suppose first that q does not lie in the image of F. Let $f_t : S^{n-1} \to \mathbb{R}^n$ be defined by $f_t(x) = F((1-t)x)$ for $0 \le t \le 1$. Then $f_0 = f$ and

$$u_t(x) = \frac{f_t(x) - q}{||f_t(x) - q||}$$

defines an homotopy from u_0 to the constant map u_1. This shows that $W(f, q) = \deg(u_0) = \deg(u_1) = 0$.

Suppose next that $F^{-1}(q) = \{p_1, \ldots, p_k\}$, and let B_i be a small open ball around p_i such that the $\bar{B}_i$'s are disjoint one from another and from the boundary of $\bar{B}^n$. Let $f_i : \partial \bar{B}_i \to \mathbb{R}^n$ be the restriction of F. Note that $W(f_i, q) = \pm 1$ according to whether F preserves or reverses orientation at p_i. On the other hand, set $X = \bar{B}^n \setminus \bigcup_{i=1}^k B_i$. The map

$$u(x) = \frac{F(x) - q}{||F(x) - q||}$$

is well-defined and smooth on X. By Problem 30, $\deg u|_{\partial X} = 0$. It follows that

$$\begin{aligned} W(f, q) &= \deg u|_{\partial \bar{B}^n} \\ &= \sum_{i=1}^k \deg u|_{\partial \bar{B}_i} \\ &= \sum_{i=1}^k W(f_i, q) \\ &= \sum_{i=1}^k \operatorname{sgn}(\det dF_{p_i}), \end{aligned}$$

as we wished. □

A map $f : S^n \to \mathbb{R}^{n+1}$ will be called *odd* or *antipode-preserving* if $f(-x) = -f(x)$ for all $x \in S^n$, where $-x$ denotes the antipodal point of x.

4.6.2 Theorem (Borsuk–Ulam) *A smooth odd map $f : S^n \to S^n$ has odd degree.*

Proof We proceed by induction on n. The initial case $n = 1$ is Problem 34. Next assume the result true for $n - 1$ and let $f : S^n \to S^n$ be a smooth odd map.

Let $g : S^{n-1} \to S^n$ be the restriction of f to the equator. By Sard's theorem (Remark 1.4.15), there is $q \in S^n$ which is a regular value of both f and g. This means that q is not in the image of g (by dimensional reasons) and the oriented number of preimages of q under f is the degree d of f.

By composing f with a rotation, we may assume that q is the north pole. Since f is odd (and f does not hit q along the equator), the south pole $-q$ is also a regular value of f, and f hits q in the southern hemisphere as many times as it hits $-q$ in the northern hemisphere S^n_+. Let f_+ denote the restriction of f to S^+_n. Now d is the oriented number of preimages of $\{\pm q\}$ under f_+. Another way is to consider the orthogonal projection to the equatorial plane

$\pi : S^n_+ \to \bar{B}^n$ and note that d is the oriented number of preimages of 0 under $\pi \circ f_+$. Since 0 does not lie in the image of $\pi \circ g$, Lemma 4.6.1 implies that $d = W(\pi \circ g, 0) = \deg(\frac{\pi \circ g}{||\pi \circ g||})$ which, by the induction hypothesis, is odd, as $\frac{\pi \circ g}{||\pi \circ g||} : S^{n-1} \to S^{n-1}$ is an odd map. □

4.6.3 Corollary *Let $f_1, \ldots, f_n$ be smooth functions on S^n. Then there is a pair of antipodal points $\pm p \in S^n$ such that*

$$f_1(p) = f_1(-p), \ldots, f_n(p) = f_n(-p).$$

Proof Let $f : S^n \to \mathbb{R}^n$ have components f_i and suppose, to the contrary, that $g(x) = f(x) - f(-x)$ never vanishes. Then $h : S^n \to S^n$ defined by

$$h(x) = \left(\frac{g(x)}{||g(x)||}, 0 \right)$$

is a smooth odd map that never hits the points $(0, \ldots, 0, \pm 1) \in S^n$. By Corollary 4.5.7, $\deg h = 0$, contradicting Theorem 4.6.2. □

A popular illustration of Corollary 4.6.3 in case $n = 2$ is that if a balloon is deflated and laid flat on a table then at least two antipodal points end up on top one of another.

4.7 Maxwell's Equations

Maxwell's equations are a set of partial differential equations that, together with the Lorentz force law, form the foundation of classical electrodynamics, classical optics, and electric circuits. These fields in turn underlie modern electrical and communications technologies. Maxwell's equations describe how electric and magnetic fields are generated and altered by each other and by charges and currents. They are named after the Scottish physicist and mathematician James Clerk Maxwell who published an early form of those equations between 1861 and 1862.

The *electric field*

$$\vec{E}(t) = (E_1, E_2, E_3)$$

and the *magnetic field*

$$\vec{B}(t) = (B_1, B_2, B_3)$$

are vector fields on $\mathbb{R}^3$. *Maxwell's equations* are

$$\operatorname{div} \vec{E} = 4\pi\rho$$
$$\operatorname{div} \vec{B} = 0$$

$$\operatorname{curl} \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\operatorname{curl} \vec{B} = \frac{\partial \vec{E}}{\partial t} + 4\pi \vec{J}$$

where ρ is the *electric charge density* and $\vec{J} = (J_1, J_2, J_3)$ is the *electric current density*.

Minkowski spacetime is $\mathbb{R}^4$ with coordinates (t, x_1, x_2, x_3) and an inner product of signature $(-+++)$. The *electromagnetic field* is $F \in \Omega^2(\mathbb{R}^4)$ given by

$$F = (E_1 dx_1 + E_2 dx_2 + E_3 dx_3) \wedge dt + B_1 dx_2 \wedge dx_3 + B_2 dx_3 \wedge dx_1 + B_3 dx_1 \wedge dx_2.$$

We use the Hodge star (Problem 11 in ▶ Chap. 2) to write

$$*F = -(B_1 dx_1 + B_2 dx_2 + B_3 dx_3) \wedge dt + E_1 dx_2 \wedge dx_3 + E_2 dx_3 \wedge dx_1 + E_3 dx_1 \wedge dx_2.$$

The *source* is $\mathcal{J} \in \Omega^3(\mathbb{R}^4)$ given by

$$\begin{aligned} \mathcal{J} &= *(-\rho\, dt + J_1\, dx_1 + J_2\, dx_2 + J_3\, dx_3) \\ &= \rho dx_1 \wedge dx_2 \wedge dx_3 \\ &\quad -dt \wedge (J_1 dx_2 \wedge dx_3 + J_2 dx_3 \wedge dx_1 + J_3 dx_1 \wedge dx_2). \end{aligned} \tag{4.7.1}$$

Now Maxwell's equations are equivalent to

$$\begin{aligned} dF &= 0 \\ d * F &= 4\pi \mathcal{J}. \end{aligned} \tag{4.7.2}$$

The second equation says in particular that $\mathcal{J}$ is exact, thus $d\mathcal{J} = 0$. Computing $d\mathcal{J}$ from (4.7.1) we get the *law of conservation of charge*

$$\frac{d\rho}{dt} + \operatorname{div} \vec{J} = 0.$$

Integrating throughout over a compact domain W in $\mathbb{R}^3$ with smooth boundary, and using the divergence theorem (see Problem 15), we obtain

$$\int_{\partial W} (\vec{J} \cdot \vec{n})\, dA = -\frac{d}{dt} \int_W \rho\, dx_1 dx_2 dx_3.$$

The left-hand side represents the total amount of charge flowing outwards through the surface ∂W per unit time. The right-hand side represents the amount by which the

charge is decreasing inside the region W per unit time. In other words, charge does not disappear into or is created of out of nothingness—it decreases in a region of space only because it flows into other regions. This is an important test of Maxwell's equations since all experimental evidence points to charge conservation.

In the coordinate-free form (4.7.2), Maxwell's equations can be easily generalized to Lorentzian manifolds. A *Lorentzian manifold* is a smooth n-manifold M equipped with a symmetric tensor field g of type $(0, 2)$ such that g_p is an inner product on T_pM with signature $(-+\cdots+)$. The case $n = 4$ is particularly important, as it is the case of General Relativity. Note that on a Lorentzian manifold M, it makes sense to consider the pointwise Hodge star $*_p : \Lambda^k(T_pM) \to \Lambda^{n-k}(T_pM)$ for $k = 1, \ldots, n$ and $p \in M$. Maxwell's equations on M have the same form (4.7.2), where $F \in \Omega^2(M)$ and $\mathcal{J} \in \Omega^3(M)$. *Lorentz transformations* are diffeomorphisms of M that preserve the tensor g. It is clear that Maxwell's equations are invariant under Lorentz transformations, in the sense that given a solution $(F, \mathcal{J})$ and a Lorentz transformation f, also $(f^*F, f^*\mathcal{J})$ is a solution.

The geometrization of Maxwell's equations on the twentieth century led to a vast generalization in the form of the so-called Yang–Mills equations, which describe not only electromagnetism, but also the strong and weak nuclear forces; however, this is much beyond the scope of these modest notes (for further reading, see e.g. [Fra12]).

4.8 Problems

► *Section 4.1*

1 Let M be a smooth manifold of dimension n and let $f : M \to \mathbb{R}^{n+1}$ be an immersion. Prove that M is orientable if and only if there exists a nowhere vanishing smooth vector field X along f (see page 25) such that X_p is normal to $df_p(T_pM)$ in $\mathbb{R}^{n+1}$ for all $p \in M$.

2 Prove that $\mathbb{R}P^n$ is orientable if and only if n is odd.

3 Show that the global n-form constructed in Example 4.1.7 in the case of S^n can be given as the restriction of

$$\alpha = \sum_{i=1}^{n+1} (-1)^{i-1} x_i \, dx_1 \wedge \cdots \wedge \hat{dx_i} \wedge \cdots \wedge dx_{n+1}$$

to S^n, up to a constant multiple.

4 Prove that a parallelizable manifold is orientable.

5 Let M and N be smooth manifolds. Prove that $M \times N$ is orientable if and only if M and N are orientable.

6 (Integration on a Riemannian Manifold) Let (M, g) be a Riemannian manifold of dimension n.

a. On any coordinate neighborhood U, construct a *local orthonormal frame* $E_1, \ldots, E_n$, that is, a set of n smooth vector fields on U which is orthonormal at every point of U. (Hint: Apply the Gram–Schmidt process to the coordinate vector fields.)

b. Let $\omega_1, \ldots, \omega_n$ be the 1-forms dual to an orthonormal frame on U. This is called a *local orthonormal coframe* on U. Suppose now $\omega'_1, \ldots, \omega'_n$ is a local orthonormal coframe on U'. Prove that

$$\omega_1 \wedge \cdots \wedge \omega_n = \pm \omega'_1 \wedge \cdots \wedge \omega'_n$$

at each point of $U \cap U'$.

c. Deduce that in case M is orientable, the locally defined n-forms $\omega_1 \wedge \cdots \wedge \omega_n$ can be pieced together to yield a globally defined nowhere vanishing n-form vol_M on M satisfying

$$\mathrm{vol}_M(E_1, \ldots, E_n) = 1$$

for every positive local orthonormal frame $E_1, \ldots, E_n$. This form is called the *volume form* of the oriented Riemannian manifold M. In case M is compact, its integral is called the *volume* of M.

d. Show that for a positively oriented basis $v_1, \ldots, v_n$ of T_pM, we have

$$(\mathrm{vol}_M)_p(v_1, \ldots, v_n) = \sqrt{\det\big(g_p(v_i, v_j)\big)}.$$

Deduce that, in local coordinates $(U, \varphi = (x_1, \ldots, x_n))$,

$$\mathrm{vol}_M = \sqrt{\det(g_{ij})}\, dx_1 \wedge \cdots \wedge dx_n.$$

7 Consider the unit sphere S^n in $\mathbb{R}^{n+1}$ as a Riemannian manifold where, for each $p \in S^n$, the inner product on the tangent space T_pS^n is obtained by restriction of the standard scalar product in $\mathbb{R}^{n+1}$. Recall the n-form α on S^n given in Problem 3. Let X be the outward unit normal vector field along S^n.

a. Show that

$$\alpha_p = \iota_{X_p}(dx_1 \wedge \cdots \wedge dx_{n+1}|_p)$$

for all $p \in S^n$.

b. Deduce from (a) that α is the volume form of S^n with respect to some orientation.

c. In case $n = 2$, compute the volume of S^2.

8 (Integration on a Lie Group) Let G be a Lie group of dimension n. A differential form ω on G is called *left-invariant* if $L_g^*\omega = \omega$ for all $g \in G$. Similarly, one defines right-invariant differential forms.

a. Check that the space of left-invariant n-forms on G is one-dimensional.

Fix a non-zero left-invariant n-form ω and consider the associated orientation on G. For a compactly supported continuous function f on G, define the integral

$$\int_G f := \int_G f\omega.$$

b. Show that this integral is *left-invariant*, in the sense that $\int_G f \circ L_g = \int_G f$ for all $g \in G$.
c. Show that $R_g^*\omega$ is a left-invariant n-form for all $g \in G$. Deduce that there exists a smooth homomorphism λ from G into the multiplicative group of positive real numbers such that $\lambda(g) = \left|\frac{R_g^*\omega}{\omega}\right|$ for all $g \in G$; λ is called the *modular function*.
d. Check that $\int_G (f \circ R_g)\lambda(g) = \int_G f$ for all $g \in G$. Deduce that the integral on G is *right-invariant* if and only if $\lambda \equiv 1$; in this case, the integral on G is called *bi-invariant* and G is called *unimodular*.
e. Prove that compact Lie groups are unimodular. (Hint: Consider the image of λ.)

In case G is compact, we can, moreover, multiply ω by a non-zero constant and assume that the *total volume* $\int_G 1$ is 1.

9 Let G be a compact Lie group and denote its Lie algebra by $\mathfrak{g}$. Fix an arbitrary positive-definite inner product $\langle , \rangle_0$ on $\mathfrak{g}$.
a. Show that $\langle X, Y\rangle = \int_G b_{X,Y}$ defines a positive-definite inner product on $\mathfrak{g}$, where X, $Y \in \mathfrak{g}$, $b_{X,Y}$ is the continuous function on G given by $b_{X,Y}(g) = \langle \mathrm{Ad}_g X, \mathrm{Ad}_g Y\rangle_0$, and we use the bi-invariant integral from Problem 8.
b. Prove that Ad_g is an orthogonal transformation of $\mathfrak{g}$ with respect to $\langle , \rangle$, for all $g \in G$; we say that $\langle , \rangle$ is *Ad-invariant*.

10 Let a discrete group Γ act freely and properly on an orientable smooth manifold $\tilde{M}$ and consider the quotient smooth manifold $M = \tilde{M}/\Gamma$ as in Theorem A.4.2. Prove that M is orientable if and only if every element of Γ acts as an orientation-preserving diffeomorphism of $\tilde{M}$.

11 (Classification of Smooth 1-Manifolds) In this problem, we prove that every connected smooth manifold of dimension one (without boundary) is diffeomorphic to S^1 or $\mathbb{R}$.
a. Assume first M is orientable.
(i) Construct a nowhere zero smooth vector field X on M. (Hint: Use a partition of unity.)
(ii) Assume that X admits a non-injective maximal integral curve γ. Prove that M is diffeomorphic to S^1. (Hint: Show γ is a periodic function defined on $\mathbb{R}$ and hence induces an injective local diffeomorphism $\bar{\gamma} : S^1 \to M$; deduce the surjectivity of $\bar{\gamma}$ from the connectivity of M.)
(iii) Assume now every maximal integral curve of X is injective. Prove that M is diffeomorphic to $\mathbb{R}$. (Hint: The connectivity of M implies that there cannot be more than one maximal integral curve of X.)

b. Assume now M is arbitrary and prove it must be orientable. (Hint: Owing to Proposition A.4.5, the universal covering $\tilde{M}$ is orientable and thus, by the above, diffeomorphic to $\mathbb{R}$. Check that any orientation-reversing diffeomorphism of $\mathbb{R}$ has a fixed point and use Problem 10.)

Remark It is not hard to see that every connected 1-manifold with non-empty boundary is diffeomorphic to $[0, 1]$ or $[0, +\infty)$ (one can prove this by arguing that the *double manifold*, or manifold obtained by gluing two copies of the manifold along its boundary, is a connected 1-manifold with no boundary). Also, the classification of compact connected 2-manifolds (or surfaces) is a classical subject. In the orientable case, such a surface (without boundary) must be the sphere S^2 with a certain number $g \geq 0$ of "handles attached" (a sphere with one handle is a torus T^2, a sphere with two handles is a bitorus, and so on). In the non-orientable case, such a surface must be the projective plane $\mathbb{R}P^2$ with a number $g \geq 0$ of handles attached (here $g = 1$ gives a Klein bottle). The number g is in a sense the number of "holes" of the surface, and it is called the *genus* of the surface. The interested reader is referred to [Wal06], which discusses Morse theory from an elementary point of view and uses it to classify compact smooth 2-manifolds; alternatively, for an elementary "cut-and-paste" argument, a classic reference is [Mas91, Chap. I].

▶ *Section 4.2*

12 Let $\gamma : [a, b] \to M$ be a smooth curve, and let $\gamma(a) = p$, $\gamma(b) = q$. Show that if $\omega = df$ for a smooth function f on M, then

$$\int_a^b \gamma^*\omega = f(q) - f(p).$$

13 Let $\gamma : [a, b] \to M$ be a smooth curve, and let $h : [c, d] \to [a, b]$ be a smooth map with $h(c) = a$ and $h(d) = b$. Show that

$$\int_a^b \gamma^*\omega = \int_c^d (\gamma \circ h)^*\omega$$

for every 1-form ω on M.

14 A *closed curve* in M is a smooth map $\gamma : S^1 \to M$. For a 1-form ω on M, define the *line integral* of ω around γ as

$$\int_\gamma \omega := \int_{S^1} \gamma^*\omega.$$

a. Write the line integral in local coordinates in case the image of γ lies in a coordinate neighborhood of M.

b. Show that

$$\int_\gamma \omega = \int_0^{2\pi} (\gamma \circ h)^* \omega$$

where $h : [0, 2\pi] \to S^1$ is given by $h(t) = e^{it}$.

15 (Classical Divergence Theorem) Let S be an orientable smooth surface (2-manifold), let $f : S \to \mathbb{R}^3$ be an immersion, and let $\vec{n}$ be a unit normal vector field along f as in Problem 1. Consider the Riemannian metric induced by the immersion f, that is,

$$g_p(u, v) = df_p(u) \cdot df_p(v)$$

for all $p \in M$ and $u, v \in T_pM$.

a. Prove that the volume form (see Problem 6) of (S, g) is given by

$$dA = n_1\, dx_2 \wedge dx_3 + n_2\, dx_3 \wedge dx_1 + n_3\, dx_1 \wedge dx_2$$

where n_1, n_2, n_3 are the components of $\vec{n}$ in $\mathbb{R}^3$ and each dx_i is restricted to S.

b. Assume f is an inclusion, S is the boundary of a compact domain W in $\mathbb{R}^3$, and $\vec{F}$ is a smooth vector field on W. Show that Stokes' formula 4.2.5 specializes to the classical divergence theorem:

$$\int_S (\vec{F} \cdot \vec{n})\, dA = \int_W (\mathrm{div}\vec{F})\, dx_1 dx_2 dx_3.$$

16 Let $\alpha(r)$ denote the volume of the closed n-ball of radius r and let $\beta(r)$ denote the volume of the $(n-1)$-sphere of radius r. Use Stokes' theorem to check that $\frac{d\alpha}{dr} = \beta$.

17 Consider the 1-form $\omega = (x^2 + 7y)\, dx + (-x + y \sin y^2)\, dy$ on $\mathbb{R}^2$. Compute its integral over the boundary of the triangle of vertices $(0, 0)$, $(1, 0)$, and $(0, 2)$, where the triangle is viewed as a manifold with corners with the standard orientation of $\mathbb{R}^2$ (remove the vertices to view the domain of integration as a union of open segments).

▶ *Section 4.3*

18 Let α and β be closed differential forms. Show that $\alpha \wedge \beta$ is closed. In addition, if β is exact, show that $\alpha \wedge \beta$ is exact.

19 Let $\alpha = (2x + y \cos xy)\, dx + (x \cos xy)\, dy$ be a 1-form on $\mathbb{R}^2$. Show that α is exact by finding a smooth function f on $\mathbb{R}^2$ such that $df = \alpha$.

20 Prove that T^2 and S^2 are not diffeomorphic by using de Rham cohomology.

21 A differential form ω on a Lie group G is called *bi-invariant* if it is left- and right-invariant.

a. Let $\omega \in \Omega^k(G)$ be a left-invariant differential form. Prove that ω is bi-invariant if and only if it is *Ad-invariant*, in the sense that

$$\omega_1(\mathrm{Ad}_g X_1, \ldots, \mathrm{Ad}_g X_k) = \omega_1(X_1, \ldots, X_k)$$

for all $X_1, \ldots, X_k \in \mathfrak{g}$ and $g \in G$.

b. Prove that a bi-invariant k-form ω on G satisfies

$$\sum_{j=0}^{i-1} \omega(X_0, \ldots, X_{j-1}, [X_i, X_j], X_{j+1}, \ldots, \hat{X}_i, \ldots, X_k) + \sum_{j=i+1}^{k} \omega(X_0, \ldots, \hat{X}_i, \ldots, X_{j-1}, [X_i, X_j], X_{j+1}, \ldots, X_k) = 0$$

for all $i = 0, \ldots, k$ and $X_0, \ldots, X_k \in \mathfrak{g}$. (Hint: Take $g = \exp tX_i$ in the identity in (a) and differentiate with respect to t at $t = 0$.)

c. Prove that every bi-invariant form ω on G is closed. (Hint: Use the formula in Proposition 2.4.5(i) to compute $d\omega$; the first sum vanishes because a left-invariant form evaluated on left-invariant vector fields is constant; the second sum vanishes by a clever manipulation of the identity in (b).)

► *Section 4.4*

22

a. Prove that every closed 1-form on the open subset A in $\mathbb{R}^3$ given by

$$1 < \left(\sum_{i=1}^{3} x_i^2 \right)^{1/2} < 2$$

is exact.

b. Give an example of a 2-form on A which is closed but not exact.

c. Prove that A is not diffeomorphic to the open ball in $\mathbb{R}^3$.

23 Assume $M = \partial P$, where P is a compact orientable smooth manifold with boundary, and let $f : M \to N$ be a smooth map, where $\dim M = n$. Prove that if f extends to a smooth map $F : P \to N$, then $\int_M f^*\omega = 0$ for every closed n-form ω on N.

24 Assume M is a compact orientable smooth manifold of dimension n and $f, g : M \to N$ are homotopic maps. Prove that

$$\int_M f^*\omega = \int_M g^*\omega$$

for every closed n-form ω on N.

25 Prove that a 1-form ω on a smooth manifold M has $\int_\gamma \omega = 0$ for every closed curve γ in M if and only if it is exact. (Hint: For the 'only if' part, show that $f(p) = \int_{p_0}^{p} \omega$ is well-defined and satisfies $df = \omega$.)

26 Prove that $H^1(M) = 0$ for a simply-connected smooth manifold M. (Hint: By approximation results, a smooth manifold is simply-connected if and only if every smooth closed curve is smoothly homotopic to a point.)

27 Show that two homotopy equivalent smooth manifolds must have isomorphic fundamental groups. (Hint: The assignment of the fundamental group to a pointed topological space is functorial with respect to continuous maps between pointed topological spaces.)

28

a. Prove that $\pi_1(SL(2,\mathbb{R})) \cong \mathbb{Z}$, $\pi_1(SL(3,\mathbb{R})) \cong \pi_1(SL(4,\mathbb{R})) \cong \mathbb{Z}_2$ and $\pi_1(SL(n,\mathbb{R}))$ has order at most two for $n \geq 5$. (Hint: Apply Problem 27 to Problem 15 in ▶ Chap. 3 and use Problem 30 of that chapter.)

b. Prove that $SL(n,\mathbb{C})$ is simply-connected and $\pi_1(GL(n,\mathbb{C})) \cong \mathbb{Z}$ for all n. (Hint: Use Problems 16 and 31 in ▶ Chap. 3.)

▶ *Section 4.5*

29 Let $f : S^1 \to S^1$ be a smooth map.

a. Prove that there exists a smooth map $g : \mathbb{R} \to \mathbb{R}$ such that $f(e^{it}) = e^{ig(t)}$ and $g(t + 2\pi) = g(t) + 2\pi d$ for all $t \in \mathbb{R}$, where d is the degree of f.

b. Use part (a) to show that if $f, g : S^1 \to S^1$ have the same degree, then they are homotopic. Deduce that homotopy classes of smooth maps $S^1 \to S^1$ are classified by their degree.

30 Let $f : M \to N$ be a smooth map between oriented smooth manifolds of the same dimension where N is compact and connected. Assume M is the boundary ∂P of a compact oriented smooth manifold P, M has the induced orientation, and f extends to a smooth map $F : P \to N$. Prove that $\deg f = 0$. (Hint: Problem 23.)

31 (Fundamental Theorem of Algebra) Let $f(z) = z^k + a_{k-1}z^{k-1} + \cdots + a_0$ be a complex polynomial.

a. Consider the extended complex plane $\bar{\mathbb{C}} := \mathbb{C} \cup \{\infty\}$ and show that $z : \bar{\mathbb{C}} \setminus \{\infty\} \to \mathbb{C} \cong \mathbb{R}^2$, $\frac{1}{z} : \bar{\mathbb{C}} \setminus \{0\} \to \mathbb{C} \cong \mathbb{R}^2$ define a smooth atlas on $\bar{\mathbb{C}}$. (Hint: Use Proposition 1.2.10.) Use stereographic projection from the north and south poles to construct a diffeomorphism $S^2 \cong \bar{\mathbb{C}}$.

b. Extend f to a map $\tilde{f} : \bar{\mathbb{C}} \to \bar{\mathbb{C}}$ by putting $\tilde{f}(\infty) = \infty$. Check that $\tilde{f}$ is smooth using the atlas constructed in (a).

c. Show that $\tilde{f}$ is smoothly homotopic to $g : \bar{\mathbb{C}} \to \bar{\mathbb{C}}$ where $g(z) = z^k$. What is the degree of g?

d. Deduce from (c) that f is surjective. In particular, there exists $z_0 \in \mathbb{C}$ such that $f(z_0) = 0$.

32 Define the *Hopf map* $\pi : S^3 \to S^2$ by $\pi(z_0, z_1) = (2z_0\bar{z}_1, |z_0|^2 - |z_1|^2)$, where we view $S^3 \subset \mathbb{C}^2$ and $S^2 \subset \mathbb{C} \times \mathbb{R}$.

a. Show that the level sets of π are circles of the form $\{e^{it} \cdot p | t \in \mathbb{R}\}$ for some $p \in S^3$—these circles are called *Hopf circles* in S^3.

b. Compute the linking number of $\pi^{-1}(0, 1)$ and $\pi^{-1}(0, -1)$.

c. Deduce that any two Hopf circles in S^3 are non-trivially linked (i.e. have non-zero linking number).

33 Let M be a compact connected orientable smooth surface (2-manifold) in $\mathbb{R}^3$. Consider the Riemannian metric obtained by restriction of the scalar product of $\mathbb{R}^3$ to the tangent spaces of M.

a. According to Problem 1, there exists a smooth normal unit vector field along M in $\mathbb{R}^3$. Use the canonical parallelism in $\mathbb{R}^3$ to view this vector field as a smooth map $g : M \to S^2$ and check that it is uniquely defined, up to sign. This map is called the *Gauss map* of M.

b. For $p \in M$, the differential $dg_p : T_pM \to T_{g(p)}S^2$ where T_pM and $T_{g(p)}S^2$ can again be identified under the canonical parallelism in $\mathbb{R}^3$. The *Gaussian curvature* $\kappa(p)$ of M at p is by definition the determinant $\det(dg_p)$, and does not depend on the choice of sign in (a). Prove that

$$\kappa \operatorname{vol}_M = g^* \operatorname{vol}_{S^2}.$$

c. Use (b) and the Gauss–Bonnet theorem to deduce that the degree of the Gauss map is half the Euler characteristic of M:

$$\deg g = \frac{1}{2}\chi(M).$$

► *Section 4.6*

34 Use Problem 29(a) to show that a smooth odd map $f : S^1 \to S^1$ has odd degree.

35 Prove that there exists no antipode-preserving smooth map $f : S^n \to S^{n-1}$.

36 Let $f : S^n \to \mathbb{R}^n$ be a continuous map.

a. Use the Stone–Weierstrass theorem [Roy88, Chap. 9, § 9] to show that for all $\epsilon > 0$ there exists a smooth map $g : S^n \to \mathbb{R}^n$ such that $||g(x) - f(x)|| < \epsilon$ for all $x \in S^n$.

b. Prove that there exists a pair of antipodal points $\pm p \in S^n$ such that $f(p) = f(-p)$.

c. Deduce from part (b) the following results:

(i) **(Ham Sandwich Theorem)** Let $A_1, \ldots, A_n$ be n Lebesgue measurable sets in $\mathbb{R}^n$ [Roy88, Chap. 3]. Then there exists a hyperplane $\mathcal{H}$ simultaneously bisecting all sets into half their volumes, that is,

$$\operatorname{vol}(A_i \cap \mathcal{H}^+) = \operatorname{vol}(A_i \cap \mathcal{H}^-)$$

where $\mathcal{H}^{\pm}$ are the half-spaces defined by $\mathcal{H}$. (Hint: Consider $f : S^n \to \mathbb{R}^n$ whose ith component $f_i(x) = \mathrm{vol}(A_i \cap \mathcal{H}_x^+)$, where $\mathcal{H}_x^+$ is the half-space $\{a \in \mathbb{R}^{n+1} : \langle x', a\rangle \geq x_0\}$ and we have written $x = (x_0, x') \in \mathbb{R} \times \mathbb{R}^n = \mathbb{R}^{n+1}$; you may want to use Lebesgue dominated convergence [Roy88, Chap. 4] to prove the continuity of f_i.)

(ii) **(Lyusternik–Schnirel'man)** For any cover $\{F_1, \ldots, F_{n+1}\}$ of S^n by closed sets, there exists at least one set containing a pair of antipodal points, that is, $F_i \cap (-F_i) \neq \varnothing$ for some $i = 1, \ldots, n+1$. (Hint: Consider $f : S^n \to \mathbb{R}^n$ whose ith component $f_i(x) = \mathrm{dist}(x, F_i)$.)

▶ *Section 4.7*

37 Show that the Hodge star on Minkowski spacetime with coordinates (t, x_1, x_2, x_3) and an inner product of signature $(-+++)$ satisfies

$$*1 = -dt \wedge dx_1 \wedge dx_2 \wedge dx_3 \qquad *dx_1 = -dx_2 \wedge dx_3 \wedge dt$$

$$*dt = -dx_1 \wedge dx_2 \wedge dx_3 \qquad *dx_1 \wedge dt = dx_2 \wedge dx_3$$

and cyclic permutations on x_1, x_2, x_3. Further:

$$** = \begin{cases} -1 \text{ on } \Omega^0,\ \Omega^2,\ \Omega^4 \\ +1 \text{ on } \Omega^1,\ \Omega^3. \end{cases}$$

Covering Manifolds

In this appendix, we summarize some properties of covering spaces in the context of smooth manifolds. For more details on the topological theory of covering spaces, we refer the reader to [Mun84] or [Hat02].

A.1 Topological Coverings

Recall that a (topological) *covering* of a topological space X is another topological space $\tilde{X}$ with a continuous map $p : \tilde{X} \to X$ such that X is a union of evenly covered connected open sets, where a connected open subset U of X is called *evenly covered* if

$$p^{-1}U = \bigcup_{i \in I} \tilde{U}_i \tag{A.1.1}$$

is a disjoint union of open sets $\tilde{U}_i$ of $\tilde{X}$, each of which is mapped homeomorphically onto U under p. In particular, the fibers of p are discrete subsets of $\tilde{X}$. It also follows from the definition that $\tilde{X}$ has the Hausdorff property if X does. Further, it is usual, as we shall do, to require that X and $\tilde{X}$ be connected, and then the index set I can be taken to be the same for all evenly covered connected open sets.

A.1.2 Examples (i) $p : \mathbb{R} \to S^1$, $p(t) = e^{it}$ is a covering.
(ii) $p : S^1 \to S^1$, $p(z) = z^n$ is a covering for any non-zero integer n.
(iii) $p : \mathbb{R}^n \to T^n$, $p(t_1, \ldots, t_n) = (e^{it_1}, \ldots, e^{it_n})$ is a covering.
(iv) $p : (0, 3\pi) \to S^1$, $p(t) = e^{it}$ is a local homeomorphism which is not a covering, since $1 \in S^1$ does not admit evenly covered neighborhoods.

■

A.2 Fundamental Groups

Covering spaces are closely related to fundamental groups. The *fundamental group* $\pi_1(X, x_0)$ of a topological space X with basepoint x_0 is defined as follows. As a set, it consists of the homotopy classes of loops based at x_0. The concatenation of such loops is compatible with the equivalence relation given by homotopy, so it induces a group operation on $\pi_1(X, x_0)$ making it into a group. If X is pathwise connected, the isomorphism class of the fundamental group is independent of the choice of basepoint

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(indeed for x_0, $x_1 \in X$ and c a continuous path from x_0 to x_1, conjugation by c^{-1} induces an isomorphism from $\pi_1(X, x_0)$ to $\pi_1(X, x_1)$) and in some contexts can be thus denoted by $\pi_1(X)$. Finally, a continuous map $f : X \to Y$ between topological spaces with $f(x_0) = y_0$ induces a homomorphism $f_\# : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ so that the assignment $(X, x_0) \to \pi_1(X, x_0)$ is functorial. Of course the fundamental group is trivial if and only if the space is simply connected (recall that a topological space X is said to be *simply connected* if it is pathwise connected and every loop in X is homotopic to a constant map).

Being locally Euclidean, a smooth manifold is locally pathwise connected and locally simply connected. A connected space X with such local connectivity properties admits a simply connected covering space, which is unique up to isomorphism, called the *universal covering space* of X; here an isomorphism between coverings $p_1 : \tilde{X}_1 \to X$ and $p_2 : \tilde{X}_2 \to X$ is a homeomorphism $f : \tilde{X}_1 \to \tilde{X}_2$ such that $p_2 \circ f = p_1$. More generally, there exists a bijective correspondence between isomorphism classes of basepoint-preserving coverings $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ and subgroups of $\pi_1(X, x_0)$ given by $(\tilde{X}, \tilde{x}_0) \mapsto p_\#(\pi_1(\tilde{X}, \tilde{x}_0))$; moreover, a change of basepoint in $\tilde{X}$ corresponds to passing to a conjugate subgroup of $\pi_1(X, x_0)$.

A.2.1 Lemma *The fundamental group of a connected smooth manifold M is a countable group.*

Proof Here we strongly use the second-countability of M. It implies that we can find a countable cover $\{B_i\}$ of M by open sets, each of which is diffeomorphic to a ball in Euclidean space. Again by second-countability of M, for all i, j, the set $B_i \cap B_j$ has at most countably many path-components; fix a point in each component and denote the countable collection of points thus obtained by $\mathcal{Q}$. Finally, for all $x, x' \in \mathcal{Q}$ and all i with $x, x' \in B_i$, fix a path $\gamma^i_{x,x'}$ joining x to x' inside B_i, and denote the countable collection of paths thus obtained by $\mathcal{C}$. Take $x_0 \in \mathcal{Q}$; using the simple-connectedness of the B_i, it is now more or less clear that every loop based at x_0 is homotopic to a loop at x_0 consisting of the product of finitely many elements of $\mathcal{C}$. Hence $\pi_1(M, x_0)$ is countable. □

A.3 Smooth Coverings

Suppose $\pi : \tilde{M} \to M$ is a covering where M is a smooth manifold. Then there is a natural structure of smooth manifold on $\tilde{M}$ such that the projection π is smooth. In fact, for every chart (U, φ) of M where U is evenly covered as in (A.1.1), there is a chart $(\tilde{U}_i, \varphi \circ \pi|_{\tilde{U}_i})$ for $\tilde{M}$. This gives an atlas of $\tilde{M}$, which is smooth because, for another chart (V, ψ) of M where V is evenly covered by $\cup_{i \in I} \tilde{V}_i$ and $\tilde{U}_i \cap \tilde{V}_j \neq \varnothing$ for some i, $j \in I$, we have that the transition map

$$(\psi \circ \pi|_{\tilde{V}_j}) \circ (\varphi \circ \pi|_{\tilde{U}_i})^{-1} = \psi \circ \varphi^{-1}$$

is smooth. Further, we already know that $\tilde{M}$ is a Hausdorff space, and it is also second-countable, which we check as follows. It is possible to choose a countable basis of connected open sets for M which are evenly covered. The connected components of the preimages under π of the elements of this basis form a basis of connected open sets for $\tilde{M}$, which is countable as long as the index set I is countable, but this follows from Lemma A.2.1. Note that, around any point in $\tilde{M}$, π admits a local representation as the identity, so it is a local diffeomorphism. Indeed we have proven more: M can be covered by evenly covered neighborhoods U such that the restriction of π to a connected component of $\pi^{-1}U$ is a diffeomorphism onto U. This is the definition of a *smooth covering*. Note that a topological covering whose covering map is smooth need not be a smooth covering (e.g., $\pi : \mathbb{R} \to \mathbb{R}, \pi(x) = x^3$).

Next, we can formulate a basic result in covering theory for a smooth covering $\pi : \tilde{M} \to M$ of a smooth manifold M; see [Hat02, Propositions 1.33 and 1.34] for a proof. Fix basepoints $\tilde{p} \in \tilde{M}$, $p \in M$ such that $\pi(\tilde{p}) = p$. We say that a map $f : N \to M$ admits a *lifting* if there exists a map $\tilde{f} : N \to \tilde{M}$ such that $\pi \circ \tilde{f} = f$.

A.3.1 Theorem (Lifting Criterion) *A smooth map $f : N \to M$ mapping a given point $q \in N$ to p admits a smooth lifting $\tilde{f} : N \to \tilde{M}$ with $\tilde{f}(q) = \tilde{p}$ if and only if $f_\#(\pi_1(N, q)) \subset \pi_\#(\pi_1(\tilde{M}, \tilde{p}))$. If that is the case and N is connected, such a lifting is unique.*

Taking $f : N \to M$ to be the universal covering of M in Theorem A.3.1 shows that the universal covering of M covers any other covering of M and hence justifies its name.

A.4 Deck Transformations

For a topological covering $\pi : \tilde{X} \to X$, a *deck transformation* or *covering transformation* is an isomorphism $\tilde{X} \to \tilde{X}$, namely a homeomorphism $f : \tilde{X} \to \tilde{X}$ such that $\pi \circ f = \pi$. The deck transformations plainly form a group under composition, which we denote by $\mathrm{Aut}(\tilde{X}, \pi)$. It follows from the uniqueness of liftings that a deck transformation is uniquely determined by its action on one point. In particular, the only deck transformation admitting fixed points is the identity. Since a smooth covering map $\pi : \tilde{M} \to M$ is a local diffeomorphism, in this case the equation $\pi \circ f = \pi$ implies that deck transformations are diffeomorphisms of $\tilde{M}$.

An action of a discrete group on a topological space (resp. smooth manifold) is a homomorphism from the group to the group of homeomorphisms (resp. diffeomorphisms) of the space (resp. manifold). For a smooth manifold M, we next recall the canonical action of $\pi_1(M, p)$ on its universal covering $\tilde{M}$ by deck transformations. Note that in this case, by the lifting criterion A.3.1, given $\tilde{p}_1, \tilde{p}_2 \in \pi^{-1}(p)$, there is a (unique) deck transformation mapping $\tilde{p}_1$ to $\tilde{p}_2$.

Let γ be a loop in M based at p representing an element $[\gamma] \in \pi_1(M, p)$. Given $\tilde{p} \in \pi^{-1}(p)$, we lift γ uniquely to a path $\tilde{\gamma}$ starting at $\tilde{p}$; then $[\gamma] \cdot \tilde{p}$ is by definition the endpoint of $\tilde{\gamma}$, which sits in the fiber $\pi^{-1}(p)$. The definition independs of the choice made, namely if we change γ to a homotopic curve, we get the same result (this follows

from Theorem A.3.1, since we can lift the homotopy, as it is defined on a square and a square is simply connected). We get a map $\pi_1(M, p) \to \mathrm{Aut}(\tilde{M}, \pi)$, which is surjective because $\tilde{M}$ is path-connected; it is also injective, as its kernel consists of classes $[\gamma]$ lifting to loops in $\tilde{M}$, and $\tilde{M}$ is simply connected. Further, if $[\gamma_1]$ maps to the deck transformation τ_1 taking $\tilde{p}$ to $\tilde{p}_1$, and $[\gamma_2]$ maps to the deck transformation τ_2 taking $\tilde{p}$ to $\tilde{p}_2$, then $\gamma_1 \cdot \gamma_2$ lifts to $\tilde{\gamma}_1 \cdot (\tau_1(\tilde{\gamma}_2))$ mapping $\tilde{p}$ to $\tau_1(\tilde{\gamma}_2(1)) = \tau_1\tau_2(\tilde{p})$, that is, $[\gamma_1][\gamma_2]$ maps to $\tau_1\tau_2$. Hence $\pi_1(M, p) \to \mathrm{Aut}(\tilde{M}, \pi)$ is a homomorphism and thus a group isomorphism.

A.4.1 Examples (i) Euclidean space $\mathbb{R}^n$ is *contractible* in the sense that $f_t(x) = tx$ defines a continuous homotopy from the identity f_1 to the constant map f_0. In particular, $\mathbb{R}^n$ is simply connected.

(ii) Since $\mathbb{R}$ is simply connected, Examples A.1.2(i) is the universal covering of the circle. Every deck transformation has the form $f(x) = x + 2\pi k$ for some $k \in \mathbb{Z}$. It follows that $\pi_1(S^1) \cong \mathbb{Z}$. Similarly, Examples A.1.2(iii) is the universal covering of T^n and $\pi_1(T^n) \cong \mathbb{Z}_n$.

(iii) For the product $X \times Y$ of path-connected topological spaces X and Y, we have $\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$. In fact, the components of a loop in the product are loops in the factors, and the components of a homotopy of a loop in the product are homotopies of the loop-components.

(iv) Stereographic projection maps the sphere S^n minus one point onto $\mathbb{R}^n$, which is contractible. For $n \geq 2$, any loop in S^n misses a point of S^n and hence is homotopic to a constant. It follows that S^n is simply connected for $n \geq 2$.

■

An action of a discrete group Γ on a topological space X is called *free* if no nontrivial element of Γ has fixed points, and it is called *proper* if any two points $x, y \in X$ admit open neighborhoods $U \ni x$, $V \ni y$ such that $\{\gamma \in \Gamma \mid \gamma U \cap V \neq \varnothing\}$ is finite. The action of $\pi_1(M, p)$ on the universal covering $\tilde{M}$ by deck transformations has both properties. In fact, we have already remarked it is free. To check properness, let $\tilde{p}$, $\tilde{q} \in \tilde{M}$. If these points lie in the same orbit of $\pi_1(M, p)$ or, equivalently, the same fiber of π, the required neighborhoods are the connected components of $\pi^{-1}(U)$ containing $\tilde{p}$ and $\tilde{q}$, respectively, where U is an evenly covered neighborhood of $\pi(\tilde{p}) = \pi(\tilde{q})$. On the other hand, if $\pi(\tilde{p}) =: p \neq q := \pi(\tilde{q})$, we use the Hausdorff property of M to find disjoint evenly covered neighborhoods $U \ni p$, $V \ni q$ and then it is clear that the connected component of $\pi^{-1}(U)$ containing $\tilde{p}$ and the connected component of $\pi^{-1}(V)$ containing $\tilde{q}$ do the job.

Conversely, we have:

A.4.2 Theorem *If the group Γ acts freely and properly on a smooth manifold $\tilde{M}$, then the quotient space $M = \tilde{M}/\Gamma$ endowed with the quotient topology admits a unique structure of smooth manifold such that the projection $\pi : \tilde{M} \to M$ is a smooth covering.*

A.4.3 Remark The action of Γ on $\tilde{M}$ determines a partition into equivalence classes or *orbits*, namely $\tilde{p} \sim \tilde{q}$ if and only if $\tilde{q} = \gamma\tilde{p}$ for some $\gamma \in \Gamma$. The orbit through $\tilde{p}$ is denoted by $\Gamma(\tilde{p})$. The quotient space $\tilde{M}/\Gamma$ is also called the *orbit space*.

Proof of Theorem A.4.2 The quotient topology is defined by the condition that $U \subset M$ is open if and only if $\pi^{-1}(U)$ is open in $\tilde{M}$. In particular, for an open set $\tilde{U} \subset \tilde{M}$ we have $\pi^{-1}(\pi(\tilde{U})) = \cup_{\gamma\in\Gamma}\gamma(\tilde{U})$, a union of open sets, showing that $\pi(\tilde{U})$ is open and proving that π is an open map. In particular, π maps a countable basis of open sets in $\tilde{M}$ to a countable basis of open sets in M.

The covering property follows from the fact that Γ is proper. In fact, let $\tilde{p} \in \tilde{M}$. From the definition of properness, we can choose a neighborhood $\tilde{U} \ni \tilde{p}$ such that $\{\gamma \in \Gamma \mid \gamma\tilde{U} \cap \tilde{U} \neq \varnothing\}$ is finite. Using the Hausdorff property of $\tilde{M}$ and the freeness of Γ, we can shrink $\tilde{U}$ so that this set consists of the identity only. Now the map π identifies all disjoint homeomorphic open sets γU for $\gamma \in \Gamma$ to a single open set $\pi(U)$ in M, which is then evenly covered.

The Hausdorff property of M also follows from properness of Γ. Indeed, let $p, q \in M$, $p \neq q$. Choose $\tilde{p} \in \pi^{-1}(p)$, $\tilde{q} \in \pi^{-1}(q)$ and neighborhoods $\tilde{U} \ni \tilde{p}$, $\tilde{V} \ni \tilde{q}$ such that $\{\gamma \in \Gamma \mid \gamma\tilde{U} \cap \tilde{V} \neq \varnothing\}$ is finite; note that $\tilde{q} \notin \Gamma(\tilde{p})$, so by the Hausdorff property for $\tilde{M}$, we can shrink $\tilde{U}$ so that this set becomes empty. Since π is open, $U := \pi(\tilde{U})$ and $V := \pi(\tilde{V})$ are now disjoint neighborhoods of p and q, respectively.

Finally, we construct a smooth atlas for M. Let $p \in M$ and choose an evenly covered neighborhood $U \ni p$. Write $\pi^{-1}U = \cup_{i\in I}\tilde{U}_i$ as in (A.1.1). Fix $i \in I$; by shrinking U we can ensure that $\tilde{U}_i$ is the domain of a local chart $(\tilde{U}_i, \tilde{\varphi}_i)$ of $\tilde{M}$. Now $\varphi_i := \tilde{\varphi}_i \circ (\pi|_{\tilde{U}_i})^{-1} : U \to \mathbb{R}^n$ ($n = \dim \tilde{M}$) defines a homeomorphism onto the open set $\tilde{\varphi}_i(\tilde{U}_i)$ and thus a local chart (U, φ_i) of M. The domains of such charts cover M and it remains only to check that the transition maps are smooth. So let V be another evenly covered neighborhood of p with $\pi^{-1}V = \cup_{j\in I}\tilde{V}_j$ and associated local chart $\psi_j := \tilde{\psi}_j \circ (\pi|_{\tilde{V}_j})^{-1} : U \to \mathbb{R}^n$, where $(\tilde{V}_j, \tilde{\psi}_j)$ is a local chart of $\tilde{M}$. Then

$$\psi_j \circ \varphi_i^{-1} = \tilde{\psi}_j \circ (\pi|_{\tilde{V}_j})^{-1} \circ \pi \circ \tilde{\varphi}_i^{-1}. \tag{A.4.4}$$

However, $(\pi|_{\tilde{V}_j})^{-1} \circ \pi$ is realized by the action of a unique element $\gamma \in \Gamma$ on a neighborhood of $\tilde{p}_i = (\pi|_{\tilde{U}_i})^{-1}(p)$. Since Γ acts by diffeomorphisms, this shows that the transition map (A.4.4) is smooth and finishes the proof. □

The following result is very useful.

A.4.5 Proposition *Every simply connected smooth manifold is orientable.*

Proof Let M be a simply connected smooth manifold. Fix a point $p \in M$ and an orientation o_p on T_pM. We will show how to continuously "propagate" o_p along any smooth curve $\gamma : [0, 1] \to M$ with $\gamma(0) = p$.

If the image of γ is contained in a connected coordinate domain U of M, that is easy: let $\varphi : U \to \mathbb{R}^n$ ($n = \dim M$) be a local chart and define o_t to be the orientation of $T_{\gamma(t)}M$ that makes $(d\varphi_{\gamma(t)})^{-1} \circ d\varphi_p : T_pM \to T_{\gamma(t)}M$ orientation-preserving (note that $o_0 = o_p$).

In case γ is arbitrary, we cover its image by connected coordinate domains and use the Lebesgue number lemma ([Hat02, p. 123] or [Mun84, Lemma 27.5]) to find a partition $0 = t_0 < \cdots < t_m = 1$ of $[0, 1]$ so that $\gamma([t_{i-1}, t_i])$ is contained in a connected coordinate domain U_i, for $i = 1, \ldots, m$; we then apply the previous case and induction on i.

We end up with an orientation o_1 of $T_q M$, and we define o_q to be o_1, where $q = \gamma(1)$. Next we show that o_q is independent of the cover $\mathcal{U} = \{U_i\}_{i=1}^m$ of $\gamma([0, 1])$ as above. Let $\mathcal{V} = \{V_j\}_{j=1}^k$ be another cover of $\gamma([0, 1])$ such that $\gamma([t'_{j-1}, t'_j]) \subset V_j$ and V_j is a connected coordinate domain of M for $j = 1, \ldots, k$, where $0 = t'_0 < \cdots < t'_k = 1$ is a partition of $[0, 1]$. Denote by $o_t^{\mathcal{U}}$ and $o_t^{\mathcal{V}}$ the orientations of $T_{\gamma(t)} M$ obtained by using $\mathcal{U}$ and $\mathcal{V}$, respectively. Consider $I = \{\, t \in [0, 1] \mid o_t^{\mathcal{U}} = o_t^{\mathcal{V}} \,\}$. Of course $I \neq \varnothing$ as $0 \in I$. Further, I is open: if $\bar{t} \in I$ and $\gamma(\bar{t}) \in U_i \cap V_j$, then the connected component of $\gamma^{-1}(U_i \cap V_j)$ containing $\bar{t}$ is also contained in I. Similarly, one sees that I closed. Hence $I = [0, 1]$.

Finally, we show that $o_q = o_q^{\gamma}$ is independent of the curve γ joining p to q. Given a homotopy $\{\gamma_s\}_{s \in [0,1]}$ of $\gamma = \gamma_0$ keeping the endpoints fixed and $s_0 \in [0, 1]$, we find a partition $0 = t_0 < t_1 < \cdots < t_m = 1$ and $\epsilon > 0$ so that $\gamma_s([t_{i-1}, t_i])$ is contained in a connected coordinate domain U_i for all $s \in (s_0 - \epsilon, s_0 + \epsilon) \cap [0, 1]$ and $i = 1, \ldots, m$; fix $s_1 \in (s_0 - \epsilon, s_0 + \epsilon) \cap [0, 1]$. We proceed by induction on $i = 1, \ldots, m$ as follows. By construction,

$$(d\varphi_{\gamma_{s_0}(t_i)})^{-1} \circ d\varphi_{\gamma_{s_0}(t_{i-1})} \quad \text{and} \quad (d\varphi_{\gamma_{s_1}(t_i)})^{-1} \circ d\varphi_{\gamma_{s_1}(t_{i-1})} \tag{A.4.6}$$

are orientation-preserving. Further, the composite transformation

$$d\varphi^{-1}_{\gamma_{s_1}(t_{i-1})} \circ d\varphi_{\gamma_{s_0}(t_{i-1})} \tag{A.4.7}$$

is orientation-preserving, trivially in case $i = 1$ and by the induction hypothesis for some $i > 1$. Combining (A.4.6) and (A.4.7), we deduce that

$$d\varphi^{-1}_{\gamma_{s_1}(t_i)} \circ d\varphi_{\gamma_{s_0}(t_i)}$$

is also orientation-preserving. Since $t_m = 1$ and $\gamma_{s_0}(1) = q = \gamma_{s_1}(1)$, we deduce that $o_q^{\gamma_s}$ is locally constant on s; hence it is constant on s. Since any two curves joining p to q are homotopic with endpoints fixed due to the simple-connectedness of M, we are done. □

Hodge Theory

Classical Hodge theory, named after the Scottish mathematician William V. D. Hodge, concerns the study of the cohomology groups of a compact oriented smooth manifold using partial differential equations. In this appendix, we explain the statement of Hodge's decomposition theorem (without proof) and derive some standard consequences.

B.1 Statement of the Hodge Decomposition Theorem

Let M be a compact smooth n-manifold. In ▶ Chap. 4, we introduced the de Rham cohomology $H^k(M) = H^k$ as the quotient vector space Z^k/B^k, where Z^k and B^k denote the subspaces of $\Omega^k(M) = \Omega^k$ consisting of closed and exact k-forms on M, respectively, for $k = 0, \dots, n$. One may ask whether there is a natural way of choosing a subspace $\mathcal{H}^k$ of Ω^k which is complementary to B^k in Z^k. In particular, the projection $Z^k \to H^k$ would induce a vector space isomorphism $\mathcal{H}^k \cong H^k$. Note that the cohomology class of $\omega \in Z^k$ is the affine subspace $[\omega] = \omega + B^k$, so choosing $\mathcal{H}^k$ is tantamount to a coherent choice of one representative from each cohomology class.

An ingenious approach that works on finite-dimensional Euclidean spaces is to choose, as representative, the point of the affine subspace $[\omega]$ which is *closest* to the origin of Z^k. However, since Z^k is infinite-dimensional, there are continuity issues (related to the lack of completeness of the ambient space, lack of closedness of the affine subspace…). Hodge showed that in case M is equipped with a Riemannian metric and an orientation, there is an alternative way of constructing $\mathcal{H}^k$ that gives the same answer as the ingenious approach. We will not delve into the proof, which requires Sobolev spaces and regularity theory of elliptic operators; instead we refer the interested reader to [War83, Chap. 6]. In the following we explain the modern statement of his result and present some well-known applications.

Let M be a compact oriented Riemannian n-manifold. We will need results from Problems 11, 12 in ▶ Chap. 2 and Problem 6 in ▶ Chap. 4. There is a unique volume form vol_M on M. Using that, we have a global Hodge star operator

$$* : \Omega^k \to \Omega^{n-k}$$

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for all k, defined pointwise by $(*\omega)_p = *_p(\omega_p)$, where

$$*_p : \Lambda^k(T_pM) \to \Lambda^{n-k}(T_pM)$$

for all $p \in M$ and $k = 0, \ldots, n$.

Since M is compact, we can define a (positive-definite) inner product on Ω^k by the formula

$$\langle \omega, \eta \rangle = \int_M \omega \wedge *\eta$$

and extend it to $\Omega^\bullet$ by requiring that $\bigoplus_{k=0}^n \Omega^k$ be an orthogonal decomposition. The operator δ on $\Omega^\bullet$, defined by

$$\delta = (-1)^{nk+n+1} * d *,$$

maps Ω^k to Ω^{k-1} for each k, and is the adjoint d^* of d on $\Omega^\bullet$, that is,

$$\langle d\alpha, \beta \rangle = \langle \alpha, \delta\beta \rangle.$$

Indeed, for $\alpha \in \Omega^{k-1}$ and $\beta \in \Omega^k$, one checks that $d\alpha \wedge *\beta - \alpha \wedge *\delta\beta = d(\alpha \wedge *\beta)$, integrates through over M, and uses Stokes' theorem.

It follows that the formula

$$d\delta + \delta d$$

defines a self-adjoint operator Δ on Ω^k ($\Delta = \Delta^*$) for each k, called the *Laplacian*. If $x_1, \ldots, x_n$ are local coordinates on $U \subset M$, a short calculation shows that for a smooth function f on U (0-form) we have $\Delta f = -\sum_{i=1}^n \partial^2 f / \partial x_i^2$.

We say that $\omega \in \Omega^\bullet$ is *harmonic* if $\Delta\omega = 0$, and we let

$$\mathcal{H}^k = \{\omega \in \Omega^k : \Delta\omega = 0\}$$

denote the subspace of harmonic k-forms. Note that $\langle \Delta\omega, \omega \rangle = ||d\omega||^2 + ||\delta\omega||^2$, so ω is harmonic if and only if $d\omega = \delta\omega = 0$ (recall that in Minkowski space, these are just Maxwell's equations (4.7.2) in the absence of sources). It is also easy to check that Δ commutes with $*$, so $*$ defines an isomorphism (in fact, an isometry) $\mathcal{H}^k \cong \mathcal{H}^{n-k}$.

In 1954 the eminent mathematician Hermann Weyl proclaimed that Hodge's book, containing the proof of the following theorem, was "one of the great landmarks in the history of science in the present century."

B.1.1 Theorem (Hodge Decomposition Theorem) *Let M be a compact oriented Riemannian manifold. Then, for each k, the space $\mathcal{H}^k$ is finite-dimensional. Moreover, there are the following orthogonal decompositions:*

$$\begin{aligned}\Omega^k &= \Delta(\Omega^k) \oplus \mathcal{H}^k \\ &= d\Omega^{k-1} \oplus \delta\Omega^{k+1} \oplus \mathcal{H}^k\end{aligned}$$

and

$$Z^k = B^k \oplus \mathcal{H}^k.$$

A few remarks about the statement are in order. First, we note that the first decomposition claims that $\operatorname{im}\Delta$ equals $(\ker\Delta)^\perp$; in particular, $\operatorname{im}\Delta$ is closed in Ω^k with respect to the topology induced from the inner product. Second, the hard part of the theorem is proving the finite-dimensionality of $\ker\Delta$ and the inclusion $(\ker\Delta)^\perp \subset \operatorname{im}\Delta$; all other statements follow. In fact, if we know $\ker\Delta$ is finite-dimensional, then $\Omega^k = (\ker\Delta)^\perp \oplus \ker\Delta$ follows from general principles. Also $\Delta = \Delta^*$ immediately implies the inclusion $\operatorname{im}\Delta \subset (\ker\Delta)^\perp$. Finally, the other two decompositions in the statement of the theorem follow from the first one. Indeed it is clear from the definition of Δ that $\operatorname{im}\Delta \subset \operatorname{im} d + \operatorname{im}\delta$. Conversely, $\operatorname{im} d \subset (\ker\delta)^\perp \subset (\ker\Delta)^\perp = \operatorname{im}\Delta$ and, similarly, $\operatorname{im}\delta \subset \operatorname{im}\Delta$; this proves $\operatorname{im}\Delta = \operatorname{im} d + \operatorname{im}\delta$. In addition, we see that $\operatorname{im} d \perp \operatorname{im}\delta$ from $\langle d\alpha, \delta\beta\rangle = \langle d^2\alpha, \beta\rangle = 0$. To derive the third decomposition from the second one, let $\omega \in Z^k$ and write $\omega = d\alpha + \delta\beta + \varphi$, where $\Delta\varphi = 0$. Then $d\omega = 0$ implies that $d\delta\beta = 0$. Since $\delta^2\beta = 0$, we deduce that $\delta\beta$ is harmonic and $\omega = d\alpha + (\delta\beta + \varphi)$.

B.1.2 Corollary *The de Rham spaces H^k of a compact orientable smooth manifold are finite-dimensional.*

Proof Every smooth manifold can be equipped with a Riemannian metric. Then $H^k \cong \mathcal{H}^k$ is finite-dimensional. □

For a compact smooth manifold M, $b_k = b_k(M) = \dim H^k(M)$ is called the kth *Betti number* of M.

B.1.3 Corollary (Smooth Poincaré Duality) *Let M be a compact oriented smooth n-manifold. Then $b_k = b_{n-k}$ for $k = 0, \ldots, n$.*

Proof Define the pairing $\smile$: $H^k \times H^{n-k} \to \mathbb{R}$ by setting

$$[\omega] \smile [\eta] = \int_M \omega \wedge \eta$$

for $\omega \in Z^k$, $\eta \in Z^{n-k}$. Note that $\smile$ is well-defined (independent of the choices of representatives in the cohomology classes) in view of Problem 18 in ▶ Chap. 4 and Stokes'

theorem. We claim that $\smile$ is non-degenerate. In fact, equip M with a Riemannian metric, let $[\omega] \in H^k$ be non-zero and choose a harmonic representative $\tilde{\omega} \in [\omega]$. Then $*\tilde{\omega}$ is closed and

$$[\omega] \smile [*\tilde{\omega}] = \int_M \tilde{\omega} \wedge *\tilde{\omega} = ||\tilde{\omega}||^2 > 0,$$

as desired. Now $H^{n-k} \cong (H^k)^*$ and the result follows. □

Corollary B.1.3 yields another proof of Proposition 4.5.3, namely $b_n = b_0 = 1$ for connected M.

B.1.4 Corollary *Let M be a compact oriented Riemannian manifold and let $\omega \in \mathcal{H}^k$. Then ω is the point of $[\omega] \in H^k$ closest to the origin of Z^k.*

Proof For all $\eta \in \Omega^{k-1}$, using that $\langle \omega, d\eta \rangle = \langle \delta\omega, \eta \rangle = 0$, we have $||\omega + d\eta||^2 = ||\omega||^2 + ||d\eta||^2 + 2\langle \omega, d\eta \rangle \geq ||\omega||^2$, with equality only in case $d\eta = 0$. □

To finish, we quote the great Michael Atiyah [Ati77]:

» The long term impact of Hodge's theory on differential geometry and analysis was substantial. In both cases it helped to shift the focus from purely local problems to global problems of geometry and analysis 'in the large'. Together with Marston Morse's work on the calculus of variations it set the stage for the new and more ambitious global approach which has dominated much of mathematics ever since.

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